

SIMATIC

Process Control System PCS 7 PowerControl Library AS Manual

Function Manual

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Legal information

Warning notice system

This manual contains notices you have to observe in order to ensure your personal safety, as well as to prevent damage to property. The notices referring to your personal safety are highlighted in the manual by a safety alert symbol, notices referring only to property damage have no safety alert symbol. These notices shown below are graded according to the degree of danger.

DANGER

indicates that death or severe personal injury **will** result if proper precautions are not taken.

WARNING

indicates that death or severe personal injury **may** result if proper precautions are not taken.

CAUTION

indicates that minor personal injury can result if proper precautions are not taken.

NOTICE

indicates that property damage can result if proper precautions are not taken.

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We have reviewed the contents of this publication to ensure consistency with the hardware and software described. Since variance cannot be precluded entirely, we cannot guarantee full consistency. However, the information in this publication is reviewed regularly and any necessary corrections are included in subsequent editions.

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Security information

Siemens provides products and solutions with industrial security functions that support the secure operation of plants, systems, machines and networks.

In order to protect plants, systems, machines and networks against cyber threats, it is necessary to implement – and continuously maintain – a holistic, state-of-the-art industrial security concept. Siemens' products and solutions constitute one element of such a concept.

Customers are responsible for preventing unauthorized access to their plants, systems, machines and networks. Such systems, machines and components should only be connected to an enterprise network or the internet if and to the extent such a connection is necessary and only when appropriate security measures (e.g. firewalls and/or network segmentation) are in place.

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Introduction

The PowerControl library contains technological objects for controlling and visualizing switchgear devices in the PCS 7 environment. The block library is based on the standard functions of the PCS 7 Advanced Process Library V9.1 (APL).

This manual is to be used together with the *PCS 7 Advanced Process Library V9.1 Function Manual* [1] as the APL standard functions are not described in this document.

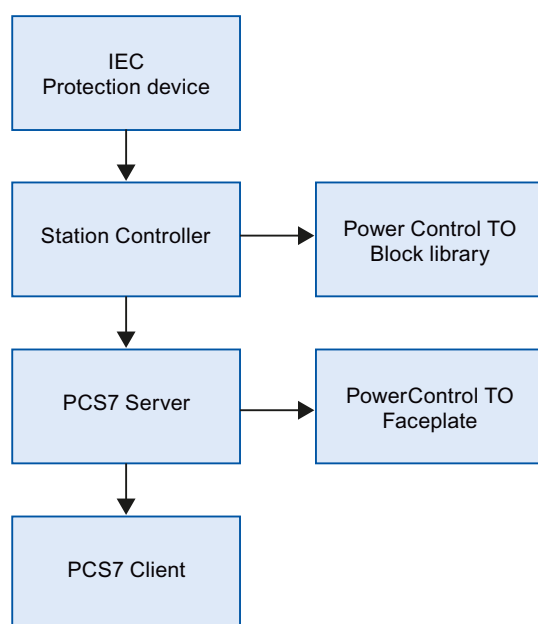
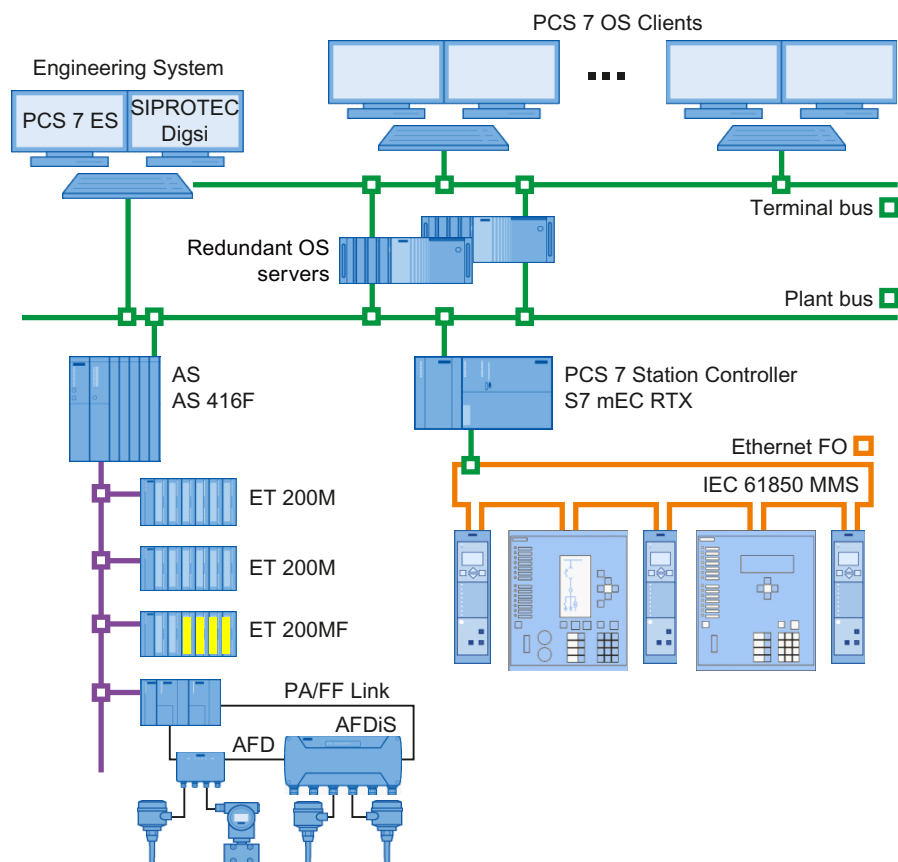
2.1 Hardware environment

The blocks of the PowerControl library can be used in different hardware environments. Separate device driver libraries are available for these environments that are not contained in the PowerControl library (PCL).

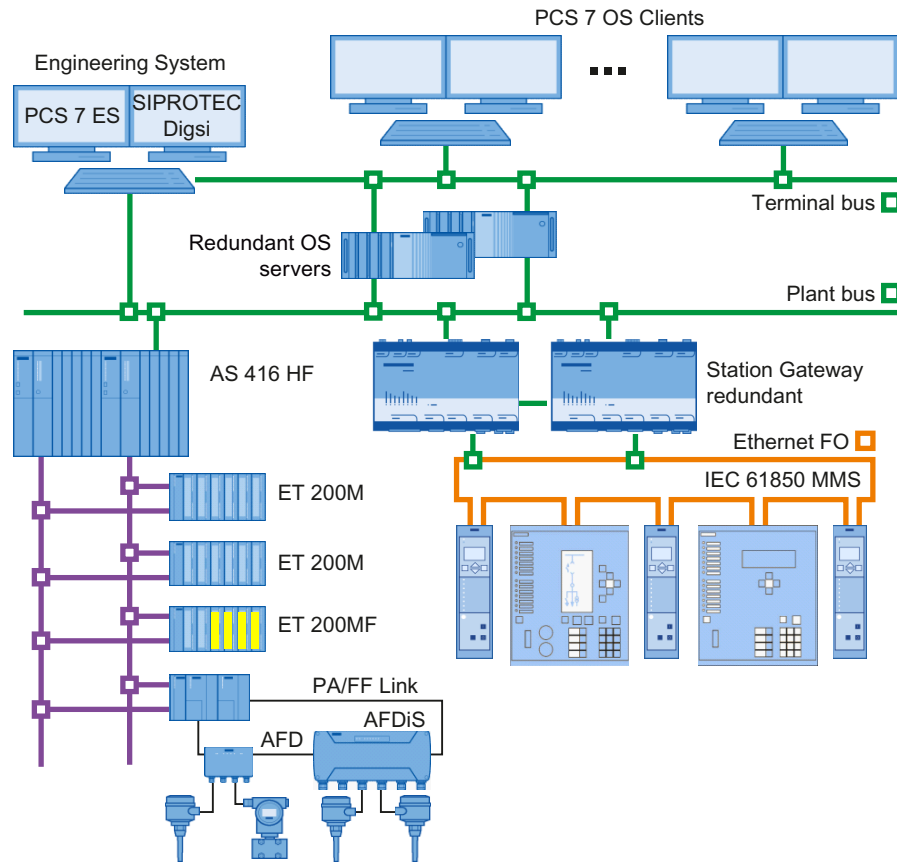
The PowerControl library can be used in the following environments:

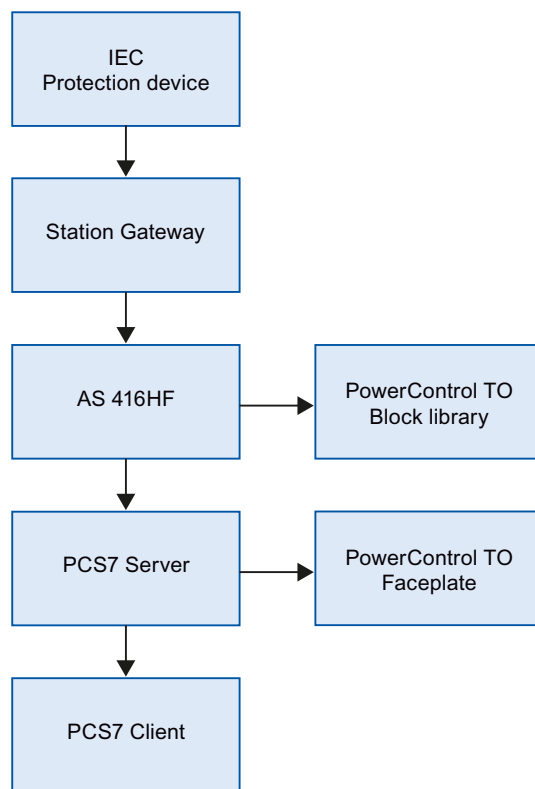
Communication log	Hardware	Device driver library
IEC 61850	Station Controller with integrated software PLC (WinAC)	SC_DRV (included in the Station Controller scope of delivery)
IEC 61850	S7 400/S7 400H with Station Gateway/redundant Station Gateway	SC_LINK (included in the Station Gateway scope of delivery)
PROFIBUS	S7 400/S7 400H with PROFIBUS interface	PowerControl PROFIBUS DP driver
-	S7 400/S7 400H with PROFIBUS interface	DataCollector blocks (included in the PCL scope of delivery)

2.1.1 Station Controller



2.1.2 Station Gateway





2.2 Technological function

The following functions are supported:

- Supply / machine (motor, generator)
- Transformer
- Synchronization device
- Line protection
- Busbar

2.3 Siprotec devices

All elements of the PCL were defined so that they can be used very flexibly for all types of protection devices.

The basic setting and the signal range were, however, modeled after the following SIPROTEC devices:

- Supply / machine:
 - Siprotec 7SJ61x
 - Siprotec 7SJ62x
 - Siprotec 7SJ63x
 - Siprotec 7SJ64x
 - Siprotec 7SJ80x
 - Siprotec 7SJ81x
 - Siprotec 6MD63
 - Siprotec 6MD66
 - Siprotec 7UM62
- Transformer:
 - Siprotec 7UT612
 - Siprotec 7UT613
 - Siprotec 7UT63x
- Synchronization device
 - Siprotec 7VE6xx
 - Siprotec 7VK61x
- Line protection:
 - Siprotec 7SD
 - Siprotec 7SA

PLC block types

3.1 Function block numbers

Note

Some older versions of S7-400 CPU 412, CPU 414 and CPU 416 do not support the preset FB number range of the Power Control Library. The FB numbers must be adapted in the Simatic Manager for these controllers. Refer to the "S7-400 instruction list" document for more detailed information on the relevant CPU types.

3.1.1 Technological blocks of the PowerControl library

Technological function	Block name	FB No.
INFEED/OUTLET/MOTOR/GENERATOR	PC_FEEDER	FB 2500
INFEED/OUTLET/MOTOR/GENERATOR on a redundant busbar (2 circuit breakers)	PC_FEEDER_2CB*	FB2501
TRANSFORMER	PC_TRAFO	FB 2502
SYNCHRONIZATION	PC_SYNCHRO	FB 2503
LINE	PC_LINE	FB 2504
BUSBAR	PC_BUSBAR	FB 2505
MEASURING POINT	PC_MEAS	FB 2506

*The PC_FEEDER_2CB block is an extension of the PC_FEEDER block. The interfaces to the driver blocks are identical to the interfaces of the PC_FEEDER block.

3.1.2 DataCollector blocks

Technological function	Block name	FB No.
FEEDER	DC_FEEDER	FB 2560
TRANSFORMER	DC_TRAFO	FB 2562
SYNCHRO	DC_SYNCHRO	FB 2563
LINE	DC_LINE	FB 2564

3.1.3 PROFIBUS DP driver blocks

Note

To use the PROFIBUS driver blocks, the following values need to be configured correctly:

- "FACTOR_E" input of the PC_FEEDER is set to 1 (No calculations)
- UN_kV, IN_A should be configured as per the used IED

The inputs UN_kV and IN_A are of the PROFIBUS drivers and not of the PC_FEEDER.

SIPROTEC devices	Block name	FB No.
7SJ6X	7SJ6X_FEEDER	FB 2580
7SA522, 7SA6	7SAX_LINE	FB 2581
7SD5, 7SD610	7SDX_LINE	FB 2582
7SJ80, 7SK80	7SJK80_FEEDER	FB 2583
7UM62	7UM62_FEEDER	FB 2584
7UT612	7UT612_TRAFO	FB 2585
7VE61, 7VE63	7VE6X_SYNC	FB 2586
7UT613, 7UT633, 7UT635	7UT6X_TRAFO	FB 2587
6MD662-664	6MD66X_FEEDER	FB 2588
7SAX	7SA522, 7SA6	FB 2597

3.1.4 Driver blocks of the SC_DRV Station Controller library

Technological function	Block name	FB No.
SUPPLY	SC_FEEDER	FB 2520
TRANSFORMER	SC_TRAFO	FB 2522
SYNCHRONIZATION	SC_SYNCHRO	FB 2523
LINE	SC_LINE	FB 2524

3.1.5 Driver blocks of the SG_LINK Station Gateway library

Technological function	Block name	FB No.
SUPPLY	SG_FEEDER	FB 2540
TRANSFORMER	SG_TRAFO	FB 2542
SYNCHRONIZATION	SG_SYNCHRO	FB 2543
LINE	SG_LINE	FB 2544

3.2 General block functions

3.2.1 Configuration of the faceplate and the units

The representation of the technology object faceplates (e.g. switch representation for PC_FEEDER) is configured directly at the function block in the CFC editor. You do this with the input structures FP_Visu and FP_Units.

The following settings can be configured in these two structures:

- Switch representation with PC_FEEDER/PC_FEEDER_2B
- Display of the interlocks in the faceplate with PC_FEEDER/PC_FEEDER_2B
- Texts of the switch positions with PC_FEEDER/PC_FEEDER_2B
- User views
- Units

A detailed description of the functionalities is available in the section of the respective technological object.

FP_Visu – General faceplate configuration

Structure element	Description	Use
Display of the switch objects PC_FEEDER/PC_FEEDER_2CB		
Field_01_Code	Symbol code for field 1	PC_FEEDER/PC_FEEDER_2CB
Field_02_Code	Symbol code for field 2	PC_FEEDER/PC_FEEDER_2CB
Field_03_Code	Symbol code for field 3	PC_FEEDER/PC_FEEDER_2CB
Field_04_Code	Symbol code for field 4	PC_FEEDER/PC_FEEDER_2CB
Field_05_Code	Symbol code for field 5	PC_FEEDER/PC_FEEDER_2CB
Field_06_Code	Symbol code for field 6	PC_FEEDER/PC_FEEDER_2CB
Field_07_Code	Symbol code for field 7 (connection field)	PC_FEEDER/PC_FEEDER_2CB
Field_08_Code	Symbol code for field 8 (connection field)	PC_FEEDER/PC_FEEDER_2CB
Field_09_Code	Symbol code for field 9 (connection field)	PC_FEEDER/PC_FEEDER_2CB
Field_10_Code	Symbol code for field 10 (connection field)	PC_FEEDER/PC_FEEDER_2CB
FieldDescriptor_01	Short description field (1)	PC_FEEDER/PC_FEEDER_2CB
FieldDescriptor_02	Short description field (2)	PC_FEEDER/PC_FEEDER_2CB
FieldDescriptor_03	Short description field (3)	PC_FEEDER/PC_FEEDER_2CB
FieldDescriptor_04	Short description field (4)	PC_FEEDER/PC_FEEDER_2CB
FieldDescriptor_05	Short description field (5)	PC_FEEDER/PC_FEEDER_2CB
FieldDescriptor_06	Short description field (6)	PC_FEEDER/PC_FEEDER_2CB
Q0_Field	Field number of switch Q0 (1-6)	PC_FEEDER
Q0_SubField	Subfield number of switch Q0 (1-2) (first or second switch in the field)	PC_FEEDER

3.2 General block functions

Structure element	Description	Use
Q1_Field	Field number of switch Q1 (1-6)	PC_FEEDER/PC_FEEDER_2CB
Q1_SubField	Subfield number of switch Q1 (1-2) (first or second switch in the field)	PC_FEEDER/PC_FEEDER_2CB
Q2_Field	Field number of switch Q2 (1-6)	PC_FEEDER/PC_FEEDER_2CB
Q2_SubField	Subfield number of switch Q2 (1-2) (first or second switch in the field)	PC_FEEDER/PC_FEEDER_2CB
Q8_Field	Field number of switch Q8 (1-6)	PC_FEEDER/PC_FEEDER_2CB
Q8_SubField	Subfield number of switch Q8 (1-2) (first or second switch in the field)	PC_FEEDER/PC_FEEDER_2CB
Q9_Field	Field number of switch Q9 (1-6)	PC_FEEDER
Q9_SubField	Subfield number of switch Q9 (1-2) (first or second switch in the field)	PC_FEEDER
Q01_Field	Field number of switch Q01 (1-6)	PC_FEEDER_2CB
Q01_SubField	Subfield number of switch Q01 (1-2) (first or second switch in the field)	PC_FEEDER_2CB
Q02_Field	Field number of switch Q02 (1-6)	PC_FEEDER_2CB
Q02_SubField	Subfield number of switch Q02 (1-2) (first or second switch in the field)	PC_FEEDER_2CB
Switch short descriptions PC_FEEDER/PC_FEEDER_2CB		
Q0_DESC	Switch description Q0 (e.g. CB)	PC_FEEDER
Q1_DESC	Switch description Q1 (e.g. DS)	PC_FEEDER/PC_FEEDER_2CB
Q2_DESC	Switch description Q2 (e.g. DS2)	PC_FEEDER/PC_FEEDER_2CB
Q8_DESC	Switch description Q8 (e.g. ES)	PC_FEEDER/PC_FEEDER_2CB
Q9_DESC	Switch description Q9 (e.g. DS3)	PC_FEEDER
Q01_DESC	Switch description Q01 (e.g. CB1)	PC_FEEDER_2CB
Q02_DESC	Switch description Q02 (e.g. CB2)	PC_FEEDER_2CB
Display of the interlocks in the faceplate with PC_FEEDER/PC_FEEDER_2CB		
Ilck_Displ	1 = Interlocks are displayed	PC_FEEDER/PC_FEEDER_2CB
Texts for the switch positions PC_FEEDER/PC_FEEDER_2CB		
Q0_TextOpen	Text Q0 open	PC_FEEDER
Q0_TextClosed	Text Q0 closed	PC_FEEDER
Q1_TextOpen	Text Q1 open	PC_FEEDER/PC_FEEDER_2CB
Q1_TextClosed	Text Q1 closed	PC_FEEDER/PC_FEEDER_2CB
Q2_TextOpen	Text Q2 open	PC_FEEDER/PC_FEEDER_2CB
Q2_TextClosed	Text Q2 closed	PC_FEEDER/PC_FEEDER_2CB
Q8_TextOpen	Text Q8 open	PC_FEEDER/PC_FEEDER_2CB
Q8_TextClosed	Text Q8 closed	PC_FEEDER/PC_FEEDER_2CB
Q9_TextOpen	Text Q9 open	PC_FEEDER
Q9_TextClosed	Text Q9 closed	PC_FEEDER
Q01_TextOpen	Text Q01 open	PC_FEEDER_2CB
Q01_TextClosed	Text Q01 closed	PC_FEEDER_2CB
Q02_TextOpen	Text Q02 open	PC_FEEDER_2CB
Q02_TextClosed	Text Q02 closed	PC_FEEDER_2CB

Structure element	Description	Use
User views		
CustomFaceplate	File name of the user view	PC_FEEDER/PC_FEEDER_2CB
CustomFaceplate_1	File name of the first user view	PC_LINE/PC_MEAS/PC_SYNC/ PC_TRAFO
CustomFaceplate_2	File name of the second user view	PC_LINE/PC_SYNC/ PC_TRAFO
Interlock display		
Q0_IlckText_1	Description of Q0 interlock field 1	PC_FEEDER/PC_FEEDER_2CB
Q0_IlckText_2	Description of Q0 interlock field 2	PC_FEEDER/PC_FEEDER_2CB
Q1_IlckText_1	Description of Q1 interlock field 1	PC_FEEDER/PC_FEEDER_2CB
Q1_IlckText_2	Description of Q1 interlock field 2	PC_FEEDER/PC_FEEDER_2CB
Q2_IlckText_1	Description of Q2 interlock field 1	PC_FEEDER/PC_FEEDER_2CB
Q2_IlckText_2	Description of Q2 interlock field 2	PC_FEEDER/PC_FEEDER_2CB
Q8_IlckText_1	Description of Q8 interlock field 1	PC_FEEDER/PC_FEEDER_2CB
Q8_IlckText_2	Description of Q8 interlock field 2	PC_FEEDER/PC_FEEDER_2CB
Q9_IlckText_1	Description of Q9 interlock field 1	PC_FEEDER/PC_FEEDER_2CB
Q9_IlckText_2	Description of Q9 interlock field 2	PC_FEEDER/PC_FEEDER_2CB
Display of execution level		
Q0_OA_Text_Btn1	Description of button 1 of execution level of button Q0	PC_FEEDER/PC_FEEDER_2CB
Q0_OA_Text_Btn2	Description of button 2 of execution level of button Q0	PC_FEEDER/PC_FEEDER_2CB
Q1_OA_Text_Btn1	Description of button 1 of execution level of button Q1	PC_FEEDER/PC_FEEDER_2CB
Q1_OA_Text_Btn2	Description of button 2 of execution level of button Q1	PC_FEEDER/PC_FEEDER_2CB
Q2_OA_Text_Btn1	Description of button 1 of execution level of button Q2	PC_FEEDER/PC_FEEDER_2CB
Q2_OA_Text_Btn2	Description of button 2 of execution level of button Q2	PC_FEEDER/PC_FEEDER_2CB
Q8_OA_Text_Btn1	Description of button 1 of execution level of button Q8	PC_FEEDER/PC_FEEDER_2CB
Q8_OA_Text_Btn2	Description of button 2 of execution level of button Q8	PC_FEEDER/PC_FEEDER_2CB
Q9_OA_Text_Btn1	Description of button 1 of execution level of button Q9	PC_FEEDER/PC_FEEDER_2CB
Q9_OA_Text_Btn2	Description of button 2 of execution level of button Q9	PC_FEEDER/PC_FEEDER_2CB
Texts for the interlock fields PC_FEEDER/ PC_FEEDER_2CB		
Q0_IlckText_1	Text for Q0 interlock field 1	PC_FEEDER
Q0_IlckText_2	Text for Q0 interlock field 1	PC_FEEDER
Q1_IlckText_1	Text for Q0 interlock field 1	PC_FEEDER/PC_FEEDER_2CB
Q1_IlckText_2	Text for Q0 interlock field 1	PC_FEEDER/PC_FEEDER_2CB

Structure element	Description	Use
Q2_IlckText_1	Text for Q0 interlock field 1	PC_FEEDER/PC_FEEDER_2CB
Q2_IlckText_2	Text for Q0 interlock field 1	PC_FEEDER/PC_FEEDER_2CB
Q8_IlckText_1	Text for Q0 interlock field 1	PC_FEEDER/PC_FEEDER_2CB
Q8_IlckText_2	Text for Q0 interlock field 1	PC_FEEDER/PC_FEEDER_2CB
Q9_IlckText_1	Text for Q0 interlock field 1	PC_FEEDER
Q9_IlckText_2	Text for Q0 interlock field 1	PC_FEEDER
Q01_IlckText_1	Text for Q01 interlock field 1	PC_FEEDER_2CB
Q01_IlckText_2	Text for Q01 interlock field 2	PC_FEEDER_2CB
Q02_IlckText_1	Text for Q02 interlock field 1	PC_FEEDER_2CB
Q02_IlckText_2	Text for Q02 interlock field 2	PC_FEEDER_2CB
Texts for the switch control panel PC_FEEDER/PC_FEEDER_2CB		
Q0_OA_Text_Btn1	Text for button 1 in the Q0 switch control panel	PC_FEEDER
Q0_OA_Text_Btn2	Text for button 2 in the Q0 switch control panel	PC_FEEDER
Q1_OA_Text_Btn1	Text for button 1 in the Q1 switch control panel	PC_FEEDER/PC_FEEDER_2CB
Q1_OA_Text_Btn2	Text for button 2 in the Q1 switch control panel	PC_FEEDER/PC_FEEDER_2CB
Q2_OA_Text_Btn1	Text for button 1 in the Q2 switch control panel	PC_FEEDER/PC_FEEDER_2CB
Q2_OA_Text_Btn2	Text for button 2 in the Q2 switch control panel	PC_FEEDER/PC_FEEDER_2CB
Q8_OA_Text_Btn1	Text for button 1 in the Q8 switch control panel	PC_FEEDER/PC_FEEDER_2CB
Q8_OA_Text_Btn2	Text for button 2 in the Q8 switch control panel	PC_FEEDER/PC_FEEDER_2CB
Q9_OA_Text_Btn1	Text for button 1 in the Q9 switch control panel	PC_FEEDER
Q9_OA_Text_Btn2	Text for button 2 in the Q9 switch control panel	PC_FEEDER
Q01_OA_Text_Btn1	Text for button 1 in the Q01 switch control panel	PC_FEEDER_2CB
Q01_OA_Text_Btn2	Text for button 2 in the Q01 switch control panel	PC_FEEDER_2CB
Q02_OA_Text_Btn1	Text for button 1 in the Q02 switch control panel	PC_FEEDER_2CB
Q02_OA_Text_Btn2	Text for button 2 in the Q02 switch control panel	PC_FEEDER_2CB
Texts for measured value display		
Text_Power	Texts for active power	PC_FEEDER/PC_FEEDER_2CB/ PC_MEAS/PC_TRAFO
Text_Q	Texts for reactive power	PC_FEEDER/PC_FEEDER_2CB/ PC_MEAS/PC_TRAFO

Structure element	Description	Use
Text_S	Texts for apparent power	PC_FEEDER/PC_FEEDER_2CB/ PC_MEAS/PC_TRAFO
Text_Voltage	Texts for voltage display	PC_FEEDER/PC_FEEDER_2CB/ PC_MEAS/PC_TRAFO
Text_Current	Texts for current display	PC_FEEDER/PC_FEEDER_2CB/ PC_MEAS/PC_TRAFO

FP_Units – displayed units in the faceplates

Structure element	Description	Use
Unit_Power	Active power unit	PC_FEEDER/PC_FEEDER_2CB/ PC_MEAS/PC_TRAFO
Unit_Q	Reactive power unit	PC_FEEDER/PC_FEEDER_2CB/ PC_MEAS/PC_TRAFO
Unit_S	Apparent power unit	PC_FEEDER/PC_FEEDER_2CB/ PC_MEAS/PC_TRAFO
Unit_Voltage	Voltage unit	PC_FEEDER/PC_FEEDER_2CB/ PC_MEAS/PC_SYNC/PC_TRAFO
Unit_Current	Current unit	PC_FEEDER/PC_FEEDER_2CB/ PC_LINE/PC_MEAS/PC_TRAFO

Example:

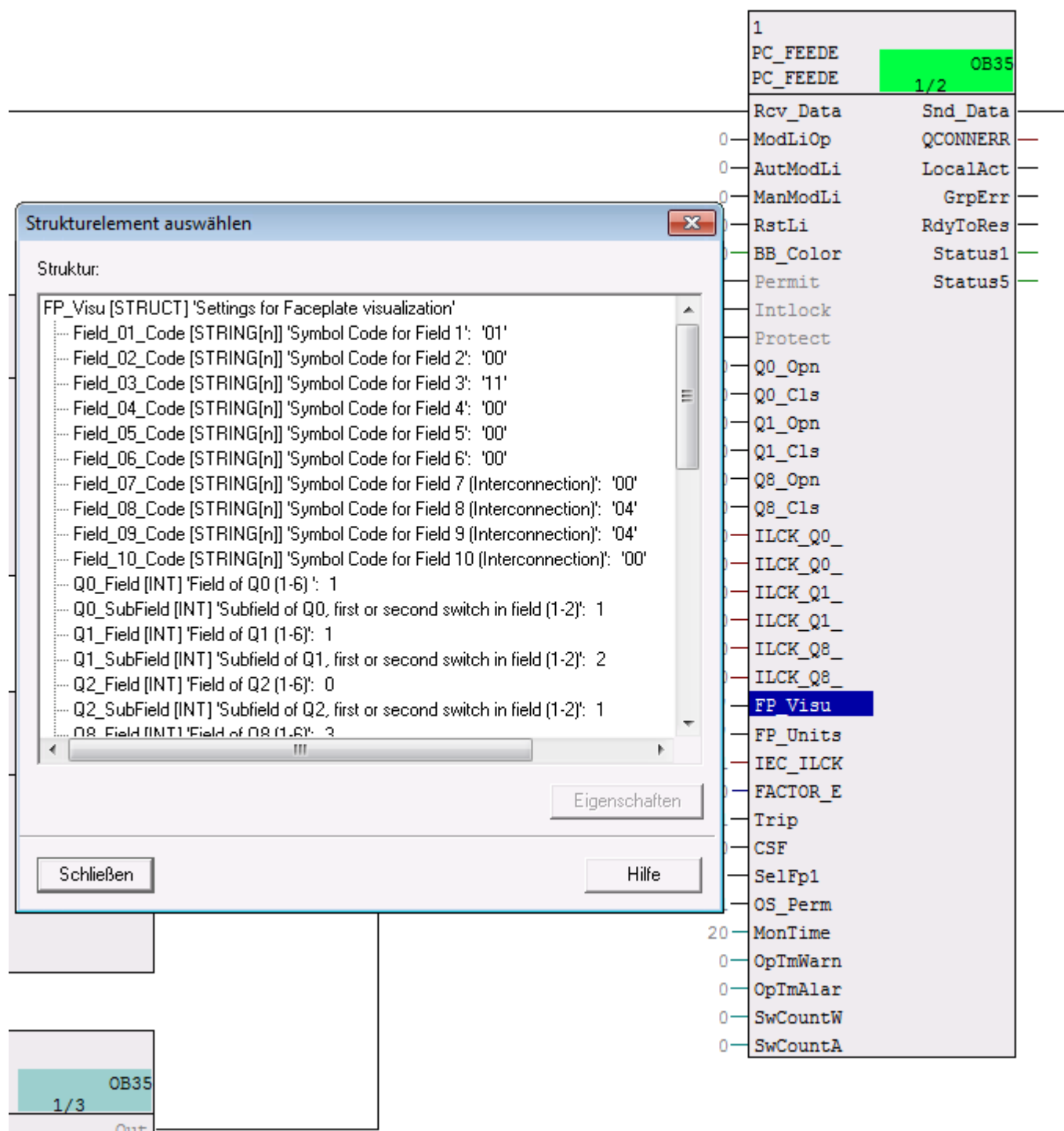


Figure 3-1 FP_Visu

3.2.2 Alarms and messages

Time stamp:

The alarms and messages of a switchgear device are output by the technological block of the PCL with the time stamp of the device. If the time stamps of the protective device are not available, they are generated in the device driver block.

Both IEC61850 solutions, Station Gateway and Station Controller, make time stamps of the devices available.

If the devices are connected through PROFIBUS, the transfer of the time stamps depend on the PROFIBUS assignment used.

3.2.3 Local operator authorization

Local operator authorization

The local operator authorization can be set for each specific instance; in other words, block instances can be enabled or disabled for use on an operator station independently of one another.

To be able to use the functionality, you first need to switch the corresponding input parameters at the block in the CFC to visible.

523	OosLi.Value	IN	BOOL	0	<...	☐	☐	☐	☐	☐	☒	☐	
521	OosOp	IN	BOOL	0		☐	☐	☐	☐	1=Oos Mode: Oos Mode by Operator	☒	☐	Kei
767	OpSt_In	IN	DWORD	16...		☐	☐	☐	☐	Enabled operator stations	☒	☐	Kei
1272	OpSt_Out	OUT	DWORD	16...		☐	☐	☐	☐	Enabled operator stations	☒	☐	Kei
1202	OpTm	OUT	DINT	0		☐	☐	☐	☐	Operation hours	☒	☐	Kei
1252	OpTmAlarm	OUT	BOOL	0		☐	☐	☐	☐	1 = Operation Time Alarm active	☒	☐	

Figure 3-2 Switch block parameters to visible

In addition, the functionality must be enabled via **Feature.Bit24**.

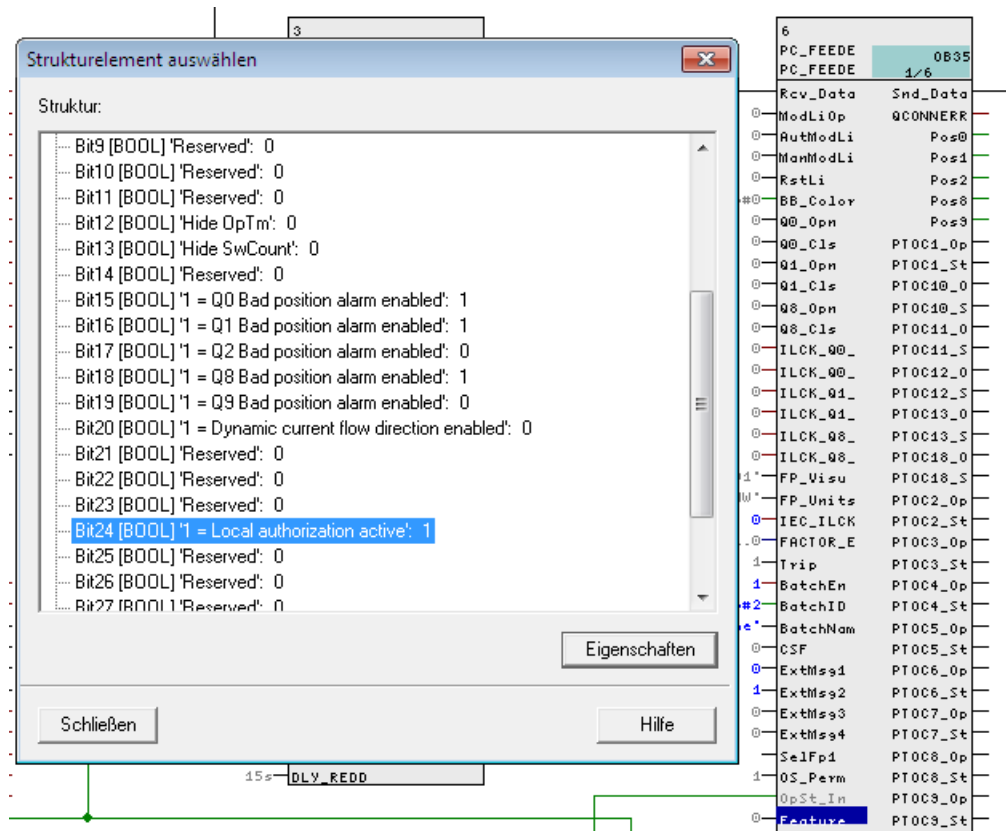


Figure 3-3 Local operator authorization - Feature bit

Position the OpStations block of the Advanced Process Library (APL) in the CFC and switch the corresponding inputs to visible. Then interconnect the **Out** output with the **OpSt_In** input of the PCL block.

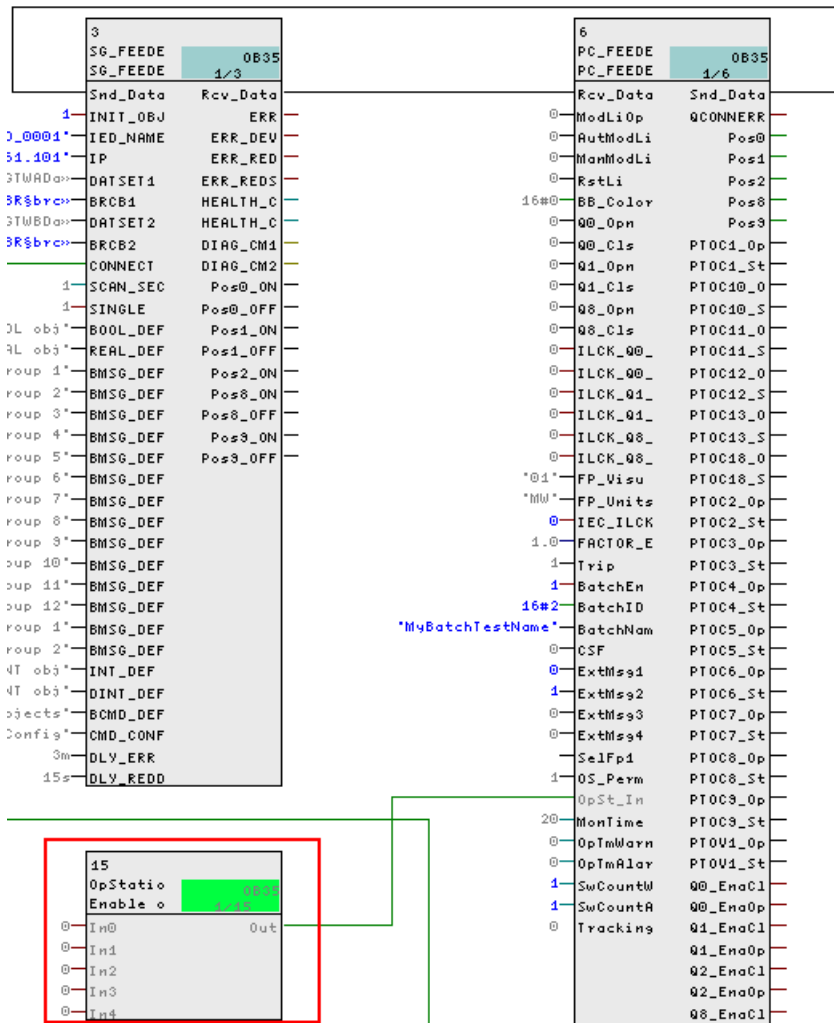


Figure 3-4 Local operator authorization - Block interconnection

Faceplates

4.1 General faceplate functions

4.1.1 Displaying auxiliary values

The standard view of the faceplates includes the APL standard function "Show auxiliary values" (see APL documentation).

The display elements for display of the auxiliary values are shown in the faceplate as soon as at least one of the two block inputs UserAna1 or UserAna2 was interconnected.

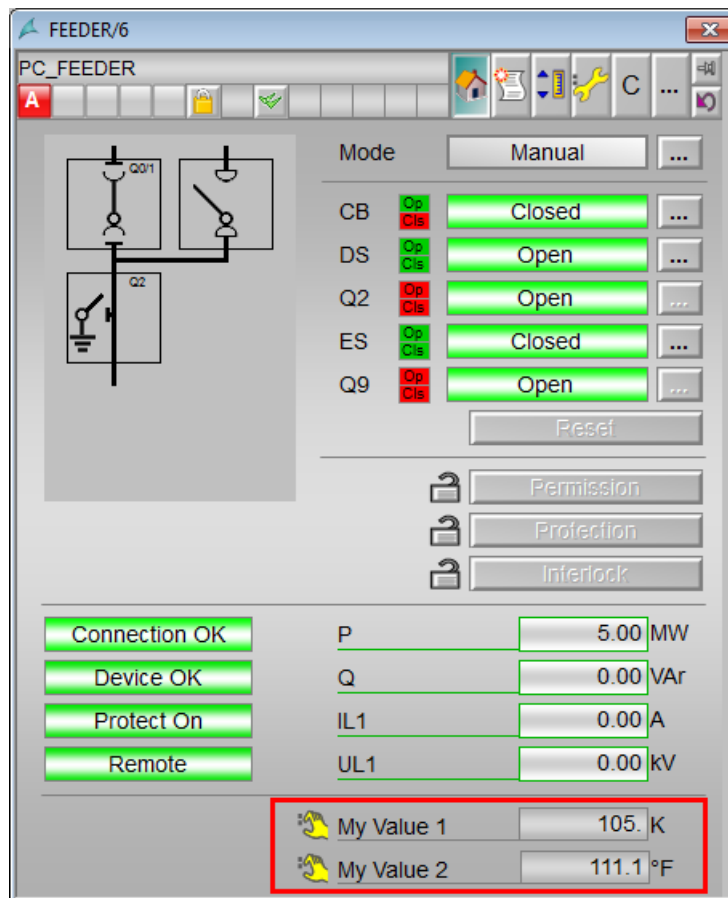


Figure 4-1 Display of auxiliary values in the standard view

4.1.2 User views

All technological objects provide the option of creating instance-specific user views. A new WinCC picture (*.pdl) with the required data is created for this and the name of the picture is configured at the *FP_Visu* structure, *CustomFaceplate* element of the function block. This user view can then be opened using the "C" button in the toolbar of the faceplate.

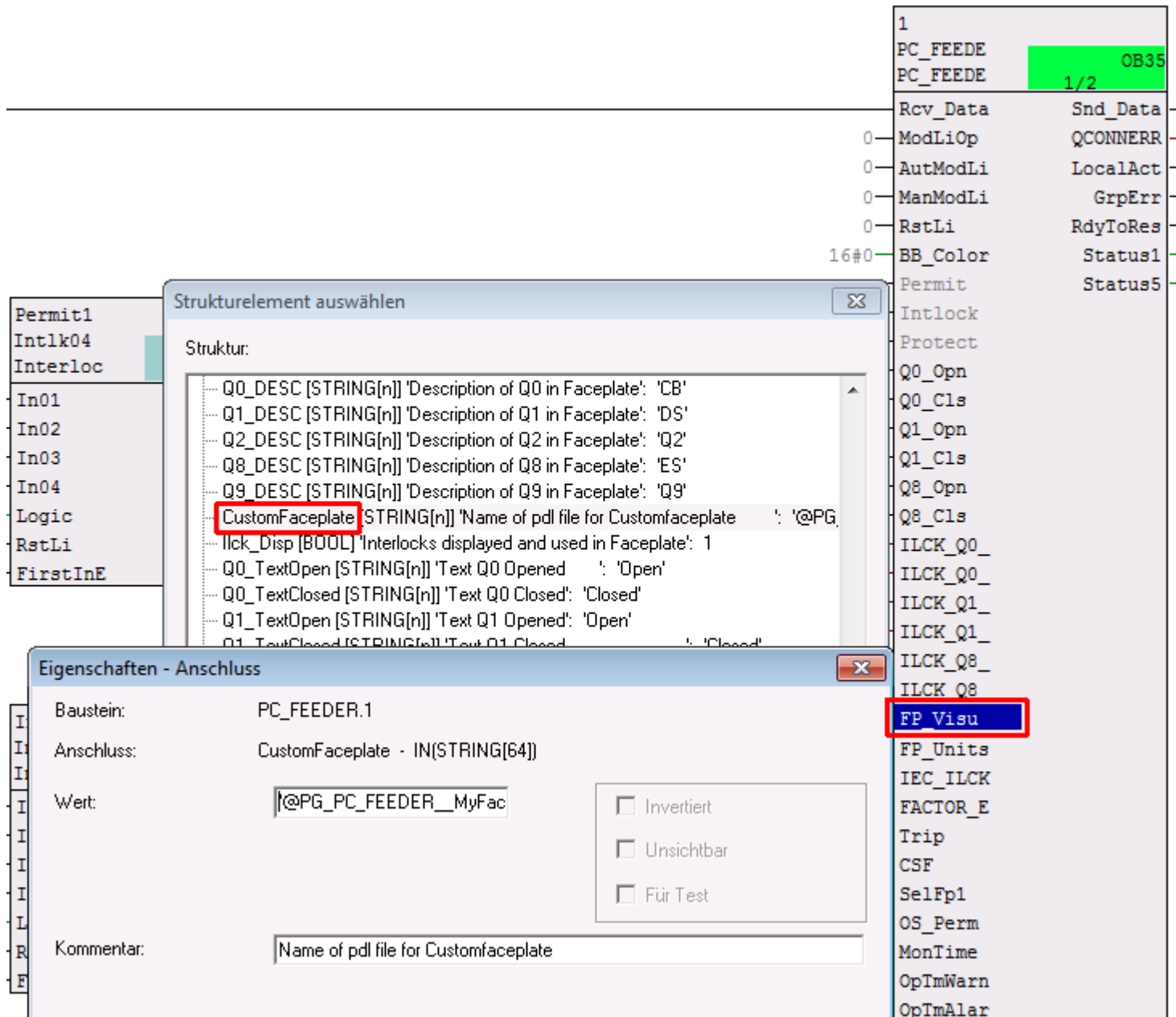


Figure 4-2 Assigning parameters for the user view

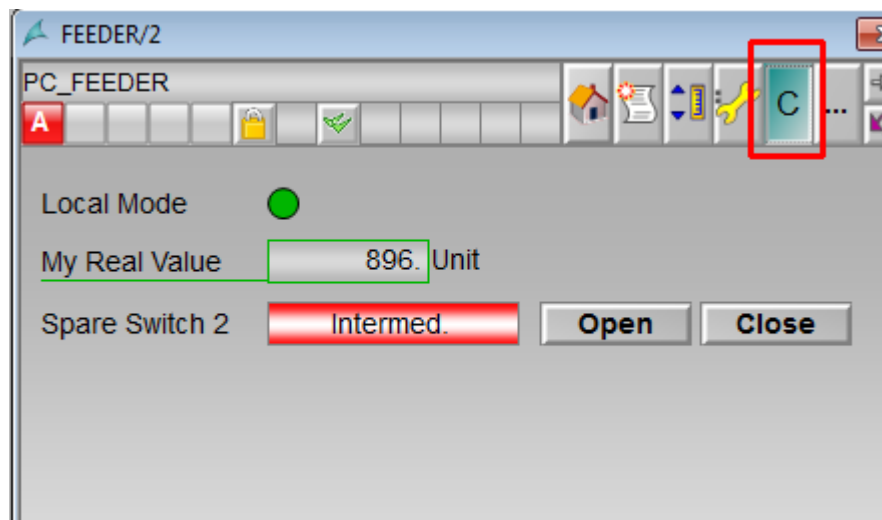


Figure 4-3 Example of a user view

4.1.2.1 Creating a user view/templates

The following steps must be performed to create a functioning user view:

- The scope of delivery of the PCS 7 PowerControl library includes a .pdl file with example elements for the user view. This file is located in the GraCS folder of the WinCC project and has the file name "**@PowerControl_CustomFP_Templates.pdl**".
- As a basis for a new user view, this file should be copied and renamed.
- The newly assigned file name is then configured at the function block (structure *FP_Visu*, structure element *CustomFaceplate*).

Note

If you are using a server/client constellation, it is necessary to copy the corresponding .pdl file of the user-defined faceplate to the associated GraCS directory on the client.

Note

The use of user views requires advanced knowledge of the WinCC Graphics Designer and the functionalities of the PCS 7 Advanced Process Library (APL). The example elements in the "**@PowerControl_CustomFP_Templates.pdl**" file were especially created for the use in user views. They must, however, still be mapped to the corresponding tags/WinCC tags. You can also add your own elements. In doing so, you have to keep in mind that elements copied from other faceplates do not automatically work in the user view and may have to be adapted.

The following figure shows the templates for the user views:

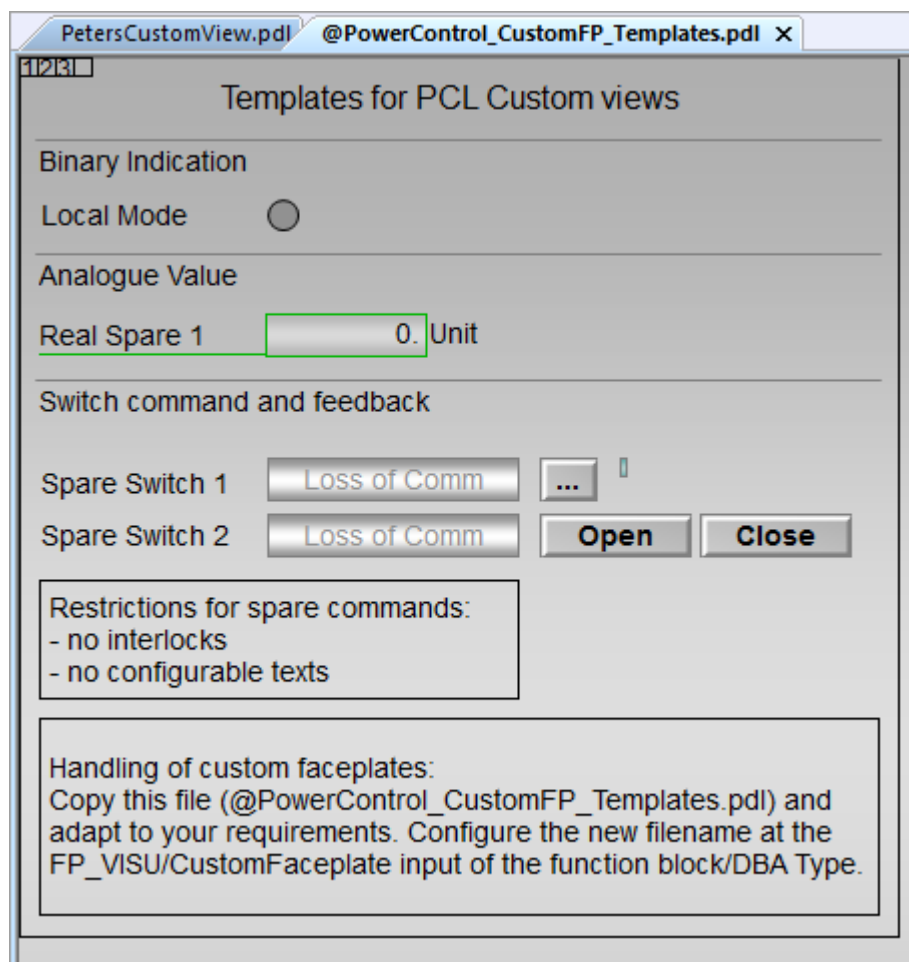


Figure 4-4 User view templates

4.1.3 Memo view

The faceplates include the APL standard function "Memo view" (see APL documentation).

The view can be called in the toolbar with the memo button.

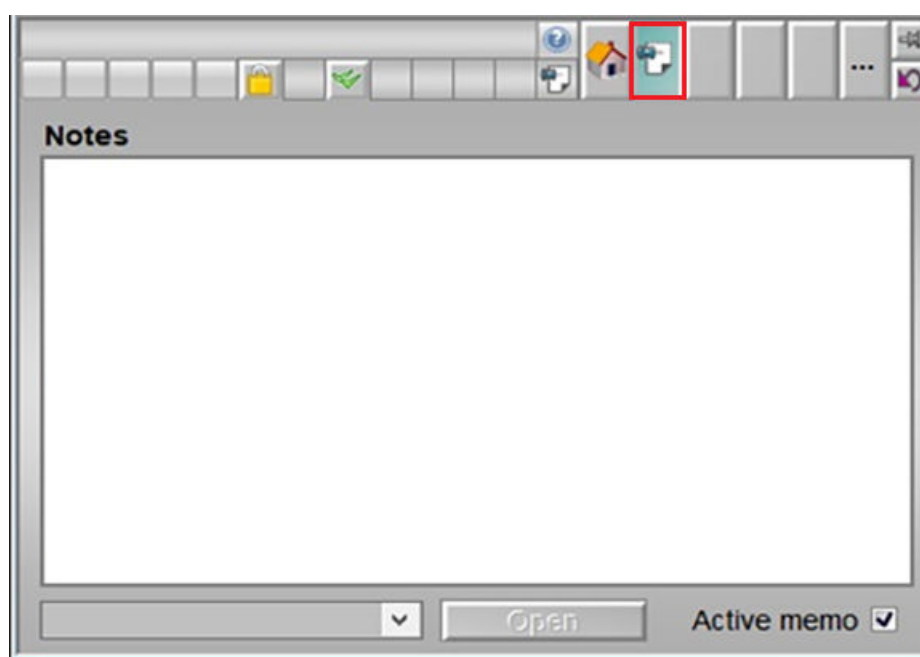


Figure 4-5 Memo view

Block symbols

5.1 General block symbol functions

5.1.1 Symbol representation of the block symbol with PC_FEEDER/PC_FEEDER_2CB

The symbols that is shown in the process picture for PC_FEEDER and PC_FEEDER_2CB can be selected using the "SymbolCode" symbol attribute.

Because symbols 1-3 of the PC_FEEDER/PC_FEEDER_2CB are displayed dynamically (depending on the switch setting), the attributes "Pos0" (switch setting of the first displayed switch) and "Pos1" (switch setting of the second displayed switch) must be correctly linked.

5.1.2 Display of the current flow direction

The current flow direction in the switchgear can be displayed by means of a direction arrow next to the symbol. It is configured with the "DirectionUp" (upward current flow) and "DirectionDown" (downward current flow) attributes. If both attributes are set to "No", the arrow is hidden.

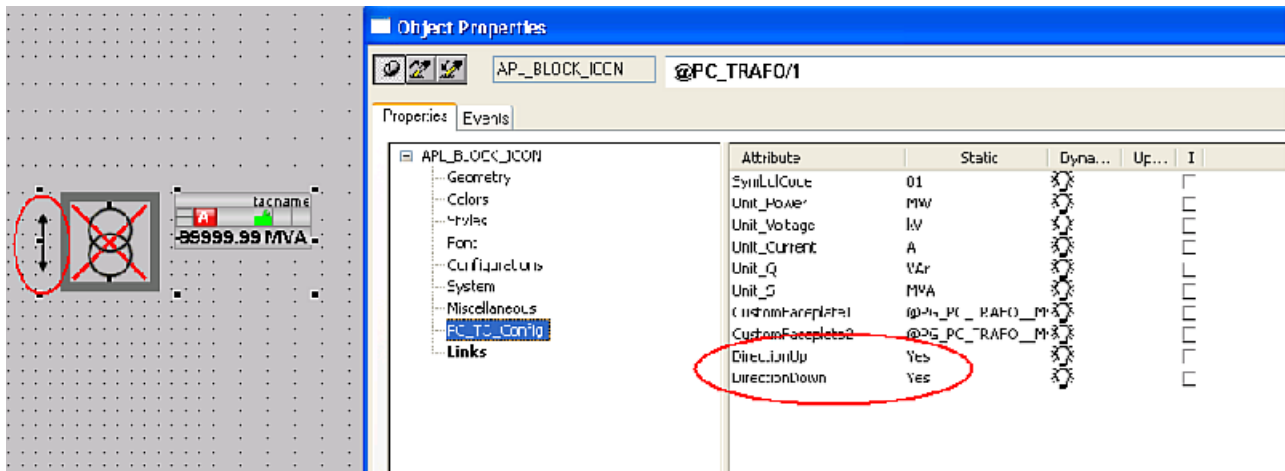


Figure 5-1 Display of the current flow direction

5.2 Configuration of the block symbol

5.2.1 General attributes

Configuration of the group:

Attribute	Description	Start value
AV_Format_Power	Formatting of the power values	##0.00
AV_Format_Current	Formatting of the power values	##0.00
AV_Format_Voltage	Formatting of the voltage values	##0.00

Colors of the group:

Attribute	Description	Start value
BorderColor	Border color of the symbol	0xC0C0C0

5.2.2 PC_FEEDER attributes

Links:

Attribute	Description	Start value
Pos0	Switch position of the first switch in the symbol (see chapter Configuring the symbols (Page 82))	Pos0
Pos1	Switch position of the second switch in the symbol (see chapter Configuring the symbols (Page 82))	Pos1

See also

Configuration of the switches (Page 85)

5.2.3 PC_BUSBAR attributes

With the PC_BUSBAR technological object, there are additional attributes with which you can configure the switch arrangement and labeling in the faceplate and in the symbol.

PC_TO_Config group:

Attribute	Description	Start value
SymbolCode	Symbol number for display in the process picture (see Appendix)	02
Unit_Voltage	Unit for the voltage for display in the symbol	kV
Unit_Frequency	Unit for the frequency for display in the symbol	Hz
Busbarwidth	Width of the busbar	140
Thickness	Thickness of the busbar	3/20

PC_FEEDER

The PC_FEEDER object is used for all feed-ins or branches of a busbar. It can also be configured as a circuit breaker, grounding switch or disconnecter. In addition, it is used for switchable machine contactors.

PC_FEEDER consists of the following objects:

- PLC block PC_FEEDER
- WinCC block symbols
- WinCC faceplate PC_FEEDER with different views
- Device driver blocks (not part of the PCL)

6.1 PLC block PC_FEEDER

6.1.1 Description of PLC block PC_FEEDER

The PC_FEEDER (FB 2500) function block represents the feeder object in the PLC. The block is the interface between the device driver block and the remaining PCS 7 environment.

The block offers the following functions:

- Detecting data from the device driver block
- Providing data for the block symbol and the faceplate
- Controlling switches of the switchgear device
- APL modes Out of service, Local, Manual, Automatic
- Interlocking functions
- Handling of operator authorizations
- Configuration of the faceplate
- Various APL standard functions

6.1.2 Interfaces

The PLC block PC_FEEDER offers the following interfaces:

1. **Device driver blocks**
The device driver block is connected to the PC_FEEDER block through the input "Rcv_Data" and the output "Snd_Data".
2. **Block symbol**
The block symbol receives data from the block through the S7 connection between the PLC and WinCC.
3. **Faceplate**
The faceplate receives data from the block through the S7 connection between the PLC and WinCC.
4. **APL interlock blocks**
The interlock blocks can be connected through the inputs "Permit", "IntLock" and "Protect".
5. **Logic of the S7 program (for controlling modes, switches, etc.)**
The PC_FEEDER block can receive commands from the CFC logic and makes data available to other blocks.

6.1.3 Configuration

Use the CFC editor to install the block in a cyclic interrupt OB (OB30 to OB38). The block is also installed automatically in the startup OB (OB 100).

6.1.4 Operating modes

The PC_FEEDER block supports the APL standard operating modes.

The block can be operated in the following operating modes:

Operating mode	Description	Permitted commands	Display
Out of service	The device is taken out of service by the faceplate or by the CFC logic	No	<i>OosAct.Value</i> = true
Local operating mode	The device signals local operating mode (for example, the key-operated switch on the device is in the "local" position)	No	<i>LocalAct.Value</i> = true
Manual mode	The device is controlled through the faceplate	Only from faceplate	<i>ManAct.Value</i> = true
Automatic mode	The device is controlled through the connected logic in the CFC	Only by block inputs	<i>AutAct.Value</i> = true

For a detailed description of the operating modes refer to the "PCS 7 Advanced Process Library Function Manual" [1].

6.1.5 Switch commands

The PC_FEEDER block sends commands from the faceplate (in manual operation) or from other connected blocks (in automatic mode) to the device driver block. The following commands are implemented:

- Q0 (circuit breaker)
- Q1
- Q2
- Q8
- Q9
- Free command 1
- Free command 2

Feedback monitoring:

If a command is sent to the device and the feedback (position) is not received within the configured time (MonTime input), a discrepancy message is generated.

6.1.6 Interlocks

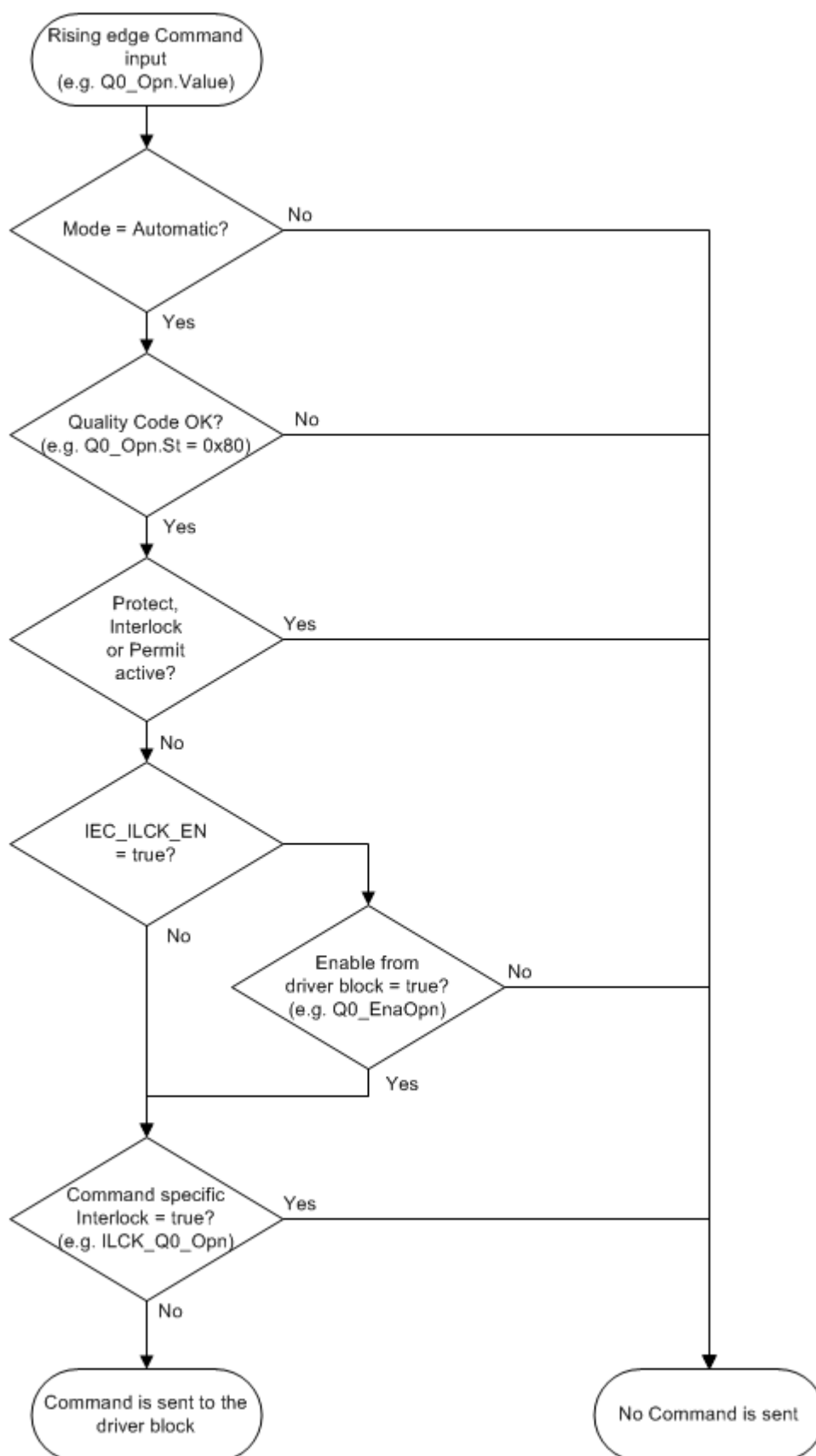
In addition to the general APL interlock functions (Permit, Interlock and Protect), PC_FEEDER offers the following two interlock mechanisms:

- Switch-specific interlock signals that come from the device driver block (for example, through IEC 61850)
- Switch-specific interlock signals that are connected with the PLC block

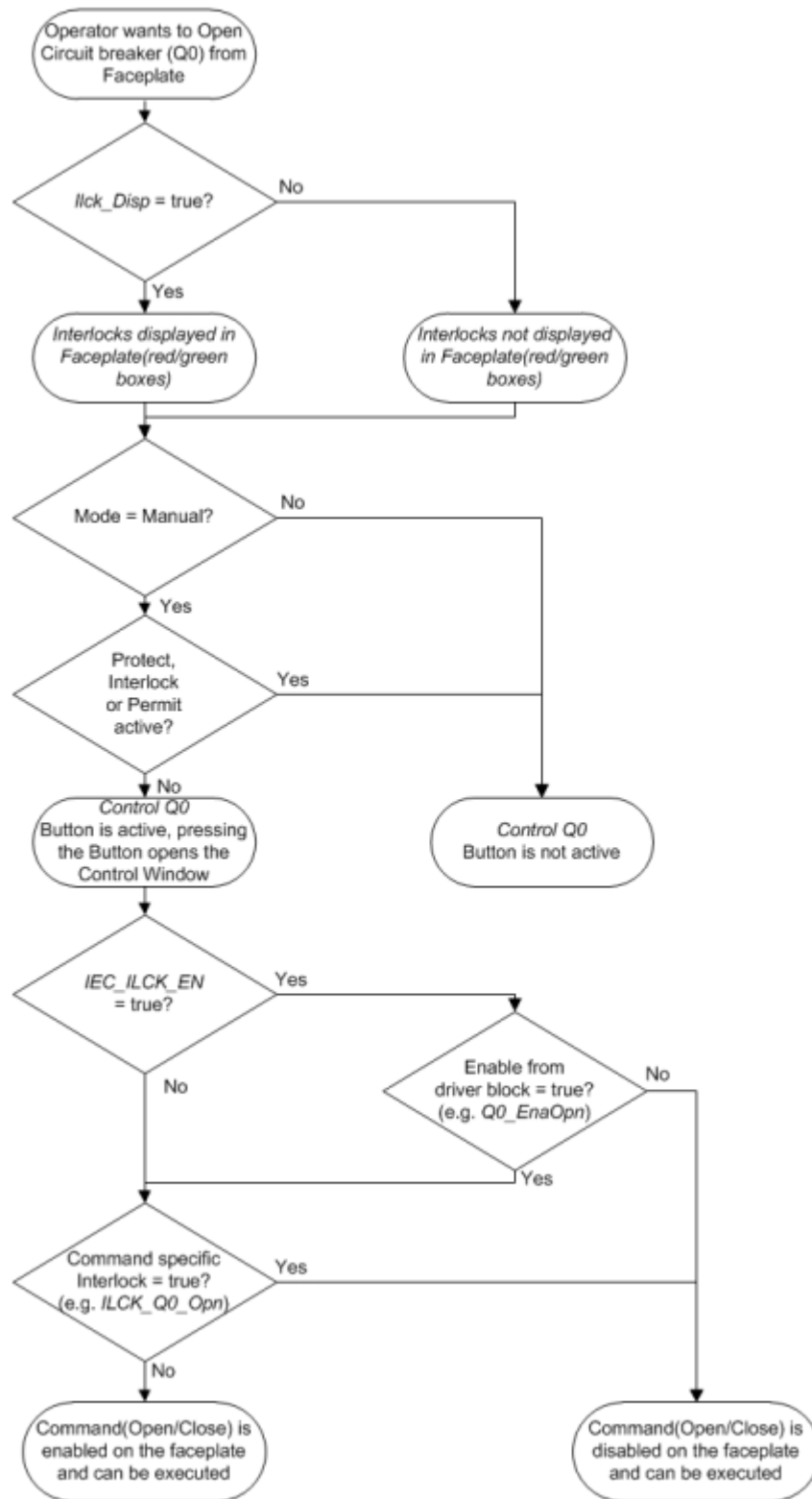
Important information: The APL interlocks Permit, Interlock and Protect only interlock the close command of Q0 (circuit breaker).

Important information: If an APL interlock "Interlock" or "Protect" is activated, an open command is sent immediately to Q0 (circuit breaker).

The following logic is used to interlock a command linked in the CFC Editor:



The following logic is used for interlocking the commands output in the faceplate:



Deactivating individual interlocks:

You can deactivate the implemented block interlocks through the input parameters Intlock, Protect and Permit.

If you want to deactivate the block interlock, you have to specify the following parameters:

- *Perm_En* = 0 or Permit.ST = 16#FF: The Permit input parameter does not have any effect.
- *Prot_En* = 0 or Protect.ST = 16#FF: The Protect input parameter does not have any effect.
- *Intl_En* = 0 or Intlock.ST = 16#FF: The Intlock input parameter does not have any effect.

Adapting the display text of individual interlocks

The parameter structure of the FP_Visu input offers the option to specify corresponding texts for the interlock display of the switches and the labels of the buttons in the Operation Area via the block. Due to the field size, the texts are limited as follows:

- Buttons - max. 8 characters
- Interlocks - max. 3 characters

The FP_Visu structure element can be opened at the block and the desired texts can be specified.

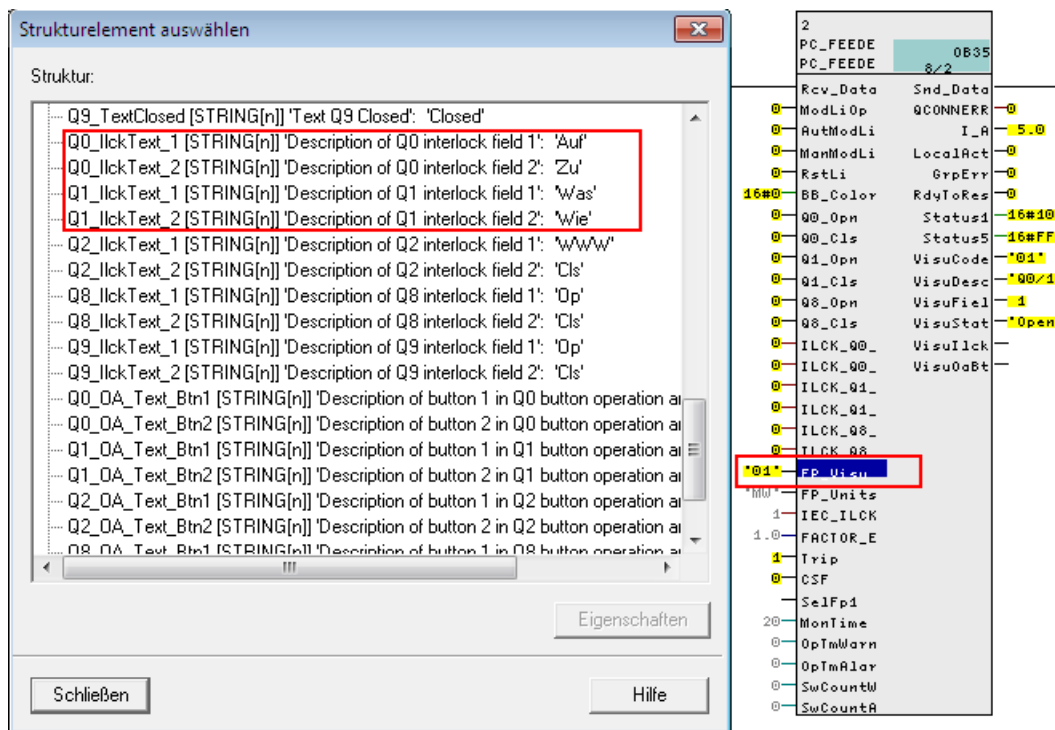


Figure 6-1 FP_Visu_InterlockText

The texts at the faceplate are applied correspondingly in Runtime.

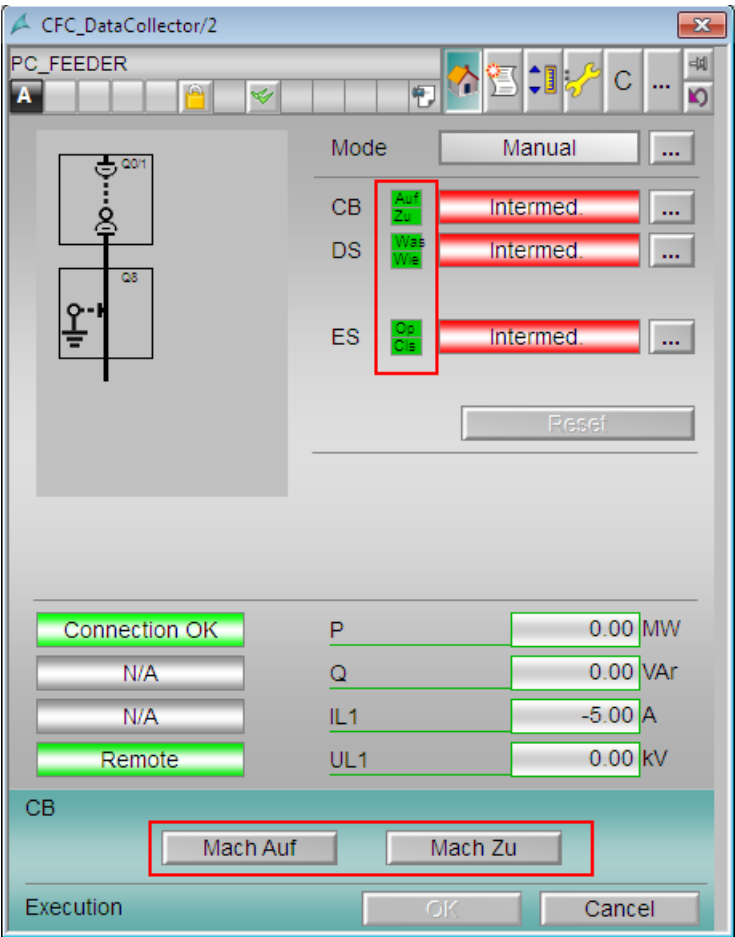


Figure 6-2 Faceplate InterlockText

Tracking the interlock condition using jump marks

When configured accordingly, the interlock conditions of the individual operating areas can be reached directly from the faceplate.

To be able to use the functionality, you first need to switch the corresponding input parameters at the block in the CFC to visible.

General I/Os												
#	Name	Type	Value	In...	A...	F...	F...	S...	T...	Comment	Invisible	W...
734	SelFp1	ANY		CF...	☐	☐	☐	☐	☐	Select Faceplate 1	☐	☐
735	SelFpIckQ0CIs	ANY		CF...	☐	☐	☐	☐	☐	Select Faceplate Interlock Q0 closed	☐	☐
736	SelFpIckQ0Opn	ANY		CF...	☐	☐	☐	☐	☐	Select Faceplate Interlock Q0 opened	☐	☒
737	SelFpIckQ1CIs	ANY		CF...	☐	☐	☐	☐	☐	Select Faceplate Interlock Q1 closed	☐	☒
738	SelFpIckQ1Opn	ANY		CF...	☐	☐	☐	☐	☐	Select Faceplate Interlock Q1 opened	☐	☒
739	SelFpIckQ2CIs	ANY		CF...	☐	☐	☐	☐	☐	Select Faceplate Interlock Q2 closed	☐	☒
740	SelFpIckQ2Opn	ANY		CF...	☐	☐	☐	☐	☐	Select Faceplate Interlock Q2 opened	☐	☒
741	SelFpIckQ8CIs	ANY		CF...	☐	☐	☐	☐	☐	Select Faceplate Interlock Q8 closed	☐	☒
742	SelFpIckQ8Opn	ANY		CF...	☐	☐	☐	☐	☐	Select Faceplate Interlock Q8 opened	☐	☒
743	SelFpIckQ9CIs	ANY		CF...	☐	☐	☐	☐	☐	Select Faceplate Interlock Q9 closed	☐	☒
744	SelFpIckQ9Opn	ANY		CF...	☐	☐	☐	☐	☐	Select Faceplate Interlock Q9 opened	☐	☒
827	Snd_Data	STRUCT		CF...	☐	☐	☐	☐	☐	Connect to SG/_SC_FEEDER	☐	☒
863	Snd_Data.BatchId	DWO...	16#0000...		☐	☐	☐	☐	☐		☒	☐

Figure 6-3 Interlock Block Jump

Then interconnect the parameter used for the interlock to the corresponding input parameter.

In the faceplate, when configured accordingly, the interlock display can thereby be used as jump mark to the faceplate of the calling block.

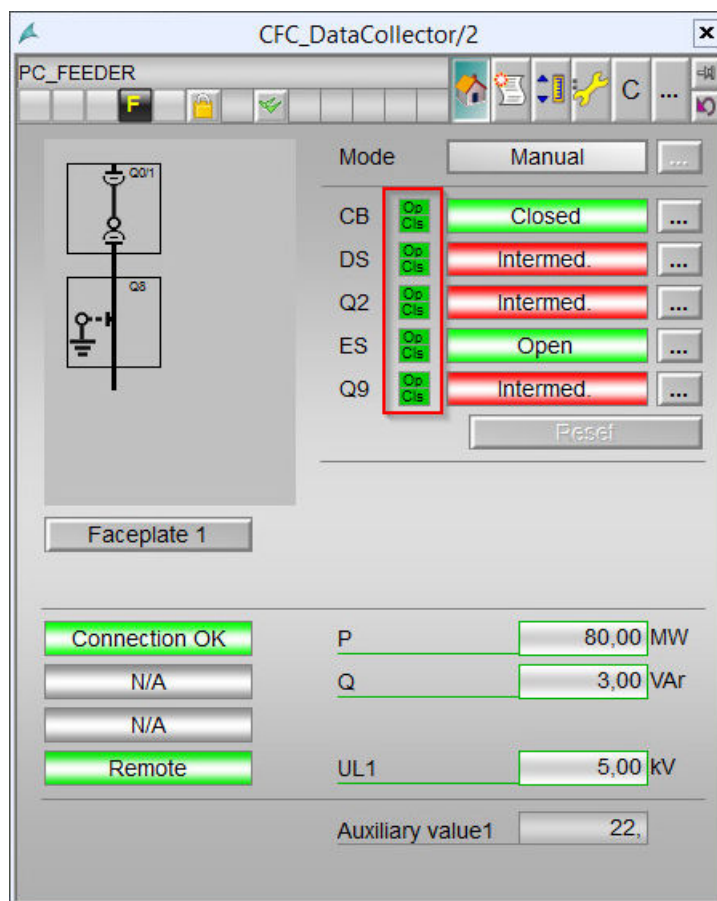


Figure 6-4 Interlock Block Jump Faceplate

6.1.7 Behavior of external tripping signals

External trippings from the CFC logic can be connected with the "Trip" input. The "Trip" input functions like the "Protect" interlock and opens Q0, if it was closed beforehand.

Similarly to the interlocks, the trigger signal is low-active:

Trip.Value = 1 → Not interlocked

Trip.Value = 0 → Interlocked, "Open Q0" command is sent

After a trigger condition the block has to be reset (see chapter Resetting the block with an interlock or external trigger (Page 47)).

6.1.8 Resetting the block with an interlock or external trigger

This block offers the APL standard function "Resetting the block in case of interlocks or errors".

After a "Protect" interlock or an external trip the block has to be reset so that a close command can be sent to the circuit breaker Q0.

In automatic mode resetting is carried out through a rising edge at the input "RstLi".

In manual mode resetting is only required if the function bit 31 (ResetinManual) is set and resetting is carried out by using the "Reset" command button in the standard view of the faceplate.

6.1.9 Tracking mode

Tracking mode is used to set the physically correct switch positions directly at the faceplate while the connection to the IED is lost. This means the current switch position can be reproduced even while the connection is lost.

6.1.9.1 Using tracking mode

By default, the use of tracking mode is enabled. To lock or release the use of tracking mode, set the corresponding bit at the OS_Perm block I/O as follows:

- 0 = Operator cannot shift to tracking mode
- 1 = Operator can shift to tracking mode

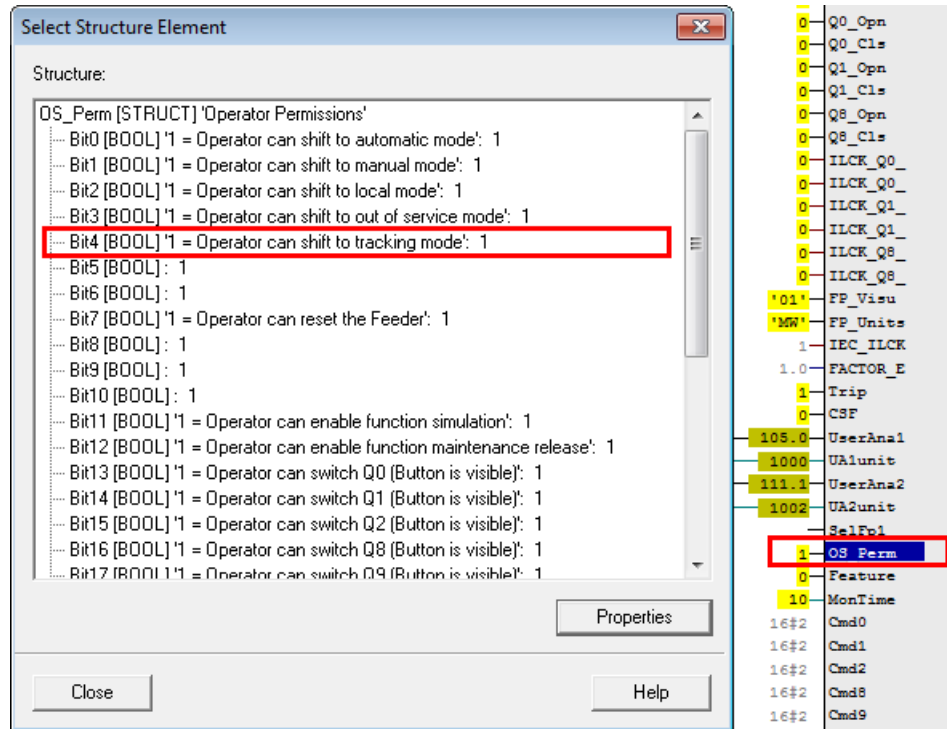


Figure 6-5 Lock/enable tracking mode at the block

6.1.9.2 Setting the operator authorization level

Tracking mode uses the APL standard function "Operator authorizations - User administration in the PCS 7 OS". This means the operator authorization level of the tracking mode can be set for specific instances with the object properties at the block I/O TrackingMode.

Note

By default, the operator authorization level for tracking mode is pre-assigned the value 3 (highest order operator inputs).

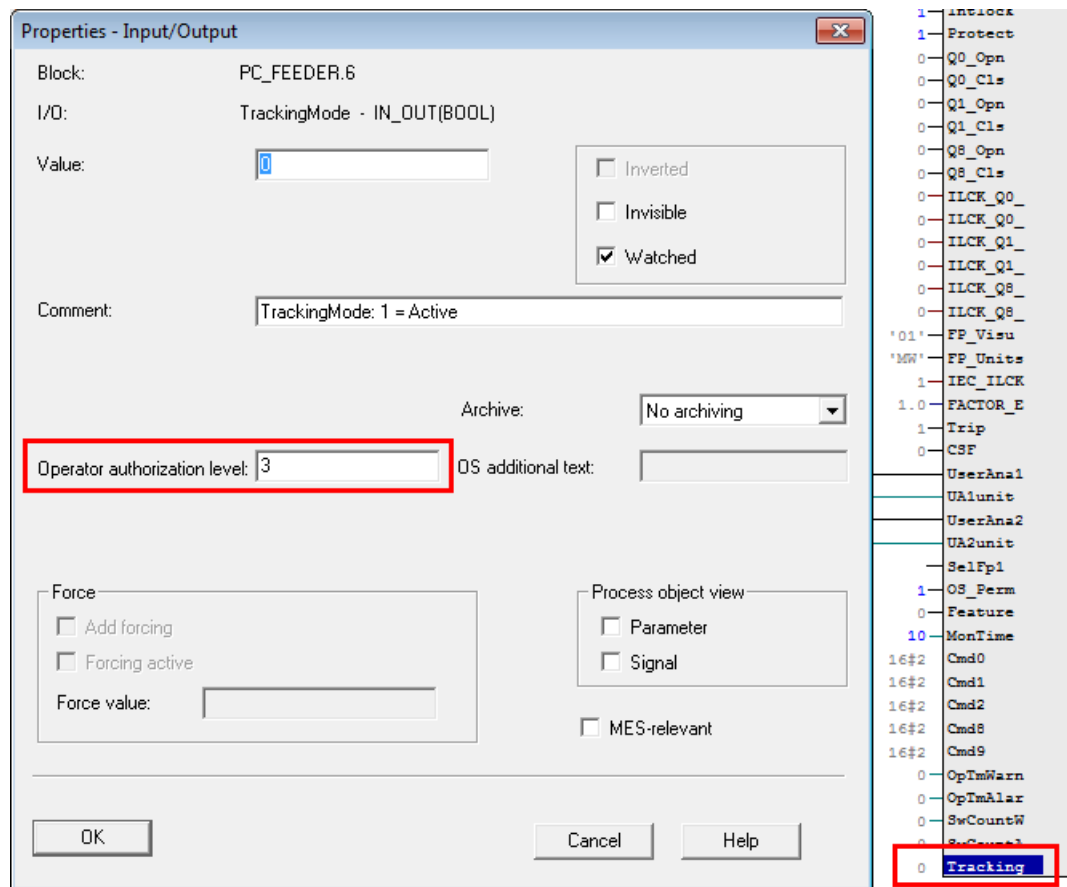


Figure 6-6 Setting the operator authorization level for tracking mode

6.1.10 Group error

This block provides the standard APL function "Outputting group errors".

The following parameters are taken into consideration when forming the group error:

- CSF
- trip

The group error output is *GrpErr.Value*.

6.1.11 Forming the group status for interlocks

This block contains the standard APL function "Forming the group status for interlock information" (see APL documentation).

6.1.12 Responses that can be configured with the "Feature" parameter

An overview of all the responses provided by the `Feature` parameter is contained in the PCS 7 APL documentation.

The following function is available for this block at the corresponding bits:

Bit	Function
0	Not used
1	OosLi can switch in "Out of service"
2-8	Not used
9	Not used
10	Not used
11	Not used
12	1 = Hide operating hours in the faceplate 0 (default) = Show operating hours in the faceplate
13	1 = Hide operating cycle hours in the faceplate 0 (default) = Show operating cycle hours in the faceplate
14	Not used
15-19	Activate alarm functionality with delay time for switch 0 = Alarm deactivated 1 = Alarm activated
20-30	Not used
31	Activate resetting of interlocks in Manual mode (see APL manual) 0 = (default) No resetting of interlocks in Manual mode 1 = Resetting of interlocks in Manual mode

6.1.13 Operator authorizations

This block includes the APL standard function "Operator authorizations".

The block provides the following authorizations for the `OS_Perm` parameter:

Bit	Function
0	1 = Operator may switch to "Automatic mode"
1	1 = Operator may switch to "Manual mode"
2	1 = Operator may switch to "Local operating mode"
3	1 = Operator may switch to "Out of service" mode
4	1 = Operator may switch to "Tracking mode"
5-6	Not used
7	1 = Operator may reset block
8-10	Not used
12	1 = Operator may activate release for maintenance

Bit	Function
13	1 = Button for Q0 is visible (PC_FEEDER) 1 = Button for Q01 is visible (PC_FEEDER_2CB)
14	1 = Button for Q1 is visible
15	1 = Button for Q2 is visible
16	1 = Button for Q8 is visible
17	1 = Button for Q9 is visible (PC_FEEDER) 1 = Button for Q02 is visible (PC_FEEDER_2CB)
18-31	Not used

For operator authorization of the buttons, the block also includes the APL standard function "Operator authorizations - User administration in the PCS 7 OS". This means the operator authorization level of the individual buttons can be set for specific instances with the object properties at the corresponding connectors at the block.

Depending on the assigned operator authorization level and the set operator authorizations in the OS user administration, the buttons can either be operated in the associated faceplate or they are locked for operation.

The following assignment applies:

- PC_FEEDER

Button	Block I/O
Q0 (Open / Close)	Q0_Opn / Q0_Cls
Q1 (Open / Close)	Q1_Opn / Q1_Cls
Q2 (Open / Close)	Q2_Opn / Q2_Cls
Q8 (Open / Close)	Q8_Opn / Q8_Cls
Q9 (Open / Close)	Q9_Opn / Q9_Cls
Tracking	TrackingMode

- PC_FEEDER_2CB

Button	Block I/O
Q01 (Open / Close)	Q01_Opn / Q01_Cls
Q02 (Open / Close)	Q02_Opn / Q02_Cls
Q1 (Open / Close)	Q1_Opn / Q1_Cls
Q2 (Open / Close)	Q2_Opn / Q2_Cls
Q8 (Open / Close)	Q8_Opn / Q8_Cls
Tracking	TrackingMode

6.1.14 Assignment of the status word for Status1 parameter

Status bit	Parameter
0	Occupied
1	BatchEn
2	Not used
3	OosAct.Value
4	OosLi.Value
5	0 = ManAct.Value 1 = AutoAct.Value
6	LocalAct.Value
7	0: Open padlock in block symbol 1: Closed padlock in block symbol
8	Not used (always true)
9	Not used
10	Not used
11	Not used
12	Not used
13	Not used
14	Not used
15	Not used
16	1 = Intlock is active
17	1 = Permit is active
18	1 = Protect is active
19	Trip.Value
20	Not used
21	Not used
22	Not used
23	"Interlock" button is activated
24	Not used
25	Not used
26	Bypass information from previous function block
27	Not used
28	Not used
29	Not used
30	"Enable" button is activated
31	"Protect" button is activated

6.1.15 Assignment of the status word for Status5 parameter

The Status5 parameter is used to show / hide values in the faceplate. The values are displayed depending on the quality code of the Station Gateway / Station Controller.

The values are displayed when the quality code is $\neq$ 0x00.

Status bit	Value	Description
0	I_A	Current phase 1
1	I_B	Current phase 2
2	I_C	Current phase 3
3	U_A	Voltage phase 1 – ground
4	U_B	Voltage phase 2 – ground
5	U_C	Voltage phase 3 – ground
6	U_AB	Voltage phase 1 – phase 2
7	U_BC	Voltage phase 2 – phase 3
8	U_CA	Voltage phase 3 – phase 1
9	U	Voltage
10	P	Power
11	Q	Reactive power
12	S	Apparent power
13	F	Frequency
14	PF	Cosine Phi
15	WpForward	Power counter P forwards
15	WqForward	Power counter Q forwards
15	WpReverse	Power counter P reverse
15	WqReverse	Power counter P reverse
19	Optm	Operating hours
20	SwCount	Operating cycles
21	-	Not used
22	-	Not used
23	-	Not used
24	-	Not used
25	-	Not used
26	-	Not used
27	-	Not used
28	-	Not used
29	-	Not used
30	-	Not used
31	-	Not used

6.1.16 Message response

6.1.16.1 Alarm delay with invalid switch positions

You can monitor the states of the switches with the PC_FEEDER block. In this case an alarm message is output when the status of a switch does not change at least for the time interval of a specified delay time.

To do so, activate the alarm functionality for the switches to be monitored at the corresponding bits of the `Feature` input parameter.

Set the delay time between reaching the invalid switch position and output of the alarm message at the `MonTime` input parameter. If the switch position changes during the specified delay time to a valid state, an alarm message is not output. If the switch changes to another invalid state while the timer is running, the timer is reset to the delay time.

6.1.17 Messages

Message instance	Message identifier	Message class	Event
MsgEvId_1	SIG_1	Alarm - high	\$\$BlockComment\$\$ trip Ip
	SIG_2	Warning - high	\$\$BlockComment\$\$ excitation Ip
	SIG_3	Alarm - high	\$\$BlockComment\$\$ trip 67-1 rectified overcurrent stage I >
	SIG_4	Warning - high	\$\$BlockComment\$\$ excitation 67-1 rectified overcurrent stage I >
	SIG_5	Alarm - high	\$\$BlockComment\$\$ trip 67-2 rectified overcurrent stage I >>
	SIG_6	Warning - high	\$\$BlockComment\$\$ excitation 67-2 rectified overcurrent stage I >>
	SIG_7	Alarm - high	\$\$BlockComment\$\$ trip 67N-1 rectified overcurrent stage IE >
	SIG_8	Warning - high	\$\$BlockComment\$\$ excitation 67N-1 rectified overcurrent stage IE >
MsgEvId_2	SIG_1	Alarm - high	\$\$BlockComment\$\$ trip 67N-2 rectified overcurrent stage IE >>
	SIG_2	Warning - high	\$\$BlockComment\$\$ excitation 67N-2 rectified overcurrent stage IE >>
	SIG_3	Alarm - high	\$\$BlockComment\$\$ trip 50-3 overcurrent stage
	SIG_4	Warning - high	\$\$BlockComment\$\$ excitation 50-3 overcurrent stage
	SIG_5	Alarm - high	\$\$BlockComment\$\$ trip 51N overcurrent stage
	SIG_6	Warning - high	\$\$BlockComment\$\$ excitation 51N overcurrent stage
	SIG_7	Alarm - high	\$\$BlockComment\$\$ trip 67-TOC overcurrent stage
	SIG_8	Warning - high	\$\$BlockComment\$\$ excitation 67-TOC overcurrent stage

Message instance	Message identifier	Message class	Event
MsgEvId_3	SIG_1	Alarm - high	\$\$BlockComment\$\$ trip 67N-TOC overcurrent stage
	SIG_2	Warning - high	\$\$BlockComment\$\$ excitation 67N-TOC overcurrent stage
	SIG_3	Alarm - high	\$\$BlockComment\$\$ trip 46-TOC overcurrent stage
	SIG_4	Warning - high	\$\$BlockComment\$\$ excitation 46-TOC overcurrent stage
	SIG_5	Alarm - high	\$\$BlockComment\$\$ trip 50-1 overcurrent stage I >
	SIG_6	Warning - high	\$\$BlockComment\$\$ excitation 50-1 overcurrent stage I >
	SIG_7	Alarm - high	\$\$BlockComment\$\$ trip 50-2 overcurrent stage I >>
	SIG_8	Warning - high	\$\$BlockComment\$\$ excitation 50-2 overcurrent stage I >>
MsgEvId_4	SIG_1	Alarm - high	\$\$BlockComment\$\$ trip 50N-1 overcurrent stage IE >
	SIG_2	Warning - high	\$\$BlockComment\$\$ excitation 50N-1 overcurrent stage IE >
	SIG_3	Alarm - high	\$\$BlockComment\$\$ trip 50N-2 overcurrent stage IE >>
	SIG_4	Warning - high	\$\$BlockComment\$\$ excitation 50N-2 overcurrent stage IE >>
	SIG_5	Warning - high	\$\$BlockComment\$\$ Excitation device
	SIG_6	Alarm - high	\$\$BlockComment\$\$ Tripping disconnector fault T1
	SIG_7	Alarm - high	\$\$BlockComment\$\$ trip disconnector fault T2
	SIG_8	Warning - high	\$\$BlockComment\$\$ Excitation disconnector fault
MsgEvId_5	SIG_1	Alarm - high	\$\$BlockComment\$\$ General tripping device
	SIG_2	Alarm - high	\$\$BlockComment\$\$ trip differential protection IDIFF>
	SIG_3	Warning - high	\$\$BlockComment\$\$ excitation differential protection IDIFF>
	SIG_4	Alarm - high	\$\$BlockComment\$\$ trip differential protection IDIFF>>
	SIG_5	Warning - high	\$\$BlockComment\$\$ excitation differential protection IDIFF>>
	SIG_6	Warning - high	\$\$BlockComment\$\$ excitation impedance protection Z1
	SIG_7	Warning - high	\$\$BlockComment\$\$ excitation impedance protection Z2
	SIG_8	Warning - high	\$\$BlockComment\$\$ excitation impedance protection Z1B

Message instance	Message identifier	Message class	Event
MsgEvld_6	SIG_1	Warning - high	\$\$BlockComment\$\$ general excitation impedance protection
	SIG_2	Alarm - high	\$\$BlockComment\$\$ trip impedance protection Z1
	SIG_3	Alarm - high	\$\$BlockComment\$\$ trip impedance protection Z2
	SIG_4	Alarm - high	\$\$BlockComment\$\$ trip impedance protection Z1B
	SIG_5	Alarm - high	\$\$BlockComment\$\$ general trip impedance protection
	SIG_6	Alarm - high	\$\$BlockComment\$\$ trip underexcitation class 1
	SIG_7	Alarm - high	\$\$BlockComment\$\$ trip underexcitation class 2
	SIG_8	Alarm - high	\$\$BlockComment\$\$ trip underexcitation class 3
MsgEvld_7	SIG_1	Alarm - high	\$\$BlockComment\$\$ trip overvoltage protection U>
	SIG_2	Warning - high	\$\$BlockComment\$\$ excitation overvoltage protection U>
	SIG_3	Alarm - high	\$\$BlockComment\$\$ trip overvoltage protection U>>
	SIG_4	Warning - high	\$\$BlockComment\$\$ excitation overvoltage protection U>>
	SIG_5	Alarm - high	\$\$BlockComment\$\$ trip frequency 1 below limit
	SIG_6	Warning - high	\$\$BlockComment\$\$ excitation frequency 1 below limit
	SIG_7	Alarm - high	\$\$BlockComment\$\$ trip frequency 2 below limit
	SIG_8	Warning - high	\$\$BlockComment\$\$ excitation frequency 2 below limit
MsgEvld_8	SIG_1	Alarm - high	\$\$BlockComment\$\$ trip undervoltage protection U<
	SIG_2	Warning - high	\$\$BlockComment\$\$ excitation undervoltage protection U<
	SIG_3	Alarm - high	\$\$BlockComment\$\$ trip undervoltage protection U<<
	SIG_4	Warning - high	\$\$BlockComment\$\$ excitation undervoltage protection U<<
	SIG_5	Alarm - high	\$\$BlockComment\$\$ trip cut-in protection
	SIG_6	Warning - high	\$\$BlockComment\$\$ excitation cut-in protection
	SIG_7	Alarm - high	\$\$BlockComment\$\$ trip reverse power protection
	SIG_8	Alarm - high	\$\$BlockComment\$\$ Tripping coupling 1

Message instance	Message identifier	Message class	Event
MsgEvId_9	SIG_1	Warning - high	\$\$BlockComment\$\$ Excitation coupling 1
	SIG_2	Status message - AS	\$\$BlockComment\$\$ Parameter set A active
	SIG_3	Status message - AS	\$\$BlockComment\$\$ Parameter set B active
	SIG_4	Status message - AS	\$\$BlockComment\$\$ Parameter set C active
	SIG_5	Status message - AS	\$\$BlockComment\$\$ Parameter set D active
	SIG_6	Alarm - high	\$\$BlockComment\$\$ Fault group message
	SIG_7	Warning - high	\$\$BlockComment\$\$ Local mode
	SIG_8	Warning - high	\$\$BlockComment\$\$ Warning group message
MsgEvId_10	SIG_1	Status message - AS	\$\$BlockComment\$\$ Interlock: Enable Q0 close
	SIG_2	Status message - AS	\$\$BlockComment\$\$ Interlock: Enable Q0 open
	SIG_3	Status message - AS	\$\$BlockComment\$\$ Interlock: Enable Q1 close
	SIG_4	Status message - AS	\$\$BlockComment\$\$ Interlock: Enable Q1 open
	SIG_5	Status message - AS	\$\$BlockComment\$\$ Interlock: Enable Q2 close
	SIG_6	Status message - AS	\$\$BlockComment\$\$ Interlock: Enable Q2 open
	SIG_7	Status message - AS	\$\$BlockComment\$\$ Interlock: Enable Q8 close
	SIG_8	Status message - AS	\$\$BlockComment\$\$ Interlock: Enable Q8 open
MsgEvId_11	SIG_1	Status message - AS	\$\$BlockComment\$\$ Interlock: Enable Q9 close
	SIG_2	Status message - AS	\$\$BlockComment\$\$ Interlock: Enable Q9 open
	SIG_3	Alarm - high	\$\$BlockComment\$\$ IEC Alarm 1
	SIG_4	Alarm - high	\$\$BlockComment\$\$ IEC Alarm 2
	SIG_5	Alarm - high	\$\$BlockComment\$\$ IEC Alarm 3
	SIG_6	Alarm - high	\$\$BlockComment\$\$ IEC Alarm 4
	SIG_7	Alarm - high	\$\$BlockComment\$\$ IEC Alarm 5
	SIG_8	Alarm - high	\$\$BlockComment\$\$ IEC Alarm 6

Message instance	Message identifier	Message class	Event
MsgEvld_12	SIG_1	Alarm - high	\$\$BlockComment\$\$ IEC Alarm 7
	SIG_2	Alarm - high	\$\$BlockComment\$\$ IEC Alarm 8
	SIG_3	Alarm - high	\$\$BlockComment\$\$ IEC Alarm 9
	SIG_4	Alarm - high	\$\$BlockComment\$\$ IEC Alarm 10
	SIG_5	Alarm - high	\$\$BlockComment\$\$ IEC Alarm 11
	SIG_6	Alarm - high	\$\$BlockComment\$\$ IEC Alarm 12
	SIG_7	Alarm - high	\$\$BlockComment\$\$ IEC Alarm 13
	SIG_8	Alarm - high	\$\$BlockComment\$\$ IEC Alarm 14
MsgEvld_d_1	SIG_1	Status message - AS	\$\$BlockComment\$\$ Circuit breaker Q0 closed
	SIG_2	Status message - AS	\$\$BlockComment\$\$ Circuit breaker Q0 open
	SIG_3	Status message - AS	\$\$BlockComment\$\$ Disconnecter Q1 closed
	SIG_4	Status message - AS	\$\$BlockComment\$\$ Disconnecter Q1 open
	SIG_5	Status message - AS	\$\$BlockComment\$\$ Switch Q2 closed
	SIG_6	Status message - AS	\$\$BlockComment\$\$ Switch Q2 open
	SIG_7	Status message - AS	\$\$BlockComment\$\$ Grounding Q8 closed
	SIG_8	Status message - AS	\$\$BlockComment\$\$ Grounding Q8 open
MsgEvld_d_2	SIG_1	Status message - AS	\$\$BlockComment\$\$ Switch Q9 closed
	SIG_2	Status message - AS	\$\$BlockComment\$\$ Switch Q9 open
	SIG_3	Status message - AS	\$\$BlockComment\$\$ Switch 1 closed
	SIG_4	Status message - AS	\$\$BlockComment\$\$ Switch 1 open
	SIG_5	Status message - AS	\$\$BlockComment\$\$ Switch 2 closed
	SIG_6	Status message - AS	\$\$BlockComment\$\$ Switch 2 open
	SIG_7	Status message - AS	\$\$BlockComment\$\$ Switch 3 closed
	SIG_8	Status message - AS	\$\$BlockComment\$\$ Switch 3 open

Message instance	Message identifier	Message class	Event
MsgEvid_device	SIG_1	AS process control message - error	\$\$BlockComment\$\$ Error device IED name:@1%s@ -> IP address: @2%s@
	SIG_2	AS control system message - fault	\$\$BlockComment\$\$ Redundancy failure
	SIG_3	AS process control message - error	\$\$BlockComment\$\$ PROFIBUS mapping alarm
	SIG_4	PLC Process Control Message - Error	\$\$BlockComment\$\$ Message buffer overflow in SIPROTEC
	SIG_5	Alarm - high	
	SIG_6	Alarm - high	
	SIG_7	Alarm - high	
	SIG_8	Alarm - high	
MsgEvidlocal1	SIG_1	Alarm - high	\$\$BlockComment\$\$ Error feedback Q0
	SIG_2	Alarm - high	\$\$BlockComment\$\$ Error feedback Q1
	SIG_3	Alarm - high	\$\$BlockComment\$\$ Error feedback Q2
	SIG_4	Alarm - high	\$\$BlockComment\$\$ Error feedback Q8
	SIG_5	Alarm - high	\$\$BlockComment\$\$ Error feedback Q9
	SIG_6	AS process control message - error	\$\$BlockComment\$\$ Connection fault
	SIG_7	Alarm - high	\$\$BlockComment\$\$ External trip signal
	SIG_8	AS process control message - error	\$\$BlockComment\$\$ AS process control message - external error
MsgEvidlocal2	SIG_1	Warning - high	\$\$BlockComment\$\$ Maintenance: Operating cycle warning (@3%d@)
	SIG_2	Alarm - high	\$\$BlockComment\$\$ Maintenance: Operating cycle alarm (@3%d@)
	SIG_3	Warning - high	\$\$BlockComment\$\$ Maintenance: Operating time warning (@4%d@)
	SIG_4	Alarm - high	\$\$BlockComment\$\$ Maintenance: Operating time alarm (@4%d@)
	SIG_5	Alarm - high	\$\$BlockComment\$\$ External message 1
	SIG_6	Alarm - high	\$\$BlockComment\$\$ External message 2
	SIG_7	Alarm - high	\$\$BlockComment\$\$ External message 3
	SIG_8	Alarm - high	\$\$BlockComment\$\$ External message 4

Message instance	Message identifier	Message class	Event
MsgEvldlocal3	SIG_1	Alarm - high	\$\$BlockComment\$\$ Q0 switch position invalid
	SIG_2	Alarm - high	\$\$BlockComment\$\$ Q1 switch position invalid
	SIG_3	Alarm - high	\$\$BlockComment\$\$ Q2 switch position invalid
	SIG_4	Alarm - high	\$\$BlockComment\$\$ Q8 switch position invalid
	SIG_5	Alarm - high	\$\$BlockComment\$\$ Q9 switch position invalid
	SIG_6	Alarm - high	
	SIG_7	Alarm - high	
	SIG_8	Alarm - high	

6.1.18 I/O list

6.1.18.1 Inputs

I/O name	Description	Value	Data type
Rcv_Data	Connect with SG_/SC_FEEDER		STRUCT
ModLiOp	1 = Interconnect/Auto, 0 = Manual: Input for auto / manual commands		STRUCT
ModLiOp.Value		0	BOOL
ModLiOp.ST		16#80	BYTE
AutModOp	1 = Automatic mode: Automatic mode by operator	0	BOOL
ManModOp	1 = Manual mode: Manual mode by operator	1	BOOL
AutModLi	1 = Automatic mode: Automatic mode via link or SFC		STRUCT
AutModLi.Value		0	BOOL
AutModLi.ST		16#80	BYTE
ManModLi	1 = Manual mode: Manual mode via link or SFC		STRUCT
ManModLi.Value		0	BOOL
ManModLi.ST		16#80	BYTE
OosOp	1 = Out of service: Out of service mode via operator	0	BOOL
OosLi	1 = Out of service: Out of service mode via field signal		STRUCT
OosLi.Value		0	BOOL
OosLi.ST		16#80	BYTE
RstOp	Reset signal operator	0	BOOL
RstLi	Linked reset signal		STRUCT

I/O name	Description	Value	Data type
RstLi.Value	Value	1	BOOL
RstLi.ST	Signal state	16#80	BYTE
BB_Color	Connection to PC_BUSBAR	16#0	DWORD
Permit	1=Authorization is OK, 0=Activation of the motor is not permitted		STRUCT
Permit.Value	Value		BOOL
Permit.ST	Signal state		BYTE
Perm_En	1=Authorization activated, 0=Authorization deactivated	1	BOOL
Intlock	0=Motor is locked, 1=Motor is not locked		STRUCT
Intlock.Value	Value		BOOL
Intlock.ST	Signal state		BYTE
Intl_En	1=Interlock activated, 0=Interlock deactivated	1	BOOL
Protect	0=Protection is active, 1=Protection is not active		STRUCT
Protect.Value	Value		BOOL
Protect.ST	Signal state		BYTE
Prot_En	1=Protection activated, 0=Protection deactivated	1	BOOL
Q0_Opn	Rising edge: Command Q0 off		STRUCT
Q0_Opn.Value		1	BOOL
Q0_Opn.ST		16#80	BYTE
Q0_Cls	Rising edge: Command Q0 on		STRUCT
Q0_Cls.Value		1	BOOL
Q0_Cls.ST		16#80	BYTE
Q1_Opn	Rising edge: Command Q1 off		STRUCT
Q1_Opn.Value		0	BOOL
Q1_Opn.ST		16#80	BYTE
Q1_Cls	Rising edge: Command Q1 on		STRUCT
Q1_Cls.Value		0	BOOL
Q1_Cls.ST		16#80	BYTE
Q2_Opn	Rising edge: Command Q2 off		STRUCT
Q2_Opn.Value		0	BOOL
Q2_Opn.ST		16#80	BYTE
Q2_Cls	Rising edge: Command Q2 on		STRUCT
Q2_Cls.Value		0	BOOL
Q2_Cls.ST		16#80	BYTE
Q8_Opn	Rising edge: Command Q8 off		STRUCT
Q8_Opn.Value		0	BOOL
Q8_Opn.ST		16#80	BYTE
Q8_Cls	Rising edge: Command Q8 on		STRUCT
Q8_Cls.Value		0	BOOL
Q8_Cls.ST		16#80	BYTE
Q9_Opn	Rising edge: Command Q9 off		STRUCT

I/O name	Description	Value	Data type
Q9_Opn.Value		0	BOOL
Q9_Opn.ST		16#80	BYTE
Q9_Cls	Rising edge: Command Q9 on		STRUCT
Q9_Cls.Value		0	BOOL
Q9_Cls.ST		16#80	BYTE
CMD_sp1_Opn	Rising edge: Free command 1 off		STRUCT
CMD_sp1_Opn.Value		0	BOOL
CMD_sp1_Opn.ST		16#80	BYTE
CMD_sp1_Cls	Rising edge: Free command 1 on		STRUCT
CMD_sp1_Cls.Value		0	BOOL
CMD_sp1_Cls.ST		16#80	BYTE
CMD_sp2_Opn	Rising edge: Free command 2 off		STRUCT
CMD_sp2_Opn.Value		0	BOOL
CMD_sp2_Opn.ST		16#80	BYTE
CMD_sp2_Cls	Rising edge: Free command 2 on		STRUCT
CMD_sp2_Cls.Value		0	BOOL
CMD_sp2_Cls.ST		16#80	BYTE
CMD_sp3_Opn	Rising edge: Free command 3 off		STRUCT
CMD_sp3_Opn.Value		0	BOOL
CMD_sp3_Opn.ST		16#80	BYTE
CMD_sp3_Cls	Rising edge: Free command 3 on		STRUCT
CMD_sp3_Cls.Value		0	BOOL
CMD_sp3_Cls.ST		16#80	BYTE
ILCK_Q0_Cls	Interlock: Interlock Q0 close	0	BOOL
ILCK_Q0_Opn	Interlock: Interlock Q0 open	0	BOOL
ILCK_Q1_Cls	Interlock: Interlock Q1 close	0	BOOL
ILCK_Q1_Opn	Interlock: Interlock Q1 open	0	BOOL
ILCK_Q2_Cls	Interlock: Interlock Q2 close	0	BOOL
ILCK_Q2_Opn	Interlock: Interlock Q2 open	0	BOOL
ILCK_Q8_Cls	Interlock: Interlock Q8 close	0	BOOL
ILCK_Q8_Opn	Interlock: Interlock Q8 open	0	BOOL
ILCK_Q9_Cls	Interlock: Interlock Q9 close	0	BOOL
ILCK_Q9_Opn	Interlock: Interlock Q9 open	0	BOOL
FP_Visu	Structure for configuration of the faceplate, see also Configuration of the faceplate and the units (Page 19)		STRUCT
FP_Units	Structure for configuration of the units in the faceplate, see also Configuration of the faceplate and the units (Page 19)		STRUCT
IEC_ILCK_EN	Interlock signals of IEC activated	0	BOOL
FACTOR_ENERGY	Factor for displaying energy values	1.0	REAL
TRIP	1= State, 0= Trip signal		STRUCT
TRIP.Value	Value	1	BOOL
TRIP.ST	Signal state	16#80	BYTE

I/O name	Description	Value	Data type
MsgLock			STRUCT
MsgLock.Value		0	BOOL
MsgLock.ST		16#80	BYTE
SampleTime	Sampling time [s]	0.1	REAL
MsgEvid_1	Message event ID	16#00000002	DWORD
MsgEvid_2	Message event ID	16#0000000F	DWORD
MsgEvid_3	Message event ID	16#00000010	DWORD
MsgEvid_4	Message event ID	16#00000011	DWORD
MsgEvid_5	Message event ID	16#00000012	DWORD
MsgEvid_6	Message event ID	16#00000013	DWORD
MsgEvid_7	Message event ID	16#00000014	DWORD
MsgEvid_8	Message event ID	16#00000018	DWORD
MsgEvid_9	Message event ID	16#0000001A	DWORD
MsgEvid_10	Message event ID	16#0000001B	DWORD
MsgEvid_11	Message event ID	16#0000001C	DWORD
MsgEvid_12	Message event ID	16#00000022	DWORD
MsgEvidlocal1	Message event ID	16#00000020	DWORD
MsgEvidlocal2	Message event ID	16#00000021	DWORD
MsgEvidlocal3	Message event ID	16#00000023	DWORD
MsgEvid_d_1	Message event ID	16#0000001D	DWORD
MsgEvid_d_2	Message event ID	16#0000001E	DWORD
MsgEvid_device	Message event ID	16#0000001F	DWORD
BatchEn	Activate remote operation of controller to batch recipe	0	BOOL
BatchID	Current batch ID (number)	16#00000000	DWORD
BatchName	Current batch name	"	STRING[32]
StepNo	Batch step number	16#00000000	DWORD
Occupied	Occupied by batch	0	BOOL
CSF			STRUCT
CSF.Value		0	BOOL
CSF.ST		16#80	BYTE
ExtMsg1	External message 1		STRUCT
ExtMsg1.Value	Value	0	BOOL
ExtMsg1.ST	Signal state	16#80	BYTE
ExtMsg2	External message 2		STRUCT
ExtMsg2.Value	Value	0	BOOL
ExtMsg2.ST	Signal state	16#80	BYTE
ExtMsg3	External message 3		STRUCT
ExtMsg3.Value	Value	0	BOOL
ExtMsg3.ST	Signal state	16#80	BYTE
ExtMsg4	External message 4		STRUCT
ExtMsg4.Value	Value	0	BOOL
ExtMsg4.ST	Signal state	16#80	BYTE

I/O name	Description	Value	Data type
UserAna1	User analog input 1		STRUCT
UserAna1.Value	Value	0.0	REAL
UserAna1.ST	Signal state	16#FF	BYTE
UA1unit	Unit of measurement user analog input 1	0	INT
UserAna2	User analog input 2		STRUCT
UserAna2.Value	Value	0.0	REAL
UserAna2.ST	Signal state	16#FF	BYTE
UA2unit	Unit of measurement user analog input 2	0	INT
UserAna3	User analog input 3		STRUCT
UserAna3.Value	Value	0.0	REAL
UserAna3.ST	Signal state	16#FF	BYTE
UA3unit	Unit of measurement user analog input 3	0	INT
UserAna4	User analog input 4		STRUCT
UserAna4.Value	Value	0.0	REAL
UserAna4.ST	Signal state	16#FF	BYTE
UA4unit	Unit of measurement user analog input 4	0	INT
UserStatus	User status bits	16#0	BYTE
SelFp1	Select faceplate 1		ANY
SelFpllckQ0Cls	Select Faceplate Interlock Q0 closed		ANY
SelFpllckQ0Opn	Select Faceplate Interlock Q0 closed		ANY
SelFpllckQ1Cls	Select Faceplate Interlock Q1 closed		ANY
SelFpllckQ1Opn	Select Faceplate Interlock Q1 closed		ANY
SelFpllckQ2Cls	Select Faceplate Interlock Q2 closed		ANY
SelFpllckQ2Opn	Select Faceplate Interlock Q2 closed		ANY
SelFpllckQ8Cls	Select Faceplate Interlock Q8 closed		ANY
SelFpllckQ8Opn	Select Faceplate Interlock Q8 closed		ANY
SelFpllckQ9Cls	Select Faceplate Interlock Q9 closed		ANY
SelFpllckQ9Opn	Select Faceplate Interlock Q9 closed		ANY
OS_Perm	Operator authorizations		STRUCT
OS_Perm.Bit0	1 = Operator may switch to Automatic mode	1	BOOL
OS_Perm.Bit1	1 = Operator may switch to Manual mode	1	BOOL
OS_Perm.Bit2	1= Operator may switch to Local operating mode	1	BOOL
OS_Perm.Bit3	1 = Operator may switch to "Out of service" mode	1	BOOL
OS_Perm.Bit4	1 = Operator may switch to "Tracking mode"	1	BOOL
OS_Perm.Bit5		1	BOOL
OS_Perm.Bit6		1	BOOL
OS_Perm.Bit7	1 = Operator can reset block	1	BOOL
OS_Perm.Bit8		1	BOOL
OS_Perm.Bit9		1	BOOL
OS_Perm.Bit10		1	BOOL
OS_Perm.Bit11	1 = Operator can activate function simulation	1	BOOL

I/O name	Description	Value	Data type
OS_Perm.Bit12	1 = Operator can enable release for maintenance	1	BOOL
OS_Perm.Bit13	1 = Button for Q0 is visible	1	BOOL
OS_Perm.Bit14	1 = Button for Q1 is visible	1	BOOL
OS_Perm.Bit15	1 = Button for Q2 is visible	1	BOOL
OS_Perm.Bit16	1 = Button for Q8 is visible	1	BOOL
OS_Perm.Bit17	1 = Button for Q9 is visible	1	BOOL
OS_Perm.Bit18		1	BOOL
OS_Perm.Bit19		1	BOOL
OS_Perm.Bit20		1	BOOL
OS_Perm.Bit21		1	BOOL
OS_Perm.Bit22		1	BOOL
OS_Perm.Bit23		1	BOOL
OS_Perm.Bit24		1	BOOL
OS_Perm.Bit25		1	BOOL
OS_Perm.Bit26		1	BOOL
OS_Perm.Bit27		1	BOOL
OS_Perm.Bit28		1	BOOL
OS_Perm.Bit29		1	BOOL
OS_Perm.Bit30		1	BOOL
OS_Perm.Bit31		1	BOOL
OpSt_In	Input parameter for local operator authorization, to be interconnected with the <code>Out</code> output parameter of the upstream block <code>OpStations</code>	16#00	DWORD
Feature	State of different functions		STRUCT
Feature.Bit0	Reserved	0	BOOL
Feature.Bit1	1 = OosLi may switch to "Out of service" mode	0	BOOL
Feature.Bit2	Reserved	0	BOOL
Feature.Bit3	Reserved	0	BOOL
Feature.Bit4	Reserved	0	BOOL
Feature.Bit5	Reserved	0	BOOL
Feature.Bit6	Reserved	0	BOOL
Feature.Bit7	Reserved	0	BOOL
Feature.Bit8	Reserved	0	BOOL
Feature.Bit9	Reserved	0	BOOL
Feature.Bit10	Reserved	0	BOOL
Feature.Bit11	Reserved	0	BOOL
Feature.Bit12	Hide OpTm	0	BOOL
Feature.Bit13	Hide SwCount	0	BOOL
Feature.Bit14	Reserved	0	BOOL
Feature.Bit15	1 = Q0 status alarm active	0	BOOL
Feature.Bit16	1 = Q1 status alarm active	0	BOOL
Feature.Bit17	1 = Q2 status alarm active	0	BOOL

I/O name	Description	Value	Data type
Feature.Bit18	1 = Q8 status alarm active	0	BOOL
Feature.Bit19	1 = Q9 status alarm active	0	BOOL
Feature.Bit20	1 = Dynamic active power direction enabled:	0	BOOL
Feature.Bit21	1 = Dynamic active power direction: negative power up; 0 = negative power down	0	BOOL
Feature.Bit22	Reserved	0	BOOL
Feature.Bit23	Reserved	0	BOOL
Feature.Bit24	1 = Local authorization enabled	0	BOOL
Feature.Bit25	1 = Suppression of all messages if MsgLock = 1	0	BOOL
Feature.Bit26	Reserved	0	BOOL
Feature.Bit27	Reserved	0	BOOL
Feature.Bit28	Reserved	0	BOOL
Feature.Bit29	Reserved	0	BOOL
Feature.Bit30	Reserved	0	BOOL
Feature.Bit31	1 = Resetting of interlocks in Manual mode	0	BOOL
MonTime	Time in seconds until Timeout (for control switch)	20	INT

6.1.18.2 Inputs / outputs

I/O name	Description	Value	Data type
Cmd0	Control position circuit breaker Q0	16#02	BYTE
Cmd1	Control position disconnector Q1	16#02	BYTE
Cmd2	Control position switch Q2	16#02	BYTE
Cmd8	Control position ground Q8	16#02	BYTE
Cmd9	Control position switch Q9	16#02	BYTE
Cmdsp1	Control position free command 1	16#02	BYTE
Cmdsp2	Control position free command 2	16#02	BYTE
Cmdsp3	Control position free command 3	16#02	BYTE
OpTmWarnLimit	Warning limit operating hours	0	DINT
OpTmAlarmLimit	Alarm limit operating hours	0	DINT
SwCountWarnLimit	Warning limit operating cycles	0	DINT
SwCountAlarmLimit	Alarm limit operating cycles	0	DINT

6.1.18.3 Outputs

I/O name	Description	Value	Data type
ENO		0	BOOL
Snd_Data	Connect with SG_/SC_FEEDER		STRUCT
QCONNERR		1	BOOL

I/O name	Description	Value	Data type
F	Frequency		STRUCT
F.Value		0.0	REAL
F.ST		16#FF	BYTE
I_A	Current Phase A		STRUCT
I_A.Value		0.0	REAL
I_A.ST		16#FF	BYTE
I_B	Current Phase B		STRUCT
I_B.Value		0.0	REAL
I_B.ST		16#FF	BYTE
I_C	Current Phase C		STRUCT
I_C.Value		0.0	REAL
I_C.ST		16#FF	BYTE
P	Power		STRUCT
P.Value		0.0	REAL
P.ST		16#FF	BYTE
PF	Cos Phi - power factor		STRUCT
PF.Value		0.0	REAL
PF.ST		16#FF	BYTE
Q	Reactive power		STRUCT
Q.Value		0.0	REAL
Q.ST		16#FF	BYTE
S	Apparent power		STRUCT
S.Value		0.0	REAL
S.ST		16#FF	BYTE
U	Neutral voltage		STRUCT
U.Value		0.0	REAL
U.ST		16#FF	BYTE
U_A	Voltage Phase A		STRUCT
U_A.Value		0.0	REAL
U_A.ST		16#FF	BYTE
U_AB	Voltage A-B		STRUCT
U_AB.Value		0.0	REAL
U_AB.ST		16#FF	BYTE
U_B	Voltage Phase B		STRUCT
U_B.Value		0.0	REAL
U_B.ST		16#FF	BYTE
U_BC	Voltage B-C		STRUCT
U_BC.Value		0.0	REAL
U_BC.ST		16#FF	BYTE
U_C	Voltage Phase C		STRUCT
U_C.Value		0.0	REAL
U_C.ST		16#FF	BYTE
U_CA	Voltage C-A		STRUCT

I/O name	Description	Value	Data type
U_CA.Value		0.0	REAL
U_CA.ST		16#FF	BYTE
REAL_sp1	Free REAL 1		STRUCT
REAL_sp1.Value		0.0	REAL
REAL_sp1.ST		16#FF	BYTE
REAL_sp2	Free REAL 2		STRUCT
REAL_sp2.Value		0.0	REAL
REAL_sp2.ST		16#FF	BYTE
REAL_sp3	Free REAL 3		STRUCT
REAL_sp3.Value		0.0	REAL
REAL_sp3.ST		16#FF	BYTE
REAL_sp4	Free REAL 4		STRUCT
REAL_sp4.Value		0.0	REAL
REAL_sp4.ST		16#FF	BYTE
REAL_sp5	Free REAL 5		STRUCT
REAL_sp5.Value		0.0	REAL
REAL_sp5.ST		16#FF	BYTE
Pos0	Position circuit breaker Q0	16#00000000	DWORD
Pos1	Position disconnecter Q1	16#00000000	DWORD
Pos2	Position switch Q2	16#00000000	DWORD
Pos8	Position ground Q8	16#00000000	DWORD
Pos9	Position switch Q9	16#00000000	DWORD
Pos_CMD_sp1	Position free command 1	16#00000000	DWORD
Pos_CMD_sp2	Position free command 2	16#00000000	DWORD
Pos_CMD_sp3	Position free command 3	16#00000000	DWORD
Beh	Behavior	0	INT
Dev_OK	Device OK	0	INT
GndFdir	Ground fault direction	0	INT
INT_sp1	Free INT 1		STRUCT
INT_sp1.Value		0	INT
INT_sp1.ST		16#FF	BYTE
INT_sp2	Free INT 2		STRUCT
INT_sp2.Value		0	INT
INT_sp2.ST		16#FF	BYTE
INT_sp3	Free INT 3		STRUCT
INT_sp3.Value		0	INT
INT_sp3.ST		16#FF	BYTE
INT_sp4	Free INT 4		STRUCT
INT_sp4.Value		0	INT
INT_sp4.ST		16#FF	BYTE
INT_sp5	Free INT 5		STRUCT
INT_sp5.Value		0	INT
INT_sp5.ST		16#FF	BYTE

I/O name	Description	Value	Data type
PTOC1_Op	Tripping 51 overcurrent stage		STRUCT
PTOC1_Op.Value		0	BOOL
PTOC1_Op.ST		16#FF	BYTE
PTOC1_Str	Excitation 51 overcurrent stage		STRUCT
PTOC1_Str.Value		0	BOOL
PTOC1_Str.ST		16#FF	BYTE
PTOC10_Op	Tripping 67-1 rectified overcurrent stage I >		STRUCT
PTOC10_Op.Value		0	BOOL
PTOC10_Op.ST		16#FF	BYTE
PTOC10_Str	Excitation 67-1 rectified overcurrent stage I >		STRUCT
PTOC10_Str.Value		0	BOOL
PTOC10_Str.ST		16#FF	BYTE
PTOC11_Op	Tripping 67-2 rectified overcurrent stage I >>		STRUCT
PTOC11_Op.Value		0	BOOL
PTOC11_Op.ST		16#FF	BYTE
PTOC11_Str	Excitation 67-2 rectified overcurrent stage I >>		STRUCT
PTOC11_Str.Value		0	BOOL
PTOC11_Str.ST		16#FF	BYTE
PTOC12_Op	Tripping 67N-1 rectified overcurrent stage IE >		STRUCT
PTOC12_Op.Value		0	BOOL
PTOC12_Op.ST		16#FF	BYTE
PTOC12_Str	Excitation 67N-1 rectified overcurrent stage IE >		STRUCT
PTOC12_Str.Value		0	BOOL
PTOC12_Str.ST		16#FF	BYTE
PTOC13_Op	Tripping 67N-2 rectified overcurrent stage IE >>		STRUCT
PTOC13_Op.Value		0	BOOL
PTOC13_Op.ST		16#FF	BYTE
PTOC13_Str	Excitation 67N-2 rectified overcurrent stage IE >>		STRUCT
PTOC13_Str.Value		0	BOOL
PTOC13_Str.ST		16#FF	BYTE
PTOC18_Op	Tripping 50-3 overcurrent stage		STRUCT
PTOC18_Op.Value		0	BOOL
PTOC18_Op.ST		16#FF	BYTE
PTOC18_Str	Excitation 50-3 overcurrent stage		STRUCT
PTOC18_Str.Value		0	BOOL
PTOC18_Str.ST		16#FF	BYTE
PTOC2_Op	Tripping 51N overcurrent stage		STRUCT
PTOC2_Op.Value		0	BOOL
PTOC2_Op.ST		16#FF	BYTE
PTOC2_Str	Excitation 51N overcurrent stage		STRUCT
PTOC2_Str.Value		0	BOOL
PTOC2_Str.ST		16#FF	BYTE

I/O name	Description	Value	Data type
PTOC3_Op	Tripping 67-TOC overcurrent stage		STRUCT
PTOC3_Op.Value		0	BOOL
PTOC3_Op.ST		16#FF	BYTE
PTOC3_Str	Excitation 67-TOC overcurrent stage		STRUCT
PTOC3_Str.Value		0	BOOL
PTOC3_Str.ST		16#FF	BYTE
PTOC4_Op	Tripping 67N-TOC overcurrent stage		STRUCT
PTOC4_Op.Value		0	BOOL
PTOC4_Op.ST		16#FF	BYTE
PTOC4_Str	Excitation 67N-TOC overcurrent stage		STRUCT
PTOC4_Str.Value		0	BOOL
PTOC4_Str.ST		16#FF	BYTE
PTOC5_Op	Tripping 46-TOC overcurrent stage		STRUCT
PTOC5_Op.Value		0	BOOL
PTOC5_Op.ST		16#FF	BYTE
PTOC5_Str	Excitation 46-TOC overcurrent stage		STRUCT
PTOC5_Str.Value		0	BOOL
PTOC5_Str.ST		16#FF	BYTE
PTOC6_Op	Tripping 50-1 overcurrent stage I >		STRUCT
PTOC6_Op.Value		0	BOOL
PTOC6_Op.ST		16#FF	BYTE
PTOC6_Str	Excitation 50-1 overcurrent stage I >		STRUCT
PTOC6_Str.Value		0	BOOL
PTOC6_Str.ST		16#FF	BYTE
PTOC7_Op	Tripping 50-2 overcurrent stage I >>		STRUCT
PTOC7_Op.Value		0	BOOL
PTOC7_Op.ST		16#FF	BYTE
PTOC7_Str	Excitation 50-2 overcurrent stage I >>		STRUCT
PTOC7_Str.Value		0	BOOL
PTOC7_Str.ST		16#FF	BYTE
PTOC8_Op	Tripping 50N-1 overcurrent stage IE >		STRUCT
PTOC8_Op.Value		0	BOOL
PTOC8_Op.ST		16#FF	BYTE
PTOC8_Str	Excitation 50N-1 overcurrent stage IE >		STRUCT
PTOC8_Str.Value		0	BOOL
PTOC8_Str.ST		16#FF	BYTE
PTOC9_Op	Tripping 50N-2 overcurrent stage IE >>		STRUCT
PTOC9_Op.Value		0	BOOL
PTOC9_Op.ST		16#FF	BYTE
PTOC9_Str	Excitation 50N-2 overcurrent stage IE >>		STRUCT
PTOC9_Str.Value		0	BOOL
PTOC9_Str.ST		16#FF	BYTE
PU_Relay	Excitation combined signal		STRUCT

I/O name	Description	Value	Data type
PU_Relay.Value		0	BOOL
PU_Relay.ST		16#FF	BYTE
RBRF1_OpEx	Tripping disconnecter fault T1		STRUCT
RBRF1_OpEx.Value		0	BOOL
RBRF1_OpEx.ST		16#FF	BYTE
RBRF1_OpIn	Tripping disconnecter fault T2		STRUCT
RBRF1_OpIn.Value		0	BOOL
RBRF1_OpIn.ST		16#FF	BYTE
RBRF1_Str	Excitation disconnecter fault		STRUCT
RBRF1_Str.Value		0	BOOL
RBRF1_Str.ST		16#FF	BYTE
TR_Relay	Tripping combined signal		STRUCT
TR_Relay.Value		0	BOOL
TR_Relay.ST		16#FF	BYTE
PDIF1_Op	Tripping differential protection IDIFF>		STRUCT
PDIF1_Op.Value		0	BOOL
PDIF1_Op.ST		16#FF	BYTE
PDIF1_Str	Excitation differential protection IDIFF>		STRUCT
PDIF1_Str.Value		0	BOOL
PDIF1_Str.ST		16#FF	BYTE
PDIF2_Op	Tripping differential protection IDIFF>>		STRUCT
PDIF2_Op.Value		0	BOOL
PDIF2_Op.ST		16#FF	BYTE
PDIF2_Str	Excitation differential protection IDIFF>>		STRUCT
PDIF2_Str.Value		0	BOOL
PDIF2_Str.ST		16#FF	BYTE
PDIS1_Str	Excitation impedance protection Z1		STRUCT
PDIS1_Str.Value		0	BOOL
PDIS1_Str.ST		16#FF	BYTE
PDIS2_Str	Excitation impedance protection Z2		STRUCT
PDIS2_Str.Value		0	BOOL
PDIS2_Str.ST		16#FF	BYTE
PDIS3_Str	Excitation impedance protection Z1B		STRUCT
PDIS3_Str.Value		0	BOOL
PDIS3_Str.ST		16#FF	BYTE
PDIS4_Str	General excitation impedance protection		STRUCT
PDIS4_Str.Value		0	BOOL
PDIS4_Str.ST		16#FF	BYTE
PDIS1_Op	Tripping impedance protection Z1		STRUCT
PDIS1_Op.Value		0	BOOL
PDIS1_Op.ST		16#FF	BYTE
PDIS2_Op	Tripping impedance protection Z2		STRUCT
PDIS2_Op.Value		0	BOOL

I/O name	Description	Value	Data type
PDIS2_Op.ST		16#FF	BYTE
PDIS3_Op	Tripping impedance protection Z1B		STRUCT
PDIS3_Op.Value		0	BOOL
PDIS3_Op.ST		16#FF	BYTE
PDIS4_Op	General tripping impedance protection		STRUCT
PDIS4_Op.Value		0	BOOL
PDIS4_Op.ST		16#FF	BYTE
PDUP1_Op	Tripping under-excitation protection class 1		STRUCT
PDUP1_Op.Value		0	BOOL
PDUP1_Op.ST		16#FF	BYTE
PDUP2_Op	Tripping under-excitation protection class 2		STRUCT
PDUP2_Op.Value		0	BOOL
PDUP2_Op.ST		16#FF	BYTE
PDUP3_Op	Tripping under-excitation protection class 3		STRUCT
PDUP3_Op.Value		0	BOOL
PDUP3_Op.ST		16#FF	BYTE
PTOV1_Op	Tripping overvoltage protection U>		STRUCT
PTOV1_Op.Value		0	BOOL
PTOV1_Op.ST		16#FF	BYTE
PTOV1_Str	Excitation overvoltage protection U>		STRUCT
PTOV1_Str.Value		0	BOOL
PTOV1_Str.ST		16#FF	BYTE
PTOV2_Op	Tripping overvoltage protection U>>		STRUCT
PTOV2_Op.Value		0	BOOL
PTOV2_Op.ST		16#FF	BYTE
PTOV2_Str	Excitation overvoltage protection U>>		STRUCT
PTOV2_Str.Value		0	BOOL
PTOV2_Str.ST		16#FF	BYTE
PTUF1_Op	Tripping frequency 1 below limit		STRUCT
PTUF1_Op.Value		0	BOOL
PTUF1_Op.ST		16#FF	BYTE
PTUF1_Str	Excitation frequency 1 below limit		STRUCT
PTUF1_Str.Value		0	BOOL
PTUF1_Str.ST		16#FF	BYTE
PTUF2_Op	Tripping frequency 2 below limit		STRUCT
PTUF2_Op.Value		0	BOOL
PTUF2_Op.ST		16#FF	BYTE
PTUF2_Str	Excitation frequency 2 below limit		STRUCT
PTUF2_Str.Value		0	BOOL
PTUF2_Str.ST		16#FF	BYTE
PTUV1_Op	Tripping undervoltage protection U<		STRUCT
PTUV1_Op.Value		0	BOOL
PTUV1_Op.ST		16#FF	BYTE

I/O name	Description	Value	Data type
PTUV1_Str	Excitation undervoltage protection U<		STRUCT
PTUV1_Str.Value		0	BOOL
PTUV1_Str.ST		16#FF	BYTE
PTUV2_Op	Tripping undervoltage protection U<<		STRUCT
PTUV2_Op.Value		0	BOOL
PTUV2_Op.ST		16#FF	BYTE
PTUV2_Str	Excitation undervoltage protection U<<		STRUCT
PTUV2_Str.Value		0	BOOL
PTUV2_Str.ST		16#FF	BYTE
PVOC2_Op	Tripping cut-in protection		STRUCT
PVOC2_Op.Value		0	BOOL
PVOC2_Op.ST		16#FF	BYTE
PVOC2_Str	Excitation cut-in protection		STRUCT
PVOC2_Str.Value		0	BOOL
PVOC2_Str.ST		16#FF	BYTE
PDOP1_Op	Tripping reverse power protection		STRUCT
PDOP1_Op.Value		0	BOOL
PDOP1_Op.ST		16#FF	BYTE
GAPC1_Op	Tripping coupling 1		STRUCT
GAPC1_Op.Value		0	BOOL
GAPC1_Op.ST		16#FF	BYTE
GAPC1_Str	Excitation coupling 1		STRUCT
GAPC1_Str.Value		0	BOOL
GAPC1_Str.ST		16#FF	BYTE
PARSET_A	Parameter set A	0	BOOL
PARSET_B	Parameter set B	0	BOOL
PARSET_C	Parameter set C	0	BOOL
PARSET_D	Parameter set D	0	BOOL
Error	Group message		STRUCT
Error.Value		0	BOOL
Error.ST		16#FF	BYTE
Loc	Local mode	0	BOOL
Warning	General warning		STRUCT
Warning.Value		0	BOOL
Warning.ST		16#FF	BYTE
Q0_EnaCls	Interlock: Enable Q0 close	0	BOOL
Q0_EnaOpn	Interlock: Enable Q0 open	0	BOOL
Q1_EnaCls	Interlock: Enable Q1 close	0	BOOL
Q1_EnaOpn	Interlock: Enable Q1 open	0	BOOL
Q2_EnaCls	Interlock: Enable Q2 close	0	BOOL
Q2_EnaOpn	Interlock: Enable Q2 open	0	BOOL
Q8_EnaCls	Interlock: Enable Q8 close	0	BOOL
Q8_EnaOpn	Interlock: Enable Q8 open	0	BOOL

6.1 PLC block PC_FEEDER

I/O name	Description	Value	Data type
Q9_EnaCls	Interlock: Enable Q9 close	0	BOOL
Q9_EnaOpn	Interlock: Enable Q9 open	0	BOOL
ALM_sp1	Free ALM 1		STRUCT
ALM_sp1.Value		0	BOOL
ALM_sp1.ST		16#FF	BYTE
ALM_sp2	Free ALM 2		STRUCT
ALM_sp2.Value		0	BOOL
ALM_sp2.ST		16#FF	BYTE
ALM_sp3	Free ALM 3		STRUCT
ALM_sp3.Value		0	BOOL
ALM_sp3.ST		16#FF	BYTE
ALM_sp4	Free ALM 4		STRUCT
ALM_sp4.Value		0	BOOL
ALM_sp4.ST		16#FF	BYTE
ALM_sp5	Free ALM 5		STRUCT
ALM_sp5.Value		0	BOOL
ALM_sp5.ST		16#FF	BYTE
ALM_sp6	Free ALM 6		STRUCT
ALM_sp6.Value		0	BOOL
ALM_sp6.ST		16#FF	BYTE
ALM_sp7	Free ALM 7		STRUCT
ALM_sp7.Value		0	BOOL
ALM_sp7.ST		16#FF	BYTE
ALM_sp8	Free ALM 8		STRUCT
ALM_sp8.Value		0	BOOL
ALM_sp8.ST		16#FF	BYTE
ALM_sp9	Free ALM 9		STRUCT
ALM_sp9.Value		0	BOOL
ALM_sp9.ST		16#FF	BYTE
ALM_sp10	Free ALM 10		STRUCT
ALM_sp10.Value		0	BOOL
ALM_sp10.ST		16#FF	BYTE
ALM_sp11	Free ALM 11		STRUCT
ALM_sp11.Value		0	BOOL
ALM_sp11.ST		16#FF	BYTE
ALM_sp12	Free ALM 12		STRUCT
ALM_sp12.Value		0	BOOL
ALM_sp12.ST		16#FF	BYTE
ALM_sp13	Free ALM 13		STRUCT
ALM_sp13.Value		0	BOOL
ALM_sp13.ST		16#FF	BYTE
ALM_sp14	Free ALM 14		STRUCT
ALM_sp14.Value		0	BOOL

I/O name	Description	Value	Data type
ALM_sp14.ST		16#FF	BYTE
SwCount	Operating cycles Q0	0	DINT
OpTm	Operating hours	0	DINT
WpForward	Supply active energy		STRUCT
WpForward.Value		0.0	REAL
WpForward.ST		16#FF	BYTE
WpReverse	Demand active energy		STRUCT
WpReverse.Value		0.0	REAL
WpReverse.ST		16#FF	BYTE
WqForward	Supply reactive energy		STRUCT
WpForward.Value		0.0	REAL
WqForward.ST		16#FF	BYTE
WqReverse	Demand reactive energy		STRUCT
WpReverse.Value		0.0	REAL
WqReverse.ST		16#FF	BYTE
DINT_sp1	DINT value 8 (free)		STRUCT
DINT_sp1.Value		0	DINT
DINT_sp1.ST		16#FF	BYTE
DINT_sp2	DINT value 8 (free)		STRUCT
DINT_sp2.Value		0	DINT
DINT_sp2.ST		16#FF	BYTE
BOOL_sp1	Free BOOL		STRUCT
BOOL_sp1.Value		0	BOOL
BOOL_sp1.ST		16#FF	BYTE
BOOL_sp2	Free BOOL		STRUCT
BOOL_sp2.Value		0	BOOL
BOOL_sp2.ST		16#FF	BYTE
BOOL_sp3	Free BOOL		STRUCT
BOOL_sp3.Value		0	BOOL
BOOL_sp3.ST		16#FF	BYTE
BOOL_sp4	Free BOOL		STRUCT
BOOL_sp4.Value		0	BOOL
BOOL_sp4.ST		16#FF	BYTE
BOOL_sp5	Free BOOL		STRUCT
BOOL_sp5.Value		0	BOOL
BOOL_sp5.ST		16#FF	BYTE
BOOL_sp6	Free BOOL		STRUCT
BOOL_sp6.Value		0	BOOL
BOOL_sp6.ST		16#FF	BYTE
BOOL_sp7	Free BOOL		STRUCT
BOOL_sp7.Value		0	BOOL
BOOL_sp7.ST		16#FF	BYTE
BOOL_sp8	Free BOOL		STRUCT

I/O name	Description	Value	Data type
BOOL_sp8.Value		0	BOOL
BOOL_sp8.ST		16#FF	BYTE
BOOL_sp9	Free BOOL		STRUCT
BOOL_sp9.Value		0	BOOL
BOOL_sp9.ST		16#FF	BYTE
BOOL_sp10	Free BOOL		STRUCT
BOOL_sp10.Value		0	BOOL
BOOL_sp10.ST		16#FF	BYTE
OpTmWarning	1 = Operating time warning active	0	BOOL
OpTmAlarm	1 = Operating time alarm active	0	BOOL
SwCountWarning	1 = Operating cycle warning active	0	BOOL
SwCountAlarm	1 = Operating cycle alarm active	0	BOOL
LockAct			STRUCT
LockAct.Value		0	BOOL
LockAct.ST		16#80	BYTE
LocalAct			STRUCT
LocalAct.Value		0	BOOL
LocalAct.ST		16#80	BYTE
AutAct			STRUCT
AutAct.Value		0	BOOL
AutAct.ST		16#80	BYTE
ManAct	1 = Manual: Manual value is active		STRUCT
ManAct.Value	Value	1	BOOL
ManAct.ST	Signal state	16#80	BYTE
OosAct			STRUCT
OosAct.Value		0	BOOL
OosAct.ST		16#80	BYTE
OS_PermOut	Operator authorization: Output for OS	16#FFFFFFFF	DWORD
OS_PermLog	Operator authorization: Output for OS	16#FFFFFFFF	DWORD
OpSt_Out	Value of the input parameter OpSt_In, for further interconnection with other blocks. Bit 31 of this parameter is used by Feature bit 24	16#00	DWORD
ST_Worst	Worst signal state	16#80	BYTE
GrpErr			STRUCT
GrpErr.Value		0	BOOL
GrpErr.ST		16#80	BYTE
RdyToReset	1 = Ready to reset via RstLi or automatic commands		STRUCT
RdyToReset.Value	Value	0	BOOL
RdyToReset.ST	Signal state	16#80	BYTE
Status1	Word Status1	16#00000000	DWORD
Status3	Word Status3	16#00000000	DWORD
Status5	Word Status3	16#FFFFFFFF	DWORD

I/O name	Description	Value	Data type
MsgErrlocal1	1 = Message error occurs	0	BOOL
MsgStatlocal1	Message status	16#0000	WORD
MsgAcknlocal1	Message acknowledgement status	16#0000	WORD
MsgErrlocal2	1 = Message error occurs	0	BOOL
MsgStatlocal2	Message status	16#0000	WORD
MsgAcknlocal2	Message acknowledgement status	16#0000	WORD
MsgErrlocal3	1 = Message error occurs	0	BOOL
MsgStatlocal3	Message status	16#0000	WORD
MsgAcknlocal3	Message acknowledgement status	16#0000	WORD
VisuCodes	Display of the switches in the faceplate		STRUCT
VisuDesc	Labeling in the faceplate		STRUCT
VisuFields	Display of the switches in the faceplate		STRUCT
VisuMisc	Configuration of the units and the user view		STRUCT
VisuStates	Switch position texts		STRUCT

6.2 PC_FEEDER block icons

6.2.1 PC_FEEDER block symbols

The following block symbols are available for the PC_FEEDER object:

PC_FEEDER/1: Symbol with group display and large tag name:



Figure 6-7 PC_FEEDER/1

PC_FEEDER/2: Symbol with group display and small tag name:

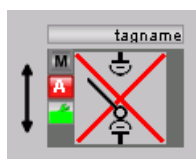


Figure 6-8 PC_FEEDER/2

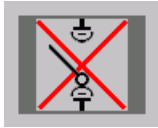
PC_FEEDER/3: Symbol without group display and with tag name:

Figure 6-9 PC_FEEDER/3

PC_FEEDER/4: 6 field symbol:

With this symbol all switches are also displayed completely in the symbol.

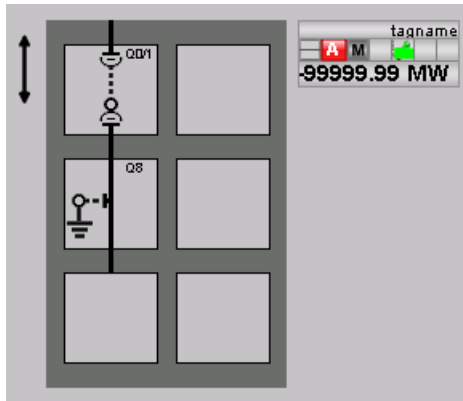


Figure 6-10 PC_FEEDER/4

PC_FEEDER/5: Rectangle symbol for overview pictures:

Additional functionalities:

- Dynamic coloring with the block input BB_Color



Figure 6-11 PC_FEEDER/5

PC_FEEDER/6: Small symbol disconnecter:

Additional functionalities:

- Dynamic coloring with the block input BB_Color
- Display closed grounding switch



Figure 6-12 PC_FEEDER/6

PC_FEEDER/7: Small symbol circuit breaker:

Additional functionalities:

- Dynamic coloring with the block input BB_Color
- Display closed grounding switch



Figure 6-13 PC_FEEDER/7

6.2.2 Dynamizing the active power direction

Dynamizing the active power direction

The active power display of the corresponding PC_FEEDER block icons can be dynamically displayed as a function of the active power P.

The latest value for the active power P is used to display the active power direction. Depending on the sign of the active power, the direction display is shown in one or the other direction or is completely hidden if the active power = 0W.

The dynamization of the active power can be enabled or disabled using **Feature.Bit20**.

- 0 = Dynamization of active power direction disabled
- 1 = Dynamization of active power direction enabled

Feature.Bit21 can be used to configure the direction of the active power in the event of a negative value

- 0 = For a negative active power value, the arrow points downwards.
- 1 = For a negative active power value, the arrow points upwards.

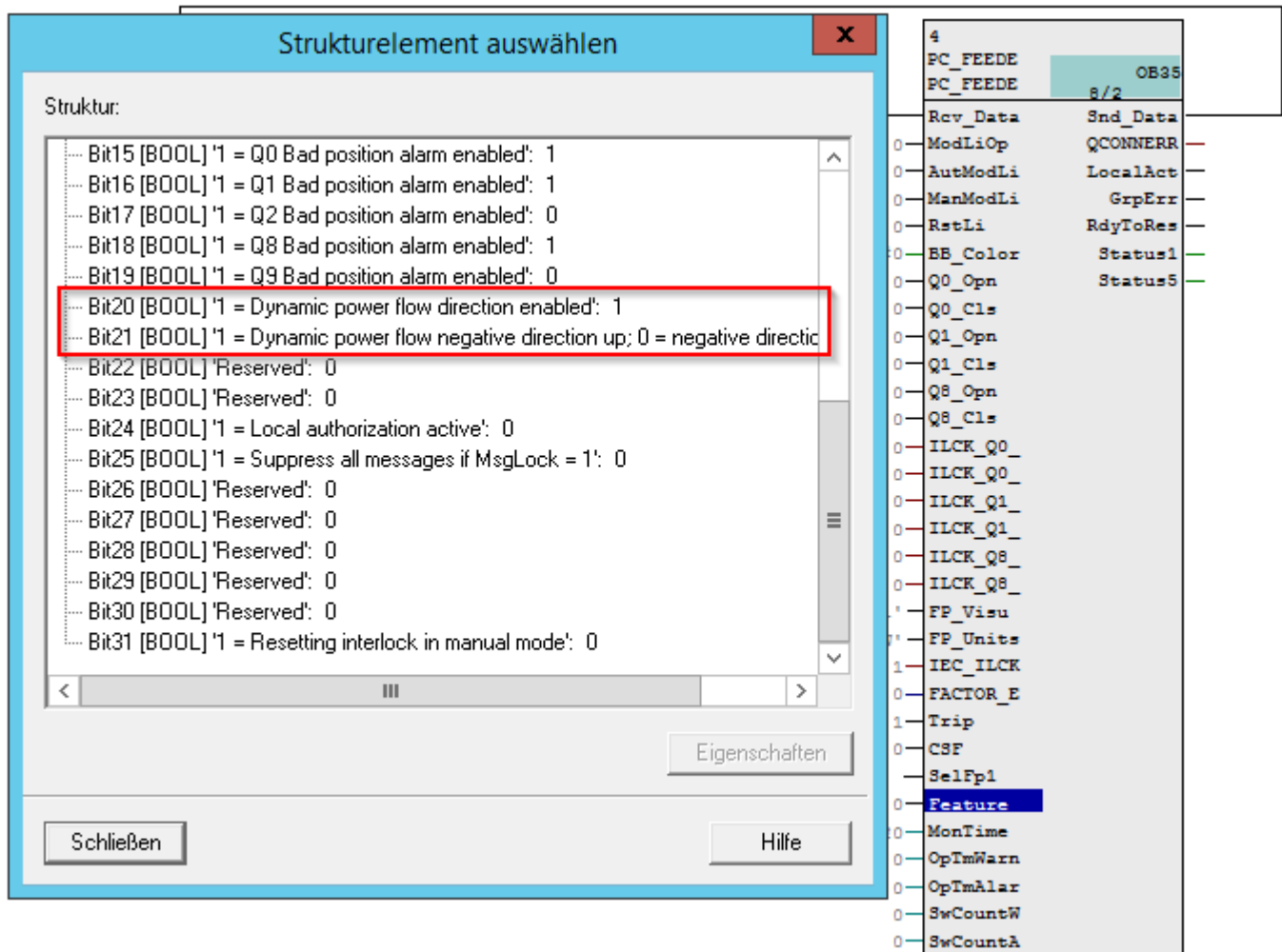


Figure 6-14 Active power direction - Feature bit PC_FEEDER

If dynamization is activated, the direction of the arrow is displayed based on the active power value and the setting at Feature.Bit21:



Figure 6-15 Active power direction - negative icon



Figure 6-16 Active power direction - positive icon

If the value of the active power = 0, the arrow is completely hidden.

6.3 PC_FEEDER faceplate

Available views:

View	Description
Default window	Switch representation, important measured values, interlocks, device status
Message view	Messages from device (messages, warnings, etc.)
Measured value window	Measured values of FEEDER type
Maintenance window	Operating hours, switches, active parameter group
User-defined window	User-specific view
Memo view	Memo view

6.3.1 Switch representation

The switches (e.g., circuit breakers, grounding switches, disconnectors) can be represented flexibly in the PC_FEEDER faceplate. Up to five switches can be shown dynamically in six fields.

The representation is carried out in the form of symbols that can be combined freely (see Appendix) arranged in six fields (fields 1-6).

There are an additional four fields (fields 7-10) which represent the connecting lines between fields 1-6.

The entire configuration is made using the "FP_Visu" structure input at the PC_FEEDER function block.

6.3.1.1 Configuring the symbols

The figure below shows the various fields in the faceplate:

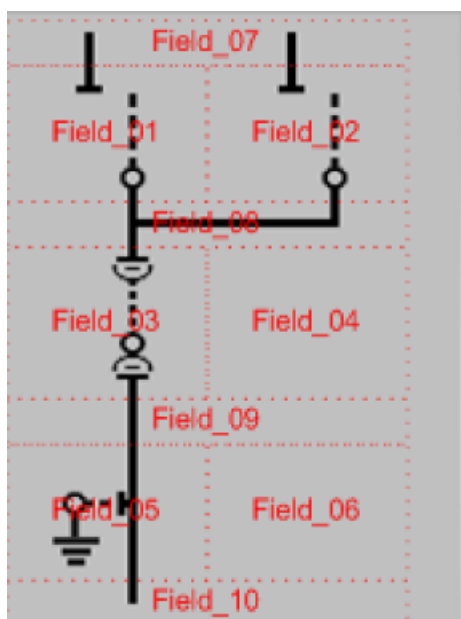


Figure 6-17 PC_FEEDER field assignment

The configuration for selecting a symbol to be displayed in a particular field is made with the "FP_Visu" structure input at the PC_FEEDER function block. Here, the symbol codes for the individual fields are configured using the structure elements Field_01_Code....Field_10_Code. The table with the symbol codes and the associated symbols is included in the appendix for this document.

The symbol codes must always be entered as two-digit numbers. The figure below shows an example of a possible symbol layout with the corresponding configuration:

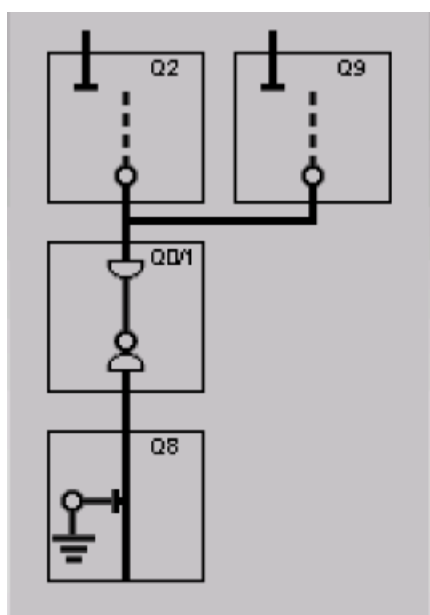


Figure 6-18 Example of a field configuration

- Field_01_Code = 04
- Field_02_Code = 04
- Field_03_Code = 01
- Field_04_Code = 00
- Field_05_Code = 11
- Field_06_Code = 00
- Field_07_Code = 03
- Field_08_Code = 02

- Field_09_Code = 04
- Field_10_Code = 01

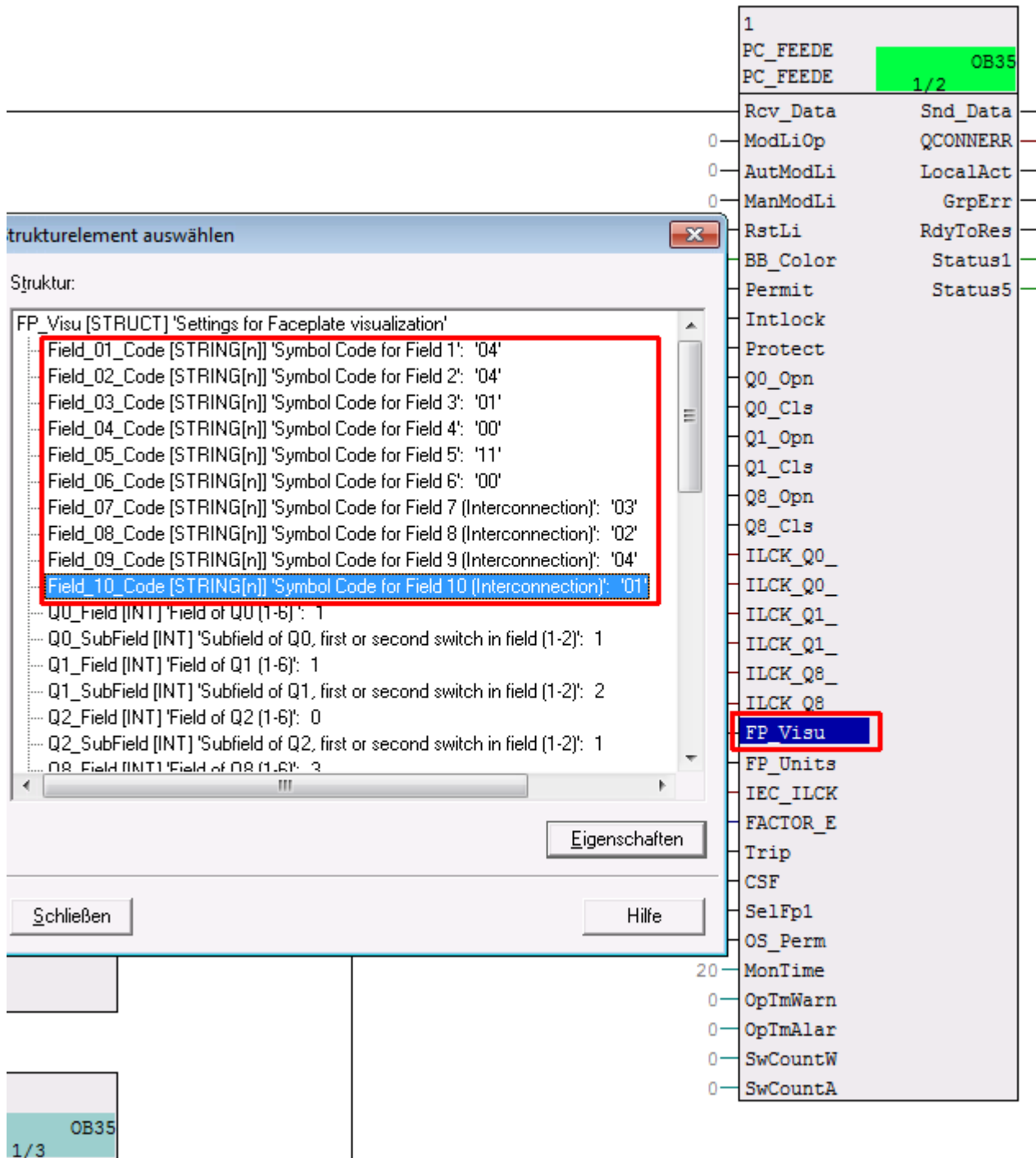


Figure 6-19 Configuration of the symbol

6.3.1.2 Configuration of the switches

The following switches can be configured in PC_FEEDER:

- Q0 (e.g., circuit breakers)
- Q1 (e.g., disconnectors)
- Q2 (e.g., additional disconnectors)
- Q8 (e.g., grounding switches)
- Q9 (e.g., additional grounding switches)

Switches can only be represented in fields 1-6. Up to two switches can be configured per field (for example, in the circuit-breaker–disconnector combination in Field 3 in the example above).

The assignment of the switches to the fields is configured with the structure elements "QX_Field" and "QX_SubField" of the "FP_Visu" structure that are available for each of the 5 switches.

The "QX_Field" attribute describes the number of the field in which the switch is located (possible values: 1-6). "QX_SubField" represents one of the two "subfields" of the "QX_Field". This attribute is required to represent two switches in one field.

"QX_SubField" may ONLY have the values 1 or 2. If, for example, as shown in the figure above, the circuit breaker (Q0) and the disconnector (Q1) are represented in field 3, the following configuration is required:

- Q0_Field = 3
- Q0_SubField = 1
- Q1_Field = 3
- Q1_SubField = 2

This means both switches are shown in field 3. Q0 is assigned to SubField 1 (Switch 1), Q1 is assigned to SubField 2 (Switch 2).

The figure below shows the complete switch configuration for the example above:

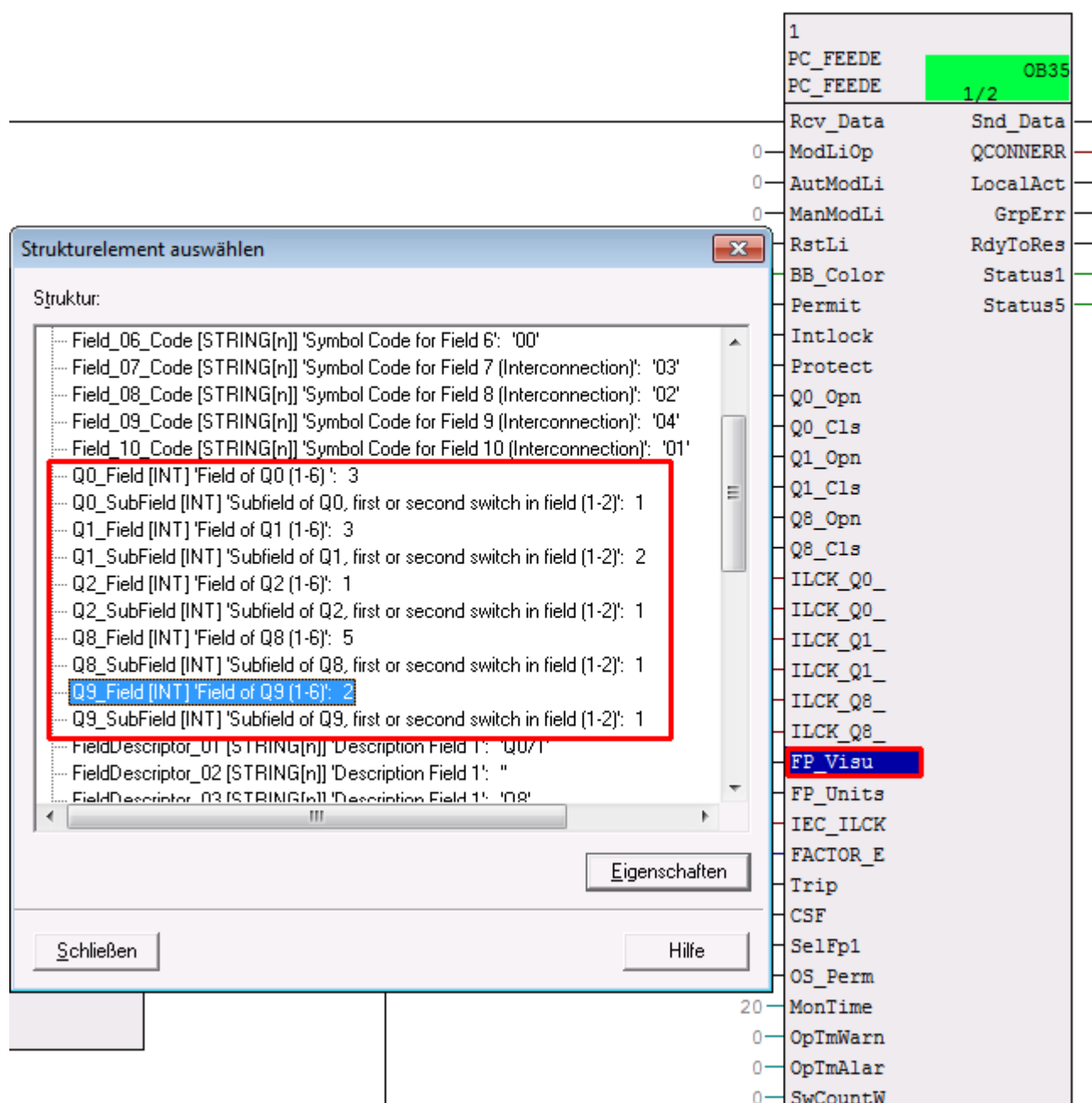


Figure 6-20 Switch configuration

Note

If the "QX_Field" attribute is set to 0, the corresponding switch is not shown in the faceplate (no display of position and control).

6.3.1.3 Labeling of the fields and switches

Labeling of the fields 1-6 can be carried out with the structure attributes "FieldDescriptor_XX" of the "FP_Visu" structure.

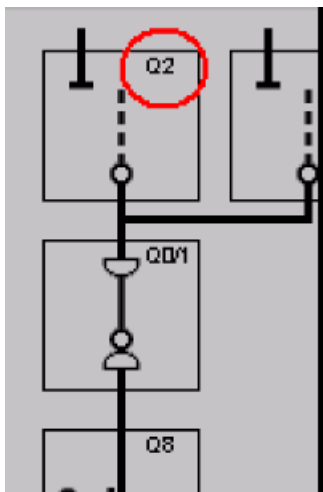


Figure 6-21 Labeling of the fields

The names of the individual switches in the faceplate are configured with the "QX_DESC" structure attributes:

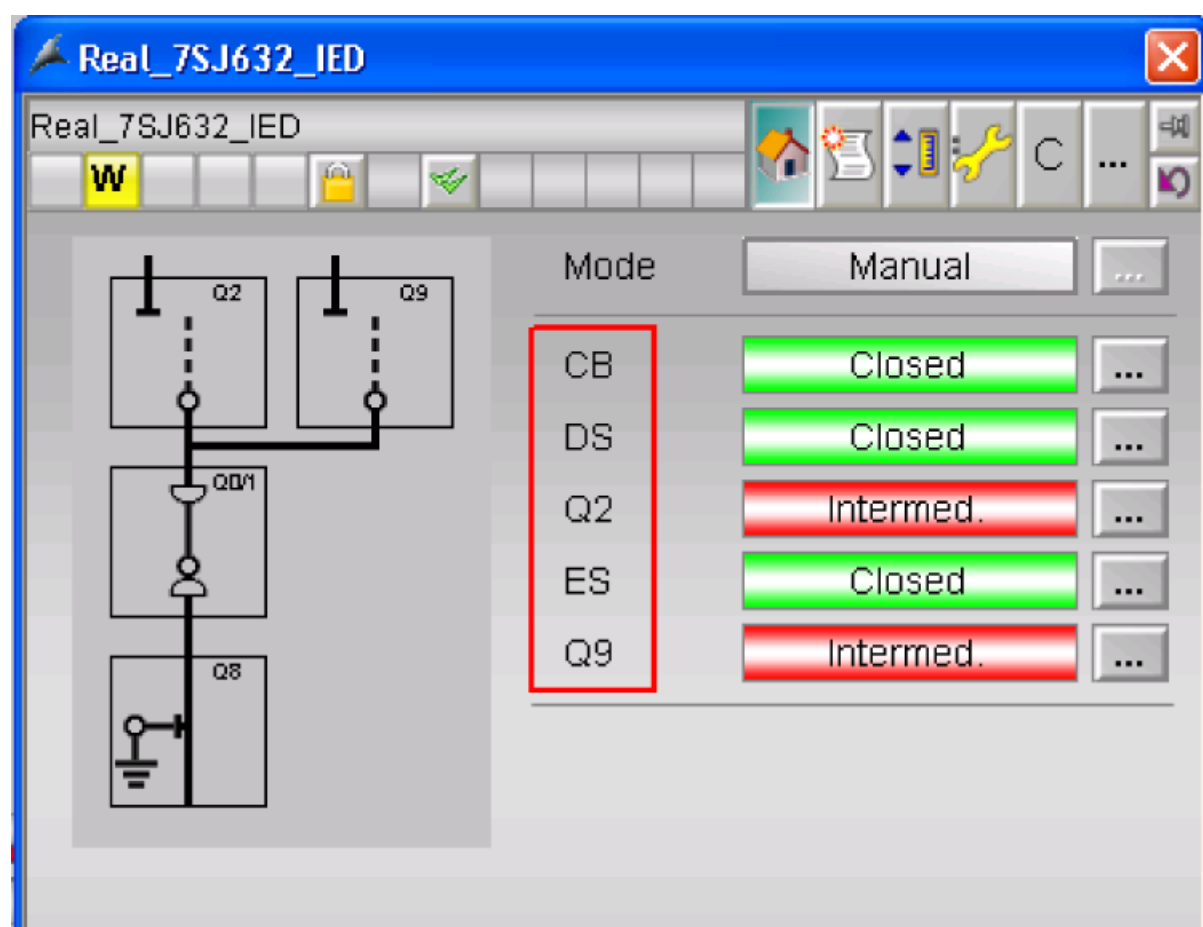


Figure 6-22 Labeling of the switches in the faceplate

6.3.1.4 Hiding buttons that are not required

If certain switches of a protective device do not have a control function or should not be controllable from the OS, the associated buttons can be hidden in the faceplate. You configure this functionality with the "OS_Perm" structure input at the PC_FEEDER function block. The following bits are available for this purpose:

Bit no.	QX switch
13	Button for Q0 is visible (PC_FEEDER) Button for Q01 is visible (PC_FEEDER_2CB)
14	Button for Q1 is visible
15	Button for Q2 is visible
16	Button for Q8 is visible
17	Button for Q9 is visible (PC_FEEDER) Button for Q02 is visible (PC_FEEDER_2CB)

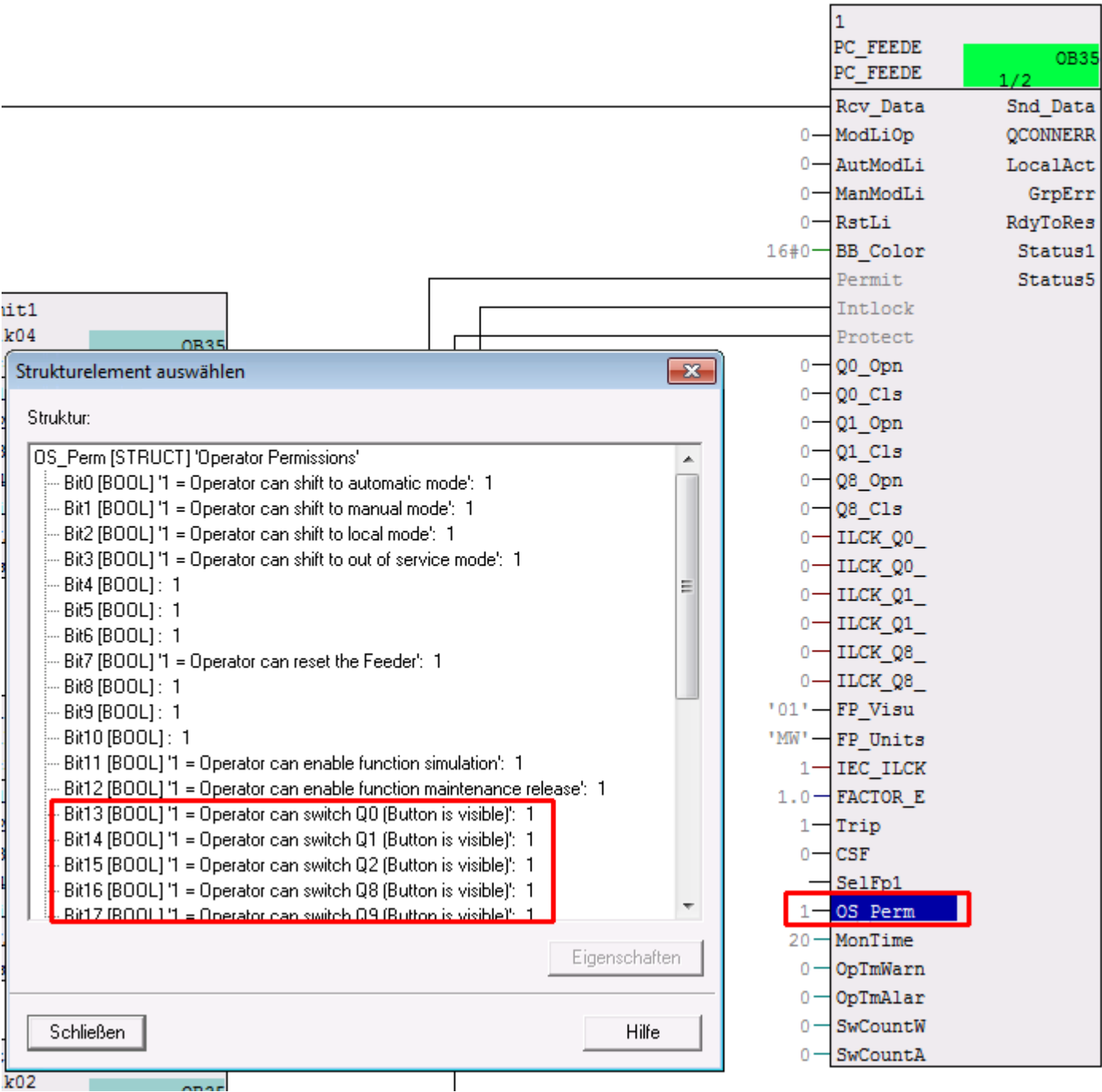


Figure 6-23 OS_Perm

The button for the grounding switch (Q8) is hidden in the example below by changing bit 16 of OS_PERM to 0:

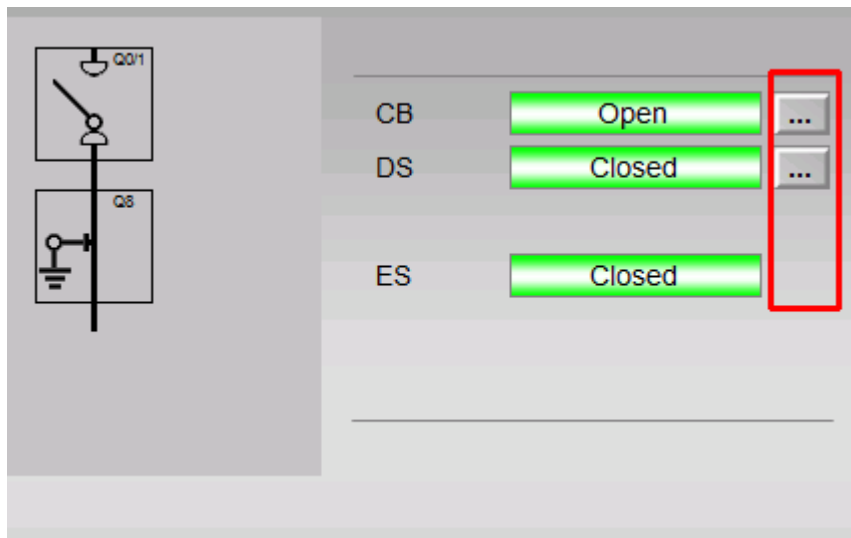


Figure 6-24 Hiding the buttons

6.3.1.5 Texts of switch settings

The feedback texts of the switch settings "open" and "closed" in the faceplate can be freely configured in the "FP_Visu" structure at the PC_FEEDER function block. This may be necessary, for example, when the carriage of a disconnector is to be displayed with "Test setting" or "Operation".

The structure elements for configuration are "QX_TextOpen" or "QX_TextClosed".

In the example below, the texts of Q1 have been adapted:

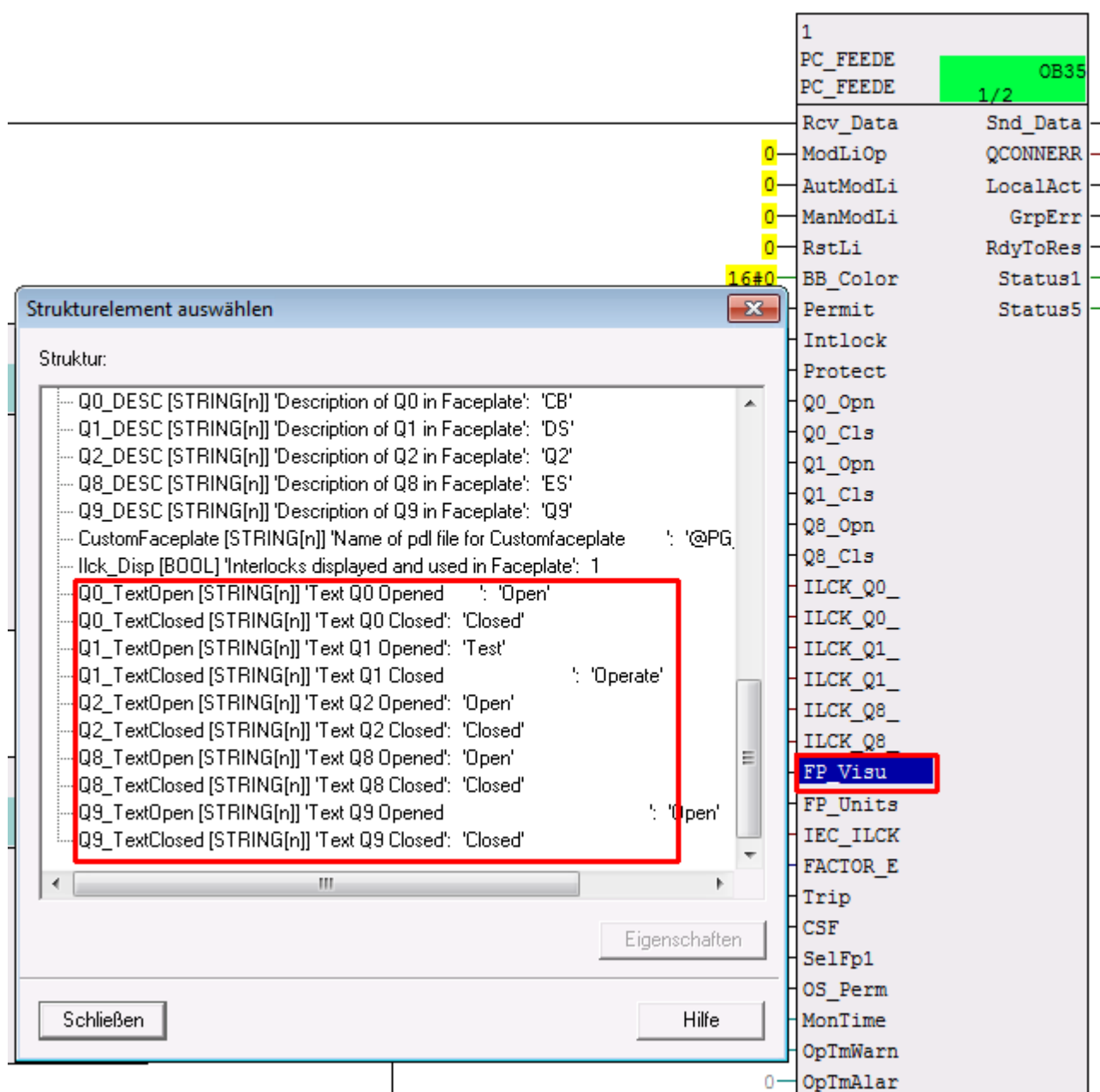


Figure 6-25 Configuration of the switch texts

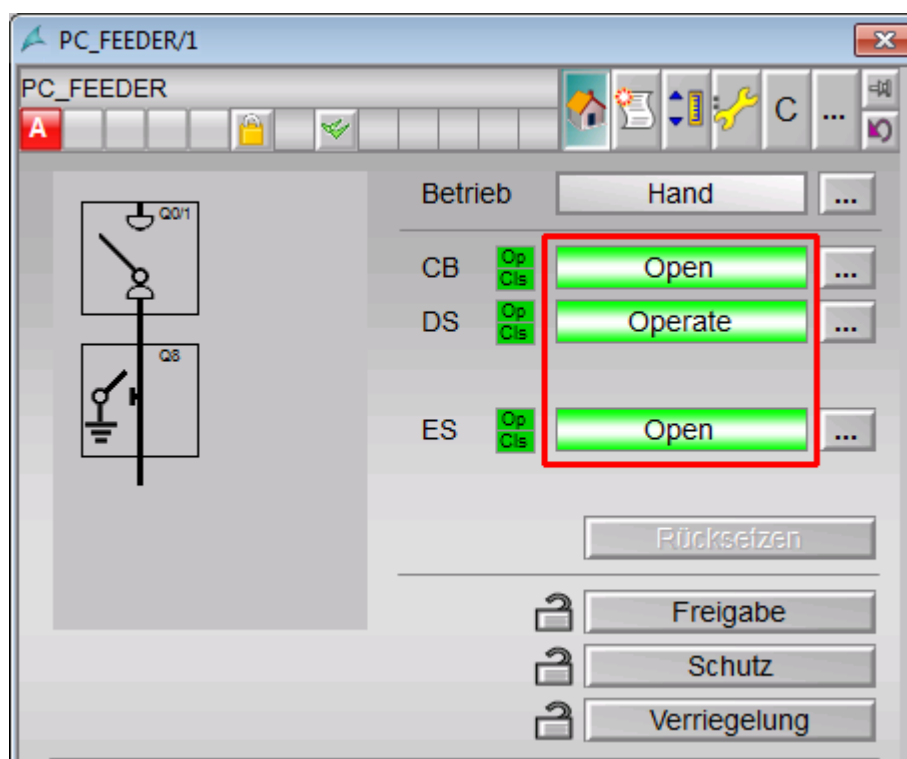


Figure 6-26 Switch texts in the faceplate

6.3.1.6 Interlock

Triggering an Open or Close command is blocked in the corresponding switch area in line with the current switch position.

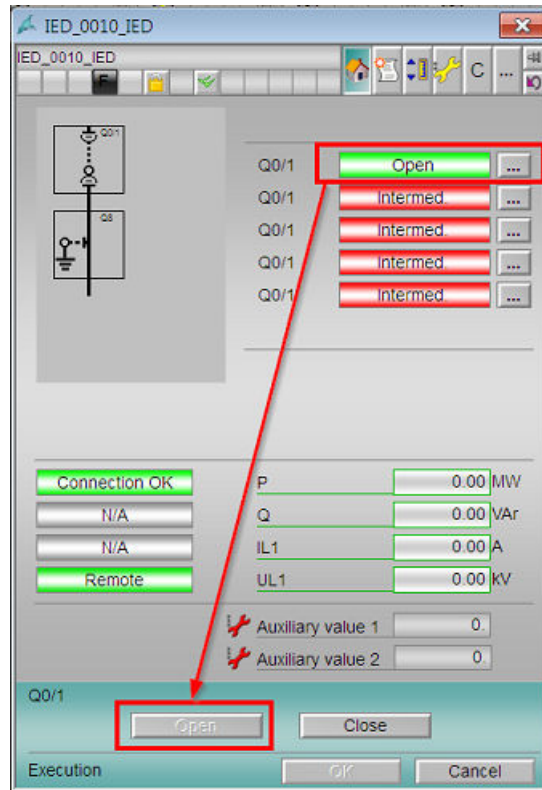


Figure 6-27 Example of an interlock

6.3.2 Standard window

6.3.2.1 Default window

The default window offers the following functions:

- Display of the device status
- Display of the current operating mode of the device
- Display of the interlock status
- Display of important measured values
- Display of switch states
- Display of auxiliary values
- Change in the operating mode of the device

- Sending of switch commands to the device
- Tracking mode of switch states while connection is lost

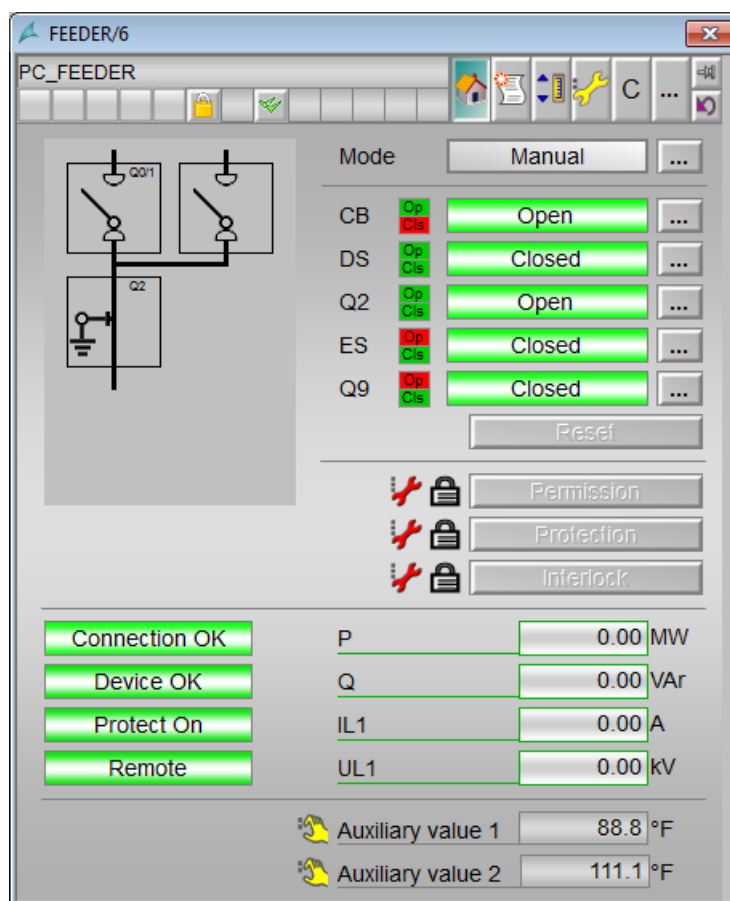


Figure 6-28 Default window PC_FEEDER

Buttons

Note: Level 5 authorizations are required to operate the buttons in this view. The number of buttons depends on the configuration.

Input box	Action	Description
Q0	Cmd0 = 0/1	Command switch Q0. 0 = Open 1 = Close
Q1	Cmd1 = 0/1	Command switch Q1. 0 = Open 1 = Close
Q2	Cmd2 = 0/1	Command switch Q2. 0 = Open 1 = Close

Input box	Action	Description
Q8	Cmd8 = 0/1	Command switch Q8. 0 = Open 1 = Close
Q9	Cmd9 = 0/1	Command switch Q9. 0 = Open 1 = Close
Tracking	Activate tracking mode	In case the connection is lost, you can use this button to activate tracking mode
Faceplate	Jump to the connected faceplate	This button can be used to jump to a different faceplate when the input SelFp1 is connected.
Mode	Change the operating mode	Change the operating mode of the block
Reset	Reset an interlock or an external tripping	Reset the block after a "Protect" interlock or an external tripping (only available if Function bit 31 is set and the block is in manual mode)
Permit/Protect/Interlock	Jump to APL interlock blocks	Allows a jump to a faceplate of APL interlock blocks

Display

Note: The displays for the auxiliary values are only shown if at least one auxiliary value was interconnected to the corresponding block input.

Display	Meaning	Tag	Description
Connection status	Communication status of IED via Ethernet.	QCONERRO	QCONERRO = 0 (Connection OK) – GREEN QCONERRO = 1 (Comm. loss) – RED
Device status 1	Status of IED	Dev_OK	Dev_OK = 1 (Device OK) – GREEN Dev_OK = 2 (Device warning) – YELLOW Dev_OK = 3 (Device message) – RED
Device status 2	Behavior of IED	Beh	Beh = 1 (Protection on) – GREEN Beh = 2 (Protection locked) – YELLOW Beh = 3 (Protection test) – RED Beh = 4 (Protection test/locked) – RED Beh = 5 (Protection off) – RED
Device status 3	Local/Remote	Loc	Loc = 1: Local Loc = 0: Remote
Switch Q0	Switch position	Pos0	Pos0 = 0 (Intermediate st.) – RED Pos0 = 1 (Open) – GREEN Pos0 = 2 (Close) – GREEN Pos0 = 3 (Faulty) – RED Pos0 = 4 (Fault) – RED

Display	Meaning	Tag	Description
Switch Q1	Switch position	Pos1	Pos1 = 0 (Intermediate st.) – RED Pos1 = 1 (Open) – GREEN Pos1 = 2 (Close) – GREEN Pos1 = 3 (Faulty) – RED Pos1 = 4 (Fault) – RED
Switch Q2	Switch position	Pos2	Pos2 = 0 (Intermediate st.) – RED Pos2 = 1 (Open) – GREEN Pos2 = 2 (Close) – GREEN Pos2 = 3 (Faulty) – RED Pos2 = 4 (Fault) – RED
Switch Q8	Switch position	Pos8	Pos8 = 0 (Intermediate st.) – RED Pos8 = 1 (Open) – GREEN Pos8 = 2 (Close) – GREEN Pos8 = 3 (Faulty) – RED Pos8 = 4 (Fault) – RED
Switch Q9	Switch position	Pos9	Pos9 = 0 (Intermediate st.) – RED Pos9 = 1 (Open) – GREEN Pos9 = 2 (Close) – GREEN Pos9 = 3 (Faulty) – RED Pos9 = 4 (Fault) – RED
P	Power	P#Value	Power
Q	Reactive power	Q#Value	Reactive power
IL1	Current L1	I_A#Value	Current L1
UL1	Voltage L1 ground	U_A#Value	Voltage L1 ground
Auxiliary value 1	User analog value 1	UserA-na1#Value	User analog value 1
Auxiliary value 2	User analog value 2	UserA-na2#Value	User analog value 2

6.3.2.2 Tracking mode

Tracking mode

If the connection to the IED is lost, you can activate tracking mode to set the physically correct switch positions directly at the faceplate. This means the current switch position can be reproduced even while the connection is lost.

Activate tracking mode

While a connection is lost, the standard view of the faceplate shows the button for activating tracking mode for the entire time the connection is lost. Once the connection is restored, the button is hidden once again and tracking mode is automatically deactivated.

If the button is pressed, tracking mode can be activated with the buttons at the bottom of the faceplate. After activation of the operating state, the buttons for setting the switch states are released again. The button for activating tracking mode is locked and turns purple.

Deactivation of tracking mode is not possible while a connection is lost.



WARNING

During the entire duration of tracking mode, the corresponding interlocks (Permit, Interlock, Protect, Q[x]_EnaCls, Q[x]_EnaOpn, ILCK_Q[x]_Cls, ILCK_Q[x]_Opn) of the individual switches are deactivated. This means all operable switches can be switched by means of the faceplate in tracking mode, regardless of the interlock settings. Also observe the warning notices in the section "Using tracking mode".

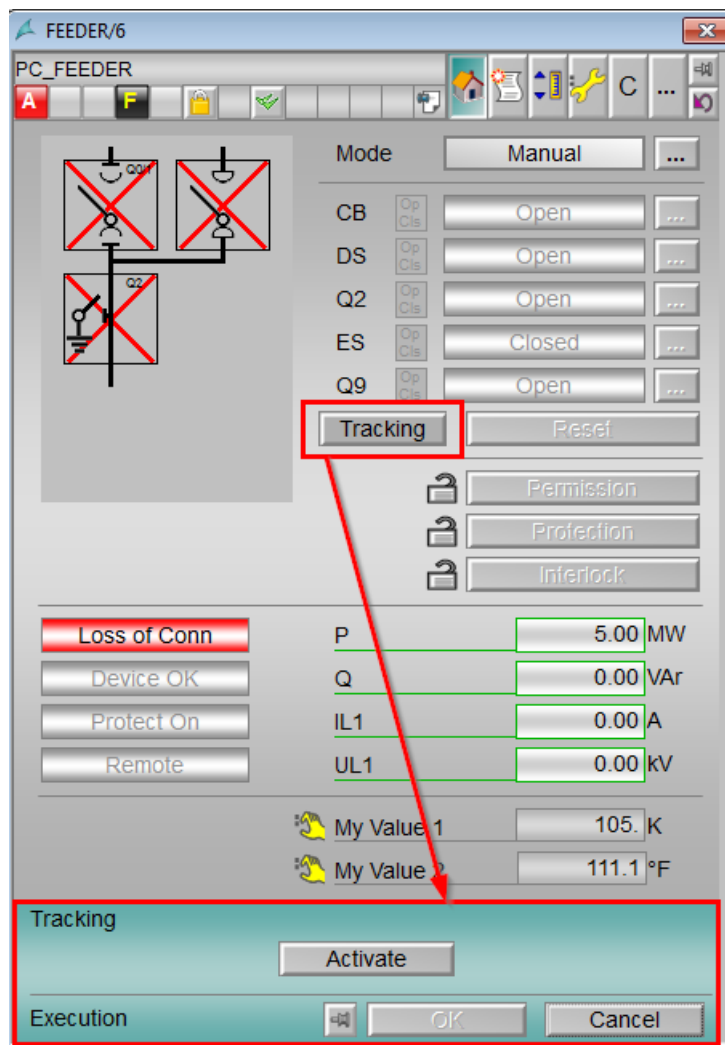


Figure 6-29 PC_FEEDER Activate tracking mode

Using tracking mode

When tracking mode was activated, the switch positions can be adjusted with the corresponding operator controls.

The set switch position is shown in the associated symbol. In addition, a purple cross at the symbol and a purple status display signal that the switch position was adjusted by means of tracking mode.

WARNING

A change of the switch position during tracking mode has effects on the corresponding output parameters of the associated block in the CFC. This means that the following output parameters have the value of the set switch position:

- Pos0 (PC_FEEDER) / Pos01 (PC_FEEDER_2CB)
- Pos1
- Pos2
- Pos8
- Pos9 (PC_FEEDER) / Pos02 (PC_FEEDER_2CB)

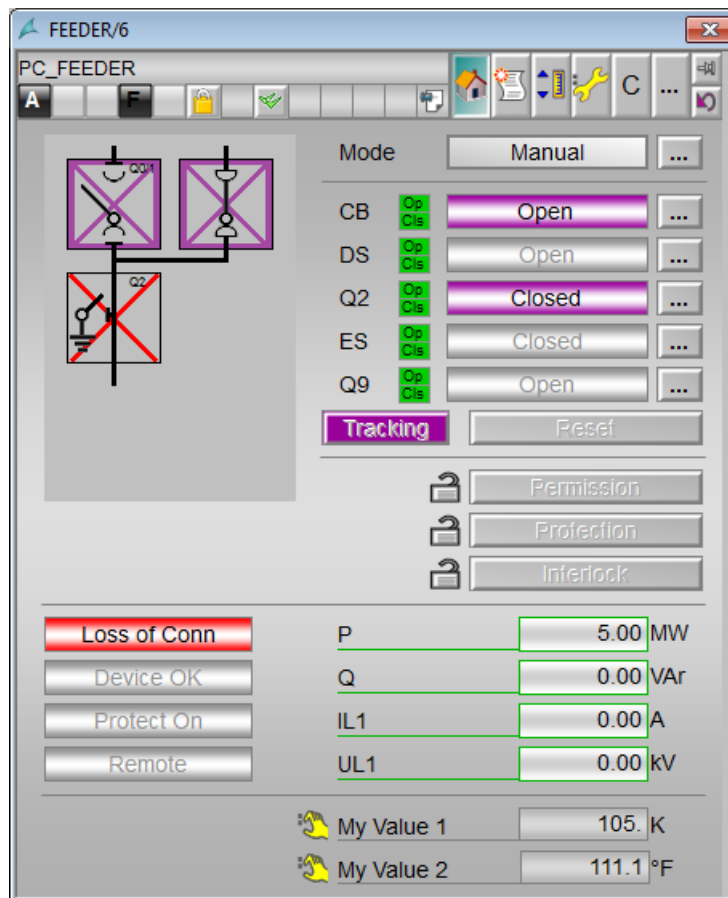


Figure 6-30 PC_FEEDER Using tracking mode

Operator authorization

Tracking mode uses the APL standard function "Operator authorizations - User administration in the PCS 7 OS". This means that the authorization for operation of the buttons selected in the figure below depends on the set operator authorization level at the block I/O as well as the operator authorizations in WinCC.

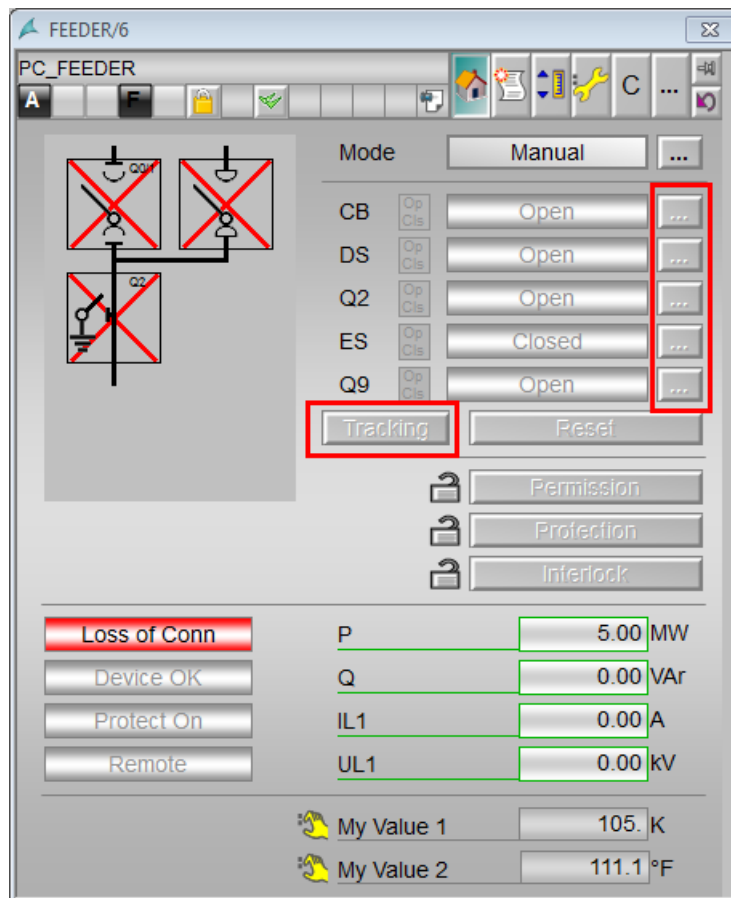


Figure 6-31 PC_FEEDER without operator authorization

Deactivate tracking mode

Once tracking mode has been activated during a loss of connection, it cannot be deactivated by the operator. Tracking mode remains activated for the entire time the connection is lost.

Once the connection has been restored, tracking mode and all associated symbols are automatically deactivated.

6.3.3 Message view

This window displays every active message and warning of the IED. The APL default message window is used.

6.3.4 Measured value window

The measured values of the IED are displayed in the measured value window.



Figure 6-32 PC_FEEDER measured value window

In the event that the signal status of all energy values (WpVorw, WqVorw, WpRückw, WqRückw) is transmitted with a value of 16#FF, the corresponding display elements are hidden. As soon as one of the energy values receives a status not equal to 16#FF, all display elements are re-displayed.

Display

Display	Meaning	Tag	Description
IL1	Current L1	I_A#Value	Current L1
IL2	Current L2	I_B#Value	Current L2
IL3	Current L3	I_C#Value	Current L3
U1-2	Voltage L1-L2	U_AB#Value	Voltage L1-L2
U2-3	Voltage L2-L3	U_BC#Value	Voltage L2-L3
U3-1	Voltage L3-L1	U_CA#Value	Voltage L3-L1
P	Power	P#Value	Power
Q	Reactive power	Q#Value	Reactive power
S	Apparent power	S#Value	Apparent power
Un	Neutral voltage	U#Value	Neutral voltage
Cos Phi	Power factor	PF#Value	Power factor
F	Frequency	F#Value	Frequency
WpVorw	Supply active energy	WpForward#Value	Supply active energy
WqVorw	Demand active energy	WpForward#Value	Demand active energy

Display	Meaning	Tag	Description
WpRückw	Supply reactive energy	WpReverse#Value	Supply reactive energy
WqRückw	Demand reactive energy	WqReverse#Value	Demand reactive energy

6.3.5 Maintenance window

The maintenance window displays the maintenance-specific information.

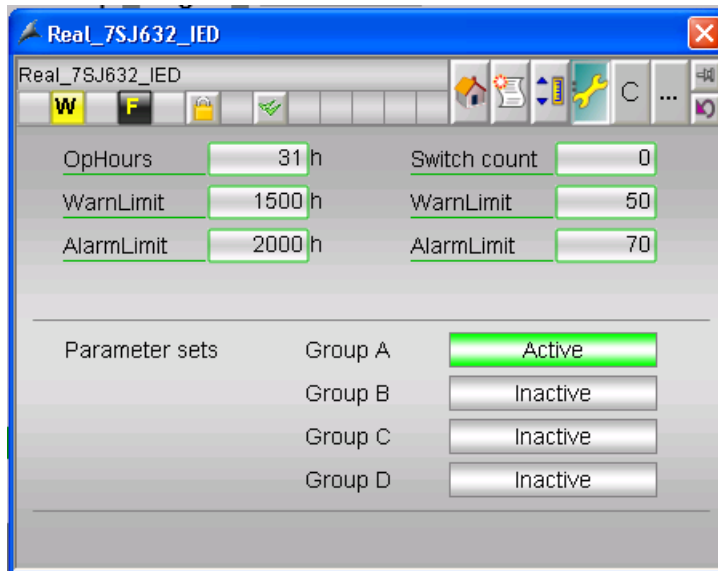


Figure 6-33 PC_FEEDER maintenance window

Display

Display	Meaning	Tag	Description
OpHours	Operating hours	OpTm	Operating hours
WarnLimit	Warning limit operating hours	OpTmWarnLimit	Warning limit operating hours
AlarmLimit	Alarm limit operating hours	OpTmAlarmLimit	Alarm limit operating hours
Operating cycles	Control number Q0	SwCount	Control number Q0
WarnLimit	Warning limit operating cycles	SwCountWarnLimit	Warning limit operating cycles
AlarmLimit	Alarm limit operating cycles	SwCountAlarmLimit	Alarm limit operating cycles
Group A	Parameter set A active	PARSET_A	Parameter set A active
Group B	Parameter set B active	PARSET_B	Parameter set B active
Group C	Parameter set C active	PARSET_C	Parameter set C active
Group D	Parameter set D active	PARSET_D	Parameter set D active

6.3.6 User view

The user-defined faceplate can be customized by the customer. See also Creating a user view/ templates (Page 31)

Example:

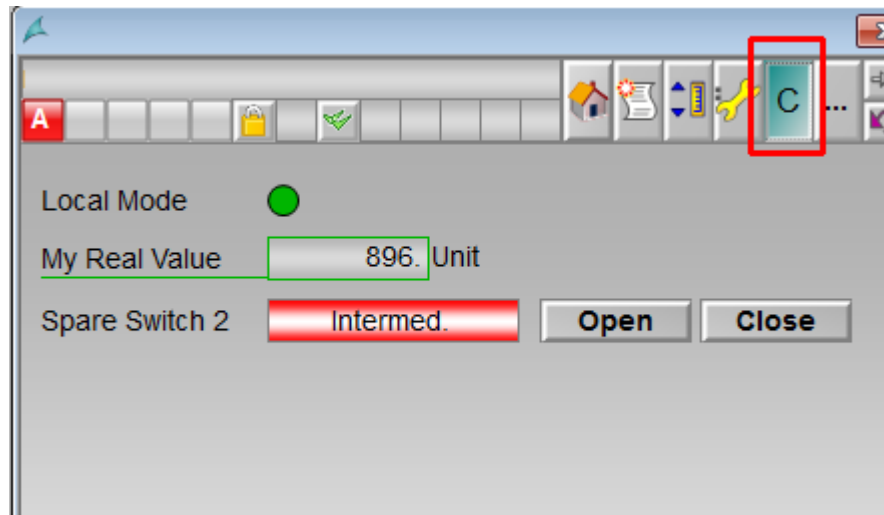


Figure 6-34 Example of user view

PC_FEEDER_2CB

The PC_FEEDER_2CB object is an extension of the PC_FEEDER object and is used for all infeeds or branches on a redundant busbar. The differences as compared to the PC_FEEDER object are:

- The PC_FEEDER_2CB block has 2 dedicated circuit breakers (Q01 and Q02)
- Each of the two circuit breakers has its own, separately acting APL interlocks (Permit, Protect, Interlock)
- The APL interlocks of the two circuit breakers are shown separately in the standard view.

The blocks of the PC_FEEDER can be used as driver blocks/DataCollector blocks; the interfaces are identical. The switch Q9 is used as the 2nd circuit breaker in the interface structures.

PC_FEEDER_2CB consists of the following objects:

- PLC block PC_FEEDER_2CB
- WinCC block icons
- WinCC faceplate PC_FEEDER_2CB with different views

This chapter only describes the properties of the PC_FEEDER_2CB object which are different to those of the PC_FEEDER object.

7.1 PLC block PC_FEEDER_2CB

7.1.1 Description of PLC block PC_FEEDER_2CB

The PC_FEEDER_2CB (FB 2501) function block represents the infeed object on a redundant busbar in the PLC. The block is the interface between the device driver block and the remaining PCS 7 environment.

The block provides the functionality of the PC_FEEDER block with the following extensions:

- 2nd dedicated circuit breaker Q02
- Extended interlock functions for the 2nd circuit breaker Q02
- Additional external trip signal for 2nd circuit breaker Q02

7.1.2 Interlocks

In addition to the interlocks of the FC_FEEDER block, the PC_FEEDER_2CB block also offers the 3 standard APL interlock functions Permit, Interlock and Protect for the 2nd circuit breaker Q02.

All parameters of the APL interlocks are equipped with corresponding prefixes for the two circuit breakers Q01 and Q02 (e.g. Q01_Intl_En for the first circuit breaker, Q02_Intl_En for the 2nd circuit breaker).

The interlock logic systems of Q01 and Q02 function completely independently of one another.

7.1.3 Behavior of external tripping signals

External trippings from the CFC logic system can be connected with the "Q01_Trip" or "Q02_Trip" input. The "Q01_Trip/Q02_Trip" inputs function like the "Protect" interlock and open Q01/Q02, if they were closed beforehand.

The logic systems of Q01 and Q02 function completely independently of one another.

7.1.4 Messages

Message instance	Message identifier	Message class	Event
MsgEvId_1	SIG_1	Alarm - high	\$\$BlockComment\$\$ trip Ip
	SIG_2	Warning - high	\$\$BlockComment\$\$ excitation Ip
	SIG_3	Alarm - high	\$\$BlockComment\$\$ trip 67-1 rectified overcurrent stage I >
	SIG_4	Warning - high	\$\$BlockComment\$\$ excitation 67-1 rectified overcurrent stage I >
	SIG_5	Alarm - high	\$\$BlockComment\$\$ trip 67-2 rectified overcurrent stage I >>
	SIG_6	Warning - high	\$\$BlockComment\$\$ excitation 67-2 rectified overcurrent stage I >>
	SIG_7	Alarm - high	\$\$BlockComment\$\$ trip 67N-1 rectified overcurrent stage IE >
	SIG_8	Warning - high	\$\$BlockComment\$\$ excitation 67N-1 rectified overcurrent stage IE >

Message instance	Message identifier	Message class	Event
MsgEvld_2	SIG_1	Alarm - high	\$\$BlockComment\$\$ trip 67N-2 rectified overcurrent stage IE >>
	SIG_2	Warning - high	\$\$BlockComment\$\$ excitation 67N-2 rectified overcurrent stage IE >>
	SIG_3	Alarm - high	\$\$BlockComment\$\$ trip 50-3 overcurrent stage
	SIG_4	Warning - high	\$\$BlockComment\$\$ excitation 50-3 overcurrent stage
	SIG_5	Alarm - high	\$\$BlockComment\$\$ trip 51N overcurrent stage
	SIG_6	Warning - high	\$\$BlockComment\$\$ excitation 51N overcurrent stage
	SIG_7	Alarm - high	\$\$BlockComment\$\$ trip 67-TOC overcurrent stage
	SIG_8	Warning - high	\$\$BlockComment\$\$ excitation 67-TOC overcurrent stage
MsgEvld_3	SIG_1	Alarm - high	\$\$BlockComment\$\$ trip 67N-TOC overcurrent stage
	SIG_2	Warning - high	\$\$BlockComment\$\$ excitation 67N-TOC overcurrent stage
	SIG_3	Alarm - high	\$\$BlockComment\$\$ trip 46-TOC overcurrent stage
	SIG_4	Warning - high	\$\$BlockComment\$\$ excitation 46-TOC overcurrent stage
	SIG_5	Alarm - high	\$\$BlockComment\$\$ trip 50-1 overcurrent stage I >
	SIG_6	Warning - high	\$\$BlockComment\$\$ excitation 50-1 overcurrent stage I >
	SIG_7	Alarm - high	\$\$BlockComment\$\$ trip 50-2 overcurrent stage I >>
	SIG_8	Warning - high	\$\$BlockComment\$\$ excitation 50-2 overcurrent stage I >>
MsgEvld_4	SIG_1	Alarm - high	\$\$BlockComment\$\$ trip 50N-1 overcurrent stage IE >
	SIG_2	Warning - high	\$\$BlockComment\$\$ excitation 50N-1 overcurrent stage IE >
	SIG_3	Alarm - high	\$\$BlockComment\$\$ trip 50N-2 overcurrent stage IE >>
	SIG_4	Warning - high	\$\$BlockComment\$\$ excitation 50N-2 overcurrent stage IE >>
	SIG_5	Warning - high	\$\$BlockComment\$\$ Excitation device
	SIG_6	Alarm - high	\$\$BlockComment\$\$ Tripping disconnecter fault T1
	SIG_7	Alarm - high	\$\$BlockComment\$\$ trip disconnecter fault T2
	SIG_8	Warning - high	\$\$BlockComment\$\$ Excitation disconnecter fault

Message instance	Message identifier	Message class	Event
MsgEvld_5	SIG_1	Alarm - high	\$\$BlockComment\$\$ General tripping device
	SIG_2	Alarm - high	\$\$BlockComment\$\$ trip differential protection IDIFF>
	SIG_3	Warning - high	\$\$BlockComment\$\$ excitation differential protection IDIFF>
	SIG_4	Alarm - high	\$\$BlockComment\$\$ trip differential protection IDIFF>>
	SIG_5	Warning - high	\$\$BlockComment\$\$ excitation differential protection IDIFF>>
	SIG_6	Warning - high	\$\$BlockComment\$\$ excitation impedance protection Z1
	SIG_7	Warning - high	\$\$BlockComment\$\$ excitation impedance protection Z2
	SIG_8	Warning - high	\$\$BlockComment\$\$ excitation impedance protection Z1B
MsgEvld_6	SIG_1	Warning - high	\$\$BlockComment\$\$ general excitation impedance protection
	SIG_2	Alarm - high	\$\$BlockComment\$\$ trip impedance protection Z1
	SIG_3	Alarm - high	\$\$BlockComment\$\$ trip impedance protection Z2
	SIG_4	Alarm - high	\$\$BlockComment\$\$ trip impedance protection Z1B
	SIG_5	Alarm - high	\$\$BlockComment\$\$ general trip impedance protection
	SIG_6	Alarm - high	\$\$BlockComment\$\$ trip underexcitation class 1
	SIG_7	Alarm - high	\$\$BlockComment\$\$ trip underexcitation class 2
	SIG_8	Alarm - high	\$\$BlockComment\$\$ trip underexcitation class 3
MsgEvld_7	SIG_1	Alarm - high	\$\$BlockComment\$\$ trip overvoltage protection U>
	SIG_2	Warning - high	\$\$BlockComment\$\$ excitation overvoltage protection U>
	SIG_3	Alarm - high	\$\$BlockComment\$\$ trip overvoltage protection U>>
	SIG_4	Warning - high	\$\$BlockComment\$\$ excitation overvoltage protection U>>
	SIG_5	Alarm - high	\$\$BlockComment\$\$ trip frequency 1 below limit
	SIG_6	Warning - high	\$\$BlockComment\$\$ excitation frequency 1 below limit
	SIG_7	Alarm - high	\$\$BlockComment\$\$ trip frequency 2 below limit
	SIG_8	Warning - high	\$\$BlockComment\$\$ excitation frequency 2 below limit

Message instance	Message identifier	Message class	Event
MsgEvId_8	SIG_1	Alarm - high	\$\$BlockComment\$\$ trip undervoltage protection U<
	SIG_2	Warning - high	\$\$BlockComment\$\$ excitation undervoltage protection U<
	SIG_3	Alarm - high	\$\$BlockComment\$\$ trip undervoltage protection U<<
	SIG_4	Warning - high	\$\$BlockComment\$\$ excitation undervoltage protection U<<
	SIG_5	Alarm - high	\$\$BlockComment\$\$ trip cut-in protection
	SIG_6	Warning - high	\$\$BlockComment\$\$ excitation cut-in protection
	SIG_7	Alarm - high	\$\$BlockComment\$\$ trip reverse power protection
	SIG_8	Alarm - high	\$\$BlockComment\$\$ Tripping coupling 1
MsgEvId_9	SIG_1	Warning - high	\$\$BlockComment\$\$ Excitation coupling 1
	SIG_2	Status message - AS	\$\$BlockComment\$\$ Parameter set A active
	SIG_3	Status message - AS	\$\$BlockComment\$\$ Parameter set B active
	SIG_4	Status message - AS	\$\$BlockComment\$\$ Parameter set C active
	SIG_5	Status message - AS	\$\$BlockComment\$\$ Parameter set D active
	SIG_6	Alarm - high	\$\$BlockComment\$\$ Fault group message
	SIG_7	Warning - high	\$\$BlockComment\$\$ Local mode
	SIG_8	Warning - high	\$\$BlockComment\$\$ Warning group message
MsgEvId_10	SIG_1	Status message - AS	\$\$BlockComment\$\$ Interlock: Enable Q01 close
	SIG_2	Status message - AS	\$\$BlockComment\$\$ Interlock: Enable Q01 open
	SIG_3	Status message - AS	\$\$BlockComment\$\$ Interlock: Enable Q1 close
	SIG_4	Status message - AS	\$\$BlockComment\$\$ Interlock: Enable Q1 open
	SIG_5	Status message - AS	\$\$BlockComment\$\$ Interlock: Enable Q2 close
	SIG_6	Status message - AS	\$\$BlockComment\$\$ Interlock: Enable Q2 open
	SIG_7	Status message - AS	\$\$BlockComment\$\$ Interlock: Enable Q8 close
	SIG_8	Status message - AS	\$\$BlockComment\$\$ Interlock: Enable Q8 open

Message instance	Message identifier	Message class	Event
MsgEvld_11	SIG_1	Status message - AS	\$\$BlockComment\$\$ Interlock: Enable Q02 close
	SIG_2	Status message - AS	\$\$BlockComment\$\$ Interlock: Enable Q02 Open
	SIG_3	Alarm - high	\$\$BlockComment\$\$ IEC Alarm 1
	SIG_4	Alarm - high	\$\$BlockComment\$\$ IEC Alarm 2
	SIG_5	Alarm - high	\$\$BlockComment\$\$ IEC Alarm 3
	SIG_6	Alarm - high	\$\$BlockComment\$\$ IEC Alarm 4
	SIG_7	Alarm - high	\$\$BlockComment\$\$ IEC Alarm 5
	SIG_8	Alarm - high	\$\$BlockComment\$\$ IEC Alarm 6
MsgEvld_12	SIG_1	Alarm - high	\$\$BlockComment\$\$ IEC Alarm 7
	SIG_2	Alarm - high	\$\$BlockComment\$\$ IEC Alarm 8
	SIG_3	Alarm - high	\$\$BlockComment\$\$ IEC Alarm 9
	SIG_4	Alarm - high	\$\$BlockComment\$\$ IEC Alarm 10
	SIG_5	Alarm - high	\$\$BlockComment\$\$ IEC Alarm 11
	SIG_6	Alarm - high	\$\$BlockComment\$\$ IEC Alarm 12
	SIG_7	Alarm - high	\$\$BlockComment\$\$ IEC Alarm 13
	SIG_8	Alarm - high	\$\$BlockComment\$\$ IEC Alarm 14
MsgEvld_d_1	SIG_1	Status message - AS	\$\$BlockComment\$\$ Circuit breaker Q01 closed
	SIG_2	Status message - AS	\$\$BlockComment\$\$ Circuit breaker Q01 opened
	SIG_3	Status message - AS	\$\$BlockComment\$\$ Disconnecter Q1 closed
	SIG_4	Status message - AS	\$\$BlockComment\$\$ Disconnecter Q1 open
	SIG_5	Status message - AS	\$\$BlockComment\$\$ Switch Q2 closed
	SIG_6	Status message - AS	\$\$BlockComment\$\$ Switch Q2 open
	SIG_7	Status message - AS	\$\$BlockComment\$\$ Grounding Q8 closed
	SIG_8	Status message - AS	\$\$BlockComment\$\$ Grounding Q8 open

Message instance	Message identifier	Message class	Event
MsgEvld_d_2	SIG_1	Status message - AS	\$\$BlockComment\$\$ Circuit breaker Q02 closed
	SIG_2	Status message - AS	\$\$BlockComment\$\$ Circuit breaker Q02 opened
	SIG_3	Status message - AS	\$\$BlockComment\$\$ Switch 1 closed
	SIG_4	Status message - AS	\$\$BlockComment\$\$ Switch 1 open
	SIG_5	Status message - AS	\$\$BlockComment\$\$ Switch 2 closed
	SIG_6	Status message - AS	\$\$BlockComment\$\$ Switch 2 open
	SIG_7	Status message - AS	\$\$BlockComment\$\$ Switch 3 closed
	SIG_8	Status message - AS	\$\$BlockComment\$\$ Switch 3 open
MsgEvld_device	SIG_1	AS process control message - error	\$\$BlockComment\$\$ Error device IED name:@1%\$@ -> IP address: @2%\$@
	SIG_2	AS control system message - fault	\$\$BlockComment\$\$ Redundancy failure
	SIG_3	AS process control message - error	\$\$BlockComment\$\$ PROFIBUS mapping alarm
	SIG_4	PLC Process Control Message - Error	\$\$BlockComment\$\$ Message buffer overflow in SIPROTEC
	SIG_5	Alarm - high	
	SIG_6	Alarm - high	
	SIG_7	Alarm - high	
	SIG_8	Alarm - high	
MsgEvldlocal1	SIG_1	Alarm - high	\$\$BlockComment\$\$ Error feedback Q01
	SIG_2	Alarm - high	\$\$BlockComment\$\$ Error feedback Q1
	SIG_3	Alarm - high	\$\$BlockComment\$\$ Error feedback Q2
	SIG_4	Alarm - high	\$\$BlockComment\$\$ Error feedback Q8
	SIG_5	Alarm - high	\$\$BlockComment\$\$ Error feedback Q02
	SIG_6	AS process control message - error	\$\$BlockComment\$\$ Connection fault
	SIG_7	Alarm - high	\$\$BlockComment\$\$ External trip signal Q01
	SIG_8	AS process control message - error	\$\$BlockComment\$\$ AS process control message - external error

Message instance	Message identifier	Message class	Event
MsgEvldlocal2	SIG_1	Warning - high	\$\$BlockComment\$\$ Maintenance: Operating cycle warning (@3%d@)
	SIG_2	Alarm - high	\$\$BlockComment\$\$ Maintenance: Operating cycle alarm (@3%d@)
	SIG_3	Warning - high	\$\$BlockComment\$\$ Maintenance: Operating time warning (@4%d@)
	SIG_4	Alarm - high	\$\$BlockComment\$\$ Maintenance: Operating time alarm (@4%d@)
	SIG_5	Alarm - high	\$\$BlockComment\$\$ External message 1
	SIG_6	Alarm - high	\$\$BlockComment\$\$ External message 2
	SIG_7	Alarm - high	\$\$BlockComment\$\$ External message 3
	SIG_8	Alarm - high	\$\$BlockComment\$\$ External trip signal Q02
MsgEvldlocal3	SIG_1	Alarm - high	\$\$BlockComment\$\$ Q01 switch position invalid
	SIG_2	Alarm - high	\$\$BlockComment\$\$ Q1 switch position invalid
	SIG_3	Alarm - high	\$\$BlockComment\$\$ Q2 switch position invalid
	SIG_4	Alarm - high	\$\$BlockComment\$\$ Q8 switch position invalid
	SIG_5	Alarm - high	\$\$BlockComment\$\$ Q02 switch position invalid
	SIG_6	Alarm - high	
	SIG_7	Alarm - high	
	SIG_8	Alarm - high	

7.1.5 I/O list

7.1.5.1 Inputs

I/O name	Description	Value	Data type
Rcv_Data	Connect with SG_/SC_FEEDER		STRUCT
ModLiOp	1 = Interconnect/Auto, 0 = Manual: Input for auto / manual commands		STRUCT
ModLiOp.Value		0	BOOL
ModLiOp.ST		16#80	BYTE
AutModOp	1 = Automatic mode: Automatic mode by operator	0	BOOL
ManModOp	1 = Manual mode: Manual mode by operator	1	BOOL
AutModLi	1 = Automatic mode: Automatic mode via link or SFC		STRUCT
AutModLi.Value		0	BOOL

I/O name	Description	Value	Data type
AutModLi.ST		16#80	BYTE
ManModLi	1 = Manual mode: Manual mode via link or SFC		STRUCT
ManModLi.Value		0	BOOL
ManModLi.ST		16#80	BYTE
OosOp	1 = Out of service: Out of service mode via operator	0	BOOL
OosLi	1 = Out of service: Out of service mode via field signal		STRUCT
OosLi.Value		0	BOOL
OosLi.ST		16#80	BYTE
RstOp	Reset signal operator	0	BOOL
RstLi	Linked reset signal		STRUCT
RstLi.Value	Value	1	BOOL
RstLi.ST	Signal state	16#80	BYTE
BB_Color	Connection to PC_BUSBAR	16#0	DWORD
Q01_Permit	1=Authorization is OK, 0=Activation not permitted		STRUCT
Q01_Permit.Value	Value		BOOL
Q01_Permit.ST	Signal state		BYTE
Q01_Perm_En	1=Authorization activated, 0=Authorization deactivated	1	BOOL
Q01_Intlock	0=Q01 is locked, 1=Q01 is not locked		STRUCT
Q01_Intlock.Value	Value		BOOL
Q01_Intlock.ST	Signal state		BYTE
Q01_Intl_En	1=Interlock activated, 0=Interlock deactivated	1	BOOL
Q01_Protect	0=Protection is active, 1=Protection is not active		STRUCT
Q01_Protect.Value	Value		BOOL
Q01_Protect.ST	Signal state		BYTE
Q01_Prot_En	1=Protection activated, 0=Protection deactivated	1	BOOL
Q02_Permit	1=Authorization is OK, 0=Activation not permitted		STRUCT
Q02_Permit.Value	Value		BOOL
Q02_Permit.ST	Signal state		BYTE
Q02_Perm_En	1=Authorization activated, 0=Authorization deactivated	1	BOOL
Q02_Intlock	0=Q02 is locked, 1=Q02 is not locked		STRUCT
Q02_Intlock.Value	Value		BOOL
Q02_Intlock.ST	Signal state		BYTE
Q02_Intl_En	1=Interlock activated, 0=Interlock deactivated	1	BOOL
Q02_Protect	0=Protection is active, 1=Protection is not active		STRUCT
Q02_Protect.Value	Value		BOOL
Q02_Protect.ST	Signal state		BYTE

I/O name	Description	Value	Data type
Q02_Prot_En	1=Protection activated, 0=Protection deactivated	1	BOOL
Q01_Opn	Rising edge: Command Q01 off		STRUCT
Q01_Opn.Value		1	BOOL
Q01_Opn.ST		16#80	BYTE
Q01_Cls	Rising edge: Command Q01 on		STRUCT
Q01_Cls.Value		1	BOOL
Q01_Cls.ST		16#80	BYTE
Q1_Opn	Rising edge: Command Q1 off		STRUCT
Q1_Opn.Value		0	BOOL
Q1_Opn.ST		16#80	BYTE
Q1_Cls	Rising edge: Command Q1 on		STRUCT
Q1_Cls.Value		0	BOOL
Q1_Cls.ST		16#80	BYTE
Q2_Opn	Rising edge: Command Q2 off		STRUCT
Q2_Opn.Value		0	BOOL
Q2_Opn.ST		16#80	BYTE
Q2_Cls	Rising edge: Command Q2 on		STRUCT
Q2_Cls.Value		0	BOOL
Q2_Cls.ST		16#80	BYTE
Q8_Opn	Rising edge: Command Q8 off		STRUCT
Q8_Opn.Value		0	BOOL
Q8_Opn.ST		16#80	BYTE
Q8_Cls	Rising edge: Command Q8 on		STRUCT
Q8_Cls.Value		0	BOOL
Q8_Cls.ST		16#80	BYTE
Q02_Opn	Rising edge: Command Q02 off		STRUCT
Q02_Opn.Value		0	BOOL
Q02_Opn.ST		16#80	BYTE
Q02_Cls	Rising edge: Command Q02 on		STRUCT
Q02_Cls.Value		0	BOOL
Q02_Cls.ST		16#80	BYTE
CMD_sp1_Opn	Rising edge: Free command 1 off		STRUCT
CMD_sp1_Opn.Value		0	BOOL
CMD_sp1_Opn.ST		16#80	BYTE
CMD_sp1_Cls	Rising edge: Free command 1 on		STRUCT
CMD_sp1_Cls.Value		0	BOOL
CMD_sp1_Cls.ST		16#80	BYTE
CMD_sp2_Opn	Rising edge: Free command 2 off		STRUCT
CMD_sp2_Opn.Value		0	BOOL
CMD_sp2_Opn.ST		16#80	BYTE
CMD_sp2_Cls	Rising edge: Free command 2 on		STRUCT
CMD_sp2_Cls.Value		0	BOOL

I/O name	Description	Value	Data type
CMD_sp2_Cls.ST		16#80	BYTE
CMD_sp3_Opn	Rising edge: Free command 3 off		STRUCT
CMD_sp3_Opn.Value		0	BOOL
CMD_sp3_Opn.ST		16#80	BYTE
CMD_sp3_Cls	Rising edge: Free command 3 on		STRUCT
CMD_sp3_Cls.Value		0	BOOL
CMD_sp3_Cls.ST		16#80	BYTE
ILCK_Q01_Cls	Interlock: Interlock Q0 close	0	BOOL
ILCK_Q01_Opn	Interlock: Interlock Q0 open	0	BOOL
ILCK_Q1_Cls	Interlock: Interlock Q1 close	0	BOOL
ILCK_Q1_Opn	Interlock: Interlock Q1 open	0	BOOL
ILCK_Q2_Cls	Interlock: Interlock Q2 close	0	BOOL
ILCK_Q2_Opn	Interlock: Interlock Q2 open	0	BOOL
ILCK_Q8_Cls	Interlock: Interlock Q8 close	0	BOOL
ILCK_Q8_Opn	Interlock: Interlock Q8 open	0	BOOL
ILCK_Q02_Cls	Interlock: Interlock Q9 close	0	BOOL
ILCK_Q02_Opn	Interlock: Interlock Q9 open	0	BOOL
FP_Visu	Configuration of the faceplate, see also AUTO-HOTSPOT		STRUCT
FP_Units	Configuration of the units in the faceplate, see also AUTOHOTSPOT		STRUCT
IEC_ILCK_EN	Interlock signals of IEC activated	0	BOOL
FACTOR_ENERGY	Factor for displaying energy values	1.0	REAL
Q01_Trip	1= State, 0= Trip signal		STRUCT
Q01_Trip.Value	Value	1	BOOL
Q01_Trip.ST	Signal state	16#80	BYTE
Q02_Trip	1= State, 0= Trip signal		STRUCT
Q02_Trip.Value	Value	1	BOOL
Q02_Trip.ST	Signal state	16#80	BYTE
MsgLock			STRUCT
MsgLock.Value		0	BOOL
MsgLock.ST		16#80	BYTE
SampleTime	Sampling time [s]	0.1	REAL
MsgEvId_1	Message event ID	16#00000002	DWORD
MsgEvId_2	Message event ID	16#0000000F	DWORD
MsgEvId_3	Message event ID	16#00000010	DWORD
MsgEvId_4	Message event ID	16#00000011	DWORD
MsgEvId_5	Message event ID	16#00000012	DWORD
MsgEvId_6	Message event ID	16#00000013	DWORD
MsgEvId_7	Message event ID	16#00000014	DWORD
MsgEvId_8	Message event ID	16#00000018	DWORD
MsgEvId_9	Message event ID	16#0000001A	DWORD
MsgEvId_10	Message event ID	16#0000001B	DWORD

I/O name	Description	Value	Data type
MsgEvId_11	Message event ID	16#0000001C	DWORD
MsgEvId_12	Message event ID	16#00000022	DWORD
MsgEvIdlocal1	Message event ID	16#00000020	DWORD
MsgEvIdlocal2	Message event ID	16#00000021	DWORD
MsgEvIdlocal3	Message event ID	16#00000023	DWORD
MsgEvId_d_1	Message event ID	16#0000001D	DWORD
MsgEvId_d_2	Message event ID	16#0000001E	DWORD
MsgEvId_device	Message event ID	16#0000001F	DWORD
BatchEn	Activate remote operation of controller to batch recipe	0	BOOL
BatchID	Current batch ID (number)	16#00000000	DWORD
BatchName	Current batch name	"	STRING[32]
StepNo	Batch step number	16#00000000	DWORD
Occupied	Occupied by batch	0	BOOL
CSF			STRUCT
CSF.Value		0	BOOL
CSF.ST		16#80	BYTE
ExtMsg1	External message 1		STRUCT
ExtMsg1.Value	Value	0	BOOL
ExtMsg1.ST	Signal state	16#80	BYTE
ExtMsg2	External message 2		STRUCT
ExtMsg2.Value	Value	0	BOOL
ExtMsg2.ST	Signal state	16#80	BYTE
ExtMsg3	External message 3		STRUCT
ExtMsg3.Value	Value	0	BOOL
ExtMsg3.ST	Signal state	16#80	BYTE
UserAna1	User analog input 1		STRUCT
UserAna1.Value	Value	0.0	REAL
UserAna1.ST	Signal state	16#FF	BYTE
UA1unit	Unit of measurement user analog input 1	0	INT
UserAna2	User analog input 2		STRUCT
UserAna2.Value	Value	0.0	REAL
UserAna2.ST	Signal state	16#FF	BYTE
UA2unit	Unit of measurement user analog input 2	0	INT
UserAna3	User analog input 3		STRUCT
UserAna3.Value	Value	0.0	REAL
UserAna3.ST	Signal state	16#FF	BYTE
UA3unit	Unit of measurement user analog input 3	0	INT
UserAna4	User analog input 4		STRUCT
UserAna4.Value	Value	0.0	REAL
UserAna4.ST	Signal state	16#FF	BYTE
UA4unit	Unit of measurement user analog input 4	0	INT
UserStatus	User status bits	16#0	BYTE

I/O name	Description	Value	Data type
SelFp1	Select faceplate 1		ANY
OS_Perm	Operator authorizations		STRUCT
OS_Perm.Bit0	1 = Operator may switch to Automatic mode	1	BOOL
OS_Perm.Bit1	1 = Operator may switch to Manual mode	1	BOOL
OS_Perm.Bit2	1 = Operator may switch to Local operating mode	1	BOOL
OS_Perm.Bit3	1 = Operator may switch to "Out of service" mode	1	BOOL
OS_Perm.Bit4	1 = Operator may switch to "Tracking mode"	1	BOOL
OS_Perm.Bit5		1	BOOL
OS_Perm.Bit6		1	BOOL
OS_Perm.Bit7	1 = Operator may reset block	1	BOOL
OS_Perm.Bit8		1	BOOL
OS_Perm.Bit9		1	BOOL
OS_Perm.Bit10		1	BOOL
OS_Perm.Bit11	1 = Operator can activate function simulation	1	BOOL
OS_Perm.Bit12	1 = Operator can enable release for maintenance	1	BOOL
OS_Perm.Bit13	1 = Button for Q01 is visible	1	BOOL
OS_Perm.Bit14	1 = Button for Q1 is visible	1	BOOL
OS_Perm.Bit15	1 = Button for Q2 is visible	1	BOOL
OS_Perm.Bit16	1 = Button for Q8 is visible	1	BOOL
OS_Perm.Bit17	1 = Button for Q02 is visible	1	BOOL
OS_Perm.Bit18		1	BOOL
OS_Perm.Bit19		1	BOOL
OS_Perm.Bit20		1	BOOL
OS_Perm.Bit21		1	BOOL
OS_Perm.Bit22		1	BOOL
OS_Perm.Bit23		1	BOOL
OS_Perm.Bit24		1	BOOL
OS_Perm.Bit25		1	BOOL
OS_Perm.Bit26		1	BOOL
OS_Perm.Bit27		1	BOOL
OS_Perm.Bit28		1	BOOL
OS_Perm.Bit29		1	BOOL
OS_Perm.Bit30		1	BOOL
OS_Perm.Bit31		1	BOOL
Feature	State of different functions		STRUCT
Feature.Bit0	Reserved	0	BOOL
Feature.Bit1	1 = OosLi may switch to "Out of service" mode	0	BOOL
Feature.Bit2	Reserved	0	BOOL
Feature.Bit3	Reserved	0	BOOL
Feature.Bit4	Reserved	0	BOOL

I/O name	Description	Value	Data type
Feature.Bit5	Reserved	0	BOOL
Feature.Bit6	Reserved	0	BOOL
Feature.Bit7	Reserved	0	BOOL
Feature.Bit8	Reserved	0	BOOL
Feature.Bit9	Reserved	0	BOOL
Feature.Bit10	Reserved	0	BOOL
Feature.Bit11	Reserved	0	BOOL
Feature.Bit12	Hide OpTm	0	BOOL
Feature.Bit13	Hide SwCount	0	BOOL
Feature.Bit14	Reserved	0	BOOL
Feature.Bit15	1 = Q01 status alarm active	0	BOOL
Feature.Bit16	1 = Q1 status alarm active	0	BOOL
Feature.Bit17	1 = Q2 status alarm active	0	BOOL
Feature.Bit18	1 = Q8 status alarm active	0	BOOL
Feature.Bit19	1 = Q02 status alarm active	0	BOOL
Feature.Bit20	1 = Dynamic active power direction enabled:	0	BOOL
Feature.Bit21	1 = Dynamic active power direction: negative power up; 0 = negative power down	0	BOOL
Feature.Bit22	Reserved	0	BOOL
Feature.Bit23	Reserved	0	BOOL
Feature.Bit24	1 = Local authorization enabled	0	BOOL
Feature.Bit25	1 = Suppression of all messages if MsgLock = 1	0	BOOL
Feature.Bit26	Reserved	0	BOOL
Feature.Bit27	Reserved	0	BOOL
Feature.Bit28	Reserved	0	BOOL
Feature.Bit29	Reserved	0	BOOL
Feature.Bit30	Reserved	0	BOOL
Feature.Bit31	1 = Resetting of interlocks in Manual mode	0	BOOL
MonTime	Time in seconds until Timeout (for control switch)	20	INT

See also

Configuration of the faceplate and the units (Page 19)

7.1.5.2 Inputs / outputs

I/O name	Description	Value	Data type
Cmd0	Control position circuit breaker Q01	16#02	BYTE
Cmd1	Control position disconnector Q1	16#02	BYTE
Cmd2	Control position switch Q2	16#02	BYTE
Cmd8	Control position ground Q8	16#02	BYTE

I/O name	Description	Value	Data type
Cmd9	Control position switch Q02	16#02	BYTE
Cmdsp1	Control position free command 1	16#02	BYTE
Cmdsp2	Control position free command 2	16#02	BYTE
Cmdsp3	Control position free command 3	16#02	BYTE
OpTmWarnLimit	Warning limit operating hours	0	DINT
OpTmAlarmLimit	Alarm limit operating hours	0	DINT
SwCountWarnLimit	Warning limit operating cycles	0	DINT
SwCountAlarmLimit	Alarm limit operating cycles	0	DINT

7.1.5.3 Outputs

I/O name	Description	Value	Data type
ENO		0	BOOL
Snd_Data	Connect with SG_/SC_FEEDER		STRUCT
QCONNERR		1	BOOL
F	Frequency		STRUCT
F.Value		0.0	REAL
F.ST		16#FF	BYTE
I_A	Current Phase A		STRUCT
I_A.Value		0.0	REAL
I_A.ST		16#FF	BYTE
I_B	Current Phase B		STRUCT
I_B.Value		0.0	REAL
I_B.ST		16#FF	BYTE
I_C	Current Phase C		STRUCT
I_C.Value		0.0	REAL
I_C.ST		16#FF	BYTE
P	Power		STRUCT
P.Value		0.0	REAL
P.ST		16#FF	BYTE
PF	Cos Phi - power factor		STRUCT
PF.Value		0.0	REAL
PF.ST		16#FF	BYTE
Q	Reactive power		STRUCT
Q.Value		0.0	REAL
Q.ST		16#FF	BYTE
S	Apparent power		STRUCT
S.Value		0.0	REAL
S.ST		16#FF	BYTE
U	Neutral voltage		STRUCT
U.Value		0.0	REAL
U.ST		16#FF	BYTE

I/O name	Description	Value	Data type
U_A	Voltage Phase A		STRUCT
U_A.Value		0.0	REAL
U_A.ST		16#FF	BYTE
U_AB	Voltage A-B		STRUCT
U_AB.Value		0.0	REAL
U_AB.ST		16#FF	BYTE
U_B	Voltage Phase B		STRUCT
U_B.Value		0.0	REAL
U_B.ST		16#FF	BYTE
U_BC	Voltage B-C		STRUCT
U_BC.Value		0.0	REAL
U_BC.ST		16#FF	BYTE
U_C	Voltage Phase C		STRUCT
U_C.Value		0.0	REAL
U_C.ST		16#FF	BYTE
U_CA	Voltage C-A		STRUCT
U_CA.Value		0.0	REAL
U_CA.ST		16#FF	BYTE
REAL_sp1	Free REAL 1		STRUCT
REAL_sp1.Value		0.0	REAL
REAL_sp1.ST		16#FF	BYTE
REAL_sp2	Free REAL 2		STRUCT
REAL_sp2.Value		0.0	REAL
REAL_sp2.ST		16#FF	BYTE
REAL_sp3	Free REAL 3		STRUCT
REAL_sp3.Value		0.0	REAL
REAL_sp3.ST		16#FF	BYTE
REAL_sp4	Free REAL 4		STRUCT
REAL_sp4.Value		0.0	REAL
REAL_sp4.ST		16#FF	BYTE
REAL_sp5	Free REAL 5		STRUCT
REAL_sp5.Value		0.0	REAL
REAL_sp5.ST		16#FF	BYTE
Pos01	Position circuit breaker Q01	16#00000000	DWORD
Pos1	Position disconnecter Q1	16#00000000	DWORD
Pos2	Position switch Q2	16#00000000	DWORD
Pos8	Position ground Q8	16#00000000	DWORD
Pos02	Position circuit breaker Q02	16#00000000	DWORD
Pos_CMD_sp1	Position free command 1	16#00000000	DWORD
Pos_CMD_sp2	Position free command 2	16#00000000	DWORD
Pos_CMD_sp3	Position free command 3	16#00000000	DWORD
Beh	Behavior	0	INT
Dev_OK	Device OK	0	INT

I/O name	Description	Value	Data type
GndFdir	Ground fault direction	0	INT
INT_sp1	Free INT 1		STRUCT
INT_sp1.Value		0	INT
INT_sp1.ST		16#FF	BYTE
INT_sp2	Free INT 2		STRUCT
INT_sp2.Value		0	INT
INT_sp2.ST		16#FF	BYTE
INT_sp3	Free INT 3		STRUCT
INT_sp3.Value		0	INT
INT_sp3.ST		16#FF	BYTE
INT_sp4	Free INT 4		STRUCT
INT_sp4.Value		0	INT
INT_sp4.ST		16#FF	BYTE
INT_sp5	Free INT 5		STRUCT
INT_sp5.Value		0	INT
INT_sp5.ST		16#FF	BYTE
PTOC1_Op	Tripping 51 overcurrent stage		STRUCT
PTOC1_Op.Value		0	BOOL
PTOC1_Op.ST		16#FF	BYTE
PTOC1_Str	Excitation 51 overcurrent stage		STRUCT
PTOC1_Str.Value		0	BOOL
PTOC1_Str.ST		16#FF	BYTE
PTOC10_Op	Tripping 67-1 rectified overcurrent stage I >		STRUCT
PTOC10_Op.Value		0	BOOL
PTOC10_Op.ST		16#FF	BYTE
PTOC10_Str	Excitation 67-1 rectified overcurrent stage I >		STRUCT
PTOC10_Str.Value		0	BOOL
PTOC10_Str.ST		16#FF	BYTE
PTOC11_Op	Tripping 67-2 rectified overcurrent stage I >>		STRUCT
PTOC11_Op.Value		0	BOOL
PTOC11_Op.ST		16#FF	BYTE
PTOC11_Str	Excitation 67-2 rectified overcurrent stage I >>		STRUCT
PTOC11_Str.Value		0	BOOL
PTOC11_Str.ST		16#FF	BYTE
PTOC12_Op	Tripping 67N-1 rectified overcurrent stage IE >		STRUCT
PTOC12_Op.Value		0	BOOL
PTOC12_Op.ST		16#FF	BYTE
PTOC12_Str	Excitation 67N-1 rectified overcurrent stage IE >		STRUCT
PTOC12_Str.Value		0	BOOL
PTOC12_Str.ST		16#FF	BYTE
PTOC13_Op	Tripping 67N-2 rectified overcurrent stage IE >>		STRUCT
PTOC13_Op.Value		0	BOOL
PTOC13_Op.ST		16#FF	BYTE

I/O name	Description	Value	Data type
PTOC13_Str	Excitation 67N-2 rectified overcurrent stage IE >>		STRUCT
PTOC13_Str.Value		0	BOOL
PTOC13_Str.ST		16#FF	BYTE
PTOC18_Op	Tripping 50-3 overcurrent stage		STRUCT
PTOC18_Op.Value		0	BOOL
PTOC18_Op.ST		16#FF	BYTE
PTOC18_Str	Excitation 50-3 overcurrent stage		STRUCT
PTOC18_Str.Value		0	BOOL
PTOC18_Str.ST		16#FF	BYTE
PTOC2_Op	Tripping 51N overcurrent stage		STRUCT
PTOC2_Op.Value		0	BOOL
PTOC2_Op.ST		16#FF	BYTE
PTOC2_Str	Excitation 51N overcurrent stage		STRUCT
PTOC2_Str.Value		0	BOOL
PTOC2_Str.ST		16#FF	BYTE
PTOC3_Op	Tripping 67-TOC overcurrent stage		STRUCT
PTOC3_Op.Value		0	BOOL
PTOC3_Op.ST		16#FF	BYTE
PTOC3_Str	Excitation 67-TOC overcurrent stage		STRUCT
PTOC3_Str.Value		0	BOOL
PTOC3_Str.ST		16#FF	BYTE
PTOC4_Op	Tripping 67N-TOC overcurrent stage		STRUCT
PTOC4_Op.Value		0	BOOL
PTOC4_Op.ST		16#FF	BYTE
PTOC4_Str	Excitation 67N-TOC overcurrent stage		STRUCT
PTOC4_Str.Value		0	BOOL
PTOC4_Str.ST		16#FF	BYTE
PTOC5_Op	Tripping 46-TOC overcurrent stage		STRUCT
PTOC5_Op.Value		0	BOOL
PTOC5_Op.ST		16#FF	BYTE
PTOC5_Str	Excitation 46-TOC overcurrent stage		STRUCT
PTOC5_Str.Value		0	BOOL
PTOC5_Str.ST		16#FF	BYTE
PTOC6_Op	Tripping 50-1 overcurrent stage I >		STRUCT
PTOC6_Op.Value		0	BOOL
PTOC6_Op.ST		16#FF	BYTE
PTOC6_Str	Excitation 50-1 overcurrent stage I >		STRUCT
PTOC6_Str.Value		0	BOOL
PTOC6_Str.ST		16#FF	BYTE
PTOC7_Op	Tripping 50-2 overcurrent stage I >>		STRUCT
PTOC7_Op.Value		0	BOOL
PTOC7_Op.ST		16#FF	BYTE

I/O name	Description	Value	Data type
PTOC7_Str	Excitation 50-2 overcurrent stage I >>		STRUCT
PTOC7_Str.Value		0	BOOL
PTOC7_Str.ST		16#FF	BYTE
PTOC8_Op	Tripping 50N-1 overcurrent stage IE >		STRUCT
PTOC8_Op.Value		0	BOOL
PTOC8_Op.ST		16#FF	BYTE
PTOC8_Str	Excitation 50N-1 overcurrent stage IE >		STRUCT
PTOC8_Str.Value		0	BOOL
PTOC8_Str.ST		16#FF	BYTE
PTOC9_Op	Tripping 50N-2 overcurrent stage IE >>		STRUCT
PTOC9_Op.Value		0	BOOL
PTOC9_Op.ST		16#FF	BYTE
PTOC9_Str	Excitation 50N-2 overcurrent stage IE >>		STRUCT
PTOC9_Str.Value		0	BOOL
PTOC9_Str.ST		16#FF	BYTE
PU_Relay	Excitation combined signal		STRUCT
PU_Relay.Value		0	BOOL
PU_Relay.ST		16#FF	BYTE
RBRF1_OpEx	Tripping disconnector fault T1		STRUCT
RBRF1_OpEx.Value		0	BOOL
RBRF1_OpEx.ST		16#FF	BYTE
RBRF1_OpIn	Tripping disconnector fault T2		STRUCT
RBRF1_OpIn.Value		0	BOOL
RBRF1_OpIn.ST		16#FF	BYTE
RBRF1_Str	Excitation disconnector fault		STRUCT
RBRF1_Str.Value		0	BOOL
RBRF1_Str.ST		16#FF	BYTE
TR_Relay	Tripping combined signal		STRUCT
TR_Relay.Value		0	BOOL
TR_Relay.ST		16#FF	BYTE
PDIF1_Op	Tripping differential protection IDIFF>		STRUCT
PDIF1_Op.Value		0	BOOL
PDIF1_Op.ST		16#FF	BYTE
PDIF1_Str	Excitation differential protection IDIFF>		STRUCT
PDIF1_Str.Value		0	BOOL
PDIF1_Str.ST		16#FF	BYTE
PDIF2_Op	Tripping differential protection IDIFF>>		STRUCT
PDIF2_Op.Value		0	BOOL
PDIF2_Op.ST		16#FF	BYTE
PDIF2_Str	Excitation differential protection IDIFF>>		STRUCT
PDIF2_Str.Value		0	BOOL
PDIF2_Str.ST		16#FF	BYTE
PDIS1_Str	Excitation impedance protection Z1		STRUCT

I/O name	Description	Value	Data type
PDIS1_Str.Value		0	BOOL
PDIS1_Str.ST		16#FF	BYTE
PDIS2_Str	Excitation impedance protection Z2		STRUCT
PDIS2_Str.Value		0	BOOL
PDIS2_Str.ST		16#FF	BYTE
PDIS3_Str	Excitation impedance protection Z1B		STRUCT
PDIS3_Str.Value		0	BOOL
PDIS3_Str.ST		16#FF	BYTE
PDIS4_Str	General excitation impedance protection		STRUCT
PDIS4_Str.Value		0	BOOL
PDIS4_Str.ST		16#FF	BYTE
PDIS1_Op	Tripping impedance protection Z1		STRUCT
PDIS1_Op.Value		0	BOOL
PDIS1_Op.ST		16#FF	BYTE
PDIS2_Op	Tripping impedance protection Z2		STRUCT
PDIS2_Op.Value		0	BOOL
PDIS2_Op.ST		16#FF	BYTE
PDIS3_Op	Tripping impedance protection Z1B		STRUCT
PDIS3_Op.Value		0	BOOL
PDIS3_Op.ST		16#FF	BYTE
PDIS4_Op	General tripping impedance protection		STRUCT
PDIS4_Op.Value		0	BOOL
PDIS4_Op.ST		16#FF	BYTE
PDUP1_Op	Tripping under-excitation protection class 1		STRUCT
PDUP1_Op.Value		0	BOOL
PDUP1_Op.ST		16#FF	BYTE
PDUP2_Op	Tripping under-excitation protection class 2		STRUCT
PDUP2_Op.Value		0	BOOL
PDUP2_Op.ST		16#FF	BYTE
PDUP3_Op	Tripping under-excitation protection class 3		STRUCT
PDUP3_Op.Value		0	BOOL
PDUP3_Op.ST		16#FF	BYTE
PTOV1_Op	Tripping overvoltage protection U>		STRUCT
PTOV1_Op.Value		0	BOOL
PTOV1_Op.ST		16#FF	BYTE
PTOV1_Str	Excitation overvoltage protection U>		STRUCT
PTOV1_Str.Value		0	BOOL
PTOV1_Str.ST		16#FF	BYTE
PTOV2_Op	Tripping overvoltage protection U>>		STRUCT
PTOV2_Op.Value		0	BOOL
PTOV2_Op.ST		16#FF	BYTE
PTOV2_Str	Excitation overvoltage protection U>>		STRUCT
PTOV2_Str.Value		0	BOOL

I/O name	Description	Value	Data type
PTOV2_Str.ST		16#FF	BYTE
PTUF1_Op	Tripping frequency 1 below limit		STRUCT
PTUF1_Op.Value		0	BOOL
PTUF1_Op.ST		16#FF	BYTE
PTUF1_Str	Excitation frequency 1 below limit		STRUCT
PTUF1_Str.Value		0	BOOL
PTUF1_Str.ST		16#FF	BYTE
PTUF2_Op	Tripping frequency 2 below limit		STRUCT
PTUF2_Op.Value		0	BOOL
PTUF2_Op.ST		16#FF	BYTE
PTUF2_Str	Excitation frequency 2 below limit		STRUCT
PTUF2_Str.Value		0	BOOL
PTUF2_Str.ST		16#FF	BYTE
PTUV1_Op	Tripping undervoltage protection U<		STRUCT
PTUV1_Op.Value		0	BOOL
PTUV1_Op.ST		16#FF	BYTE
PTUV1_Str	Excitation undervoltage protection U<		STRUCT
PTUV1_Str.Value		0	BOOL
PTUV1_Str.ST		16#FF	BYTE
PTUV2_Op	Tripping undervoltage protection U<<		STRUCT
PTUV2_Op.Value		0	BOOL
PTUV2_Op.ST		16#FF	BYTE
PTUV2_Str	Excitation undervoltage protection U<<		STRUCT
PTUV2_Str.Value		0	BOOL
PTUV2_Str.ST		16#FF	BYTE
PVOC2_Op	Tripping cut-in protection		STRUCT
PVOC2_Op.Value		0	BOOL
PVOC2_Op.ST		16#FF	BYTE
PVOC2_Str	Excitation cut-in protection		STRUCT
PVOC2_Str.Value		0	BOOL
PVOC2_Str.ST		16#FF	BYTE
PDOP1_Op	Tripping reverse power protection		STRUCT
PDOP1_Op.Value		0	BOOL
PDOP1_Op.ST		16#FF	BYTE
GAPC1_Op	Tripping coupling 1		STRUCT
GAPC1_Op.Value		0	BOOL
GAPC1_Op.ST		16#FF	BYTE
GAPC1_Str	Excitation coupling 1		STRUCT
GAPC1_Str.Value		0	BOOL
GAPC1_Str.ST		16#FF	BYTE
PARSET_A	Parameter set A	0	BOOL
PARSET_B	Parameter set B	0	BOOL
PARSET_C	Parameter set C	0	BOOL

I/O name	Description	Value	Data type
PARSET_D	Parameter set D	0	BOOL
Error	Group message		STRUCT
Error.Value		0	BOOL
Error.ST		16#FF	BYTE
Loc	Local mode	0	BOOL
Warning	General warning		STRUCT
Warning.Value		0	BOOL
Warning.ST		16#FF	BYTE
Q01_EnaCls	Interlock: Enable Q01 close	0	BOOL
Q01_EnaOpn	Interlock: Enable Q01 open	0	BOOL
Q1_EnaCls	Interlock: Enable Q1 close	0	BOOL
Q1_EnaOpn	Interlock: Enable Q1 open	0	BOOL
Q2_EnaCls	Interlock: Enable Q2 close	0	BOOL
Q2_EnaOpn	Interlock: Enable Q2 open	0	BOOL
Q8_EnaCls	Interlock: Enable Q8 close	0	BOOL
Q8_EnaOpn	Interlock: Enable Q8 open	0	BOOL
Q02_EnaCls	Interlock: Enable Q02 close	0	BOOL
Q02_EnaOpn	Interlock: Enable Q02 Open	0	BOOL
ALM_sp1	Free ALM 1		STRUCT
ALM_sp1.Value		0	BOOL
ALM_sp1.ST		16#FF	BYTE
ALM_sp2	Free ALM 2		STRUCT
ALM_sp2.Value		0	BOOL
ALM_sp2.ST		16#FF	BYTE
ALM_sp3	Free ALM 3		STRUCT
ALM_sp3.Value		0	BOOL
ALM_sp3.ST		16#FF	BYTE
ALM_sp4	Free ALM 4		STRUCT
ALM_sp4.Value		0	BOOL
ALM_sp4.ST		16#FF	BYTE
ALM_sp5	Free ALM 5		STRUCT
ALM_sp5.Value		0	BOOL
ALM_sp5.ST		16#FF	BYTE
ALM_sp6	Free ALM 6		STRUCT
ALM_sp6.Value		0	BOOL
ALM_sp6.ST		16#FF	BYTE
ALM_sp7	Free ALM 7		STRUCT
ALM_sp7.Value		0	BOOL
ALM_sp7.ST		16#FF	BYTE
ALM_sp8	Free ALM 8		STRUCT
ALM_sp8.Value		0	BOOL
ALM_sp8.ST		16#FF	BYTE
ALM_sp9	Free ALM 9		STRUCT

I/O name	Description	Value	Data type
ALM_sp9.Value		0	BOOL
ALM_sp9.ST		16#FF	BYTE
ALM_sp10	Free ALM 10		STRUCT
ALM_sp10.Value		0	BOOL
ALM_sp10.ST		16#FF	BYTE
ALM_sp11	Free ALM 11		STRUCT
ALM_sp11.Value		0	BOOL
ALM_sp11.ST		16#FF	BYTE
ALM_sp12	Free ALM 12		STRUCT
ALM_sp12.Value		0	BOOL
ALM_sp12.ST		16#FF	BYTE
ALM_sp13	Free ALM 13		STRUCT
ALM_sp13.Value		0	BOOL
ALM_sp13.ST		16#FF	BYTE
ALM_sp14	Free ALM 14		STRUCT
ALM_sp14.Value		0	BOOL
ALM_sp14.ST		16#FF	BYTE
SwCount	Operating cycles Q0	0	DINT
OpTm	Operating hours	0	DINT
WpForward	Supply active energy		STRUCT
WpForward.Value		0.0	REAL
WpForward.ST		16#FF	BYTE
WpReverse	Demand active energy		STRUCT
WpReverse.Value		0.0	REAL
WpReverse.ST		16#FF	BYTE
WqForward	Supply reactive energy		STRUCT
WpForward.Value		0.0	REAL
WqForward.ST		16#FF	BYTE
WqReverse	Demand reactive energy		STRUCT
WpReverse.Value		0.0	REAL
WqReverse.ST		16#FF	BYTE
DINT_sp1	DINT value 8 (free)		STRUCT
DINT_sp1.Value		0	DINT
DINT_sp1.ST		16#FF	BYTE
DINT_sp2	DINT value 8 (free)		STRUCT
DINT_sp2.Value		0	DINT
DINT_sp2.ST		16#FF	BYTE
BOOL_sp1	Free BOOL		STRUCT
BOOL_sp1.Value		0	BOOL
BOOL_sp1.ST		16#FF	BYTE
BOOL_sp2	Free BOOL		STRUCT
BOOL_sp2.Value		0	BOOL
BOOL_sp2.ST		16#FF	BYTE

I/O name	Description	Value	Data type
BOOL_sp3	Free BOOL		STRUCT
BOOL_sp3.Value		0	BOOL
BOOL_sp3.ST		16#FF	BYTE
BOOL_sp4	Free BOOL		STRUCT
BOOL_sp4.Value		0	BOOL
BOOL_sp4.ST		16#FF	BYTE
BOOL_sp5	Free BOOL		STRUCT
BOOL_sp5.Value		0	BOOL
BOOL_sp5.ST		16#FF	BYTE
BOOL_sp6	Free BOOL		STRUCT
BOOL_sp6.Value		0	BOOL
BOOL_sp6.ST		16#FF	BYTE
BOOL_sp7	Free BOOL		STRUCT
BOOL_sp7.Value		0	BOOL
BOOL_sp7.ST		16#FF	BYTE
BOOL_sp8	Free BOOL		STRUCT
BOOL_sp8.Value		0	BOOL
BOOL_sp8.ST		16#FF	BYTE
BOOL_sp9	Free BOOL		STRUCT
BOOL_sp9.Value		0	BOOL
BOOL_sp9.ST		16#FF	BYTE
BOOL_sp10	Free BOOL		STRUCT
BOOL_sp10.Value		0	BOOL
BOOL_sp10.ST		16#FF	BYTE
OpTmWarning	1 = Operating time warning active	0	BOOL
OpTmAlarm	1 = Operating time alarm active	0	BOOL
SwCountWarning	1 = Operating cycle warning active	0	BOOL
SwCountAlarm	1 = Operating cycle alarm active	0	BOOL
LockAct			STRUCT
LockAct.Value		0	BOOL
LockAct.ST		16#80	BYTE
LocalAct			STRUCT
LocalAct.Value		0	BOOL
LocalAct.ST		16#80	BYTE
AutAct			STRUCT
AutAct.Value		0	BOOL
AutAct.ST		16#80	BYTE
ManAct	1 = Manual: Manual value is active		STRUCT
ManAct.Value	Value	1	BOOL
ManAct.ST	Signal state	16#80	BYTE
OosAct			STRUCT
OosAct.Value		0	BOOL
OosAct.ST		16#80	BYTE

I/O name	Description	Value	Data type
OS_PermOut	Operator authorization: Output for OS	16#FFFFFFFF	DWORD
OS_PermLog	Operator authorization: Output for OS	16#FFFFFFFF	DWORD
ST_Worst	Worst signal state	16#80	BYTE
GrpErr			STRUCT
GrpErr.Value		0	BOOL
GrpErr.ST		16#80	BYTE
RdyToReset	1 = Ready to reset via RstLi or automatic commands		STRUCT
RdyToReset.Value	Value	0	BOOL
RdyToReset.ST	Signal state	16#80	BYTE
Status1	Word Status1	16#00000000	DWORD
Status3	Word Status3	16#00000000	DWORD
Status5	Word Status3	16#FFFFFFFF	DWORD
MsgErrlocal1	1 = Message error occurs	0	BOOL
MsgStatlocal1	Message status	16#0000	WORD
MsgAcknlocal1	Message acknowledgement status	16#0000	WORD
MsgErrlocal2	1 = Message error occurs	0	BOOL
MsgStatlocal2	Message status	16#0000	WORD
MsgAcknlocal2	Message acknowledgement status	16#0000	WORD
MsgErrlocal3	1=Message error occurs	0	BOOL
MsgStatlocal3	Message status	16#0000	WORD
MsgAcknlocal3	Message acknowledgement status	16#0000	WORD
VisuCodes	Display of the switches in the faceplate		STRUCT
VisuDesc	Labeling in the faceplate		STRUCT
VisuFields	Display of the switches in the faceplate		STRUCT
VisuMisc	Configuration of the units in the faceplate		STRUCT
VisuStates	Texts of switch settings in the faceplate		STRUCT

7.2 PC_FEEDER_2CB block icons

7.2.1 PC_FEEDER_2CB block symbols

The following block symbols are available for the PC_FEEDER_2CB object:

PC_FEEDER_2CB/1: Symbol with group display and large tag name:



Figure 7-1 PC_FEEDER_2CB/1

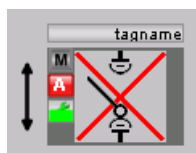
PC_FEEDER_2CB/2: Symbol with group display and small tag name:

Figure 7-2 PC_FEEDER_2CB/2

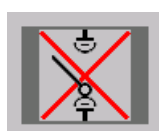
PC_FEEDER_2CB/3: Symbol without group display and with tag name:

Figure 7-3 PC_FEEDER_2CB/3

PC_FEEDER_2CB/4: 6 field symbols:

With this symbol the switches are also displayed completely in the symbol.

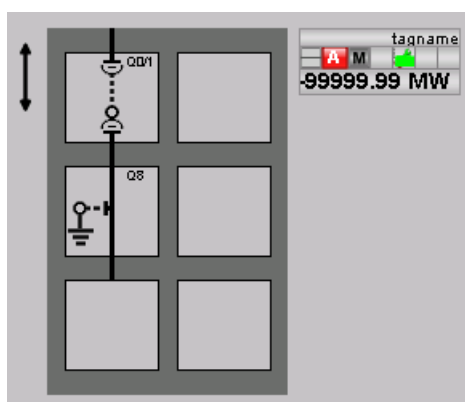


Figure 7-4 PC_FEEDER_2CB/4

PC_FEEDER_2CB/5: Rectangle symbol for overview pictures:

Additional functionalities:

- Dynamic coloring with the block input BB_Color



Figure 7-5 PC_FEEDER_2CB/5

PC_FEEDER_2CB/6: Small symbol disconnecter:

Additional functionalities:

- Dynamic coloring with the block input BB_Color
- Display closed grounding switch



Figure 7-6 PC_FEEDER_2CB/6

PC_FEEDER_2CB/7: Small symbol circuit breaker:

Additional functionalities:

- Dynamic coloring with the block input BB_Color
- Display closed grounding switch



Figure 7-7 PC_FEEDER_2CB/7

7.2.2 Dynamizing the active power direction

Dynamizing the active power direction

The active power display of the corresponding PC_FEEDER_2CB block icons can be dynamically displayed as a function of the active power P.

The latest value for the active power P is used to display the active power direction. Depending on the sign of the active power, the direction display is shown in one or the other direction or is completely hidden if the active power = 0W.

The dynamization of the active power can be enabled or disabled using **Feature.Bit20**.

- 0 = Dynamization of active power direction disabled
- 1 = Dynamization of active power direction enabled

Feature.Bit21 can be used to configure the direction of the active power in the event of a negative value

- 0 = For a negative active power value, the arrow points downwards.
- 1 = For a negative active power value, the arrow points upwards.

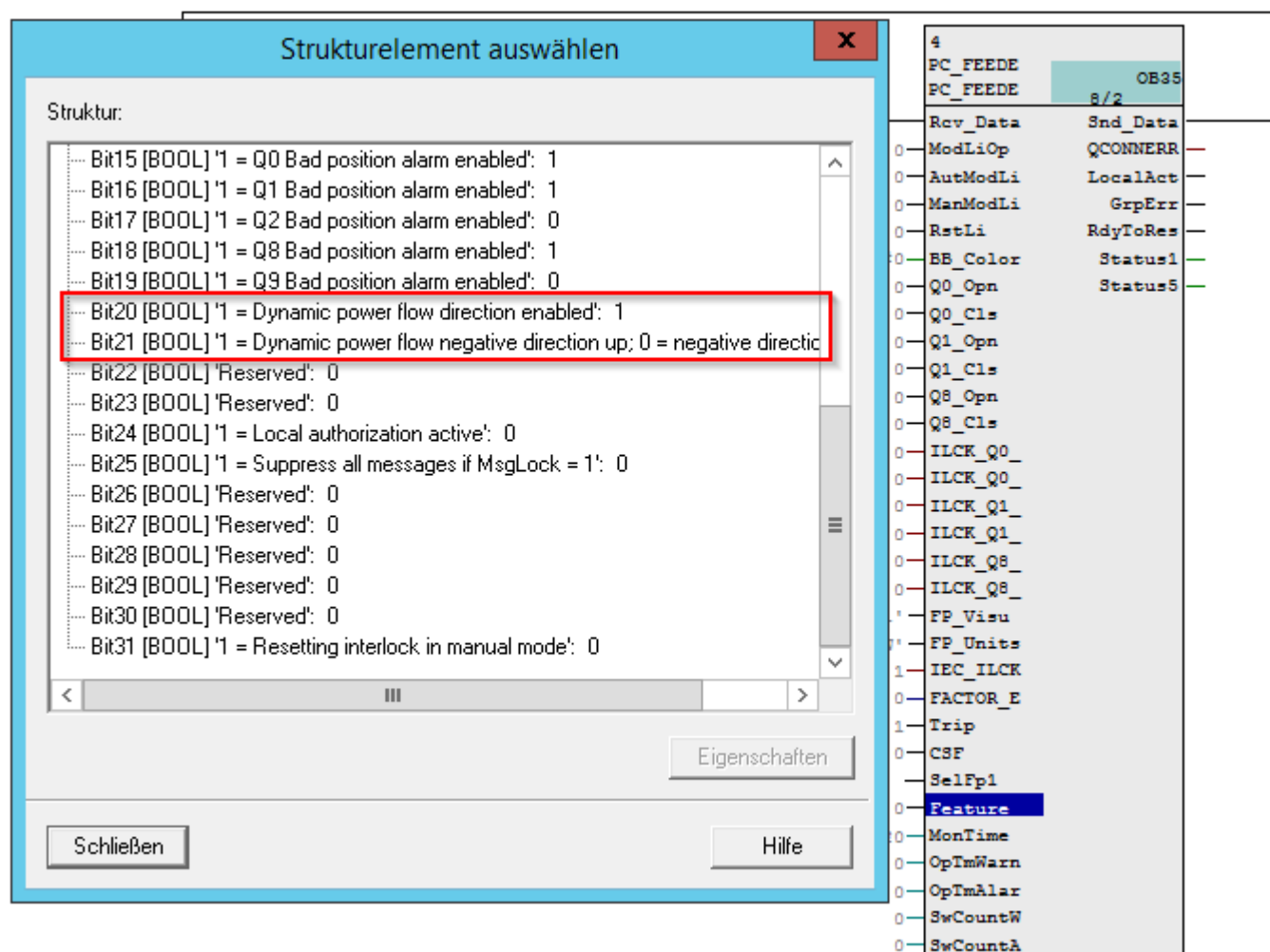


Figure 7-8 Active power direction - Feature bit PC_FEEDER

If dynamization is activated, the direction of the arrow is displayed based on the active power value and the setting at Feature.Bit21:



Figure 7-9 Active power direction - negative icon



Figure 7-10 Active power direction - positive icon

If the value of the active power = 0, the arrow is completely hidden.

7.3 PC_FEEDER_2CB faceplate

Available views:

View	Description
Default window	Switch representation, important measured values, interlocks, device status
Message view	Messages from device (messages, warnings, etc.). Identical to PC_FEEDER.
Measured value window	Measured values of FEEDER type. Identical to PC_FEEDER.
Maintenance window	Operating hours, switch, active parameter group. Identical to PC_FEEDER.
User-defined window	User-specific view. Identical to PC_FEEDER.
Memo view	Memo view

7.3.1 Standard window

7.3.1.1 Default window

The standard window offers the following functions:

- Display of the device status
- Display of the current operating mode of the device
- View of the interlock states of Q01 and Q02
- Display of important measured values
- Display of switch states
- Display of auxiliary values
- Change in the operating mode of the device

- Sending of switch commands to the device
- Tracking mode of switch states while connection is lost

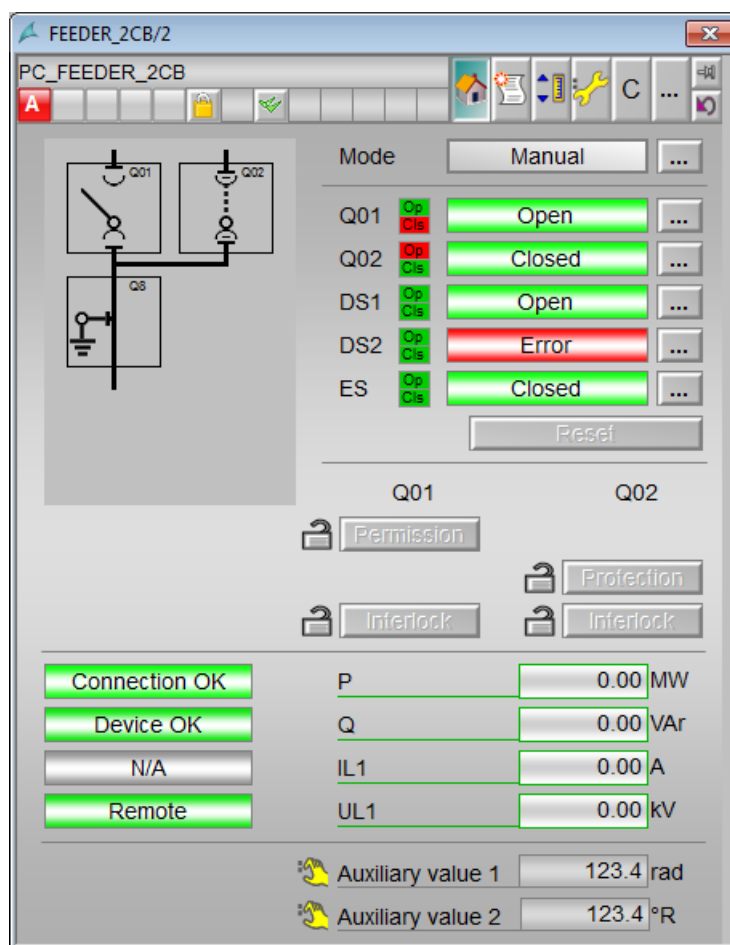


Figure 7-11 Standard window PC_FEEDER_2CB

Buttons

Note: Level 5 authorizations are required to operate the buttons in this view. The number of buttons depends on the configuration.

Input box	Action	Description
Q01	Cmd01 = 0/1	Switch Q01 command. 0 = Open 1 = Close
Q1	Cmd1 = 0/1	Command switch Q1. 0 = Open 1 = Close

Input box	Action	Description
Q2	Cmd2 = 0/1	Command switch Q2. 0 = Open 1 = Close
Q8	Cmd8 = 0/1	Command switch Q8. 0 = Open 1 = Close
Q02	Cmd9 = 0/1	Switch Q02 command. 0 = Open 1 = Close
Tracking	Activate tracking mode	In case the connection is lost, you can use this button to activate tracking mode
Faceplate	Jump to the connected faceplate	This button can be used to jump to a different faceplate when the input SelfP1 is connected.
Mode	Change the operating mode	Change the operating mode of the block
Reset	Reset an interlock or an external tripping	Reset the block after a "Protect" interlock or an external tripping (only available if Function bit 31 is set and the block is in manual mode)
Permit/Protect/Interlock	Jump to APL interlock blocks	Allows a jump to a faceplate of APL interlock blocks

Display

Note: The displays for the auxiliary values are only shown if at least one auxiliary value was interconnected to the corresponding block input.

Display	Meaning	Tag	Description
Connection status	Communication status of IED via Ethernet.	QCONERRO	QCONERRO = 0 (Connection OK) – GREEN QCONERRO = 1 (Comm. loss) – RED
Device status 1	Status of IED	Dev_OK	Dev_OK = 1 (Device OK) – GREEN Dev_OK = 2 (Device warning) – YELLOW Dev_OK = 3 (Device message) – RED
Device status 2	Behavior of IED	Beh	Beh = 1 (Protection on) – GREEN Beh = 2 (Protection locked) – YELLOW Beh = 3 (Protection test) – RED Beh = 4 (Protection test/locked) – RED Beh = 5 (Protection off) – RED
Device status 3	Local/Remote	Loc	Loc = 1: Local Loc = 0: Remote
Switch Q01	Switch position	Pos01	Pos01 = 0 (Intermediate st.) – RED Pos01 = 1 (Open) – GREEN Pos01 = 2 (Close) – GREEN Pos01 = 3 (Bad) – RED Pos01 = 4 (Error) – RED

Display	Meaning	Tag	Description
Switch Q1	Switch position	Pos1	Pos1 = 0 (Intermediate st.) – RED Pos1 = 1 (Open) – GREEN Pos1 = 2 (Close) – GREEN Pos1 = 3 (Faulty) – RED Pos1 = 4 (Fault) – RED
Switch Q2	Switch position	Pos2	Pos2 = 0 (Intermediate st.) – RED Pos2 = 1 (Open) – GREEN Pos2 = 2 (Close) – GREEN Pos2 = 3 (Faulty) – RED Pos2 = 4 (Fault) – RED
Switch Q8	Switch position	Pos8	Pos8 = 0 (Intermediate st.) – RED Pos8 = 1 (Open) – GREEN Pos8 = 2 (Close) – GREEN Pos8 = 3 (Faulty) – RED Pos8 = 4 (Fault) – RED
Switch Q02	Switch position	Pos02	Pos02 = 0 (Intermediate st.) – RED Pos02 = 1 (Open) – GREEN Pos02 = 2 (Close) – GREEN Pos02 = 3 (Bad) – RED Pos02 = 4 (Error) – RED
P	Power	P#Value	Power
Q	Reactive power	Q#Value	Reactive power
IL1	Current L1	I_A#Value	Current L1
UL1	Voltage L1 ground	U_A#Value	Voltage L1 ground
Auxiliary value 1	User analog value 1	UserA-na1#Value	User analog value 1
Auxiliary value 2	User analog value 2	UserA-na2#Value	User analog value 2

PC_TRAFO

The PC_TRAFO object is used for all transformer protection devices. The faceplate serves only for visualization. No commands are implemented in the default faceplates. However, four free commands are available in the user-defined faceplate and in the PLC faceplate.

PC_TRAFO consists of the following objects:

- PLC block PC_TRAFO
- WinCC block symbols
- WinCC faceplate PC_TRAFO with different views
- Device driver blocks (not part of the PCL)

8.1 PLC block PC_TRAFO

8.1.1 Description of PLC block PC_TRAFO

The PC_TRAFO (FB 2502) function block represents the transformer object in the PLC. The block is the interface between the device driver block and the remaining PCS 7 environment.

The block offers the following functions:

- Detecting data from the device driver block
- Providing data for the block symbol and the faceplate
- Controlling switches of the switchgear device (only free commands)
- APL operating modes Manual, Automatic
- Interlocking functions
- Handling of operator authorizations
- Various APL standard functions

8.1.2 Interfaces

The PLC block PC_TRAFO offers the following interfaces:

1. Device driver blocks

The device driver block is connected to the PC_TRAFO block through the input "Rcv_Data" and the output "Snd_Data".

2. Block symbol

The block symbol receives data from the block through the S7 connection between the PLC and WinCC.

3. Faceplate

The faceplate receives data from the block through the S7 connection between the PLC and WinCC.

4. APL interlock blocks

The interlock blocks can be connected through the "IntLock" input.

5. Logic of the S7 program (for controlling modes, switches, etc.)

The PC_TRAFO block can receive commands from the CFC logic and makes data available to other blocks.

8.1.3 Configuration

Use the CFC editor to install the block in a cyclic interrupt OB (OB30 to OB38). The block is also installed automatically in the startup OB (OB 100).

8.1.4 Operating modes

The PC_TRAFO block supports the APL standard operating modes.

The operating mode can only be changed through the interconnections on the PLC block.

The block can be operated in the following operating modes:

Operating mode	Description	Permitted commands	Display
Local operating mode	The device signals local operating mode (for example, the key-operated switch on the device is in the "local" position)	No	<i>LocalAct.Value</i> = true
Manual mode	The device is controlled through the faceplate	Only from faceplate	<i>ManAct.Value</i> = true
Automatic mode	The device is controlled through the connected logic in the CFC	Only by block inputs	<i>AutAct.Value</i> = true

For a detailed description of the operating modes refer to the "PCS 7 Advanced Process Library Function Manual" [1].

8.1.5 Switch commands

The PC_TRAFO block sends commands from the faceplate (in manual operation) or from other connected blocks (in automatic mode) to the device driver block. The following commands are implemented:

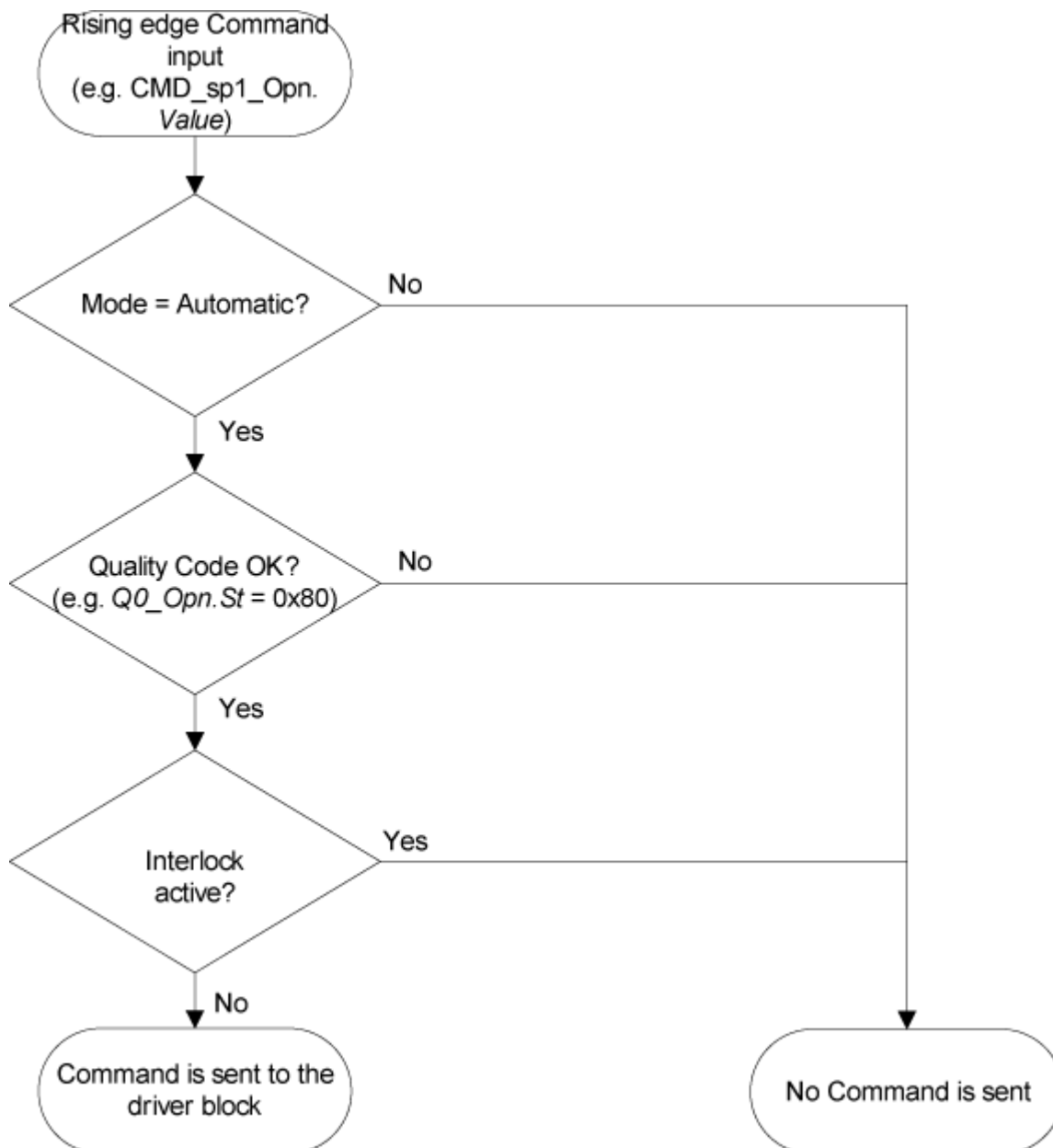
- Free command 1
- Free command 2
- Free command 3
- Free command 4

8.1.6 Interlocks

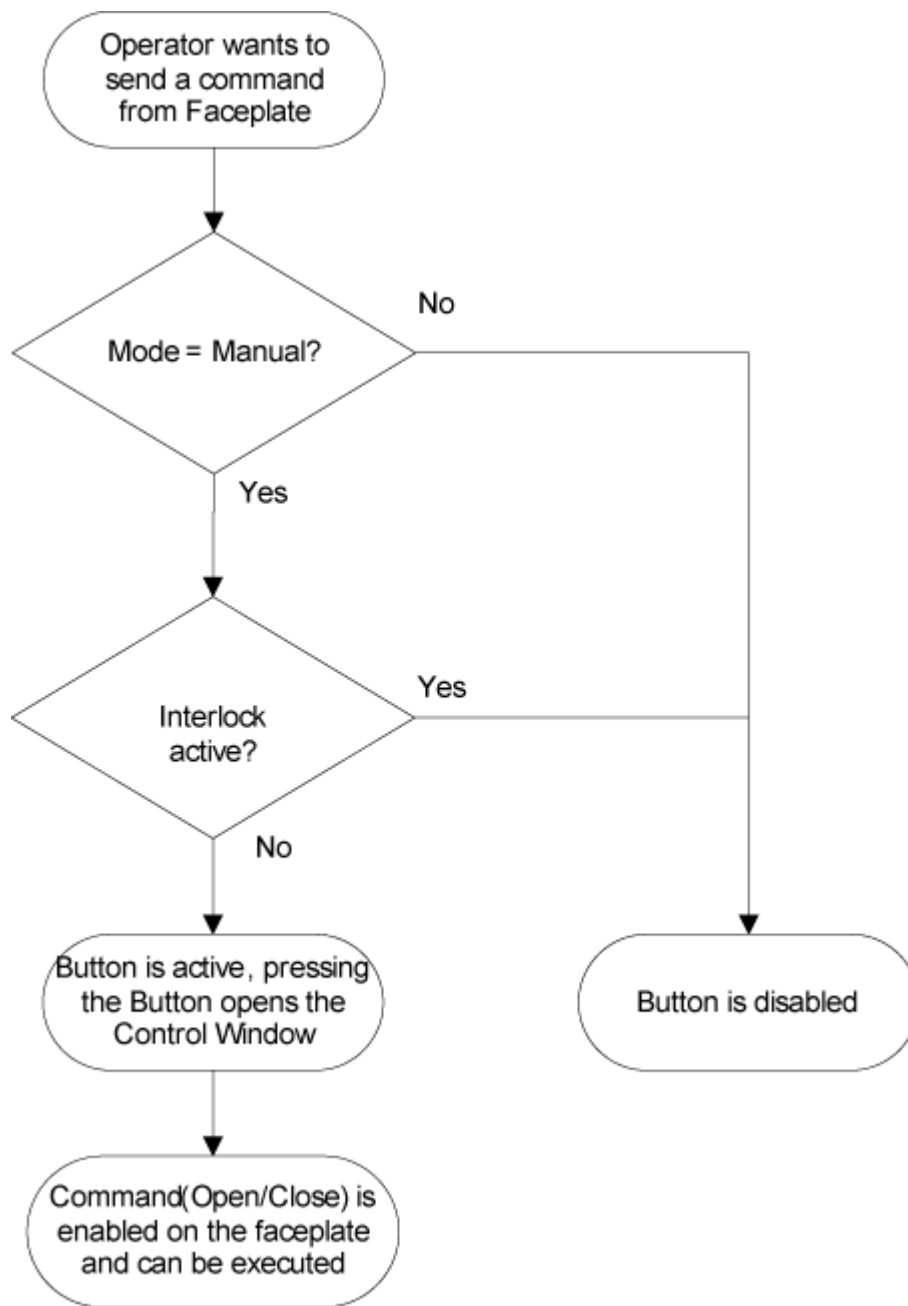
The PC_TRAFO block only provides the APL standard interlock "Interlock".

Important information: The APL interlock locks all free commands of the PC_TRAFO block but does not open any switches at activation.

The following logic is used to interlock a linked command:



The following logic is used for interlocking the commands output in the faceplate:



Deactivating individual interlocks:

You deactivate the implemented block interlock through the input parameter Intlock

If you want to deactivate the block interlock, you have to specify the following parameters:

- `Intl_En = 0` or `Intlock.ST = 16#FF`: The Intlock input parameter does not have any effect.

8.1.7 Group error

This block provides the standard APL function "Outputting group errors".

The following parameters are taken into consideration when forming the group error:

- CSF

The group error output is *GrpErr.Value*.

8.1.8 Forming the group status for interlocks

This block contains the standard APL function "Forming the group status for interlock information" (see APL documentation).

8.1.9 Responses that can be configured with the "Feature" parameter

An overview of all the responses provided by the `Feature` parameter is contained in the PCS 7 APL documentation.

The following function is available for this block at the corresponding bits:

Bit	Function
0-11	Not used
12	1 = Hide operating hours in the faceplate 0 (default) = Show operating hours in the faceplate
13	1 = Hide operating cycle hours in the faceplate 0 (default) = Show operating cycle hours in the faceplate
14-31	Not used

8.1.10 Operator authorizations

This block includes the APL standard function "Operator authorizations".

The block provides the following authorizations for the `OS_Perm` parameter:

Bit	Function
0	1 = Operator may switch to "Automatic mode"
1	1 = Operator may switch to "Manual mode"
2	1 = Operator may switch to "Local operating mode"
3-31	Not used

8.1.11 Assignment of the status word for Status1 parameter

Status bit	Parameter
0	Occupied
1	BatchEn
2	Not used
3	Not used
4	Not used
5	0 = ManAct.Value 1 = AutoAct.Value
6	LocalAct.Value
7	0: Open padlock in block symbol 1: Closed padlock in block symbol
8	Not used (always true)
9	Not used
10	Not used
11	Not used
12	Not used
13	Not used
14	Not used
15	Not used
16	1 = Intlock is active
17	Not used
18	Not used
19	Trip.Value
20	Not used
21	Not used
22	Not used
23	"Interlock" button is activated
24	Not used
25	Not used
26	Bypass information from previous function block
27	Not used
28	Not used
29	Not used
30	Not used
31	Not used

8.1.12 Assignment of the status word for Status5 parameter

The Status5 parameter is used to show / hide values in the faceplate. The values are displayed depending on the quality code of the Station Gateway / Station Controller.

The values are displayed when the quality code is $\leq 0x00$.

Status bit	Value	Description
0	I1_A	Primary current Phase 1
1	I1_B	Primary current Phase 2
2	I1_C	Primary current Phase 3
3	I2_A	Secondary current Phase 1
4	I2_B	Secondary current Phase 2
5	I2_C	Secondary current Phase 3
6	I3_A	Current 3rd. winding phase 1
7	I3_B	Current 3rd. winding phase 2
8	I3_C	Current 3rd. winding phase 3
9	Th_Trip	Thermal overload: Tripping
10	S	Apparent power
11	P	Power
12	Q	Reactive power
13	F	Frequency
14	U_AB	Voltage phase 1 – phase 2
15	U_BC	Voltage phase 2 – phase 3
16	U_CA	Voltage phase 3 – phase 1
17	U	Voltage
18	PF	Power factor
19	Optm	Operating hours
20	SwCount	Operating cycles
27	WpVorw	Power counter P forwards
27	WqVorw	Power counter Q forwards
27	WpRückw	Power counter P reverse
27	WqRückw	Power counter P reverse

8.1.13 Messages

Message instance	Message identifier	Message class	Event
MsgEvId_1	SIG_1	Alarm - high	\$\$BlockComment\$\$ Excitation overcurrent stage I >
	SIG_2	Warning - high	\$\$BlockComment\$\$ Tripping overcurrent stage I >
	SIG_3	Alarm - high	\$\$BlockComment\$\$ Excitation overcurrent stage I >>
	SIG_4	Warning - high	\$\$BlockComment\$\$ Tripping overcurrent stage I >>
	SIG_5	Alarm - high	\$\$BlockComment\$\$ Excitation overcurrent stage Ip
	SIG_6	Warning - high	\$\$BlockComment\$\$ Tripping overcurrent stage Ip
	SIG_7	Alarm - high	\$\$BlockComment\$\$ Excitation overcurrent stage 3 I 0 >
	SIG_8	Warning - high	\$\$BlockComment\$\$ Tripping overcurrent stage 3 I 0 >
MsgEvId_2	SIG_1	Alarm - high	\$\$BlockComment\$\$ Excitation overcurrent stage 3 I 0 >>
	SIG_2	Warning - high	\$\$BlockComment\$\$ Tripping overcurrent stage 3 I 0 >>
	SIG_3	Alarm - high	\$\$BlockComment\$\$ Excitation overcurrent stage 3 I 0p
	SIG_4	Warning - high	\$\$BlockComment\$\$ Tripping overcurrent stage 3 I 0p
	SIG_5	Alarm - high	\$\$BlockComment\$\$ Excitation unbalanced load protection I >
	SIG_6	Warning - high	\$\$BlockComment\$\$ Tripping unbalanced load protection I >
	SIG_7	Alarm - high	\$\$BlockComment\$\$ Excitation unbalanced load protection I >>
	SIG_8	Warning - high	\$\$BlockComment\$\$ Tripping unbalance load protection I >>

Message instance	Message identifier	Message class	Event
MsgEvid_3	SIG_1	Alarm - high	\$\$BlockComment\$\$ Excitation unbalanced load protection I2th
	SIG_2	Warning - high	\$\$BlockComment\$\$ Tripping unbalanced load protection I2th
	SIG_3	Alarm - high	\$\$BlockComment\$\$ Excitation theta overload protection
	SIG_4	Warning - high	\$\$BlockComment\$\$ Tripping theta overload protection
	SIG_5	Alarm - high	\$\$BlockComment\$\$ Tripping differential protection IDIFF>
	SIG_6	Warning - high	\$\$BlockComment\$\$ Excitation differential protection IDIFF>
	SIG_7	Alarm - high	\$\$BlockComment\$\$ Tripping differential protection IDIFF>>
	SIG_8	Warning - high	\$\$BlockComment\$\$ Excitation differential protection IDIFF>>
MsgEvid_4	SIG_1	Alarm - high	\$\$BlockComment\$\$ Excitation differential protection I REF >
	SIG_2	Warning - high	\$\$BlockComment\$\$ Tripping differential protection I REF >
	SIG_3	Alarm - high	\$\$BlockComment\$\$ Tripping coupling 1
	SIG_4	Warning - high	\$\$BlockComment\$\$ Excitation coupling 1
	SIG_5	Alarm - high	\$\$BlockComment\$\$ Excitation disconnector fault I n
	SIG_6	Alarm - high	\$\$BlockComment\$\$ Tripping disconnector fault T1
	SIG_7	Warning - high	\$\$BlockComment\$\$ Tripping disconnector fault T1
	SIG_8	Alarm - high	\$\$BlockComment\$\$ Theta overload protection alarm
MsgEvid_5	SIG_1	Warning - high	\$\$BlockComment\$\$ Excitation device
	SIG_2	Alarm - high	\$\$BlockComment\$\$ General tripping device
	SIG_3	Status message - AS	\$\$BlockComment\$\$ Parameter set A active
	SIG_4	Status message - AS	\$\$BlockComment\$\$ Parameter set B active
	SIG_5	Status message - AS	\$\$BlockComment\$\$ Parameter set C active
	SIG_6	Status message - AS	\$\$BlockComment\$\$ Parameter set D active
	SIG_7	Alarm - high	\$\$BlockComment\$\$ Fault group message
	SIG_8	Warning - high	\$\$BlockComment\$\$ Warning group message

Message instance	Message identifier	Message class	Event
MsgEvld_6	SIG_1	Warning - high	\$\$BlockComment\$\$ Local mode
	SIG_2	Alarm - high	\$\$BlockComment\$\$ IEC Alarm 1
	SIG_3	Alarm - high	\$\$BlockComment\$\$ IEC Alarm 2
	SIG_4	Alarm - high	\$\$BlockComment\$\$ IEC Alarm 3
	SIG_5	Alarm - high	\$\$BlockComment\$\$ IEC Alarm 4
	SIG_6	Alarm - high	\$\$BlockComment\$\$ IEC Alarm 5
	SIG_7	Alarm - high	\$\$BlockComment\$\$ IEC Alarm 6
	SIG_8	Alarm - high	\$\$BlockComment\$\$ IEC Alarm 7
MsgEvld_device	SIG_1	AS process control message - error	\$\$BlockComment\$\$ Error device IED name:@1%s@ -> IP address: @2%s@
	SIG_2	AS control system message - fault	\$\$BlockComment\$\$ Redundancy failure
	SIG_3	AS process control message - error	\$\$BlockComment\$\$ PROFIBUS mapping alarm
	SIG_4	Alarm - high	
	SIG_5	Alarm - high	
	SIG_6	Alarm - high	
	SIG_7	Alarm - high	
	SIG_8	Alarm - high	
MsgEvldlocal1	SIG_1	AS process control message - error	\$\$BlockComment\$\$ Connection fault
	SIG_2	Alarm - high	
	SIG_3	Alarm - high	
	SIG_4	Alarm - high	
	SIG_5	Alarm - high	
	SIG_6	Alarm - high	
	SIG_7	Alarm - high	
	SIG_8	AS process control message - error	\$\$BlockComment\$\$ AS process control message - external error

Message instance	Message identifier	Message class	Event
MsgEvidlocal2	SIG_1	Warning - high	\$\$BlockComment\$\$ Maintenance: Operating cycle warning (@3%d@)
	SIG_2	Alarm - high	\$\$BlockComment\$\$ Maintenance: Operating cycle alarm (@3%d@)
	SIG_3	Warning - high	\$\$BlockComment\$\$ Maintenance: Operating time warning (@4%d@)
	SIG_4	Alarm - high	\$\$BlockComment\$\$ Maintenance: Operating time alarm (@4%d@)
	SIG_5	Alarm - high	\$\$BlockComment\$\$ External message 1
	SIG_6	Alarm - high	\$\$BlockComment\$\$ External message 2
	SIG_7	Alarm - high	\$\$BlockComment\$\$ External message 3
	SIG_8	Alarm - high	\$\$BlockComment\$\$ External message 4
MsgEvid_d_1	SIG_1	Status message - AS	\$\$BlockComment\$\$ Switch 1 closed
	SIG_2	Status message - AS	\$\$BlockComment\$\$ Switch 1 open
	SIG_3	Status message - AS	\$\$BlockComment\$\$ Switch 2 closed
	SIG_4	Status message - AS	\$\$BlockComment\$\$ Switch 2 open
	SIG_5	Status message - AS	\$\$BlockComment\$\$ Switch 3 closed
	SIG_6	Status message - AS	\$\$BlockComment\$\$ Switch 3 open
	SIG_7	Status message - AS	\$\$BlockComment\$\$ Switch 4 closed
	SIG_8	Status message - AS	\$\$BlockComment\$\$ Switch 4 open

8.1.14 I/O list

8.1.14.1 Inputs

I/O name	Description	Value	Data type
Rcv_Data	Connect with SG /SC_TRAFO		STRUCT
ModLiOp	1 = Interconnect/Auto, 0 = Manual: Input for auto / manual commands		STRUCT
ModLiOp.Value		1	BOOL
ModLiOp.ST		16#80	BYTE
AutModOp	1 = Automatic mode: Automatic mode by operator	0	BOOL
ManModOp	1 = Manual mode: Manual mode by operator	1	BOOL
AutModLi			STRUCT

I/O name	Description	Value	Data type
AutModLi.Value		1	BOOL
AutModLi.ST		16#80	BYTE
ManModLi			STRUCT
ManModLi.Value		0	BOOL
ManModLi.ST		16#80	BYTE
BB1_Color	Connection to PC_BUSBAR for automatic color display	16#0	DWORD
BB2_Color	Connection to PC_BUSBAR for automatic color display	16#0	DWORD
FP_Visu	Configuration of the faceplate		STRUCT
FP_Units	Configuration of the units in the faceplate		STRUCT
MS_RelOp	Operator input for MS release , 1: MS release request	0	BOOL
Monitor	1=Monitoring ON, 0=Monitoring OFF: Feed-back monitoring	1	BOOL
MonTiStatic	Monitoring time of the feedback [s]	3.0	REAL
MonTiDynamic	Monitoring time of the feedback [s]	3.0	REAL
Intlock	0=Motor is locked, 1=Motor is not locked		STRUCT
Intlock.Value	Value	1	BOOL
Intlock.ST	Signal state	16#FF	BYTE
Intl_En	1=Interlock activated, 0=Interlock deactivated	1	BOOL
FACTOR_ENERGY	Factor for displaying energy values	1.0	REAL
CMD_sp1_Opn	Rising edge: Free command 1 off		STRUCT
CMD_sp1_Opn.Value		0	BOOL
CMD_sp1_Opn.ST		16#80	BYTE
CMD_sp1_Cls	Rising edge: Free command 1 on		STRUCT
CMD_sp1_Cls.Value		0	BOOL
CMD_sp1_Cls.ST		16#80	BYTE
CMD_sp2_Opn	Rising edge: Free command 2 off		STRUCT
CMD_sp2_Opn.Value		1	BOOL
CMD_sp2_Opn.ST		16#80	BYTE
CMD_sp2_Cls	Rising edge: Free command 2 on		STRUCT
CMD_sp2_Cls.Value		1	BOOL
CMD_sp2_Cls.ST		16#80	BYTE
CMD_sp3_Opn	Rising edge: Free command 3 off		STRUCT
CMD_sp3_Opn.Value		1	BOOL
CMD_sp3_Opn.ST		16#80	BYTE
CMD_sp3_Cls	Rising edge: Free command 3 on		STRUCT
CMD_sp3_Cls.Value		1	BOOL
CMD_sp3_Cls.ST		16#80	BYTE
CMD_sp4_Opn	Rising edge: Free command 4 off		STRUCT
CMD_sp4_Opn.Value		0	BOOL
CMD_sp4_Opn.ST		16#80	BYTE
CMD_sp4_Cls	Rising edge: Free command 4 on		STRUCT

I/O name	Description	Value	Data type
CMD_sp4_Cls.Value		0	BOOL
CMD_sp4_Cls.ST		16#80	BYTE
MsgLock			STRUCT
MsgLock.Value		0	BOOL
MsgLock.ST		16#80	BYTE
SampleTime	Sampling time [s]	0.1	REAL
MsgEvId_1	Message event ID	16#00000006	DWORD
MsgEvId_2	Message event ID	16#00000007	DWORD
MsgEvId_3	Message event ID	16#00000008	DWORD
MsgEvId_4	Message event ID	16#0000000A	DWORD
MsgEvId_5	Message event ID	16#0000000B	DWORD
MsgEvId_6	Message event ID	16#0000000C	DWORD
MsgEvId_d_1	Message event ID	16#00000049	DWORD
MsgEvIdlocal1	Message event ID	16#00000016	DWORD
MsgEvIdlocal2	Message event ID	16#00000017	DWORD
MsgEvId_device	Message event ID	16#0000000D	DWORD
BatchEn	Activate remote operation of controller to batch recipe	0	BOOL
BatchID	Current batch ID (number)	16#00000000	DWORD
BatchName	Current batch name	"	STRING[32]
StepNo	Batch step number	16#00000000	DWORD
Occupied	Occupied by batch	0	BOOL
CSF			STRUCT
CSF.Value		1	BOOL
CSF.ST		16#80	BYTE
ExtMsg1	External message 1		STRUCT
ExtMsg1.Value	Value	0	BOOL
ExtMsg1.ST	Signal state	16#80	BYTE
ExtMsg2	External message 2		STRUCT
ExtMsg2.Value	Value	0	BOOL
ExtMsg2.ST	Signal state	16#80	BYTE
ExtMsg3	External message 3		STRUCT
ExtMsg3.Value	Value	0	BOOL
ExtMsg3.ST	Signal state	16#80	BYTE
ExtMsg4	External message 4		STRUCT
ExtMsg4.Value	Value	0	BOOL
ExtMsg4.ST	Signal state	16#80	BYTE
UserAna1	User analog input 1		STRUCT
UserAna1.Value	Value	0.0	REAL
UserAna1.ST	Signal state	16#FF	BYTE
UA1unit	Unit of measurement user analog input 1	0	INT
UserAna2	User analog input 2		STRUCT
UserAna2.Value	Value	0.0	REAL

I/O name	Description	Value	Data type
UserAna2.ST	Signal state	16#FF	BYTE
UA2unit	Unit of measurement user analog input 2	0	INT
UserAna3	User analog input 3		STRUCT
UserAna3.Value	Value	0.0	REAL
UserAna3.ST	Signal state	16#FF	BYTE
UA3unit	Unit of measurement user analog input 3	0	INT
UserAna4	User analog input 4		STRUCT
UserAna4.Value	Value	0.0	REAL
UserAna4.ST	Signal state	16#FF	BYTE
UA4unit	Unit of measurement user analog input 4	0	INT
UserStatus	User status bits	16#0	BYTE
SelFp1	Select faceplate 1		ANY
OS_Perm	Operator authorizations		STRUCT
OS_Perm.Bit0	1 = Operator may switch to Automatic mode	1	BOOL
OS_Perm.Bit1	1 = Operator may switch to Manual mode	1	BOOL
OS_Perm.Bit2	1 = Operator may switch to Local operating mode	1	BOOL
OS_Perm.Bit3	1 = Operator may switch to "Out of service" mode	1	BOOL
OS_Perm.Bit4		1	BOOL
OS_Perm.Bit5		1	BOOL
OS_Perm.Bit6		1	BOOL
OS_Perm.Bit7		1	BOOL
OS_Perm.Bit8		1	BOOL
OS_Perm.Bit9		1	BOOL
OS_Perm.Bit10		1	BOOL
OS_Perm.Bit11	1 = Operator can activate function simulation	1	BOOL
OS_Perm.Bit12	1 = Operator can enable release for maintenance	1	BOOL
OS_Perm.Bit13		1	BOOL
OS_Perm.Bit14		1	BOOL
OS_Perm.Bit15		1	BOOL
OS_Perm.Bit16		1	BOOL
OS_Perm.Bit17		1	BOOL
OS_Perm.Bit18		1	BOOL
OS_Perm.Bit19		1	BOOL
OS_Perm.Bit20		1	BOOL
OS_Perm.Bit21		1	BOOL
OS_Perm.Bit22		1	BOOL
OS_Perm.Bit23		1	BOOL
OS_Perm.Bit24		1	BOOL
OS_Perm.Bit25		1	BOOL
OS_Perm.Bit26		1	BOOL

I/O name	Description	Value	Data type
OS_Perm.Bit27		1	BOOL
OS_Perm.Bit28		1	BOOL
OS_Perm.Bit29		1	BOOL
OS_Perm.Bit30		1	BOOL
OS_Perm.Bit31		1	BOOL
Feature	State of different functions		STRUCT
Feature.Bit0	Reserved	0	BOOL
Feature.Bit1	Reserved	0	BOOL
Feature.Bit2	Reserved	0	BOOL
Feature.Bit3	Reserved	0	BOOL
Feature.Bit4	Reserved	0	BOOL
Feature.Bit5	Reserved	0	BOOL
Feature.Bit6	Reserved	0	BOOL
Feature.Bit7	Reserved	0	BOOL
Feature.Bit8	Reserved	0	BOOL
Feature.Bit9	Reserved	0	BOOL
Feature.Bit10	Reserved	0	BOOL
Feature.Bit11	Reserved	0	BOOL
Feature.Bit12	Hide OpTm	0	BOOL
Feature.Bit13	Hide SwCount	0	BOOL
Feature.Bit14	Reserved	0	BOOL
Feature.Bit15	Reserved	0	BOOL
Feature.Bit16	Reserved	0	BOOL
Feature.Bit17	Reserved	0	BOOL
Feature.Bit18	Reserved	0	BOOL
Feature.Bit19	Reserved	0	BOOL
Feature.Bit20	1 = Dynamic active power direction enabled:	0	BOOL
Feature.Bit21	1 = Dynamic active power direction: negative power up; 0 = negative power down	0	BOOL
Feature.Bit22	Reserved	0	BOOL
Feature.Bit23	Reserved	0	BOOL
Feature.Bit24	1 = Local authorization enabled	0	BOOL
Feature.Bit25	1 = Suppression of all messages if MsgLock = 1	0	BOOL
Feature.Bit26	Reserved	0	BOOL
Feature.Bit27	Reserved	0	BOOL
Feature.Bit28	Reserved	0	BOOL
Feature.Bit29	Reserved	0	BOOL
Feature.Bit30	Reserved	0	BOOL
Feature.Bit31	Reserved	0	BOOL
MonTime	Time in seconds until Timeout (for control switch)	20	INT

8.1.14.2 Inputs / outputs

I/O name	Description	Value	Data type
Cmdsp1	Control position free command 1	16#02	BYTE
Cmdsp2	Control position free command 2	16#02	BYTE
Cmdsp3	Control position free command 3	16#02	BYTE
Cmdsp4	Control position free command 4	16#02	BYTE
OpTmWarnLimit	Warning limit operating hours	0	DINT
OpTmAlarmLimit	Alarm limit operating hours	0	DINT
SwCountWarnLimit	Warning limit operating cycles	0	DINT
SwCountAlarmLimit	Alarm limit operating cycles	0	DINT

8.1.14.3 Outputs

I/O name	Description	Value	Data type
Snd_Data	Connect with SG/_SC_TRAFO		STRUCT
QCONNERR		1	BOOL
F	Frequency		STRUCT
F.Value		0.0	REAL
F.ST		16#FF	BYTE
P	Power P		STRUCT
P.Value		0.0	REAL
P.ST		16#FF	BYTE
Q	Power Q		STRUCT
Q.Value		0.0	REAL
Q.ST		16#FF	BYTE
S	Power S		STRUCT
S.Value		0.0	REAL
S.ST		16#FF	BYTE
PF	Power factor		STRUCT
PF.Value		0.0	REAL
PF.ST		16#FF	BYTE
U	Neutral voltage		STRUCT
U.Value		0.0	REAL
U.ST		16#FF	BYTE
U_AB	Voltage Phase A to Phase B		STRUCT
U_AB.Value		0.0	REAL
U_AB.ST		16#FF	BYTE
U_BC	Voltage Phase B to Phase C		STRUCT
U_BC.Value		0.0	REAL
U_BC.ST		16#FF	BYTE
U_CA	Voltage Phase C to Phase A		STRUCT
U_CA.Value		0.0	REAL

I/O name	Description	Value	Data type
U_CA.ST		16#FF	BYTE
I1_A	Current 1 Phase A		STRUCT
I1_A.Value		0.0	REAL
I1_A.ST		16#FF	BYTE
I1_B	Current 1 Phase B		STRUCT
I1_B.Value		0.0	REAL
I1_B.ST		16#FF	BYTE
I1_C	Current 1 Phase C		STRUCT
I1_C.Value		0.0	REAL
I1_C.ST		16#FF	BYTE
I2_A	Current 2 Phase A		STRUCT
I2_A.Value		0.0	REAL
I2_A.ST		16#FF	BYTE
I2_B	Current 2 Phase B		STRUCT
I2_B.Value		0.0	REAL
I2_B.ST		16#FF	BYTE
I2_C	Current 2 Phase C		STRUCT
I2_C.Value		0.0	REAL
I2_C.ST		16#FF	BYTE
I3_A	Current 3 Phase A		STRUCT
I3_A.Value		0.0	REAL
I3_A.ST		16#FF	BYTE
I3_B	Current 3 Phase B		STRUCT
I3_B.Value		0.0	REAL
I3_B.ST		16#FF	BYTE
I3_C	Current 3 Phase C		STRUCT
I3_C.Value		0.0	REAL
I3_C.ST		16#FF	BYTE
THETA	Theta		STRUCT
THETA.Value		0.0	REAL
THETA.ST		16#FF	BYTE
THETA_L1	Theta/Theta Tripping L1		STRUCT
THETA_L1.Value		0.0	REAL
THETA_L1.ST		16#FF	BYTE
THETA_L2	Theta/Theta Tripping L2		STRUCT
THETA_L2.Value		0.0	REAL
THETA_L2.ST		16#FF	BYTE
THETA_L3	Theta/Theta Tripping L3		STRUCT
THETA_L3.Value		0.0	REAL
THETA_L3.ST		16#FF	BYTE
REAL_sp1	Free REAL 1		STRUCT
REAL_sp1.Value		0.0	REAL
REAL_sp1.ST		16#FF	BYTE

I/O name	Description	Value	Data type
REAL_sp2	Free REAL 2		STRUCT
REAL_sp2.Value		0.0	REAL
REAL_sp2.ST		16#FF	BYTE
REAL_sp3	Free REAL 3		STRUCT
REAL_sp3.Value		0.0	REAL
REAL_sp3.ST		16#FF	BYTE
REAL_sp4	Free REAL 4		STRUCT
REAL_sp4.Value		0.0	REAL
REAL_sp4.ST		16#FF	BYTE
REAL_sp5	Free REAL 5		STRUCT
REAL_sp5.Value		0.0	REAL
REAL_sp5.ST		16#FF	BYTE
Pos_CMD_sp1	Position free command 1	16#00000000	DWORD
Pos_CMD_sp2	Position free command 2	16#00000000	DWORD
Pos_CMD_sp3	Position free command 3	16#00000000	DWORD
Pos_CMD_sp4	Position free command 4	16#00000000	DWORD
Beh	Behavior	0	INT
Dev_OK	Device OK	0	INT
INT_sp1	Free INT 1		STRUCT
INT_sp1.Value		0	INT
INT_sp1.ST		16#FF	BYTE
INT_sp2	Free INT 2		STRUCT
INT_sp2.Value		0	INT
INT_sp2.ST		16#FF	BYTE
INT_sp3	Free INT 3		STRUCT
INT_sp3.Value		0	INT
INT_sp3.ST		16#FF	BYTE
INT_sp4	Free INT 4		STRUCT
INT_sp4.Value		0	INT
INT_sp4.ST		16#FF	BYTE
INT_sp5	Free INT 5		STRUCT
INT_sp5.Value		0	INT
INT_sp5.ST		16#FF	BYTE
INT_sp6	Free INT 6		STRUCT
INT_sp6.Value		0	INT
INT_sp6.ST		16#FF	BYTE
PTOC1_Str			STRUCT
PTOC1_Str.Value		0	BOOL
PTOC1_Str.ST		16#FF	BYTE
PTOC1_Op			STRUCT
PTOC1_Op.Value		0	BOOL
PTOC1_Op.ST		16#FF	BYTE
PTOC2_Str			STRUCT

I/O name	Description	Value	Data type
PTOC2_Str.Value		0	BOOL
PTOC2_Str.ST		16#FF	BYTE
PTOC2_Op			STRUCT
PTOC2_Op.Value		0	BOOL
PTOC2_Op.ST		16#FF	BYTE
PTOC3_Str			STRUCT
PTOC3_Str.Value		0	BOOL
PTOC3_Str.ST		16#FF	BYTE
PTOC3_Op			STRUCT
PTOC3_Op.Value		0	BOOL
PTOC3_Op.ST		16#FF	BYTE
PTOC7_Str			STRUCT
PTOC7_Str.Value		0	BOOL
PTOC7_Str.ST		16#FF	BYTE
PTOC7_Op			STRUCT
PTOC7_Op.Value		0	BOOL
PTOC7_Op.ST		16#FF	BYTE
PTOC8_Str			STRUCT
PTOC8_Str.Value		0	BOOL
PTOC8_Str.ST		16#FF	BYTE
PTOC8_Op			STRUCT
PTOC8_Op.Value		0	BOOL
PTOC8_Op.ST		16#FF	BYTE
PTOC9_Str			STRUCT
PTOC9_Str.Value		0	BOOL
PTOC9_Str.ST		16#FF	BYTE
PTOC9_Op			STRUCT
PTOC9_Op.Value		0	BOOL
PTOC9_Op.ST		16#FF	BYTE
PTOC12_Str			STRUCT
PTOC12_Str.Value		0	BOOL
PTOC12_Str.ST		16#FF	BYTE
PTOC12_Op			STRUCT
PTOC12_Op.Value		0	BOOL
PTOC12_Op.ST		16#FF	BYTE
PTOC13_Str			STRUCT
PTOC13_Str.Value		0	BOOL
PTOC13_Str.ST		16#FF	BYTE
PTOC13_Op			STRUCT
PTOC13_Op.Value		0	BOOL
PTOC13_Op.ST		16#FF	BYTE
PTOC15_Str			STRUCT
PTOC15_Str.Value		0	BOOL

I/O name	Description	Value	Data type
PTOC15_Str.ST		16#FF	BYTE
PTOC15_Op			STRUCT
PTOC15_Op.Value		0	BOOL
PTOC15_Op.ST		16#FF	BYTE
PTTR1_Str			STRUCT
PTTR1_Str.Value		0	BOOL
PTTR1_Str.ST		16#FF	BYTE
PTTR1_Op			STRUCT
PTTR1_Op.Value		0	BOOL
PTTR1_Op.ST		16#FF	BYTE
PDIF1_Op			STRUCT
PDIF1_Op.Value		0	BOOL
PDIF1_Op.ST		16#FF	BYTE
PDIF1_Str			STRUCT
PDIF1_Str.Value		0	BOOL
PDIF1_Str.ST		16#FF	BYTE
PDIF2_Op			STRUCT
PDIF2_Op.Value		0	BOOL
PDIF2_Op.ST		16#FF	BYTE
PDIF2_Str			STRUCT
PDIF2_Str.Value		0	BOOL
PDIF2_Str.ST		16#FF	BYTE
PDIF3_Op			STRUCT
PDIF3_Op.Value		0	BOOL
PDIF3_Op.ST		16#FF	BYTE
PDIF3_Str			STRUCT
PDIF3_Str.Value		0	BOOL
PDIF3_Str.ST		16#FF	BYTE
GAPC1_Op			STRUCT
GAPC1_Op.Value		0	BOOL
GAPC1_Op.ST		16#FF	BYTE
GAPC1_Str			STRUCT
GAPC1_Str.Value		0	BOOL
GAPC1_Str.ST		16#FF	BYTE
RBRF1_Str			STRUCT
RBRF1_Str.Value		0	BOOL
RBRF1_Str.ST		16#FF	BYTE
RBRF1_OpEx			STRUCT
RBRF1_OpEx.Value		0	BOOL
RBRF1_OpEx.ST		16#FF	BYTE
RBRF1_OpIn			STRUCT
RBRF1_OpIn.Value		0	BOOL
RBRF1_OpIn.ST		16#FF	BYTE

I/O name	Description	Value	Data type
PTTR1_AlmThm			STRUCT
PTTR1_AlmThm.Value		0	BOOL
PTTR1_AlmThm.ST		16#FF	BYTE
PU_Relay			STRUCT
PU_Relay.Value		0	BOOL
PU_Relay.ST		16#FF	BYTE
TR_Relay			STRUCT
TR_Relay.Value		0	BOOL
TR_Relay.ST		16#FF	BYTE
PARSET_A	Parameter set A	0	BOOL
PARSET_B	Parameter set B	0	BOOL
PARSET_C	Parameter set C	0	BOOL
PARSET_D	Parameter set D	0	BOOL
Error			STRUCT
Error.Value		0	BOOL
Error.ST		16#FF	BYTE
Warning			STRUCT
Warning.Value		0	BOOL
Warning.ST		16#FF	BYTE
Loc	Local	0	BOOL
ALM_sp1			STRUCT
ALM_sp1.Value		0	BOOL
ALM_sp1.ST		16#FF	BYTE
ALM_sp2			STRUCT
ALM_sp2.Value		0	BOOL
ALM_sp2.ST		16#FF	BYTE
ALM_sp3			STRUCT
ALM_sp3.Value		0	BOOL
ALM_sp3.ST		16#FF	BYTE
ALM_sp4			STRUCT
ALM_sp4.Value		0	BOOL
ALM_sp4.ST		16#FF	BYTE
ALM_sp5			STRUCT
ALM_sp5.Value		0	BOOL
ALM_sp5.ST		16#FF	BYTE
ALM_sp6			STRUCT
ALM_sp6.Value		0	BOOL
ALM_sp6.ST		16#FF	BYTE
ALM_sp7			STRUCT
ALM_sp7.Value		0	BOOL
ALM_sp7.ST		16#FF	BYTE
SwCount	Operating cycles Q0	0	DINT
OpTm	Operating hours	0	DINT

I/O name	Description	Value	Data type
WpForward	Supply active energy		STRUCT
WpForward.Value		0.0	REAL
WpForward.ST		16#FF	BYTE
WpReverse	Demand active energy		STRUCT
WpReverse.Value		0.0	REAL
WpReverse.ST		16#FF	BYTE
WqForward	Supply reactive energy		STRUCT
WqForward.Value		0.0	REAL
WqForward.ST		16#FF	BYTE
WqReverse	Demand reactive energy		STRUCT
WqReverse.Value		0.0	REAL
WqReverse.ST		16#FF	BYTE
DINT_sp1	DINT value 8 (free)		STRUCT
DINT_sp1.Value		0	DINT
DINT_sp1.ST		16#FF	BYTE
DINT_sp2	DINT value 8 (free)		STRUCT
DINT_sp2.Value		0	DINT
DINT_sp2.ST		16#FF	BYTE
BOOL_sp1	Free BOOL		STRUCT
BOOL_sp1.Value		0	BOOL
BOOL_sp1.ST		16#FF	BYTE
BOOL_sp2	Free BOOL		STRUCT
BOOL_sp2.Value		0	BOOL
BOOL_sp2.ST		16#FF	BYTE
BOOL_sp3	Free BOOL		STRUCT
BOOL_sp3.Value		0	BOOL
BOOL_sp3.ST		16#FF	BYTE
BOOL_sp4	Free BOOL		STRUCT
BOOL_sp4.Value		0	BOOL
BOOL_sp4.ST		16#FF	BYTE
BOOL_sp5	Free BOOL		STRUCT
BOOL_sp5.Value		0	BOOL
BOOL_sp5.ST		16#FF	BYTE
BOOL_sp6	Free BOOL		STRUCT
BOOL_sp6.Value		0	BOOL
BOOL_sp6.ST		16#FF	BYTE
BOOL_sp7	Free BOOL		STRUCT
BOOL_sp7.Value		0	BOOL
BOOL_sp7.ST		16#FF	BYTE
BOOL_sp8	Free BOOL		STRUCT
BOOL_sp8.Value		0	BOOL
BOOL_sp8.ST		16#FF	BYTE
BOOL_sp9	Free BOOL		STRUCT

I/O name	Description	Value	Data type
BOOL_sp9.Value		0	BOOL
BOOL_sp9.ST		16#FF	BYTE
BOOL_sp10	Free BOOL		STRUCT
BOOL_sp10.Value		0	BOOL
BOOL_sp10.ST		16#FF	BYTE
OpTmWarning	1 = Operating time warning active	0	BOOL
OpTmAlarm	1 = Operating time alarm active	0	BOOL
SwCountWarning	1 = Operating cycle warning active	0	BOOL
SwCountAlarm	1 = Operating cycle alarm active	0	BOOL
LockAct			STRUCT
LockAct.Value		0	BOOL
LockAct.ST		16#80	BYTE
LocalAct			STRUCT
LocalAct.Value		0	BOOL
LocalAct.ST		16#80	BYTE
AutAct			STRUCT
AutAct.Value		0	BOOL
AutAct.ST		16#80	BYTE
ManAct	1 = Manual: Manual value is active		STRUCT
ManAct.Value	Value	1	BOOL
ManAct.ST	Signal state	16#80	BYTE
OS_PermOut	Operator authorization: Output for OS	16#FFFFFFFF	DWORD
OS_PermLog	Operator authorization: Output for OS	16#FFFFFFFF	DWORD
ST_Worst	Worst signal state	16#80	BYTE
GrpErr			STRUCT
GrpErr.Value		0	BOOL
GrpErr.ST		16#80	BYTE
Status1	Word Status1	16#00000000	DWORD
Status3	Word Status3	16#00000000	DWORD
Status5	Word Status3	16#FFFFFFFF	DWORD
MsgErrlocal1	1 = Message error occurs	0	BOOL
MsgStatlocal1	Message status	16#0000	WORD
MsgAcknlocal1	Message acknowledgement status	16#0000	WORD
MsgErrlocal2	1 = Message error occurs	0	BOOL
MsgStatlocal2	Message status	16#0000	WORD
MsgAcknlocal2	Message acknowledgement status	16#0000	WORD
VisuCodes	Symbol display in the icon and faceplate		STRUCT
VisuMisc	Configuration of the units in the faceplate		STRUCT

8.2 PC_TRAFO block icons

8.2.1 PC_TRAFO block symbols

The following block symbols are available for the PC_TRAFO object:

PC_TRAFO/1: Symbol with group display and large tag name:



Figure 8-1 PC_TRAFO/1

PC_TRAFO/2: Symbol with group display and small tag name:

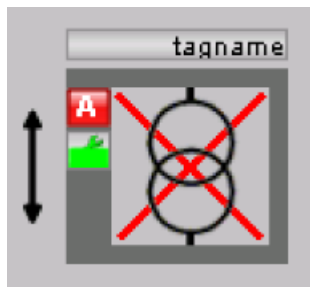


Figure 8-2 PC_TRAFO/2

PC_TRAFO/3: Small symbol with dynamic color display:

Additional functionalities:

- Dynamic color display with BB1_Color (primary side) and BB2_Color (secondary side)

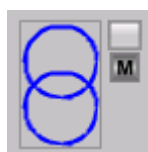


Figure 8-3 PC_TRAFO/3

8.2.2 Dynamizing the active power direction

Dynamizing the active power direction

The active power display of the corresponding PC_TRAFO block icons can be dynamically displayed as a function of the active power P.

The latest value for the active power P is used to display the active power direction. Depending on the sign of the active power, the direction display is shown in one or the other direction or is completely hidden if the active power = 0W.

The dynamization of the active power can be enabled or disabled using **Feature.Bit20**.

- 0 = Dynamization of active power direction disabled
- 1 = Dynamization of active power direction enabled

Feature.Bit21 can be used to configure the direction of the active power in the event of a negative value

- 0 = For a negative active power value, the arrow points downwards.
- 1 = For a negative active power value, the arrow points upwards.

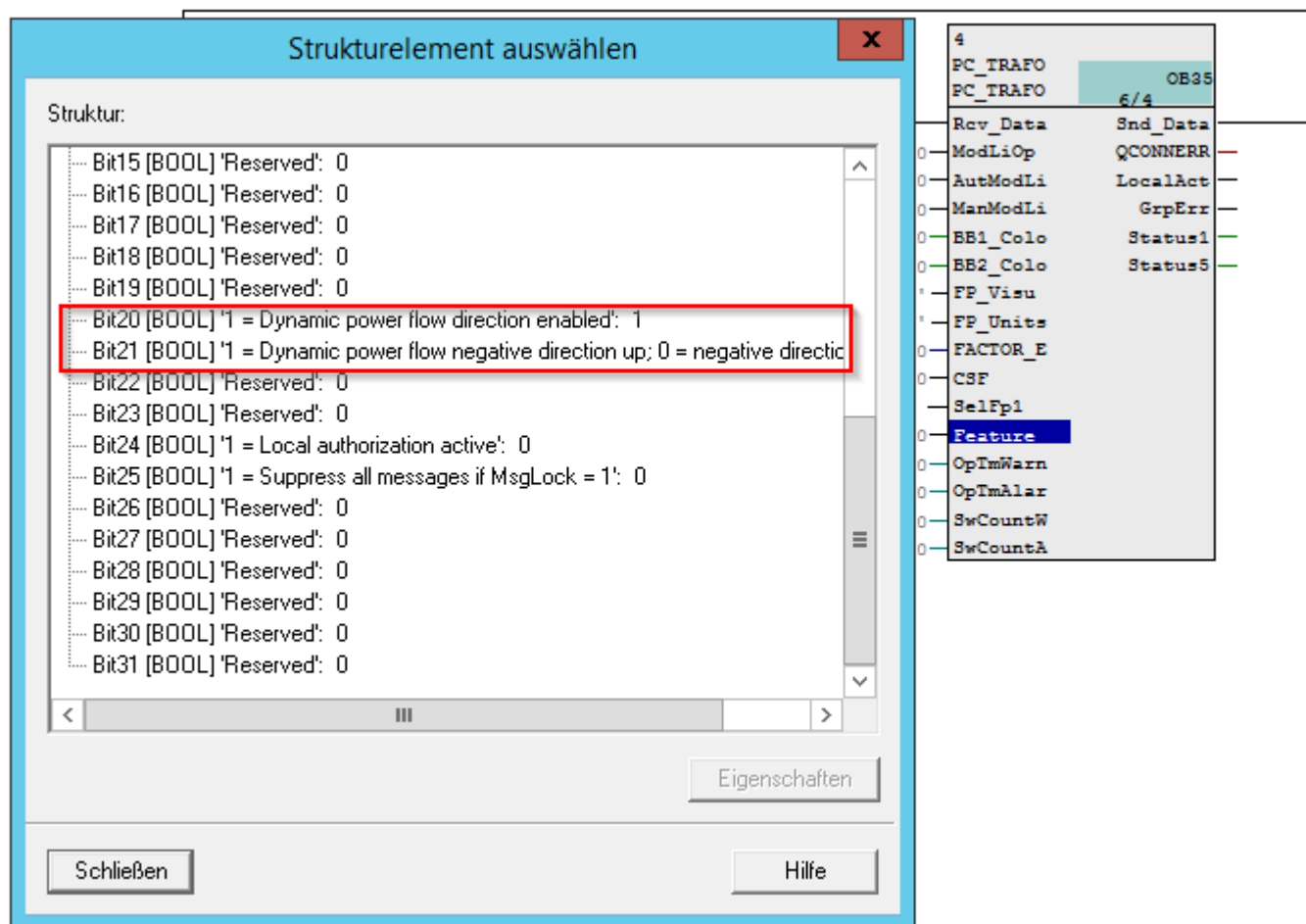


Figure 8-4 Active power direction - Feature bit PC_TRAFO

If dynamization is activated, the direction of the arrow is displayed based on the active power value and the setting at Feature.Bit21:

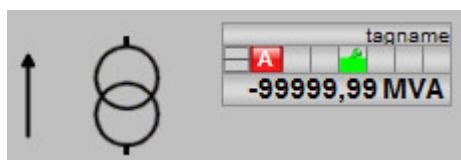


Figure 8-5 Active power direction - negative icon PC_TRAFO



Figure 8-6 Active power direction - positive icon PC_TRAFO

If the value of the active power = 0, the arrow is completely hidden.

8.3 PC_TRAFO faceplate

Available views:

View	Description
Default window	Important measured values, device status
Message view	Messages from device (messages, warnings, etc.)
Measured value window 1	Measured values
Measured value window 2	Measured values
Maintenance window	Operating hours, switches, active parameter group
User-defined window 1	User-specific view 1
User-defined window 2	User-specific view 2
Memo view	Memo view

8.3.1 Default window

The default window offers the following functions:

- Display of the device status
- Display of important measured values
- Display of auxiliary values

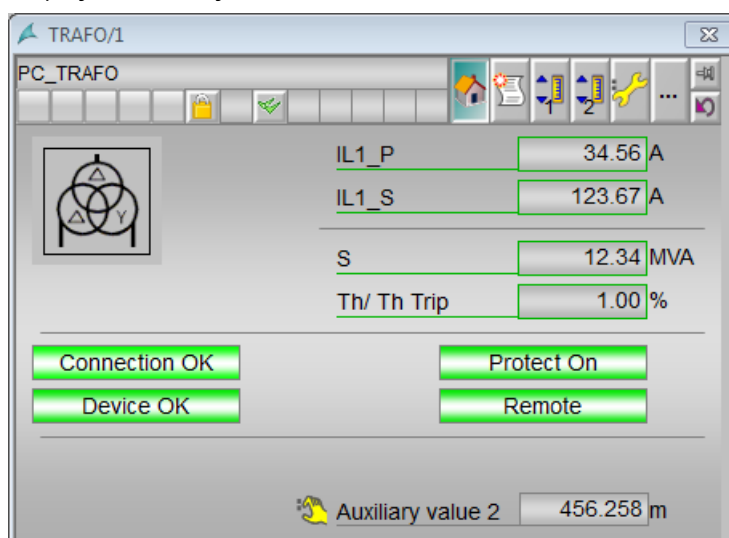


Figure 8-7 PC_TRAFO default window

Display

Note: The displays for the auxiliary values are only shown if at least one auxiliary value was interconnected to the corresponding block input.

Display	Meaning	Tag	Description
Connection status	Communication status of IED via Ethernet.	QCO-NERRO	QCONERR0 = 0 (Connection OK) – GREEN QCONERR0 = 1 (Comm. loss) – RED
Device status 1	Status of IED	Dev_OK	Dev_OK = 1 (Device OK) – GREEN Dev_OK = 2 (Device warning) – YELLOW Dev_OK = 3 (Device message) – RED
Device status 2	Behavior of IED	Beh	Beh = 1 (Protection on) – GREEN Beh = 2 (Protection locked) – YELLOW Beh = 3 (Protection test) – RED Beh = 4 (Protection test/locked) – RED Beh = 5 (Protection off) – RED
Device status 3	Local/Remote	Loc	Loc = 1: Local Loc = 0: Remote
S	Apparent power	S#Value	Apparent power
Th/Th Tripping	Theta/Theta Tripping	THE-TA#Value	Theta/Theta Tripping
IL1_P	Current L1 primary	I1_A#Value	Current L1 primary
IL1_S	Current L1 secondary	I2_A#Value	Current L1 secondary
Auxiliary value 1	User analog value 1	UserA-na1#Value	User analog value 1
Auxiliary value 2	User analog value 2	UserA-na2#Value	User analog value 2

8.3.2 Message view

This window displays every active message and warning of the IED. The APL default message window is used.

8.3.3 Measured value window 1

The measured values of the IED are displayed in the measured value window.

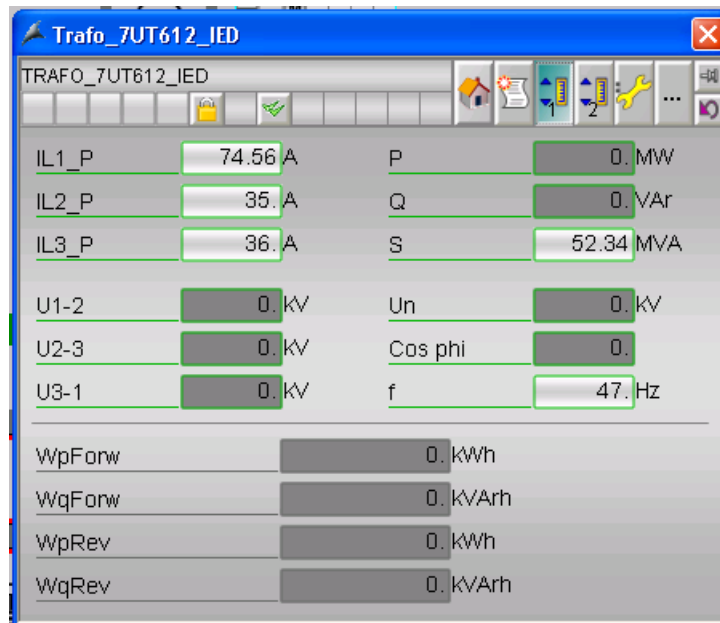


Figure 8-8 1 PC_TRAFO measured value window

In the event that the signal status of all energy values (WpVorw, WqVorw, WpRückw, WqRückw) is transmitted with a value of 16#FF, the corresponding display elements are hidden. As soon as one of the energy values receives a status not equal to 16#FF, all display elements are re-displayed.

Display

Display	Meaning	Tag	Description
IL1_P	Current L1 primary	I1_A#Value	Current L1 primary
IL2_P	Current L2 primary	I1_B#Value	Current L2 primary
IL3_P	Current L3 primary	I1_C#Value	Current L3 primary
U1-2	Voltage L1-L2	U_AB#Value	Voltage L1-L2
U2-3	Voltage L2-L3	U_BC#Value	Voltage L2-L3
U3-1	Voltage L3-L1	U_CA#Value	Voltage L3-L1
P	Power	P#Value	Power
Q	Reactive power	Q#Value	Reactive power
S	Apparent power	S#Value	Apparent power
Un	Neutral voltage	U#Value	Neutral voltage
Cos Phi	Power factor	PF#Value	Power factor
F	Frequency	F#Value	Frequency
WpVorw	Supply active energy	WpForward#Value	Supply active energy

Display	Meaning	Tag	Description
WqVorw	Demand active energy	WpForward#Value	Demand active energy
WpRückw	Supply reactive energy	WpReverse#Value	Supply reactive energy
WqRückw	Demand reactive energy	WqReverse#Value	Demand reactive energy

8.3.4 Measured value window 2

The measured values of the IED are displayed in the measured value window.

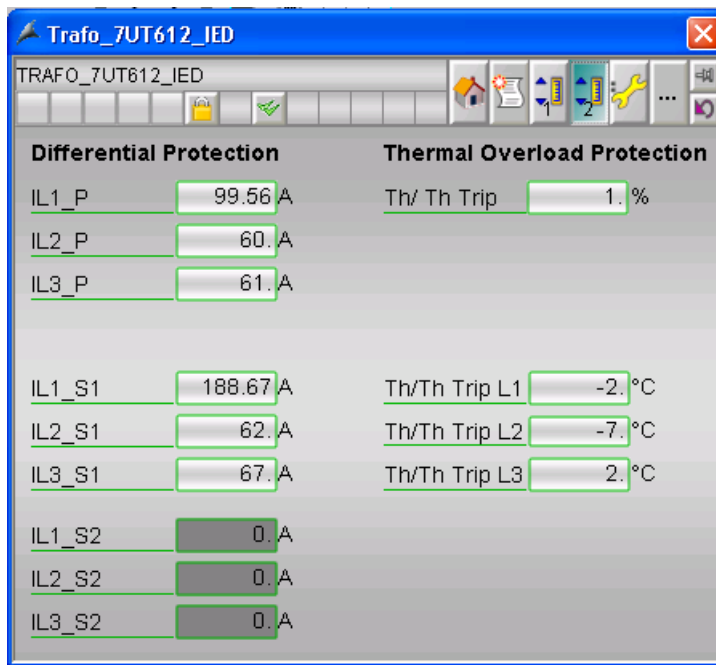


Figure 8-9 PC_TRAFO measured value window 2

Display

Display	Meaning	Tag	Description
IL1_P	Current L1 primary	I1_A#Value	Current L1 primary
IL2_P	Current L2 primary	I1_B#Value	Current L2 primary
IL3_P	Current L3 secondary 1	I1_C#Value	Current L3 secondary 1
IL1_S1	Current L1 secondary 1	I2_A#Value	Current L1 secondary 1
IL2_S1	Current L2 secondary 1	I2_B#Value	Current L2 secondary 1
IL3_S1	Current L3 secondary 2	I2_C#Value	Current L3 secondary 2

Display	Meaning	Tag	Description
IL1_S2	Current L1 secondary 2	I3_A#Value	Current L1 secondary 2
IL2_S2	Current L2 secondary 2	I3_B#Value	Current L2 secondary 2
IL3_S2	Current L3 secondary 2	I3_C#Value	Current L3 secondary 2
Th/Th Tripping	Theta/Theta Tripping	THETA#Value	Theta/Theta Tripping
Th/Th Tripping L1	Theta/Theta Tripping L1	THE-TA_L1#Value	Theta/Theta Tripping L1
Th/Th Tripping L2	Theta/Theta Tripping L2	THE-TA_L2#Value	Theta/Theta Tripping L2
Th/Th Tripping L3	Theta/Theta Tripping L3	THE-TA_L3#Value	Theta/Theta Tripping L3

8.3.5 Maintenance window

The maintenance window displays the maintenance-specific information.

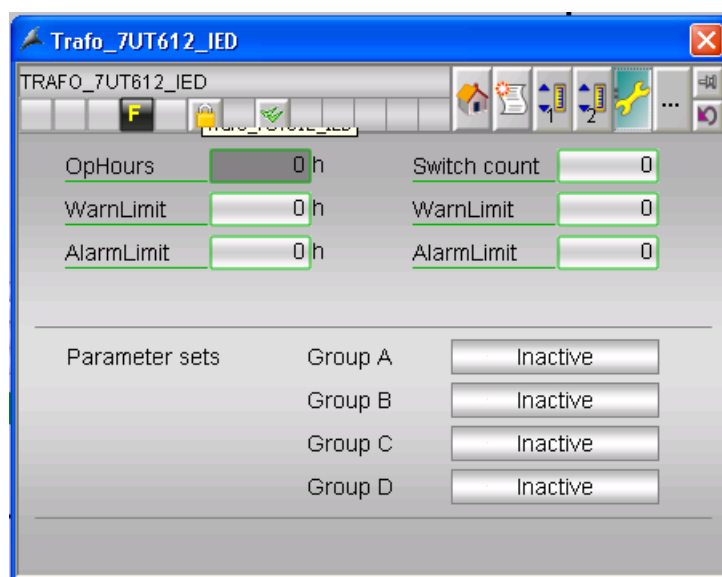


Figure 8-10 PC_TRAFO maintenance window

Display

Display	Meaning	Tag	Description
OpHours	Operating hours	OpTm	Operating hours
WarnLimit	Warning limit operating hours	OpTmWarnLimit	Warning limit operating hours
AlarmLimit	Alarm limit operating hours	OpTmAlarmLimit	Alarm limit operating hours

Display	Meaning	Tag	Description
Operating cycles	Control number Q0	SwCount	Control number Q0
WarnLimit	Warning limit operating cycles	SwCountWarnLimit	Warning limit operating cycles
AlarmLimit	Alarm limit operating cycles	SwCountAlarmLimit	Alarm limit operating cycles
Group A	Parameter set A active	PARSET_A	Parameter set A active
Group B	Parameter set B active	PARSET_B	Parameter set B active
Group C	Parameter set C active	PARSET_C	Parameter set C active
Group D	Parameter set D active	PARSET_D	Parameter set D active

8.3.6 User view

The user-defined faceplates 1 and 2 can be customized by the customer. See also Creating a user view/templates (Page 31)

Example:

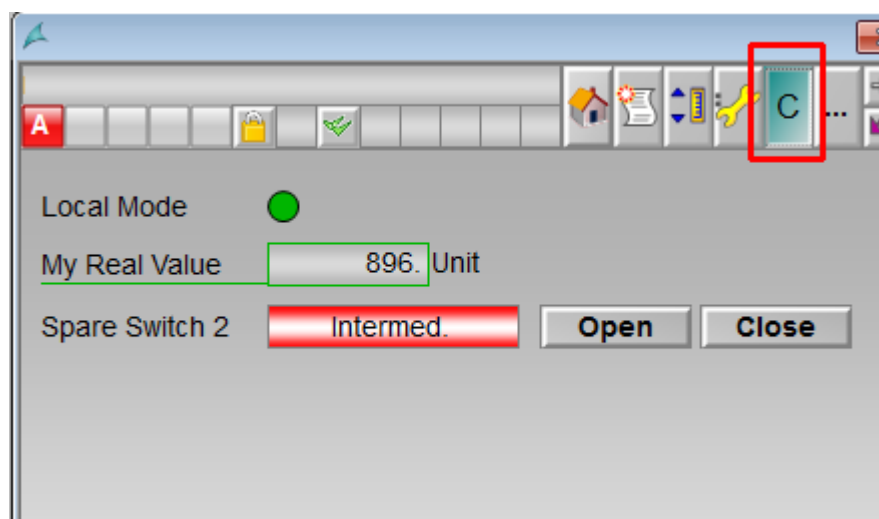


Figure 8-11 Example of user view

PC_SYNC

The PC_SYNC object is used for synchronization devices. The faceplate serves only for visualization. No commands are implemented in the default faceplates. However, eight free commands are available in the user-defined faceplate and in the PLC faceplate.

PC_SYNC consists of the following objects:

- PLC block PC_SYNC
- WinCC block symbols
- WinCC faceplate PC_SYNC with different views
- Device driver blocks (not part of the PCL)

9.1 PLC block PC_SYNC

9.1.1 Defining synchronization points

To adapt the display of synchronization points in the faceplate, the used synchronization points can be defined with the input "SyncGroups". The input has the type Byte. Each individual bit activates a synchronization point. For example, 0xFF means that all 8 synchronization points are displayed. With the value 0x03, only point 1 and point 2 are displayed.

9.1.2 Description of PLC block PC_SYNC

The PC_SYNC (FB 2503) function block represents the synchronization object in the PLC. The block is the interface between the device driver block and the remaining PCS 7 environment.

The block offers the following functions:

- Detecting data from the device driver block
- Providing data for the block symbol and the faceplate
- Controlling switches of the switchgear device (only free commands)
- APL operating modes Manual, Automatic
- Interlocking functions
- Handling of operator authorizations
- Various APL standard functions

9.1.3 Interfaces

The PLC block PC_SYNC offers the following interfaces:

1. **Device driver blocks**
The device driver block is connected to the PC_SYNC block through the input "Rcv_Data" and the output "Snd_Data".
2. **Block symbol**
The block symbol receives data from the block through the S7 connection between the PLC and WinCC.
3. **Faceplate**
The faceplate receives data from the block through the S7 connection between the PLC and WinCC.
4. **APL interlock blocks**
The interlock blocks can be connected through the "IntLock" input.
5. **Logic of the S7 program (for controlling modes, switches, etc.)**
The PC_SYNC block can receive commands from the CFC logic and makes data available to other blocks.

9.1.4 Configuration

Use the CFC editor to install the block in a cyclic interrupt OB (OB30 to OB38). The block is also installed automatically in the startup OB (OB 100).

9.1.5 Operating modes

The PC_SYNC block supports the APL standard operating modes.

The operating mode can only be changed through the interconnections on the PLC block.

The block can be operated in the following operating modes:

Operating mode	Description	Permitted commands	Display
Local operating mode	The device signals local operating mode (for example, the key-operated switch on the device is in the "local" position)	No	<i>LocalAct.Value</i> = true
Manual mode	The device is controlled through the faceplate	Only from faceplate	<i>ManAct.Value</i> = true
Automatic mode	The device is controlled through the connected logic in the CFC	Only by block inputs	<i>AutAct.Value</i> = true

For a detailed description of the operating modes refer to the "PCS 7 Advanced Process Library Function Manual" [1].

9.1.6 Switch commands

The PC_SYNC block sends commands from the faceplate (in manual operation) or from other connected blocks (in automatic mode) to the device driver block. The following commands are implemented:

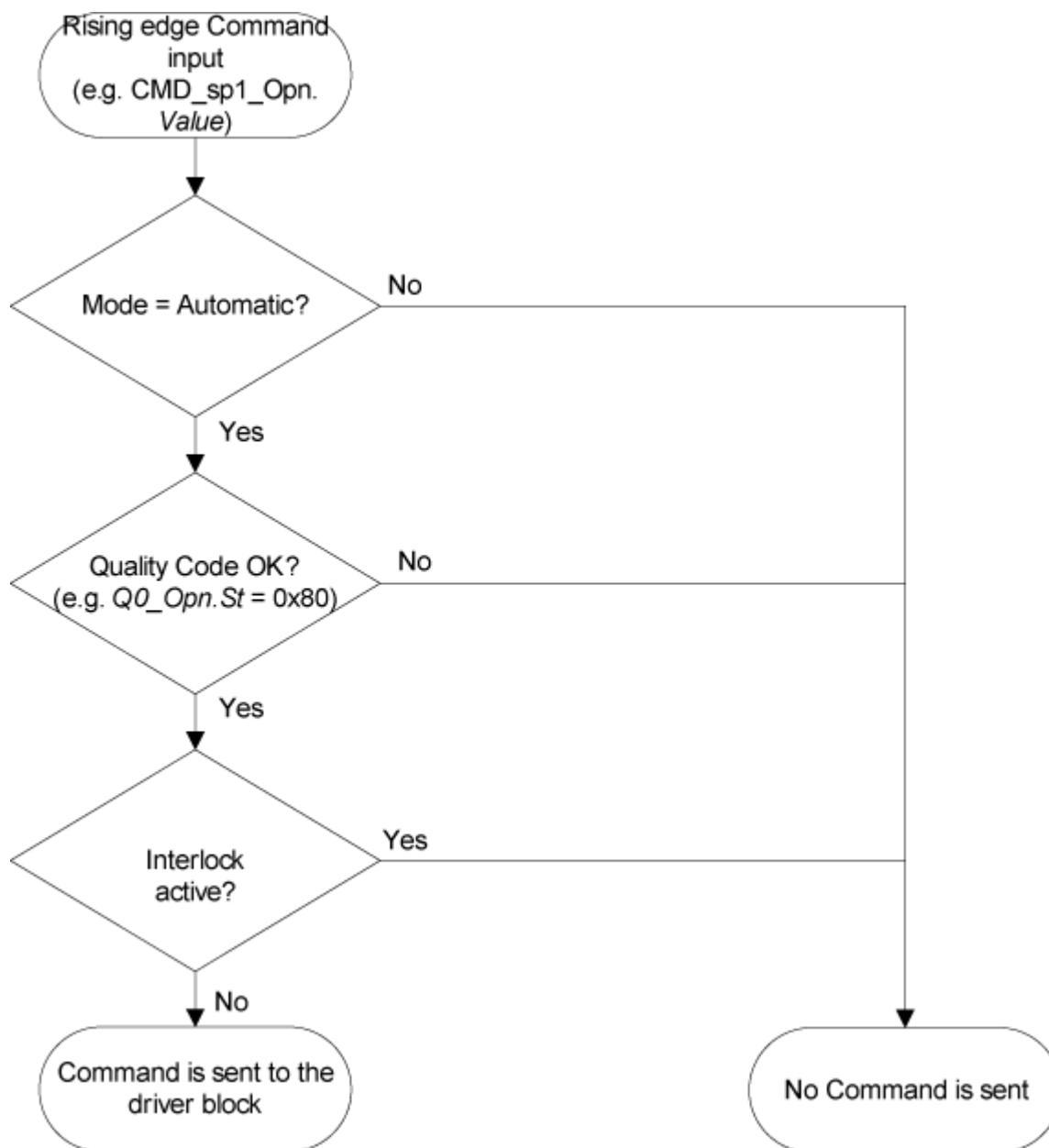
- Free command 1
- Free command 2
- Free command 3
- Free command 4
- Free command 5
- Free command 6
- Free command 7
- Free command 8

9.1.7 Interlocks

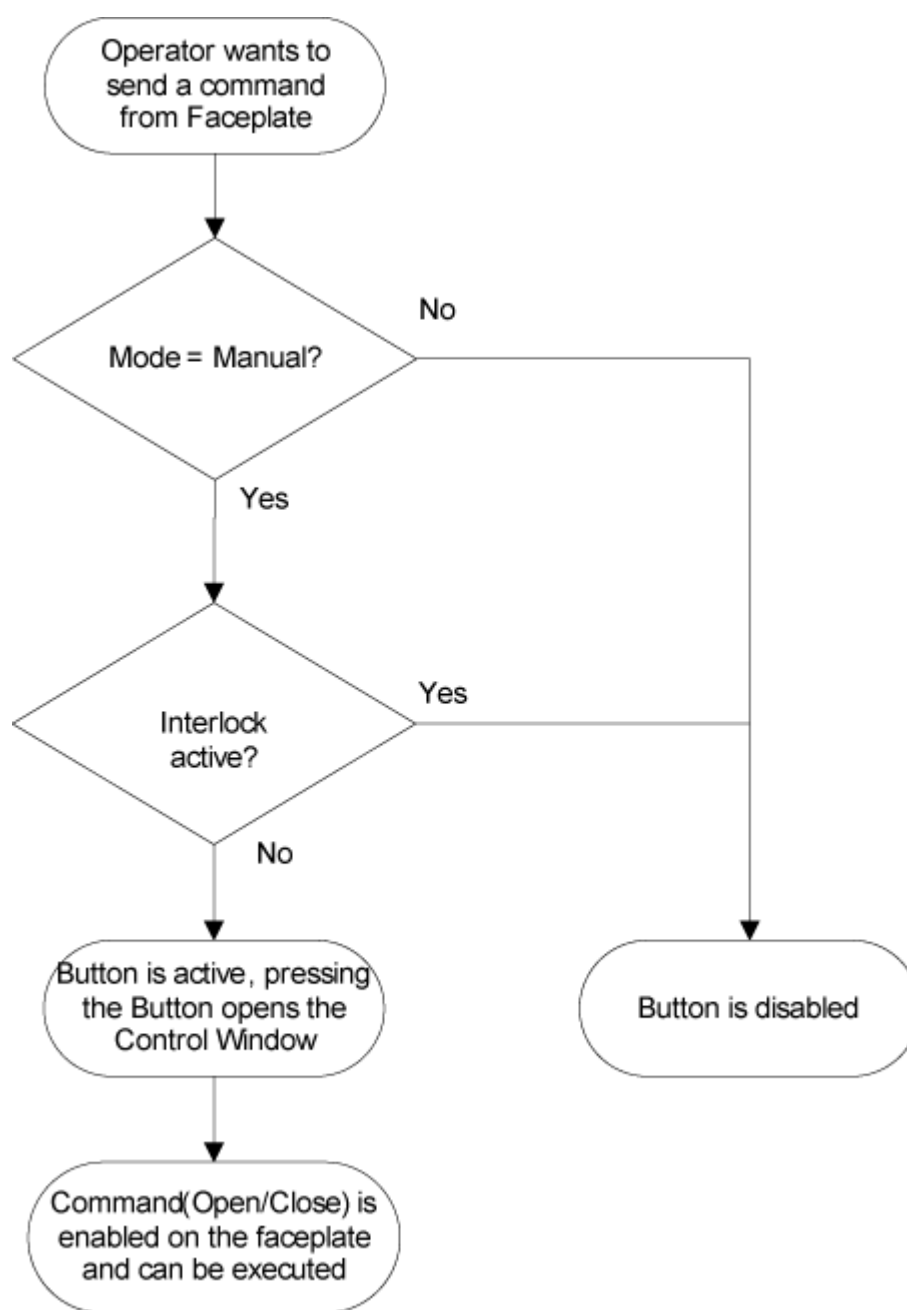
The PC_SYNC block only contains the APL standard interlock "Interlock".

Important information: The APL interlock locks all free commands of the PC_SYNC block but does not open any switches at activation.

The following logic is used to interlock a linked command:



The following logic is used for interlocking the commands output in the faceplate:



Deactivating individual interlocks:

You deactivate the implemented block interlock through the input parameter Intlock

If you want to deactivate the block interlock, you have to specify the following parameters:

- `Intl_En = 0` or `Intlock.ST = 16#FF`: The Intlock input parameter does not have any effect.

9.1.8 Group error

This block provides the standard APL function "Outputting group errors".

The following parameters are taken into consideration when forming the group error:

- CSF

The group error output is *GrpErr.Value*.

9.1.9 Forming the group status for interlocks

This block contains the standard APL function "Forming the group status for interlock information" (see APL documentation).

9.1.10 Responses that can be configured with the "Feature" parameter

An overview of all the responses provided by the `Feature` parameter is contained in the PCS 7 APL documentation.

The following function is available for this block at the corresponding bits:

Bit	Function
0-11	Not used
12	1 = Hide operating hours in the faceplate 0 (default) = Show operating hours in the faceplate
13	1 = Hide operating cycle hours in the faceplate 0 (default) = Show operating cycle hours in the faceplate
14-31	Not used

9.1.11 Operator authorizations

This block includes the APL standard function "Operator authorizations".

The block provides the following authorizations for the `OS_Perm` parameter:

Bit	Function
0	1 = Operator may switch to "Automatic mode"
1	1 = Operator may switch to "Manual mode"
2	1 = Operator may switch to "Local operating mode"
3-31	Not used

9.1.12 Assignment of the status word for Status1 parameter

Status bit	Parameter
0	Occupied
1	BatchEn
2	Not used
3	Not used
4	Not used
5	0 = ManAct.Value 1 = AutoAct.Value
6	LocalAct.Value
7	0: Open padlock in block symbol 1: Closed padlock in block symbol
8	Not used (always true)
9	Not used
10	Not used
11	Not used
12	Not used
13	Not used
14	Not used
15	Not used
16	1 = Intlock is active
17	Not used
18	Not used
19	Trip.Value
20	Not used
21	Not used
22	Not used
23	"Interlock" button is activated
24	Not used
25	Not used
26	Bypass information from previous function block
27	Not used
28	Not used
29	Not used
30	Not used
31	Not used

9.1.13 Assignment of the status word for Status5 parameter

The Status5 parameter is used to show / hide values in the faceplate. The values are displayed depending on the quality code of the Station Gateway / Station Controller.

The values are displayed when the quality code is $\neq$ 0x00.

Status bit	Value	Description
1	Synchronization point 1	All values of synchronization point 1
2	Synchronization point 2	All values of synchronization point 2
3	Synchronization point 3	All values of synchronization point 3
4	Synchronization point 4	All values of synchronization point 4
5	Synchronization point 5	All values of synchronization point 5
6	Synchronization point 6	All values of synchronization point 6
7	Synchronization point 7	All values of synchronization point 7
8	Synchronization point 8	All values of synchronization point 8
19	Optm	Operating hours
20	SwCount	Operating cycles

9.1.14 Messages

Message instance	Message identifier	Message class	Event
MsgEvId_1	SIG_1	Alarm - high	\$\$BlockComment\$\$ Excitation device
	SIG_2	Alarm - high	\$\$BlockComment\$\$ General tripping device
	SIG_3	Alarm - high	\$\$BlockComment\$\$ Tripping coupling 1
	SIG_4	Warning - high	\$\$BlockComment\$\$ Excitation coupling 1
	SIG_5	Alarm - high	\$\$BlockComment\$\$ Tripping coupling 2
	SIG_6	Warning - high	\$\$BlockComment\$\$ Excitation coupling 2
	SIG_7	Alarm - high	\$\$BlockComment\$\$ Tripping coupling 3
	SIG_8	Warning - high	\$\$BlockComment\$\$ Excitation coupling 3
MsgEvId_2	SIG_1	Alarm - high	\$\$BlockComment\$\$ Tripping coupling 4
	SIG_2	Warning - high	\$\$BlockComment\$\$ Excitation coupling 4
	SIG_3	Alarm - high	\$\$BlockComment\$\$ Tripping frequency 1 <
	SIG_4	Warning - high	\$\$BlockComment\$\$ Excitation frequency 1 <
	SIG_5	Alarm - high	\$\$BlockComment\$\$ Tripping frequency 2 <
	SIG_6	Warning - high	\$\$BlockComment\$\$ Excitation frequency 2 <
	SIG_7	Alarm - high	\$\$BlockComment\$\$ Tripping frequency 3 <
	SIG_8	Warning - high	\$\$BlockComment\$\$ Excitation frequency 3 <

Message instance	Message identifier	Message class	Event
MsgEvId_3	SIG_1	Alarm - high	\$\$BlockComment\$\$ Tripping frequency 4 <
	SIG_2	Warning - high	\$\$BlockComment\$\$ Excitation frequency 4 <
	SIG_3	Warning - high	\$\$BlockComment\$\$ Excitation disconnector fault
	SIG_4	Alarm - high	\$\$BlockComment\$\$ Tripping disconnector fault T1
	SIG_5	Alarm - high	\$\$BlockComment\$\$ trip disconnector fault T2
	SIG_6	Warning - high	\$\$BlockComment\$\$ Local mode
	SIG_7	Alarm - high	\$\$BlockComment\$\$ Fault group message
	SIG_8	Warning - high	\$\$BlockComment\$\$ Warning group message
MsgEvId_4	SIG_1	Alarm - high	\$\$BlockComment\$\$ Synchronization point 1 - voltage difference exceeded
	SIG_2	Alarm - high	\$\$BlockComment\$\$ Synchronization point 2 - voltage difference exceeded
	SIG_3	Alarm - high	\$\$BlockComment\$\$ Synchronization point 3 - voltage difference exceeded
	SIG_4	Alarm - high	\$\$BlockComment\$\$ Synchronization point 4 - voltage difference exceeded
	SIG_5	Alarm - high	\$\$BlockComment\$\$ Synchronization point 5 - voltage difference exceeded
	SIG_6	Alarm - high	\$\$BlockComment\$\$ Synchronization point 6 - voltage difference exceeded
	SIG_7	Alarm - high	\$\$BlockComment\$\$ Synchronization point 7 - voltage difference exceeded
	SIG_8	Alarm - high	\$\$BlockComment\$\$ Synchronization point 8 - voltage difference exceeded
MsgEvId_5	SIG_1	Status message - AS	\$\$BlockComment\$\$ Synchronization point 1 - measurement running
	SIG_2	Status message - AS	\$\$BlockComment\$\$ Synchronization point 2 - measurement running
	SIG_3	Status message - AS	\$\$BlockComment\$\$ Synchronization point 3 - measurement running
	SIG_4	Status message - AS	\$\$BlockComment\$\$ Synchronization point 4 - measurement running
	SIG_5	Status message - AS	\$\$BlockComment\$\$ Synchronization point 5 - measurement running
	SIG_6	Status message - AS	\$\$BlockComment\$\$ Synchronization point 6 - measurement running
	SIG_7	Status message - AS	\$\$BlockComment\$\$ Synchronization point 7 - measurement running
	SIG_8	Status message - AS	\$\$BlockComment\$\$ Synchronization point 8 - measurement running

Message instance	Message identifier	Message class	Event
MsgEvId_6	SIG_1	Warning - high	\$\$BlockComment\$\$ Synchronization point 1 - difference frequency exceeded
	SIG_2	Warning - high	\$\$BlockComment\$\$ Synchronization point 2 - difference frequency exceeded
	SIG_3	Warning - high	\$\$BlockComment\$\$ Synchronization point 3 - difference frequency exceeded
	SIG_4	Warning - high	\$\$BlockComment\$\$ Synchronization point 4 - difference frequency exceeded
	SIG_5	Warning - high	\$\$BlockComment\$\$ Synchronization point 5 - difference frequency exceeded
	SIG_6	Warning - high	\$\$BlockComment\$\$ Synchronization point 6 - difference frequency exceeded
	SIG_7	Warning - high	\$\$BlockComment\$\$ Synchronization point 7 - difference frequency exceeded
	SIG_8	Warning - high	\$\$BlockComment\$\$ Synchronization point 8 - difference frequency exceeded
MsgEvId_7	SIG_1	Warning - high	\$\$BlockComment\$\$ Synchronization point 1 - angle difference exceeded
	SIG_2	Warning - high	\$\$BlockComment\$\$ Synchronization point 2 - angle difference exceeded
	SIG_3	Warning - high	\$\$BlockComment\$\$ Synchronization point 3 - angle difference exceeded
	SIG_4	Warning - high	\$\$BlockComment\$\$ Synchronization point 4 - angle difference exceeded
	SIG_5	Warning - high	\$\$BlockComment\$\$ Synchronization point 5 - angle difference exceeded
	SIG_6	Warning - high	\$\$BlockComment\$\$ Synchronization point 6 - angle difference exceeded
	SIG_7	Warning - high	\$\$BlockComment\$\$ Synchronization point 7 - angle difference exceeded
	SIG_8	Warning - high	\$\$BlockComment\$\$ Synchronization point 8 - angle difference exceeded

Message instance	Message identifier	Message class	Event
MsgEvid_8	SIG_1	Status message - AS	\$\$BlockComment\$\$ Synchronization point 1 - close enable
	SIG_2	Status message - AS	\$\$BlockComment\$\$ Synchronization point 2 - close enable
	SIG_3	Status message - AS	\$\$BlockComment\$\$ Synchronization point 3 - close enable
	SIG_4	Status message - AS	\$\$BlockComment\$\$ Synchronization point 4 - close enable
	SIG_5	Status message - AS	\$\$BlockComment\$\$ Synchronization point 5 - close enable
	SIG_6	Status message - AS	\$\$BlockComment\$\$ Synchronization point 6 - close enable
	SIG_7	Status message - AS	\$\$BlockComment\$\$ Synchronization point 7 - close enable
	SIG_8	Status message - AS	\$\$BlockComment\$\$ Synchronization point 8 - close enable
MsgEvid_9	SIG_1	Status message - AS	\$\$BlockComment\$\$ Parameter set A active
	SIG_2	Status message - AS	\$\$BlockComment\$\$ Parameter set B active
	SIG_3	Status message - AS	\$\$BlockComment\$\$ Parameter set C active
	SIG_4	Status message - AS	\$\$BlockComment\$\$ Parameter set D active
	SIG_5	Alarm - high	\$\$BlockComment\$\$ IEC Alarm 1
	SIG_6	Alarm - high	\$\$BlockComment\$\$ IEC Alarm 2
	SIG_7	Alarm - high	\$\$BlockComment\$\$ IEC Alarm 3
	SIG_8	Alarm - high	\$\$BlockComment\$\$ IEC Alarm 4
MsgEvid_10	SIG_1	Alarm - high	\$\$BlockComment\$\$ IEC Alarm 5
	SIG_2	Alarm - high	\$\$BlockComment\$\$ IEC Alarm 6
	SIG_3	Alarm - high	\$\$BlockComment\$\$ IEC Alarm 7
	SIG_4	Alarm - high	\$\$BlockComment\$\$ IEC Alarm 8
	SIG_5	Alarm - high	\$\$BlockComment\$\$ IEC Alarm 9
	SIG_6	Alarm - high	\$\$BlockComment\$\$ IEC Alarm 10
	SIG_7	Alarm - high	\$\$BlockComment\$\$ IEC Alarm 11
	SIG_8	Alarm - high	\$\$BlockComment\$\$ IEC Alarm 12

Message instance	Message identifier	Message class	Event
MsgEvIdlocal1	SIG_1	AS process control message - error	\$\$BlockComment\$\$ Connection fault
	SIG_2	Alarm - high	
	SIG_3	Alarm - high	
	SIG_4	Alarm - high	
	SIG_5	Alarm - high	
	SIG_6	Alarm - high	
	SIG_7	Alarm - high	
	SIG_8	AS process control message - error	\$\$BlockComment\$\$ AS process control message - external error
MsgEvIdlocal2	SIG_1	Warning - high	\$\$BlockComment\$\$ Maintenance: Operating cycle warning (@3%d@)
	SIG_2	Alarm - high	\$\$BlockComment\$\$ Maintenance: Operating cycle alarm (@3%d@)
	SIG_3	Warning - high	\$\$BlockComment\$\$ Maintenance: Operating time warning (@4%d@)
	SIG_4	Alarm - high	\$\$BlockComment\$\$ Maintenance: Operating time alarm (@4%d@)
	SIG_5	Alarm - high	\$\$BlockComment\$\$ External message 1
	SIG_6	Alarm - high	\$\$BlockComment\$\$ External message 2
	SIG_7	Alarm - high	\$\$BlockComment\$\$ External message 3
	SIG_8	Alarm - high	\$\$BlockComment\$\$ External message 4
MsgEvId_d_1	SIG_1	Status message - AS	\$\$BlockComment\$\$ Switch 1 closed
	SIG_2	Status message - AS	\$\$BlockComment\$\$ Switch 1 open
	SIG_3	Status message - AS	\$\$BlockComment\$\$ Switch 2 closed
	SIG_4	Status message - AS	\$\$BlockComment\$\$ Switch 2 open
	SIG_5	Status message - AS	\$\$BlockComment\$\$ Switch 3 closed
	SIG_6	Status message - AS	\$\$BlockComment\$\$ Switch 3 open
	SIG_7	Status message - AS	\$\$BlockComment\$\$ Switch 4 closed
	SIG_8	Status message - AS	\$\$BlockComment\$\$ Switch 4 open

Message instance	Message identifier	Message class	Event
MsgEvid_device	SIG_1	AS process control message - error	\$\$BlockComment\$\$ Error device IED name:@1%s@ -> IP address: @2%s@
	SIG_2	AS control system message - fault	\$\$BlockComment\$\$ Redundancy failure
	SIG_3	AS process control message - error	\$\$BlockComment\$\$ PROFIBUS mapping alarm
	SIG_4	Alarm - high	
	SIG_5	Alarm - high	
	SIG_6	Alarm - high	
	SIG_7	Alarm - high	
	SIG_8	Alarm - high	
MsgEvid_d_2	SIG_1	Status message - AS	\$\$BlockComment\$\$ Switch 5 closed
	SIG_2	Status message - AS	\$\$BlockComment\$\$ Switch 5 open
	SIG_3	Status message - AS	\$\$BlockComment\$\$ Switch 6 closed
	SIG_4	Status message - AS	\$\$BlockComment\$\$ Switch 6 open
	SIG_5	Status message - AS	\$\$BlockComment\$\$ Switch 7 closed
	SIG_6	Status message - AS	\$\$BlockComment\$\$ Switch 7 open
	SIG_7	Status message - AS	\$\$BlockComment\$\$ Switch 8 closed
	SIG_8	Status message - AS	\$\$BlockComment\$\$ Switch 8 open

9.1.15 I/O list

9.1.15.1 Inputs

I/O name	Description	Value	Data type
Rcv_Data	Connect with SG_/SC_SYNC		STRUCT
ModLiOp	1 = Interconnect/Auto, 0 = Manual: Input for auto / manual commands		STRUCT
ModLiOp.Value		0	BOOL
ModLiOp.ST		16#80	BYTE
AutModOp	1 = Automatic mode: Automatic mode by operator	0	BOOL

I/O name	Description	Value	Data type
ManModOp	1 = Manual mode: Manual mode by operator	1	BOOL
AutModLi			STRUCT
AutModLi.Value		0	BOOL
AutModLi.ST		16#80	BYTE
ManModLi			STRUCT
ManModLi.Value		0	BOOL
ManModLi.ST		16#80	BYTE
FP_Visu	Configuration of the faceplate		STRUCT
FP_Units	Configuration of the units in the faceplate		STRUCT
MS_RelOp	Operator input for MS release , 1: MS release request	0	BOOL
Monitor	1=Monitoring ON, 0=Monitoring OFF: Feed-back monitoring	1	BOOL
MonTiStatic	Monitoring time of the feedback [s]	3.0	REAL
MonTiDynamic	Monitoring time of the feedback [s]	3.0	REAL
Intlock	0=Motor is locked, 1=Motor is not locked		STRUCT
Intlock.Value	Value	1	BOOL
Intlock.ST	Signal state	16#FF	BYTE
Intl_En	1=Interlock activated, 0=Interlock deactivated	1	BOOL
CMD_sp1_Opn	Rising edge: Free command 1 off		STRUCT
CMD_sp1_Opn.Value		0	BOOL
CMD_sp1_Opn.ST		16#80	BYTE
CMD_sp1_Cls	Rising edge: Free command 1 on		STRUCT
CMD_sp1_Cls.Value		0	BOOL
CMD_sp1_Cls.ST		16#80	BYTE
CMD_sp2_Opn	Rising edge: Free command 2 off		STRUCT
CMD_sp2_Opn.Value		0	BOOL
CMD_sp2_Opn.ST		16#80	BYTE
CMD_sp2_Cls	Rising edge: Free command 2 on		STRUCT
CMD_sp2_Cls.Value		0	BOOL
CMD_sp2_Cls.ST		16#80	BYTE
CMD_sp3_Opn	Rising edge: Free command 3 off		STRUCT
CMD_sp3_Opn.Value		0	BOOL
CMD_sp3_Opn.ST		16#80	BYTE
CMD_sp3_Cls	Rising edge: Free command 3 on		STRUCT
CMD_sp3_Cls.Value		0	BOOL
CMD_sp3_Cls.ST		16#80	BYTE
CMD_sp4_Opn	Rising edge: Free command 4 off		STRUCT
CMD_sp4_Opn.Value		0	BOOL
CMD_sp4_Opn.ST		16#80	BYTE
CMD_sp4_Cls	Rising edge: Free command 4 on		STRUCT
CMD_sp4_Cls.Value		0	BOOL

I/O name	Description	Value	Data type
CMD_sp4_Cls.ST		16#80	BYTE
CMD_sp5_Opn	Rising edge: Free command 5 off		STRUCT
CMD_sp5_Opn.Value		0	BOOL
CMD_sp5_Opn.ST		16#80	BYTE
CMD_sp5_Cls	Rising edge: Free command 5 on		STRUCT
CMD_sp5_Cls.Value		0	BOOL
CMD_sp5_Cls.ST		16#80	BYTE
CMD_sp6_Opn	Rising edge: Free command 6 off		STRUCT
CMD_sp6_Opn.Value		0	BOOL
CMD_sp6_Opn.ST		16#80	BYTE
CMD_sp6_Cls	Rising edge: Free command 6 on		STRUCT
CMD_sp6_Cls.Value		0	BOOL
CMD_sp6_Cls.ST		16#80	BYTE
CMD_sp7_Opn	Rising edge: Free command 7 off		STRUCT
CMD_sp7_Opn.Value		0	BOOL
CMD_sp7_Opn.ST		16#80	BYTE
CMD_sp7_Cls	Rising edge: Free command 7 on		STRUCT
CMD_sp7_Cls.Value		0	BOOL
CMD_sp7_Cls.ST		16#80	BYTE
CMD_sp8_Opn	Rising edge: Free command 8 off		STRUCT
CMD_sp8_Opn.Value		0	BOOL
CMD_sp8_Opn.ST		16#80	BYTE
CMD_sp8_Cls	Rising edge: Free command 8 on		STRUCT
CMD_sp8_Cls.Value		0	BOOL
CMD_sp8_Cls.ST		16#80	BYTE
MsgLock			STRUCT
MsgLock.Value		0	BOOL
MsgLock.ST		16#80	BYTE
SampleTime	Sampling time [s]	0.1	REAL
MsgEvid_1	Message event ID	16#00000037	DWORD
MsgEvid_2	Message event ID	16#00000038	DWORD
MsgEvid_3	Message event ID	16#00000039	DWORD
MsgEvid_4	Message event ID	16#0000003A	DWORD
MsgEvid_5	Message event ID	16#0000003B	DWORD
MsgEvid_6	Message event ID	16#0000003C	DWORD
MsgEvid_7	Message event ID	16#0000003D	DWORD
MsgEvid_8	Message event ID	16#0000003E	DWORD
MsgEvid_9	Message event ID	16#0000003F	DWORD
MsgEvid_10	Message event ID	16#00000040	DWORD
MsgEvidlocal1	Message event ID	16#00000041	DWORD
MsgEvidlocal2	Message event ID	16#00000042	DWORD
MsgEvid_d_1	Message event ID	16#00000043	DWORD
MsgEvid_d_2	Message event ID	16#00000056	DWORD

I/O name	Description	Value	Data type
MsgEvId_device	Message event ID	16#00000044	DWORD
BatchEn	Activate remote operation of controller to batch recipe	0	BOOL
BatchID	Current batch ID (number)	16#00000000	DWORD
BatchName	Current batch name	"	STRING[32]
StepNo	Batch step number	16#00000000	DWORD
Occupied	Occupied by batch	0	BOOL
CSF			STRUCT
CSF.Value		0	BOOL
CSF.ST		16#80	BYTE
ExtMsg1	External message 1		STRUCT
ExtMsg1.Value	Value	0	BOOL
ExtMsg1.ST	Signal state	16#80	BYTE
ExtMsg2	External message 2		STRUCT
ExtMsg2.Value	Value	0	BOOL
ExtMsg2.ST	Signal state	16#80	BYTE
ExtMsg3	External message 3		STRUCT
ExtMsg3.Value	Value	0	BOOL
ExtMsg3.ST	Signal state	16#80	BYTE
ExtMsg4	External message 4		STRUCT
ExtMsg4.Value	Value	0	BOOL
ExtMsg4.ST	Signal state	16#80	BYTE
UserAna1	User analog input 1		STRUCT
UserAna1.Value	Value	0.0	REAL
UserAna1.ST	Signal state	16#FF	BYTE
UA1unit	Unit of measurement user analog input 1	0	INT
UserAna2	User analog input 2		STRUCT
UserAna2.Value	Value	0.0	REAL
UserAna2.ST	Signal state	16#FF	BYTE
UA2unit	Unit of measurement user analog input 2	0	INT
UserAna3	User analog input 3		STRUCT
UserAna3.Value	Value	0.0	REAL
UserAna3.ST	Signal state	16#FF	BYTE
UA3unit	Unit of measurement user analog input 3	0	INT
UserAna4	User analog input 4		STRUCT
UserAna4.Value	Value	0.0	REAL
UserAna4.ST	Signal state	16#FF	BYTE
UA4unit	Unit of measurement user analog input 4	0	INT
UserStatus	User status bits	16#0	BYTE
SelfP1	Select faceplate 1		ANY
OS_Perm	Operator authorizations		STRUCT
OS_Perm.Bit0	1 = Operator may switch to Automatic mode	1	BOOL
OS_Perm.Bit1	1 = Operator may switch to Manual mode	1	BOOL

I/O name	Description	Value	Data type
OS_Perm.Bit2	1 = Operator may switch to Local operating mode	1	BOOL
OS_Perm.Bit3	1 = Operator may switch to "Out of service" mode	1	BOOL
OS_Perm.Bit4		1	BOOL
OS_Perm.Bit5		1	BOOL
OS_Perm.Bit6		1	BOOL
OS_Perm.Bit7		1	BOOL
OS_Perm.Bit8		1	BOOL
OS_Perm.Bit9		1	BOOL
OS_Perm.Bit10		1	BOOL
OS_Perm.Bit11	1 = Operator can activate function simulation	1	BOOL
OS_Perm.Bit12	1 = Operator can enable release for maintenance	1	BOOL
OS_Perm.Bit13		1	BOOL
OS_Perm.Bit14		1	BOOL
OS_Perm.Bit15		1	BOOL
OS_Perm.Bit16		1	BOOL
OS_Perm.Bit17		1	BOOL
OS_Perm.Bit18		1	BOOL
OS_Perm.Bit19		1	BOOL
OS_Perm.Bit20		1	BOOL
OS_Perm.Bit21		1	BOOL
OS_Perm.Bit22		1	BOOL
OS_Perm.Bit23		1	BOOL
OS_Perm.Bit24		1	BOOL
OS_Perm.Bit25		1	BOOL
OS_Perm.Bit26		1	BOOL
OS_Perm.Bit27		1	BOOL
OS_Perm.Bit28		1	BOOL
OS_Perm.Bit29		1	BOOL
OS_Perm.Bit30		1	BOOL
OS_Perm.Bit31		1	BOOL
Feature	State of different functions		STRUCT
Feature.Bit0	Reserved	0	BOOL
Feature.Bit1	Reserved	0	BOOL
Feature.Bit2	Reserved	0	BOOL
Feature.Bit3	Reserved	0	BOOL
Feature.Bit4	Reserved	0	BOOL
Feature.Bit5	Reserved	0	BOOL
Feature.Bit6	Reserved	0	BOOL
Feature.Bit7	Reserved	0	BOOL
Feature.Bit8	Reserved	0	BOOL

I/O name	Description	Value	Data type
Feature.Bit9	Reserved	0	BOOL
Feature.Bit10	Reserved	0	BOOL
Feature.Bit11	Reserved	0	BOOL
Feature.Bit12	Hide OpTm	0	BOOL
Feature.Bit13	Hide SwCount	0	BOOL
Feature.Bit14	Reserved	0	BOOL
Feature.Bit15	Reserved	0	BOOL
Feature.Bit16	Reserved	0	BOOL
Feature.Bit17	Reserved	0	BOOL
Feature.Bit18	Reserved	0	BOOL
Feature.Bit19	Reserved	0	BOOL
Feature.Bit20	Reserved	0	BOOL
Feature.Bit21	Reserved	0	BOOL
Feature.Bit22	Reserved	0	BOOL
Feature.Bit23	Reserved	0	BOOL
Feature.Bit24	1 = Local authorization enabled	0	BOOL
Feature.Bit25	1 = Suppression of all messages if MsgLock = 1	0	BOOL
Feature.Bit26	Reserved	0	BOOL
Feature.Bit27	Reserved	0	BOOL
Feature.Bit28	Reserved	0	BOOL
Feature.Bit29	Reserved	0	BOOL
Feature.Bit30	Reserved	0	BOOL
Feature.Bit31	Reserved	0	BOOL
MonTime	Time in seconds until Timeout (for control switch)	20	INT

9.1.15.2 Inputs / outputs

I/O name	Description	Value	Data type
Cmdsp1	Control position free command 1	16#02	BYTE
Cmdsp2	Control position free command 2	16#02	BYTE
Cmdsp3	Control position free command 3	16#02	BYTE
Cmdsp4	Control position free command 4	16#02	BYTE
Cmdsp5	Control position free command 5	16#02	BYTE
Cmdsp6	Control position free command 6	16#02	BYTE
Cmdsp7	Control position free command 7	16#02	BYTE
Cmdsp8	Control position free command 8	16#02	BYTE
OpTmWarnLimit	Warning limit operating hours	0	DINT
OpTmAlarmLimit	Alarm limit operating hours	0	DINT
SwCountWarnLimit	Warning limit operating cycles	0	DINT
SwCountAlarmLimit	Alarm limit operating cycles	0	DINT

9.1.15.3 Outputs

I/O name	Description	Value	Data type
Snd_Data	Connect with SG/_SC_SYNC		STRUCT
QCONNERR		1	BOOL
P1_U1	P1_U1		STRUCT
P1_U1.Value		0.0	REAL
P1_U1.ST		16#FF	BYTE
P1_U2	P1_U2		STRUCT
P1_U2.Value		0.0	REAL
P1_U2.ST		16#FF	BYTE
P2_U1	P2_U1		STRUCT
P2_U1.Value		0.0	REAL
P2_U1.ST		16#FF	BYTE
P2_U2	P2_U2		STRUCT
P2_U2.Value		0.0	REAL
P2_U2.ST		16#FF	BYTE
P3_U1	P3_U1		STRUCT
P3_U1.Value		0.0	REAL
P3_U1.ST		16#FF	BYTE
P3_U2	P3_U2		STRUCT
P3_U2.Value		0.0	REAL
P3_U2.ST		16#FF	BYTE
P4_U1	P4_U1		STRUCT
P4_U1.Value		0.0	REAL
P4_U1.ST		16#FF	BYTE
P4_U2	P4_U2		STRUCT
P4_U2.Value		0.0	REAL
P4_U2.ST		16#FF	BYTE
P5_U1	P5_U1		STRUCT
P5_U1.Value		0.0	REAL
P5_U1.ST		16#FF	BYTE
P5_U2	P5_U2		STRUCT
P5_U2.Value		0.0	REAL
P5_U2.ST		16#FF	BYTE
P6_U1	P6_U1		STRUCT
P6_U1.Value		0.0	REAL
P6_U1.ST		16#FF	BYTE
P6_U2	P6_U2		STRUCT
P6_U2.Value		0.0	REAL
P6_U2.ST		16#FF	BYTE
P7_U1	P7_U1		STRUCT
P7_U1.Value		0.0	REAL

I/O name	Description	Value	Data type
P7_U1.ST		16#FF	BYTE
P7_U2	P7_U2		STRUCT
P7_U2.Value		0.0	REAL
P7_U2.ST		16#FF	BYTE
P8_U1	P8_U1		STRUCT
P8_U1.Value		0.0	REAL
P8_U1.ST		16#FF	BYTE
P8_U2	P8_U2		STRUCT
P8_U2.Value		0.0	REAL
P8_U2.ST		16#FF	BYTE
P1_F1	P1_F1		STRUCT
P1_F1.Value		0.0	REAL
P1_F1.ST		16#FF	BYTE
P1_F2	P1_F2		STRUCT
P1_F2.Value		0.0	REAL
P1_F2.ST		16#FF	BYTE
P2_F1	P2_F1		STRUCT
P2_F1.Value		0.0	REAL
P2_F1.ST		16#FF	BYTE
P2_F2	P2_F2		STRUCT
P2_F2.Value		0.0	REAL
P2_F2.ST		16#FF	BYTE
P3_F1	P3_F1		STRUCT
P3_F1.Value		0.0	REAL
P3_F1.ST		16#FF	BYTE
P3_F2	P3_F2		STRUCT
P3_F2.Value		0.0	REAL
P3_F2.ST		16#FF	BYTE
P4_F1	P4_F1		STRUCT
P4_F1.Value		0.0	REAL
P4_F1.ST		16#FF	BYTE
P4_F2	P4_F2		STRUCT
P4_F2.Value		0.0	REAL
P4_F2.ST		16#FF	BYTE
P5_F1	P5_F1		STRUCT
P5_F1.Value		0.0	REAL
P5_F1.ST		16#FF	BYTE
P5_F2	P5_F2		STRUCT
P5_F2.Value		0.0	REAL
P5_F2.ST		16#FF	BYTE
P6_F1	P6_F1		STRUCT
P6_F1.Value		0.0	REAL
P6_F1.ST		16#FF	BYTE

I/O name	Description	Value	Data type
P6_F2	P6_F2		STRUCT
P6_F2.Value		0.0	REAL
P6_F2.ST		16#FF	BYTE
P7_F1	P7_F1		STRUCT
P7_F1.Value		0.0	REAL
P7_F1.ST		16#FF	BYTE
P7_F2	P7_F2		STRUCT
P7_F2.Value		0.0	REAL
P7_F2.ST		16#FF	BYTE
P8_F1	P8_F1		STRUCT
P8_F1.Value		0.0	REAL
P8_F1.ST		16#FF	BYTE
P8_F2	P8_F2		STRUCT
P8_F2.Value		0.0	REAL
P8_F2.ST		16#FF	BYTE
P1_DU	P1_DU		STRUCT
P1_DU.Value		0.0	REAL
P1_DU.ST		16#FF	BYTE
P2_DU	P2_DU		STRUCT
P2_DU.Value		0.0	REAL
P2_DU.ST		16#FF	BYTE
P3_DU	P3_DU		STRUCT
P3_DU.Value		0.0	REAL
P3_DU.ST		16#FF	BYTE
P4_DU	P4_DU		STRUCT
P4_DU.Value		0.0	REAL
P4_DU.ST		16#FF	BYTE
P5_DU	P5_DU		STRUCT
P5_DU.Value		0.0	REAL
P5_DU.ST		16#FF	BYTE
P6_DU	P6_DU		STRUCT
P6_DU.Value		0.0	REAL
P6_DU.ST		16#FF	BYTE
P7_DU	P7_DU		STRUCT
P7_DU.Value		0.0	REAL
P7_DU.ST		16#FF	BYTE
P8_DU	P8_DU		STRUCT
P8_DU.Value		0.0	REAL
P8_DU.ST		16#FF	BYTE
P1_DF	P1_DF		STRUCT
P1_DF.Value		0.0	REAL
P1_DF.ST		16#FF	BYTE
P2_DF	P2_DF		STRUCT

I/O name	Description	Value	Data type
P2_DF.Value		0.0	REAL
P2_DF.ST		16#FF	BYTE
P3_DF	P3_DF		STRUCT
P3_DF.Value		0.0	REAL
P3_DF.ST		16#FF	BYTE
P4_DF	P4_DF		STRUCT
P4_DF.Value		0.0	REAL
P4_DF.ST		16#FF	BYTE
P5_DF	P5_DF		STRUCT
P5_DF.Value		0.0	REAL
P5_DF.ST		16#FF	BYTE
P6_DF	P6_DF		STRUCT
P6_DF.Value		0.0	REAL
P6_DF.ST		16#FF	BYTE
P7_DF	P7_DF		STRUCT
P7_DF.Value		0.0	REAL
P7_DF.ST		16#FF	BYTE
P8_DF	P8_DF		STRUCT
P8_DF.Value		0.0	REAL
P8_DF.ST		16#FF	BYTE
P1_DANG	P1_DANG		STRUCT
P1_DANG.Value		0.0	REAL
P1_DANG.ST		16#FF	BYTE
P2_DANG	P2_DANG		STRUCT
P2_DANG.Value		0.0	REAL
P2_DANG.ST		16#FF	BYTE
P3_DANG	P3_DANG		STRUCT
P3_DANG.Value		0.0	REAL
P3_DANG.ST		16#FF	BYTE
P4_DANG	P4_DANG		STRUCT
P4_DANG.Value		0.0	REAL
P4_DANG.ST		16#FF	BYTE
P5_DANG	P5_DANG		STRUCT
P5_DANG.Value		0.0	REAL
P5_DANG.ST		16#FF	BYTE
P6_DANG	P6_DANG		STRUCT
P6_DANG.Value		0.0	REAL
P6_DANG.ST		16#FF	BYTE
P7_DANG	P7_DANG		STRUCT
P7_DANG.Value		0.0	REAL
P7_DANG.ST		16#FF	BYTE
P8_DANG	P8_DANG		STRUCT
P8_DANG.Value		0.0	REAL

I/O name	Description	Value	Data type
P8_DANG.ST		16#FF	BYTE
REAL_sp1	Free REAL 1		STRUCT
REAL_sp1.Value		0.0	REAL
REAL_sp1.ST		16#FF	BYTE
REAL_sp2	Free REAL 2		STRUCT
REAL_sp2.Value		0.0	REAL
REAL_sp2.ST		16#FF	BYTE
REAL_sp3	Free REAL 3		STRUCT
REAL_sp3.Value		0.0	REAL
REAL_sp3.ST		16#FF	BYTE
REAL_sp4	Free REAL 4		STRUCT
REAL_sp4.Value		0.0	REAL
REAL_sp4.ST		16#FF	BYTE
Pos_CMD_sp1	Position free command 1	16#00000000	DWORD
Pos_CMD_sp2	Position free command 2	16#00000000	DWORD
Pos_CMD_sp3	Position free command 3	16#00000000	DWORD
Pos_CMD_sp4	Position free command 4	16#00000000	DWORD
Pos_CMD_sp5	Position free command 5	16#00000000	DWORD
Pos_CMD_sp6	Position free command 6	16#00000000	DWORD
Pos_CMD_sp7	Position free command 7	16#00000000	DWORD
Pos_CMD_sp8	Position free command 8	16#00000000	DWORD
Beh	Behavior	0	INT
Dev_OK	Device OK	0	INT
INT_sp1	Free INT 1		STRUCT
INT_sp1.Value		0	INT
INT_sp1.ST		16#FF	BYTE
INT_sp2	Free INT 2		STRUCT
INT_sp2.Value		0	INT
INT_sp2.ST		16#FF	BYTE
INT_sp3	Free INT 3		STRUCT
INT_sp3.Value		0	INT
INT_sp3.ST		16#FF	BYTE
INT_sp4	Free INT 4		STRUCT
INT_sp4.Value		0	INT
INT_sp4.ST		16#FF	BYTE
INT_sp5	Free INT 5		STRUCT
INT_sp5.Value		0	INT
INT_sp5.ST		16#FF	BYTE
INT_sp6	Free INT 6		STRUCT
INT_sp6.Value		0	INT
INT_sp6.ST		16#FF	BYTE
TR_Relay	Excitation relay		STRUCT
TR_Relay.Value		0	BOOL

I/O name	Description	Value	Data type
TR_Relay.ST		16#FF	BYTE
PU_Relay	General tripping relay		STRUCT
PU_Relay.Value		0	BOOL
PU_Relay.ST		16#FF	BYTE
GAPC1_Op	Tripping coupling 1		STRUCT
GAPC1_Op.Value		0	BOOL
GAPC1_Op.ST		16#FF	BYTE
GAPC1_Str	Excitation coupling 1		STRUCT
GAPC1_Str.Value		0	BOOL
GAPC1_Str.ST		16#FF	BYTE
GAPC2_Op	Tripping coupling 2		STRUCT
GAPC2_Op.Value		0	BOOL
GAPC2_Op.ST		16#FF	BYTE
GAPC2_Str	Excitation coupling 2		STRUCT
GAPC2_Str.Value		0	BOOL
GAPC2_Str.ST		16#FF	BYTE
GAPC3_Op	Tripping coupling 3		STRUCT
GAPC3_Op.Value		0	BOOL
GAPC3_Op.ST		16#FF	BYTE
GAPC3_Str	Excitation coupling 3		STRUCT
GAPC3_Str.Value		0	BOOL
GAPC3_Str.ST		16#FF	BYTE
GAPC4_Op	Tripping coupling 4		STRUCT
GAPC4_Op.Value		0	BOOL
GAPC4_Op.ST		16#FF	BYTE
GAPC4_Str	Excitation coupling 4		STRUCT
GAPC4_Str.Value		0	BOOL
GAPC4_Str.ST		16#FF	BYTE
PTUF1_Op	Tripping frequency 1 <		STRUCT
PTUF1_Op.Value		0	BOOL
PTUF1_Op.ST		16#FF	BYTE
PTUF1_Str	Excitation frequency 1 <		STRUCT
PTUF1_Str.Value		0	BOOL
PTUF1_Str.ST		16#FF	BYTE
PTUF2_Op	Tripping frequency 2 <		STRUCT
PTUF2_Op.Value		0	BOOL
PTUF2_Op.ST		16#FF	BYTE
PTUF2_Str	Excitation frequency 2 <		STRUCT
PTUF2_Str.Value		0	BOOL
PTUF2_Str.ST		16#FF	BYTE
PTUF3_Op	Tripping frequency 3 <		STRUCT
PTUF3_Op.Value		0	BOOL
PTUF3_Op.ST		16#FF	BYTE

I/O name	Description	Value	Data type
PTUF3_Str	Excitation frequency 3 <		STRUCT
PTUF3_Str.Value		0	BOOL
PTUF3_Str.ST		16#FF	BYTE
PTUF4_Op	Tripping frequency 4 <		STRUCT
PTUF4_Op.Value		0	BOOL
PTUF4_Op.ST		16#FF	BYTE
PTUF4_Str	Excitation frequency 4 <		STRUCT
PTUF4_Str.Value		0	BOOL
PTUF4_Str.ST		16#FF	BYTE
RBRF1_Str	Excitation disconnecter fault		STRUCT
RBRF1_Str.Value		0	BOOL
RBRF1_Str.ST		16#FF	BYTE
RBRF1_OpEx	Tripping disconnecter fault T1		STRUCT
RBRF1_OpEx.Value		0	BOOL
RBRF1_OpEx.ST		16#FF	BYTE
RBRF1_OpIn	Tripping disconnecter fault T2		STRUCT
RBRF1_OpIn.Value		0	BOOL
RBRF1_OpIn.ST		16#FF	BYTE
Loc	Local mode	0	BOOL
Error	Fault group message		STRUCT
Error.Value		0	BOOL
Error.ST		16#FF	BYTE
Warning	Warning group message		STRUCT
Warning.Value		0	BOOL
Warning.ST		16#FF	BYTE
P1_U_NOK	Synchronization point 1 - voltage difference exceeded	0	BOOL
P2_U_NOK	Synchronization point 2 - voltage difference exceeded	0	BOOL
P3_U_NOK	Synchronization point 3 - voltage difference exceeded	0	BOOL
P4_U_NOK	Synchronization point 4 - voltage difference exceeded	0	BOOL
P5_U_NOK	Synchronization point 5 - voltage difference exceeded	0	BOOL
P6_U_NOK	Synchronization point 6 - voltage difference exceeded	0	BOOL
P7_U_NOK	Synchronization point 7 - voltage difference exceeded	0	BOOL
P8_U_NOK	Synchronization point 8 - voltage difference exceeded	0	BOOL
P1_MEAS	Synchronization point 1 - measurement running	0	BOOL
P2_MEAS	Synchronization point 2 - measurement running	0	BOOL

I/O name	Description	Value	Data type
P3_MEAS	Synchronization point 3 - measurement running	0	BOOL
P4_MEAS	Synchronization point 4 - measurement running	0	BOOL
P5_MEAS	Synchronization point 5 - measurement running	0	BOOL
P6_MEAS	Synchronization point 6 - measurement running	0	BOOL
P7_MEAS	Synchronization point 7 - measurement running	0	BOOL
P8_MEAS	Synchronization point 8 - measurement running	0	BOOL
P1_F_NOK	Synchronization point 1 - difference frequency exceeded	0	BOOL
P2_F_NOK	Synchronization point 2 - difference frequency exceeded	0	BOOL
P3_F_NOK	Synchronization point 3 - difference frequency exceeded	0	BOOL
P4_F_NOK	Synchronization point 4 - difference frequency exceeded	0	BOOL
P5_F_NOK	Synchronization point 5 - difference frequency exceeded	0	BOOL
P6_F_NOK	Synchronization point 6 - difference frequency exceeded	0	BOOL
P7_F_NOK	Synchronization point 7 - difference frequency exceeded	0	BOOL
P8_F_NOK	Synchronization point 8 - difference frequency exceeded	0	BOOL
P1_ANG_NOK	Synchronization point 1 - angle difference exceeded	0	BOOL
P2_ANG_NOK	Synchronization point 2 - angle difference exceeded	0	BOOL
P3_ANG_NOK	Synchronization point 3 - angle difference exceeded	0	BOOL
P4_ANG_NOK	Synchronization point 4 - angle difference exceeded	0	BOOL
P5_ANG_NOK	Synchronization point 5 - angle difference exceeded	0	BOOL
P6_ANG_NOK	Synchronization point 6 - angle difference exceeded	0	BOOL
P7_ANG_NOK	Synchronization point 7 - angle difference exceeded	0	BOOL
P8_ANG_NOK	Synchronization point 8 - angle difference exceeded	0	BOOL
P1_CL_EN	Synchronization point 1 - close enable	0	BOOL
P2_CL_EN	Synchronization point 2 - close enable	0	BOOL
P3_CL_EN	Synchronization point 3 - close enable	0	BOOL
P4_CL_EN	Synchronization point 4 - close enable	0	BOOL

I/O name	Description	Value	Data type
P5_CL_EN	Synchronization point 5 - close enable	0	BOOL
P6_CL_EN	Synchronization point 6 - close enable	0	BOOL
P7_CL_EN	Synchronization point 7 - close enable	0	BOOL
P8_CL_EN	Synchronization point 8 - close enable	0	BOOL
PARSET_A	Parameter set A active	0	BOOL
PARSET_B	Parameter set B active	0	BOOL
PARSET_C	Parameter set C active	0	BOOL
PARSET_D	Parameter set D active	0	BOOL
ALM_sp1	Free alarm 1		STRUCT
ALM_sp1.Value		0	BOOL
ALM_sp1.ST		16#FF	BYTE
ALM_sp2	Free alarm 2		STRUCT
ALM_sp2.Value		0	BOOL
ALM_sp2.ST		16#FF	BYTE
ALM_sp3	Free alarm 3		STRUCT
ALM_sp3.Value		0	BOOL
ALM_sp3.ST		16#FF	BYTE
ALM_sp4	Free alarm 4		STRUCT
ALM_sp4.Value		0	BOOL
ALM_sp4.ST		16#FF	BYTE
ALM_sp5	Free alarm 5		STRUCT
ALM_sp5.Value		0	BOOL
ALM_sp5.ST		16#FF	BYTE
ALM_sp6	Free alarm 6		STRUCT
ALM_sp6.Value		0	BOOL
ALM_sp6.ST		16#FF	BYTE
ALM_sp7	Free alarm 7		STRUCT
ALM_sp7.Value		0	BOOL
ALM_sp7.ST		16#FF	BYTE
ALM_sp8	Free alarm 8		STRUCT
ALM_sp8.Value		0	BOOL
ALM_sp8.ST		16#FF	BYTE
ALM_sp9	Free alarm 9		STRUCT
ALM_sp9.Value		0	BOOL
ALM_sp9.ST		16#FF	BYTE
ALM_sp10	Free alarm 10		STRUCT
ALM_sp10.Value		0	BOOL
ALM_sp10.ST		16#FF	BYTE
ALM_sp11	Free alarm 11		STRUCT
ALM_sp11.Value		0	BOOL
ALM_sp11.ST		16#FF	BYTE
ALM_sp12	Free alarm 12		STRUCT
ALM_sp12.Value		0	BOOL

I/O name	Description	Value	Data type
ALM_sp12.ST		16#FF	BYTE
SwCount	Operating cycles Q0	0	DINT
OpTm	Operating hours	0	DINT
DINT_sp1	DINT value (free)		STRUCT
DINT_sp1.Value		0	DINT
DINT_sp1.ST		16#FF	BYTE
DINT_sp2	DINT value (free)		STRUCT
DINT_sp2.Value		0	DINT
DINT_sp2.ST		16#FF	BYTE
DINT_sp3	DINT value (free)		STRUCT
DINT_sp3.Value		0	DINT
DINT_sp3.ST		16#FF	BYTE
DINT_sp4	DINT value (free)		STRUCT
DINT_sp4.Value		0	DINT
DINT_sp4.ST		16#FF	BYTE
DINT_sp5	DINT value (free)		STRUCT
DINT_sp5.Value		0	DINT
DINT_sp5.ST		16#FF	BYTE
DINT_sp6	DINT value (free)		STRUCT
DINT_sp6.Value		0	DINT
DINT_sp6.ST		16#FF	BYTE
P1_U2_DOWN	Point 1 Decrease U2	0	BOOL
P2_U2_DOWN	Point 2 Decrease U2	0	BOOL
P3_U2_DOWN	Point 3 Decrease U2	0	BOOL
P4_U2_DOWN	Point 4 Decrease U2	0	BOOL
P5_U2_DOWN	Point 5 Decrease U2	0	BOOL
P6_U2_DOWN	Point 6 Decrease U2	0	BOOL
P7_U2_DOWN	Point 7 Decrease U2	0	BOOL
P8_U2_DOWN	Point 8 Decrease U2	0	BOOL
P1_U2_UP	Point 1 Increase U2	0	BOOL
P2_U2_UP	Point 2 Increase U2	0	BOOL
P3_U2_UP	Point 3 Increase U2	0	BOOL
P4_U2_UP	Point 4 Increase U2	0	BOOL
P5_U2_UP	Point 5 Increase U2	0	BOOL
P6_U2_UP	Point 6 Increase U2	0	BOOL
P7_U2_UP	Point 7 Increase U2	0	BOOL
P8_U2_UP	Point 8 Increase U2	0	BOOL
P1_F2_DOWN	Point 1 Decrease F2	0	BOOL
P2_F2_DOWN	Point 2 Decrease F2	0	BOOL
P3_F2_DOWN	Point 3 Decrease F2	0	BOOL
P4_F2_DOWN	Point 4 Decrease F2	0	BOOL
P5_F2_DOWN	Point 5 Decrease F2	0	BOOL
P6_F2_DOWN	Point 6 Decrease F2	0	BOOL

I/O name	Description	Value	Data type
P7_F2_DOWN	Point 7 Decrease F2	0	BOOL
P8_F2_DOWN	Point 8 Decrease F2	0	BOOL
P1_F2_UP	Point 1 Increase F2	0	BOOL
P2_F2_UP	Point 2 Increase F2	0	BOOL
P3_F2_UP	Point 3 Increase F2	0	BOOL
P4_F2_UP	Point 4 Increase F2	0	BOOL
P5_F2_UP	Point 5 Increase F2	0	BOOL
P6_F2_UP	Point 6 Increase F2	0	BOOL
P7_F2_UP	Point 7 Increase F2	0	BOOL
P8_F2_UP	Point 8 Increase F2	0	BOOL
BOOL_sp1	Free		STRUCT
BOOL_sp1.Value		0	BOOL
BOOL_sp1.ST		16#80	BYTE
BOOL_sp2	Free		STRUCT
BOOL_sp2.Value		0	BOOL
BOOL_sp2.ST		16#80	BYTE
BOOL_sp3	Free		STRUCT
BOOL_sp3.Value		0	BOOL
BOOL_sp3.ST		16#80	BYTE
BOOL_sp4	Free		STRUCT
BOOL_sp4.Value		0	BOOL
BOOL_sp4.ST		16#80	BYTE
BOOL_sp5	Free		STRUCT
BOOL_sp5.Value		0	BOOL
BOOL_sp5.ST		16#80	BYTE
BOOL_sp6	Free		STRUCT
BOOL_sp6.Value		0	BOOL
BOOL_sp6.ST		16#80	BYTE
BOOL_sp7	Free		STRUCT
BOOL_sp7.Value		0	BOOL
BOOL_sp7.ST		16#80	BYTE
BOOL_sp8	Free		STRUCT
BOOL_sp8.Value		0	BOOL
BOOL_sp8.ST		16#80	BYTE
BOOL_sp9	Free		STRUCT
BOOL_sp9.Value		0	BOOL
BOOL_sp9.ST		16#80	BYTE
BOOL_sp10	Free		STRUCT
BOOL_sp10.Value		0	BOOL
BOOL_sp10.ST		16#80	BYTE
BOOL_sp11	Free		STRUCT
BOOL_sp11.Value		0	BOOL
BOOL_sp11.ST		16#80	BYTE

I/O name	Description	Value	Data type
BOOL_sp12	Free		STRUCT
BOOL_sp12.Value		0	BOOL
BOOL_sp12.ST		16#80	BYTE
BOOL_sp13	Free		STRUCT
BOOL_sp13.Value		0	BOOL
BOOL_sp13.ST		16#80	BYTE
BOOL_sp14	Free		STRUCT
BOOL_sp14.Value		0	BOOL
BOOL_sp14.ST		16#80	BYTE
BOOL_sp15	Free		STRUCT
BOOL_sp15.Value		0	BOOL
BOOL_sp15.ST		16#80	BYTE
BOOL_sp16	Free		STRUCT
BOOL_sp16.Value		0	BOOL
BOOL_sp16.ST		16#80	BYTE
BOOL_sp17	Free		STRUCT
BOOL_sp17.Value		0	BOOL
BOOL_sp17.ST		16#80	BYTE
BOOL_sp18	Free		STRUCT
BOOL_sp18.Value		0	BOOL
BOOL_sp18.ST		16#80	BYTE
OpTmWarning	1 = Operating time warning active	0	BOOL
OpTmAlarm	1 = Operating time alarm active	0	BOOL
SwCountWarning	1 = Operating cycle warning active	0	BOOL
SwCountAlarm	1 = Operating cycle alarm active	0	BOOL
LockAct			STRUCT
LockAct.Value		0	BOOL
LockAct.ST		16#80	BYTE
LocalAct			STRUCT
LocalAct.Value		0	BOOL
LocalAct.ST		16#80	BYTE
AutAct			STRUCT
AutAct.Value		0	BOOL
AutAct.ST		16#80	BYTE
ManAct	1 = Manual: Manual value is active		STRUCT
ManAct.Value	Value	1	BOOL
ManAct.ST	Signal state	16#80	BYTE
OS_PermOut	Operator authorization: Output for OS	16#FFFFFFFF	DWORD
OS_PermLog	Operator authorization: Output for OS	16#FFFFFFFF	DWORD
ST_Worst	Worst signal state	16#80	BYTE
GrpErr			STRUCT
GrpErr.Value		0	BOOL
GrpErr.ST		16#80	BYTE

I/O name	Description	Value	Data type
Status1	Word Status1	16#00000000	DWORD
Status3	Word Status3	16#00000000	DWORD
Status5	Word Status3	16#FFFFFFF	DWORD
MsgErrlocal1	1 = Message error occurs	0	BOOL
MsgStatlocal1	Message status	16#0000	WORD
MsgAcknlocal1	Message acknowledgement status	16#0000	WORD
MsgErrlocal2	1 = Message error occurs	0	BOOL
MsgStatlocal2	Message status	16#0000	WORD
MsgAcknlocal2	Message acknowledgement status	16#0000	WORD
VisuMisc	Configuration of the units in the faceplate		STRUCT

9.2 PC_SYNC block symbols

The following block symbols are available for the PC_SYNC object:

PC_SYNC/1: Symbol with group display and tag name:

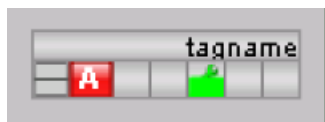


Figure 9-1 PC_SYNC/1

9.3 PC_SYNC faceplate

Available views:

View	Description
Default window	Important measured values, device status
Message view	Messages from device (messages, warnings, etc.)
Measured value window 1	Measured values
Measured value window 2	Measured values
Maintenance window	Operating hours, switches, active parameter group
User-defined window 1	User-specific view 1
User-defined window 2	User-specific view 2
Memo view	Memo view

9.3.1 Default window

The default window offers the following functions:

- Display of the states of all synchronization points
- Display of important measured values of synchronization point 1
- Display of auxiliary values
- Device status

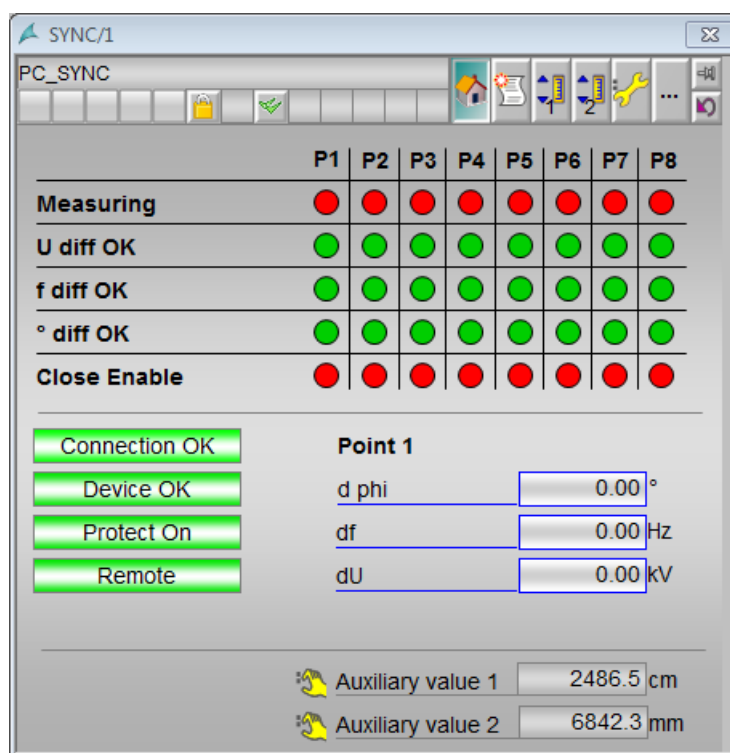


Figure 9-2 PC_SYNC default window

Display

Note: The displays for the auxiliary values are only shown if at least one auxiliary value was interconnected to the corresponding block input.

Display	Meaning	Tag	Description
Connection status	Communication status of IED via Ethernet.	QCON- NERRO	QCONERR0 = 0 (Connection OK) – GREEN QCONERR0 = 1 (Comm. loss) – RED
Device status 1	Status of IED	Dev_OK	Dev_OK = 1 (Device OK) – GREEN Dev_OK = 2 (Device warning) – YELLOW Dev_OK = 3 (Device message) – RED

Display	Meaning	Tag	Description
Device status 2	Behavior of IED	Beh	Beh = 1 (Protection on) – GREEN Beh = 2 (Protection locked) – YELLOW Beh = 3 (Protection test) – RED Beh = 4 (Protection test/locked) – RED Beh = 5 (Protection off) – RED
Device status 3	Local/Remote	Loc	Loc = 1: Local Loc = 0: Remote
D Phi	Point 1 angle difference	P1_DANG#Value	Point 1 angle difference
D f	Point 1 frequency difference	P1_DF#Value	Point 1 frequency difference
D U	Point 1 voltage difference	P1_DU#Value	Point 1 voltage difference
Measured value	Measured value	PX_MEAS*	Measured value
U Diff OK	Differential voltage OK	PX_U_NO K*	Differential voltage OK
F Diff OK	Differential frequency OK	PX_F_NOK*	Differential frequency OK
° Diff OK	Differential angle OK	PX_ANG_NOK*	Differential angle OK
Close enable	Close enable	PX_CL_EN*	Close enable
Auxiliary value 1	User analog value 1	UserA-na1#Value	User analog value 1
Auxiliary value 2	User analog value 2	UserA-na2#Value	User analog value 2

*: X = Synchronization point no. (1-8)

9.3.2 Message view

This window displays every active message and warning of the IED. The APL default message window is used.

9.3.3 Measured value window 1

The measured values of synchronization point 1-4 are displayed in the measured value window.

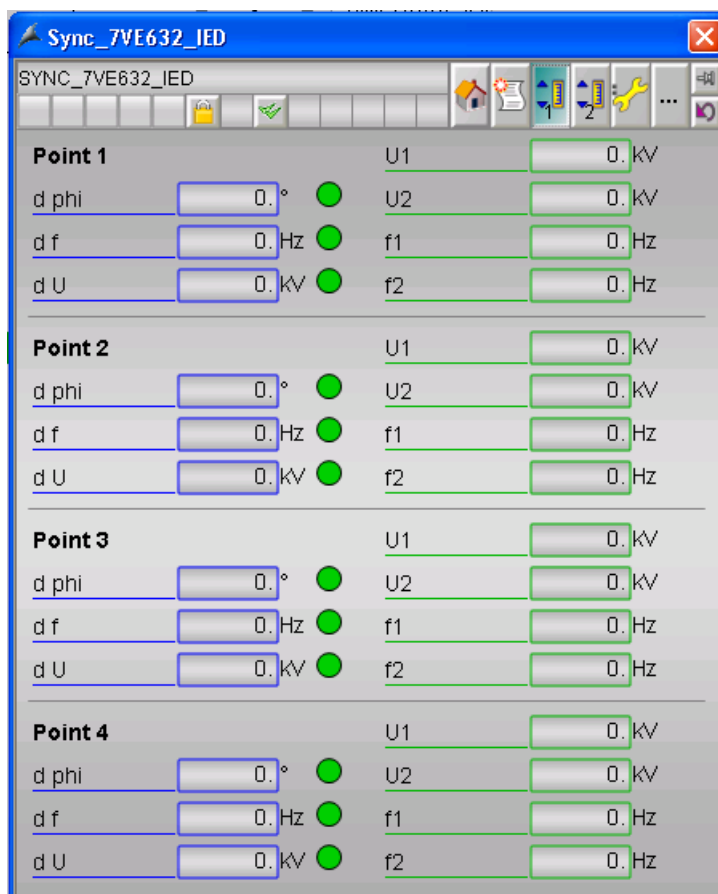


Figure 9-3 PC_SYNC measured value window 1

Display

Display	Meaning	Tag	Description
U1	Voltage 1	PX_U1#Value*	Voltage 1
U2	Voltage 2	PX_U2#Value*	Voltage 2
F1	Frequency 1	PX_F1#Value*	Frequency 1
F2	Frequency 2	PX_F2#Value*	Frequency 2
D Phi	Angle difference	PX_DANG#Value*	Point 1 angle difference
D f	Frequency difference	PX_DF#Value*	Point 1 frequency difference

Display	Meaning	Tag	Description
D U	Voltage difference	PX_DU#Value*	Point 1 voltage difference
Measured value	Measured value	PX_MEAS*	Measured value
U Diff OK	Differential voltage OK	PX_U_NOK*	Differential voltage OK
F Diff OK	Differential frequency OK	PX_F_NOK*	Differential frequency OK
° Diff OK	Differential angle OK	PX_ANG_NO*K	Differential angle OK
Close enable	Close enable	PX_CL_EN*	Close enable

*: X = Synchronization point No. (1-8)

9.3.4 Measured value window 2

The measured values of synchronization point 5-8 are displayed in the measured value window.

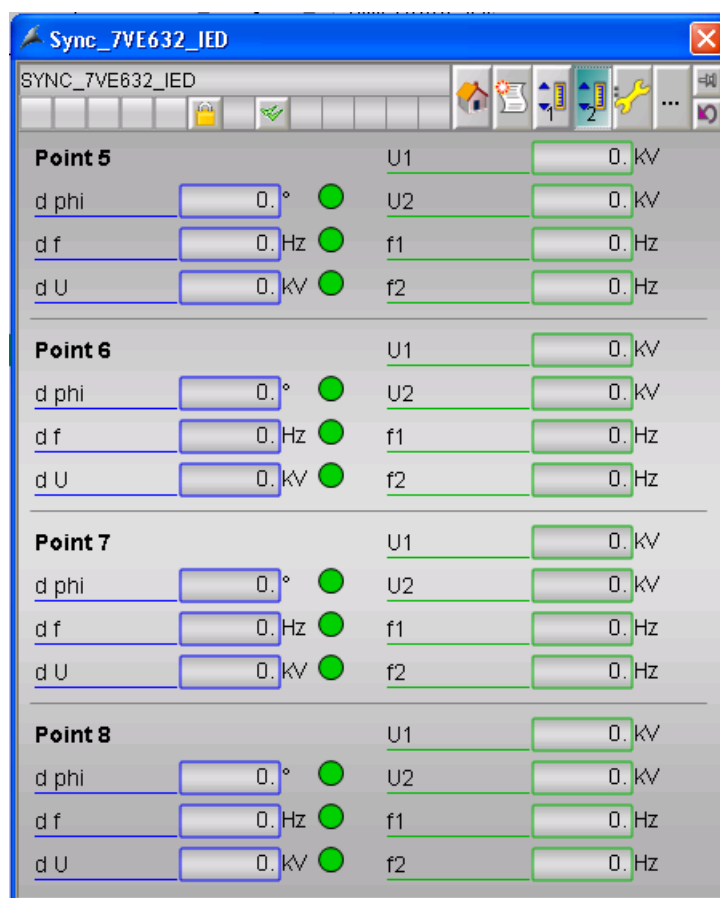


Figure 9-4 PC_SYNC measured value window 2

Display

Display	Meaning	Tag	Description
U1	Voltage 1	PX_U1#Value*	Voltage 1
U2	Voltage 2	PX_U2#Value*	Voltage 2
F1	Frequency 1	PX_F1#Value*	Frequency 1
F2	Frequency 2	PX_F2#Value*	Frequency 2
D Phi	Angle difference	PX_DANG#Value*	Point 1 angle difference
D f	Frequency difference	PX_DF#Value*	Point 1 frequency difference
D U	Voltage difference	PX_DU#Value*	Point 1 voltage difference
Measured value	Measured value	PX_MEAS*	Measured value
U Diff OK	Differential voltage OK	PX_U_NOK*	Differential voltage OK
F Diff OK	Differential frequency OK	PX_F_NOK*	Differential frequency OK
° Diff OK	Differential angle OK	PX_ANG_NO*K	Differential angle OK
Close enable	Close enable	PX_CL_EN*	Close enable

*: X = Synchronization point No. (1-8)

9.3.5 Maintenance window

The maintenance window displays the maintenance-specific information.

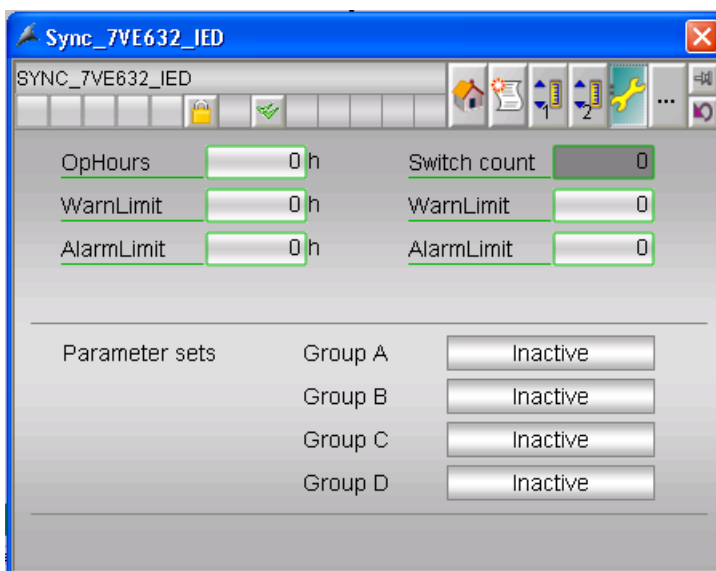


Figure 9-5 PC_SYNC maintenance window

Display

Display	Meaning	Tag	Description
OpHours	Operating hours	OpTm	Operating hours
WarnLimit	Warning limit operating hours	OpTm-WarnLimit	Warning limit operating hours
AlarmLimit	Alarm limit operating hours	OpTmA-larmLimit	Alarm limit operating hours
Operating cycles	Control number Q0	SwCount	Control number Q0
WarnLimit	Warning limit operating cycles	SwCount-WarnLimit	Warning limit operating cycles
AlarmLimit	Alarm limit operating cycles	SwCountAlarmLimit	Alarm limit operating cycles
Group A	Parameter set A active	PARSET_A	Parameter set A active
Group B	Parameter set B active	PARSET_B	Parameter set B active
Group C	Parameter set C active	PARSET_C	Parameter set C active
Group D	Parameter set D active	PARSET_D	Parameter set D active

9.3.6 User view

The user-defined faceplates 1 and 2 can be customized by the customer. See also Creating a user view/templates (Page 31)

Example:

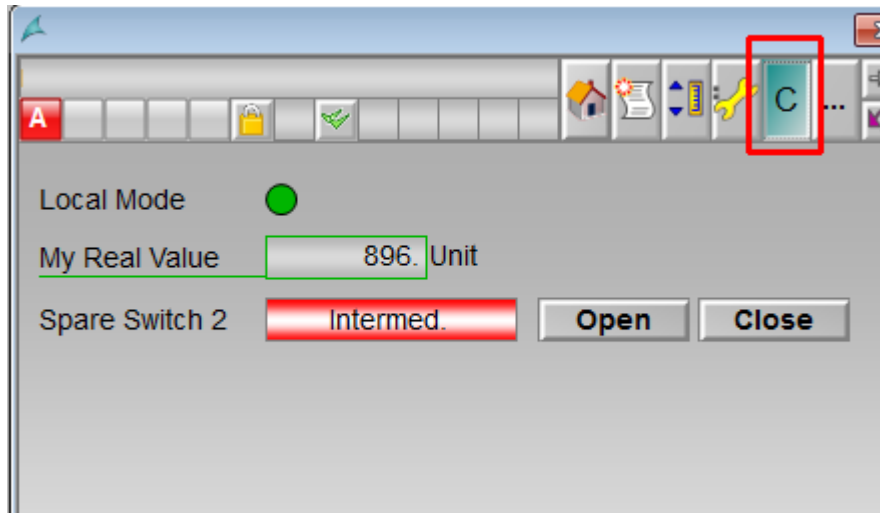


Figure 9-6 Example of user view

PC_LINE

The PC_LINE object is used for all differential and distance protection devices. The faceplate serves only for visualization. No commands are implemented in the default faceplates. However, four free commands are available in the user-defined faceplate and in the PLC faceplate.

PC_LINE consists of the following objects:

- PLC block PC_LINE
- WinCC block symbols
- WinCC faceplate PC_LINE with different views
- Device driver blocks (not part of the PCL)

10.1 PLC block PC_LINE

10.1.1 Description of PLC block PC_LINE

The PC_LINE (FB 2504) function block represents the line object in the PLC. The block is the interface between the device driver block and the remaining PCS 7 environment.

The block offers the following functions:

- Detecting data from the device driver block
- Providing data for the block symbol and the faceplate
- Controlling switches of the switchgear device (only free commands)
- APL operating modes Manual, Automatic
- Interlocking functions
- Handling of operator authorizations
- Various APL standard functions

10.1.2 Interfaces

The PLC block PC_LINE offers the following interfaces:

1. **Device driver blocks**

The device driver block is connected to the PC_LINE block through the input "Rcv_Data" and the output "Snd_Data".

2. **Block symbol**

The block symbol receives data from the block through the S7 connection between the PLC and WinCC.

3. Faceplate

The faceplate receives data from the block through the S7 connection between the PLC and WinCC.

4. APL interlock blocks

The interlock blocks can be connected through the "IntLock" input.

5. Logic of the S7 program (for controlling modes, switches, etc.)

The PC_LINE block can receive commands from the CFC logic and makes data available to other blocks.

10.1.3 Configuration

Use the CFC editor to install the block in a cyclic interrupt OB (OB30 to OB38). The block is also installed automatically in the startup OB (OB 100).

10.1.4 Operating modes

The PC_LINE block supports the APL standard operating modes.

The operating mode can only be changed through the interconnections on the PLC block.

The block can be operated in the following operating modes:

Operating mode	Description	Permitted commands	Display
Local operating mode	The device signals local operating mode (for example, the key-operated switch on the device is in the "local" position)	No	<i>LocalAct.Value</i> = true
Manual mode	The device is controlled through the faceplate	Only from faceplate	<i>ManAct.Value</i> = true
Automatic mode	The device is controlled through the connected logic in the CFC	Only by block inputs	<i>AutAct.Value</i> = true

For a detailed description of the operating modes refer to the "PCS 7 Advanced Process Library Function Manual" [1].

10.1.5 Switch commands

The PC_LINE block sends commands from the faceplate (in manual operation) or from other connected blocks (in automatic mode) to the device driver block. The following commands are implemented:

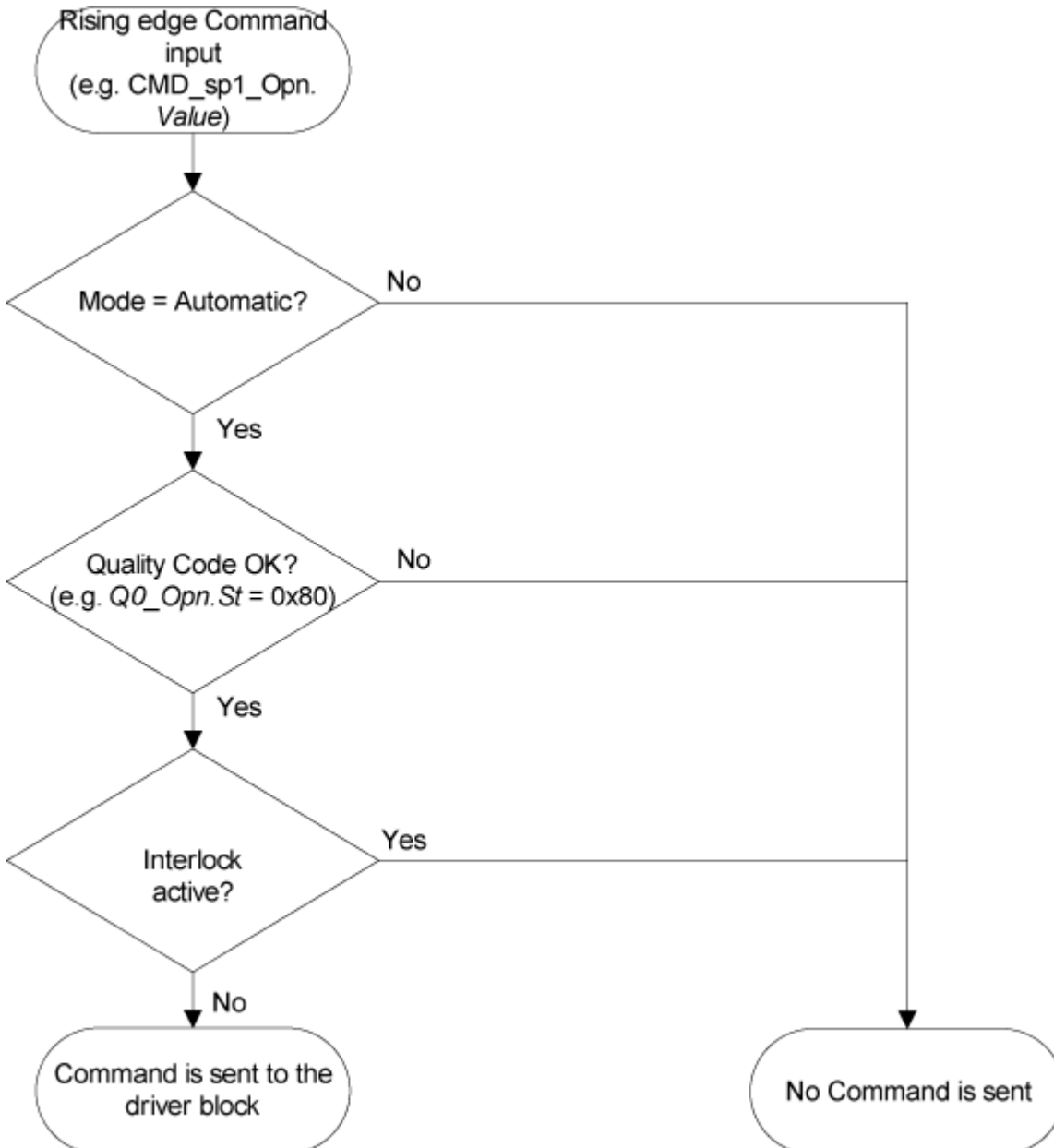
- Free command 1
- Free command 2
- Free command 3
- Free command 4

10.1.6 Interlocks

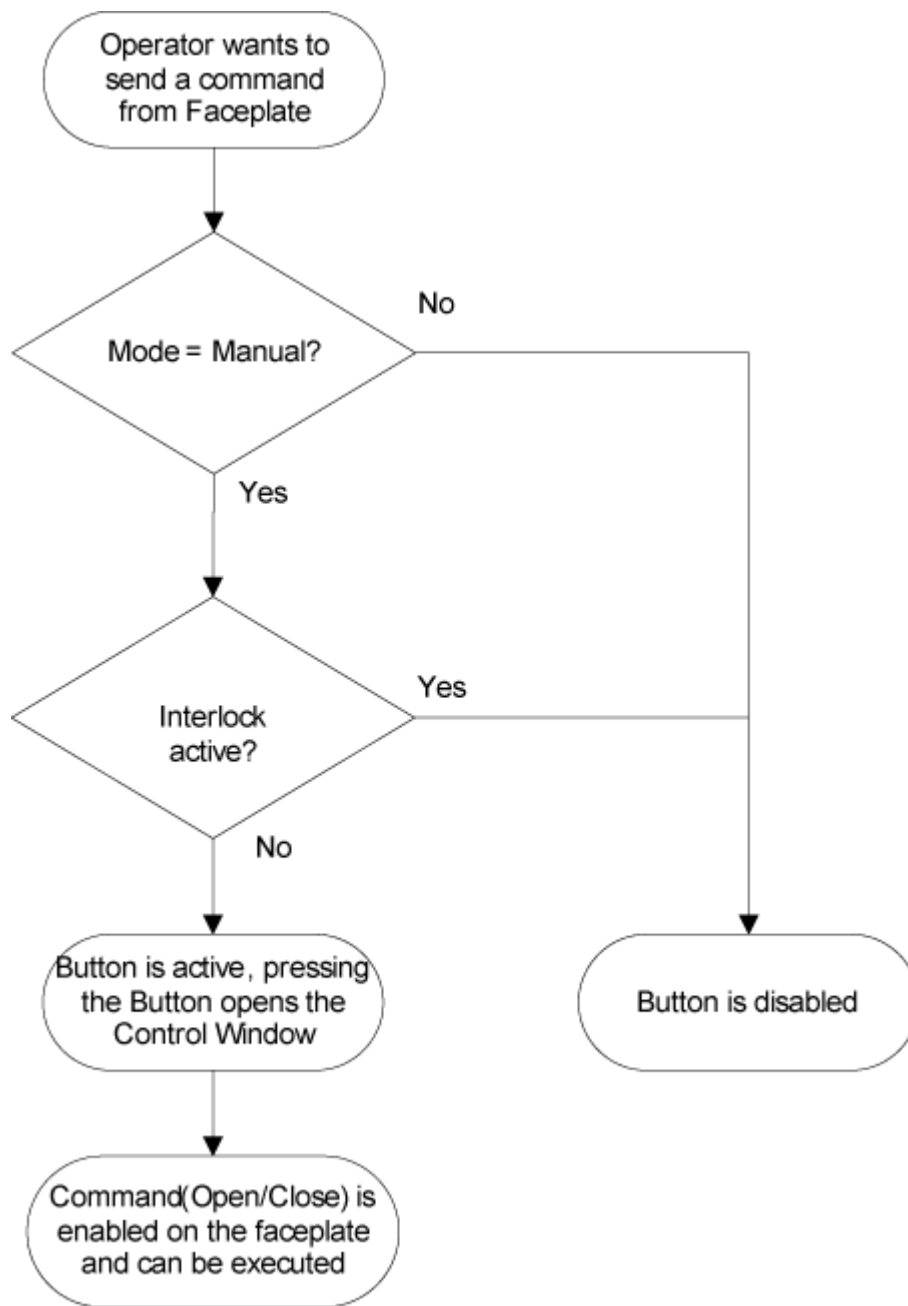
The PC_LINE block only provides the APL standard interlock "Interlock".

Important information: The APL interlock locks all free commands of the PC_LINE block but does not open any switches at activation.

The following logic is used to interlock a linked command:



The following logic is used for interlocking the commands output in the faceplate:



Deactivating individual interlocks:

You deactivate the implemented block interlock through the input parameter Intlock

If you want to deactivate the block interlock, you have to specify the following parameters:

- `Intl_En = 0` or `Intlock.ST = 16#FF`: The Intlock input parameter does not have any effect.

10.1.7 Group error

This block provides the standard APL function "Outputting group errors".

The following parameters are taken into consideration when forming the group error:

- CSF

The group error output is *GrpErr.Value*.

10.1.8 Forming the group status for interlocks

This block contains the standard APL function "Forming the group status for interlock information" (see APL documentation).

10.1.9 Responses that can be configured with the "Feature" parameter

An overview of all the responses provided by the `Feature` parameter is contained in the PCS 7 APL documentation.

The following function is available for this block at the corresponding bits:

Bit	Function
0-11	Not used
12	1 = Hide operating hours in the faceplate 0 (default) = Show operating hours in the faceplate
13	1 = Hide operating cycle hours in the faceplate 0 (default) = Show operating cycle hours in the faceplate
14-31	Not used

10.1.10 Operator authorizations

This block includes the APL standard function "Operator authorizations".

The block provides the following authorizations for the `OS_Perm` parameter:

Bit	Function
0	1 = Operator may switch to "Automatic mode"
1	1 = Operator may switch to "Manual mode"
2	1 = Operator may switch to "Local operating mode"
3-31	Not used

10.1.11 Assignment of the status word for Status1 parameter

Status bit	Parameter
0	Occupied
1	BatchEn
2	Not used
3	Not used
4	Not used
5	0 = ManAct.Value 1 = AutoAct.Value
6	LocalAct.Value
7	0: Open padlock in block symbol 1: Closed padlock in block symbol
8	Not used (always true)
9	Not used
10	Not used
11	Not used
12	Not used
13	Not used
14	Not used
15	Not used
16	1 = Intlock is active
17	Not used
18	Not used
19	Trip.Value
20	Not used
21	Not used
22	Not used
23	"Interlock" button is activated
24	Not used
25	Not used
26	Bypass information from previous function block
27	Not used
28	Not used
29	Not used
30	Not used
31	Not used

10.1.12 Assignment of the status word for Status5 parameter

The Status5 parameter is used to show / hide values in the faceplate. The values are displayed depending on the quality code of the Station Gateway / Station Controller.

The values are displayed when the quality code is <> 0x00.

Status bit	Value	Description
0	I_A	Current phase 1
1	I_B	Current phase 2
2	I_C	Current phase 3
3	IDIFFL1	Idiff phase 1
4	IDIFFL2	Idiff phase 2
5	IDIFFL3	Idiff phase 3
6	IRESTL1	Irest phase 1
7	IRESTL2	Irest phase 2
8	IRESTL3	Irest phase 3
9	I30	3I0
10	F	Frequency
11	P1A_MIN	Availability minutes Point 1
12	P1A_H	Availability hours Point 1
13	P2A_MIN	Availability minutes Point 2
14	P2A_H	Availability hours Point 2
19	Optm	Operating hours
20	SwCount	Operating cycles

10.1.13 Messages

Message instance	Message identifier	Message class	Event
MsgEvId_1	SIG_1	Warning - high	\$\$BlockComment\$\$ Excitation device
	SIG_2	Alarm - high	\$\$BlockComment\$\$ General tripping device
	SIG_3	Alarm - high	\$\$BlockComment\$\$ Tripping IDMT Phase Ip
	SIG_4	Warning - high	\$\$BlockComment\$\$ Excitation IDMT Phase Ip
	SIG_5	Alarm - high	\$\$BlockComment\$\$ Tripping DMT I >
	SIG_6	Warning - high	\$\$BlockComment\$\$ Excitation DMT I >
	SIG_7	Alarm - high	\$\$BlockComment\$\$ Tripping DMT I >>
	SIG_8	Warning - high	\$\$BlockComment\$\$ Excitation DMT I >>

Message instance	Message identifier	Message class	Event
MsgEvld_2	SIG_1	Alarm - high	\$\$BlockComment\$\$ Tripping impedance protection Z1
	SIG_2	Warning - high	\$\$BlockComment\$\$ excitation impedance protection Z1
	SIG_3	Alarm - high	\$\$BlockComment\$\$ trip impedance protection Z2
	SIG_4	Warning - high	\$\$BlockComment\$\$ excitation impedance protection Z2
	SIG_5	Alarm - high	\$\$BlockComment\$\$ trip impedance protection Z1B
	SIG_6	Warning - high	\$\$BlockComment\$\$ excitation impedance protection Z1B
	SIG_7	Alarm - high	\$\$BlockComment\$\$ general trip impedance protection
	SIG_8	Warning - high	\$\$BlockComment\$\$ general excitation impedance protection
MsgEvld_3	SIG_1	Alarm - high	\$\$BlockComment\$\$ Tripping differential protection IDIFF>
	SIG_2	Warning - high	\$\$BlockComment\$\$ Excitation differential protection IDIFF>
	SIG_3	Alarm - high	\$\$BlockComment\$\$ Tripping differential protection IDIFF>>
	SIG_4	Warning - high	\$\$BlockComment\$\$ Excitation differential protection IDIFF>>
	SIG_5	Warning - high	\$\$BlockComment\$\$ Excitation disconnector fault
	SIG_6	Alarm - high	\$\$BlockComment\$\$ Tripping disconnector fault T1
	SIG_7	Alarm - high	\$\$BlockComment\$\$ Tripping disconnector fault T1
	SIG_8	Warning - high	\$\$BlockComment\$\$ Local mode
MsgEvld_4	SIG_1	Alarm - high	\$\$BlockComment\$\$ Fault group message
	SIG_2	Warning - high	\$\$BlockComment\$\$ Warning group message
	SIG_3	Status message - AS	\$\$BlockComment\$\$ Parameter set A active
	SIG_4	Status message - AS	\$\$BlockComment\$\$ Parameter set B active
	SIG_5	Status message - AS	\$\$BlockComment\$\$ Parameter set C active
	SIG_6	Status message - AS	\$\$BlockComment\$\$ Parameter set D active
	SIG_7	Alarm - high	\$\$BlockComment\$\$ IEC Alarm 1
	SIG_8	Alarm - high	\$\$BlockComment\$\$ IEC Alarm 2

Message instance	Message identifier	Message class	Event
MsgEvid_5	SIG_1	Alarm - high	\$\$BlockComment\$\$ IEC Alarm 3
	SIG_2	Alarm - high	\$\$BlockComment\$\$ IEC Alarm 4
	SIG_3	Alarm - high	\$\$BlockComment\$\$ IEC Alarm 5
	SIG_4	Alarm - high	\$\$BlockComment\$\$ IEC Alarm 6
	SIG_5	Alarm - high	\$\$BlockComment\$\$ IEC Alarm 7
	SIG_6	Alarm - high	\$\$BlockComment\$\$ IEC Alarm 8
	SIG_7	Alarm - high	\$\$BlockComment\$\$ IEC Alarm 9
	SIG_8	Alarm - high	\$\$BlockComment\$\$ IEC Alarm 10
MsgEvidlocal1	SIG_1	AS process control message - error	\$\$BlockComment\$\$ Connection fault
	SIG_2	Alarm - high	
	SIG_3	Alarm - high	
	SIG_4	Alarm - high	
	SIG_5	Alarm - high	
	SIG_6	Alarm - high	
	SIG_7	Alarm - high	
	SIG_8	AS process control message - error	\$\$BlockComment\$\$ AS process control message - external error
MsgEvidlocal2	SIG_1	Warning - high	\$\$BlockComment\$\$ Maintenance: Operating cycle warning (@3%d@)
	SIG_2	Alarm - high	\$\$BlockComment\$\$ Maintenance: Operating cycle alarm (@3%d@)
	SIG_3	Warning - high	\$\$BlockComment\$\$ Maintenance: Operating time warning (@4%d@)
	SIG_4	Alarm - high	\$\$BlockComment\$\$ Maintenance: Operating time alarm (@4%d@)
	SIG_5	Alarm - high	\$\$BlockComment\$\$ External message 1
	SIG_6	Alarm - high	\$\$BlockComment\$\$ External message 2
	SIG_7	Alarm - high	\$\$BlockComment\$\$ External message 3
	SIG_8	Alarm - high	\$\$BlockComment\$\$ External message 4

Message instance	Message identifier	Message class	Event
MsgEvld_d_1	SIG_1	Status message - AS	\$\$BlockComment\$\$ Switch 1 closed
	SIG_2	Status message - AS	\$\$BlockComment\$\$ Switch 1 open
	SIG_3	Status message - AS	\$\$BlockComment\$\$ Switch 2 closed
	SIG_4	Status message - AS	\$\$BlockComment\$\$ Switch 2 open
	SIG_5	Status message - AS	\$\$BlockComment\$\$ Switch 3 closed
	SIG_6	Status message - AS	\$\$BlockComment\$\$ Switch 3 open
	SIG_7	Status message - AS	\$\$BlockComment\$\$ Switch 4 closed
	SIG_8	Status message - AS	\$\$BlockComment\$\$ Switch 4 open
MsgEvld_device	SIG_1	AS process control message - error	\$\$BlockComment\$\$ Error device IED name:@1%\$@ -> IP address: @2%\$@
	SIG_2	AS control system message - fault	\$\$BlockComment\$\$ Redundancy failure
	SIG_3	AS process control message - error	\$\$BlockComment\$\$ PROFIBUS mapping alarm
	SIG_4	Alarm - high	
	SIG_5	Alarm - high	
	SIG_6	Alarm - high	
	SIG_7	Alarm - high	
	SIG_8	Alarm - high	

10.1.14 I/O list

10.1.14.1 Inputs

I/O name	Description	Value	Data type
Rcv_Data	Connect with SG/_SC_LINE		STRUCT
ModLiOp	1 = Interconnect/Auto, 0 = Manual: Input for auto / manual commands		STRUCT
ModLiOp.Value		0	BOOL
ModLiOp.ST		16#80	BYTE
AutModOp	1 = Automatic mode: Automatic mode by operator	0	BOOL

I/O name	Description	Value	Data type
ManModOp	1 = Manual mode: Manual mode by operator	1	BOOL
AutModLi			STRUCT
AutModLi.Value		0	BOOL
AutModLi.ST		16#80	BYTE
ManModLi			STRUCT
ManModLi.Value		0	BOOL
ManModLi.ST		16#80	BYTE
FP_Visu	Configuration of the faceplate		STRUCT
FP_Units	Configuration of the units in the faceplate		STRUCT
MS_RelOp	Operator input for MS release , 1: MS release request	0	BOOL
Monitor	1=Monitoring ON, 0=Monitoring OFF: Feed-back monitoring	1	BOOL
MonTiStatic	Monitoring time of the feedback [s]	3.0	REAL
MonTiDynamic	Monitoring time of the feedback [s]	3.0	REAL
Intlock	0=Motor is locked, 1=Motor is not locked		STRUCT
Intlock.Value	Value	1	BOOL
Intlock.ST	Signal state	16#FF	BYTE
Intl_En	1=Interlock activated, 0=Interlock deactivated	1	BOOL
CMD_sp1_Opn	Rising edge: Free command 1 off		STRUCT
CMD_sp1_Opn.Value		0	BOOL
CMD_sp1_Opn.ST		16#80	BYTE
CMD_sp1_Cls	Rising edge: Free command 1 on		STRUCT
CMD_sp1_Cls.Value		0	BOOL
CMD_sp1_Cls.ST		16#80	BYTE
CMD_sp2_Opn	Rising edge: Free command 2 off		STRUCT
CMD_sp2_Opn.Value		0	BOOL
CMD_sp2_Opn.ST		16#80	BYTE
CMD_sp2_Cls	Rising edge: Free command 2 on		STRUCT
CMD_sp2_Cls.Value		0	BOOL
CMD_sp2_Cls.ST		16#80	BYTE
CMD_sp3_Opn	Rising edge: Free command 3 off		STRUCT
CMD_sp3_Opn.Value		0	BOOL
CMD_sp3_Opn.ST		16#80	BYTE
CMD_sp3_Cls	Rising edge: Free command 3 on		STRUCT
CMD_sp3_Cls.Value		0	BOOL
CMD_sp3_Cls.ST		16#80	BYTE
CMD_sp4_Opn	Rising edge: Free command 4 off		STRUCT
CMD_sp4_Opn.Value		0	BOOL
CMD_sp4_Opn.ST		16#80	BYTE
CMD_sp4_Cls	Rising edge: Free command 4 on		STRUCT
CMD_sp4_Cls.Value		0	BOOL

I/O name	Description	Value	Data type
CMD_sp4_Cls.ST		16#80	BYTE
MsgLock			STRUCT
MsgLock.Value		0	BOOL
MsgLock.ST		16#80	BYTE
SampleTime	Sampling time [s]	0.1	REAL
MsgEvId_1	Message event ID	16#0000004D	DWORD
MsgEvId_2	Message event ID	16#0000004E	DWORD
MsgEvId_3	Message event ID	16#0000004F	DWORD
MsgEvId_4	Message event ID	16#00000050	DWORD
MsgEvId_5	Message event ID	16#00000051	DWORD
MsgEvIdlocal1	Message event ID	16#00000052	DWORD
MsgEvIdlocal2	Message event ID	16#00000053	DWORD
MsgEvId_d_1	Message event ID	16#00000054	DWORD
MsgEvId_device	Message event ID	16#00000055	DWORD
BatchEn	Activate remote operation of controller to batch recipe	0	BOOL
BatchID	Current batch ID (number)	16#00000000	DWORD
BatchName	Current batch name	"	STRING[32]
StepNo	Batch step number	16#00000000	DWORD
Occupied	Occupied by batch	0	BOOL
CSF			STRUCT
CSF.Value		0	BOOL
CSF.ST		16#80	BYTE
ExtMsg1	External message 1		STRUCT
ExtMsg1.Value	Value	0	BOOL
ExtMsg1.ST	Signal state	16#80	BYTE
ExtMsg2	External message 2		STRUCT
ExtMsg2.Value	Value	0	BOOL
ExtMsg2.ST	Signal state	16#80	BYTE
ExtMsg3	External message 3		STRUCT
ExtMsg3.Value	Value	0	BOOL
ExtMsg3.ST	Signal state	16#80	BYTE
ExtMsg4	External message 4		STRUCT
ExtMsg4.Value	Value	0	BOOL
ExtMsg4.ST	Signal state	16#80	BYTE
UserAna1	User analog input 1		STRUCT
UserAna1.Value	Value	0.0	REAL
UserAna1.ST	Signal state	16#FF	BYTE
UA1unit	Unit of measurement user analog input 1	0	INT
UserAna2	User analog input 2		STRUCT
UserAna2.Value	Value	0.0	REAL
UserAna2.ST	Signal state	16#FF	BYTE
UA2unit	Unit of measurement user analog input 2	0	INT

I/O name	Description	Value	Data type
UserAna3	User analog input 3		STRUCT
UserAna3.Value	Value	0.0	REAL
UserAna3.ST	Signal state	16#FF	BYTE
UA3unit	Unit of measurement user analog input 3	0	INT
UserAna4	User analog input 4		STRUCT
UserAna4.Value	Value	0.0	REAL
UserAna4.ST	Signal state	16#FF	BYTE
UA4unit	Unit of measurement user analog input 4	0	INT
UserStatus	User status bits	16#0	BYTE
SelFp1	Select faceplate 1		ANY
OS_Perm	Operator authorizations		STRUCT
OS_Perm.Bit0	1 = Operator may switch to Automatic mode	1	BOOL
OS_Perm.Bit1	1 = Operator may switch to Manual mode	1	BOOL
OS_Perm.Bit2	1 = Operator may switch to Local operating mode	1	BOOL
OS_Perm.Bit3	1 = Operator may switch to "Out of service" mode	1	BOOL
OS_Perm.Bit4		1	BOOL
OS_Perm.Bit5		1	BOOL
OS_Perm.Bit6		1	BOOL
OS_Perm.Bit7		1	BOOL
OS_Perm.Bit8		1	BOOL
OS_Perm.Bit9		1	BOOL
OS_Perm.Bit10		1	BOOL
OS_Perm.Bit11	1 = Operator can activate function simulation	1	BOOL
OS_Perm.Bit12	1 = Operator can enable release for maintenance	1	BOOL
OS_Perm.Bit13		1	BOOL
OS_Perm.Bit14		1	BOOL
OS_Perm.Bit15		1	BOOL
OS_Perm.Bit16		1	BOOL
OS_Perm.Bit17		1	BOOL
OS_Perm.Bit18		1	BOOL
OS_Perm.Bit19		1	BOOL
OS_Perm.Bit20		1	BOOL
OS_Perm.Bit21		1	BOOL
OS_Perm.Bit22		1	BOOL
OS_Perm.Bit23		1	BOOL
OS_Perm.Bit24		1	BOOL
OS_Perm.Bit25		1	BOOL
OS_Perm.Bit26		1	BOOL
OS_Perm.Bit27		1	BOOL
OS_Perm.Bit28		1	BOOL

I/O name	Description	Value	Data type
OS_Perm.Bit29		1	BOOL
OS_Perm.Bit30		1	BOOL
OS_Perm.Bit31		1	BOOL
Feature	State of different functions		STRUCT
Feature.Bit0	Reserved	0	BOOL
Feature.Bit1	Reserved	0	BOOL
Feature.Bit2	Reserved	0	BOOL
Feature.Bit3	Reserved	0	BOOL
Feature.Bit4	Reserved	0	BOOL
Feature.Bit5	Reserved	0	BOOL
Feature.Bit6	Reserved	0	BOOL
Feature.Bit7	Reserved	0	BOOL
Feature.Bit8	Reserved	0	BOOL
Feature.Bit9	Reserved	0	BOOL
Feature.Bit10	Reserved	0	BOOL
Feature.Bit11	Reserved	0	BOOL
Feature.Bit12	Hide OpTm	0	BOOL
Feature.Bit13	Hide SwCount	0	BOOL
Feature.Bit14	Reserved	0	BOOL
Feature.Bit15	Reserved	0	BOOL
Feature.Bit16	Reserved	0	BOOL
Feature.Bit17	Reserved	0	BOOL
Feature.Bit18	Reserved	0	BOOL
Feature.Bit19	Reserved	0	BOOL
Feature.Bit20	Reserved	0	BOOL
Feature.Bit21	Reserved	0	BOOL
Feature.Bit22	Reserved	0	BOOL
Feature.Bit23	Reserved	0	BOOL
Feature.Bit24	1 = Local authorization enabled	0	BOOL
Feature.Bit25	1 = Suppression of all messages if MsgLock = 1	0	BOOL
Feature.Bit26	Reserved	0	BOOL
Feature.Bit27	Reserved	0	BOOL
Feature.Bit28	Reserved	0	BOOL
Feature.Bit29	Reserved	0	BOOL
Feature.Bit30	Reserved	0	BOOL
Feature.Bit31	Reserved	0	BOOL
MonTime	Time in seconds until Timeout (for control switch)	20	INT

10.1.14.2 Inputs / outputs

I/O name	Description	Value	Data type
Cmdsp1	Control position free command 1	16#02	BYTE
Cmdsp2	Control position free command 2	16#02	BYTE
Cmdsp3	Control position free command 3	16#02	BYTE
Cmdsp4	Control position free command 4	16#02	BYTE
OpTmWarnLimit	Warning limit operating hours	0	DINT
OpTmAlarmLimit	Alarm limit operating hours	0	DINT
SwCountWarnLimit	Warning limit operating cycles	0	DINT
SwCountAlarmLimit	Alarm limit operating cycles	0	DINT

10.1.14.3 Outputs

I/O name	Description	Value	Data type
Snd_Data	Connect with SG/_SC_LINE		STRUCT
QCONNERR		1	BOOL
F	Frequency		STRUCT
F.Value		0.0	REAL
F.ST		16#FF	BYTE
I_A	Current Phase A		STRUCT
I_A.Value		0.0	REAL
I_A.ST		16#FF	BYTE
I_B	Current Phase B		STRUCT
I_B.Value		0.0	REAL
I_B.ST		16#FF	BYTE
I_C	Current Phase C		STRUCT
I_C.Value		0.0	REAL
I_C.ST		16#FF	BYTE
P	Power		STRUCT
P.Value		0.0	REAL
P.ST		16#FF	BYTE
PF	Cos Phi - power factor		STRUCT
PF.Value		0.0	REAL
PF.ST		16#FF	BYTE
Q	Reactive power		STRUCT
Q.Value		0.0	REAL
Q.ST		16#FF	BYTE
S	Apparent power		STRUCT
S.Value		0.0	REAL
S.ST		16#FF	BYTE
U	Neutral voltage		STRUCT
U.Value		0.0	REAL

I/O name	Description	Value	Data type
U.ST		16#FF	BYTE
U_A	Voltage Phase A		STRUCT
U_A.Value		0.0	REAL
U_A.ST		16#FF	BYTE
U_AB	Voltage A-B		STRUCT
U_AB.Value		0.0	REAL
U_AB.ST		16#FF	BYTE
U_B	Voltage Phase B		STRUCT
U_B.Value		0.0	REAL
U_B.ST		16#FF	BYTE
U_BC	Voltage B-C		STRUCT
U_BC.Value		0.0	REAL
U_BC.ST		16#FF	BYTE
U_C	Voltage Phase C		STRUCT
U_C.Value		0.0	REAL
U_C.ST		16#FF	BYTE
U_CA	Voltage C-A		STRUCT
U_CA.Value		0.0	REAL
U_CA.ST		16#FF	BYTE
IDIFFL1	Idiff L1		STRUCT
IDIFFL1.Value		0.0	REAL
IDIFFL1.ST		16#FF	BYTE
IDIFFL2	Idiff L2		STRUCT
IDIFFL2.Value		0.0	REAL
IDIFFL2.ST		16#FF	BYTE
IDIFFL3	Idiff L3		STRUCT
IDIFFL3.Value		0.0	REAL
IDIFFL3.ST		16#FF	BYTE
IDIFF3IO	Diff 3IO		STRUCT
IDIFF3IO.Value		0.0	REAL
IDIFF3IO.ST		16#FF	BYTE
IRESTL1	Irest L1		STRUCT
IRESTL1.Value		0.0	REAL
IRESTL1.ST		16#FF	BYTE
IRESTL2	Irest L2		STRUCT
IRESTL2.Value		0.0	REAL
IRESTL2.ST		16#FF	BYTE
IRESTL3	Irest L3		STRUCT
IRESTL3.Value		0.0	REAL
IRESTL3.ST		16#FF	BYTE
I30	3IO		STRUCT
I30.Value		0.0	REAL
I30.ST		16#FF	BYTE

I/O name	Description	Value	Data type
REAL_sp1	Free REAL 1		STRUCT
REAL_sp1.Value		0.0	REAL
REAL_sp1.ST		16#FF	BYTE
REAL_sp2	Free REAL 2		STRUCT
REAL_sp2.Value		0.0	REAL
REAL_sp2.ST		16#FF	BYTE
REAL_sp3	Free REAL 3		STRUCT
REAL_sp3.Value		0.0	REAL
REAL_sp3.ST		16#FF	BYTE
REAL_sp4	Free REAL 4		STRUCT
REAL_sp4.Value		0.0	REAL
REAL_sp4.ST		16#FF	BYTE
REAL_sp5	Free REAL 5		STRUCT
REAL_sp5.Value		0.0	REAL
REAL_sp5.ST		16#FF	BYTE
REAL_sp6	Free REAL 6		STRUCT
REAL_sp6.Value		0.0	REAL
REAL_sp6.ST		16#FF	BYTE
REAL_sp7	Free REAL 7		STRUCT
REAL_sp7.Value		0.0	REAL
REAL_sp7.ST		16#FF	BYTE
REAL_sp8	Free REAL 8		STRUCT
REAL_sp8.Value		0.0	REAL
REAL_sp8.ST		16#FF	BYTE
Pos_CMD_sp1	Position free command 1	16#00000000	DWORD
Pos_CMD_sp2	Position free command 2	16#00000000	DWORD
Pos_CMD_sp3	Position free command 3	16#00000000	DWORD
Pos_CMD_sp4	Position free command 4	16#00000000	DWORD
Beh	Behavior	0	INT
Dev_OK	Device OK	0	INT
INT_sp1	Free INT 1		STRUCT
INT_sp1.Value		0	INT
INT_sp1.ST		16#FF	BYTE
INT_sp2	Free INT 2		STRUCT
INT_sp2.Value		0	INT
INT_sp2.ST		16#FF	BYTE
INT_sp3	Free INT 3		STRUCT
INT_sp3.Value		0	INT
INT_sp3.ST		16#FF	BYTE
INT_sp4	Free INT 4		STRUCT
INT_sp4.Value		0	INT
INT_sp4.ST		16#FF	BYTE
INT_sp5	Free INT 5		STRUCT

I/O name	Description	Value	Data type
INT_sp5.Value		0	INT
INT_sp5.ST		16#FF	BYTE
INT_sp6	Free INT 6		STRUCT
INT_sp6.Value		0	INT
INT_sp6.ST		16#FF	BYTE
TR_Relay	Excitation relay		STRUCT
TR_Relay.Value		0	BOOL
TR_Relay.ST		16#FF	BYTE
PU_Relay	General tripping relay		STRUCT
PU_Relay.Value		0	BOOL
PU_Relay.ST		16#FF	BYTE
PTOC1_Op	Tripping IDMT Phase Ip		STRUCT
PTOC1_Op.Value		0	BOOL
PTOC1_Op.ST		16#FF	BYTE
PTOC1_Str	Excitation IDMT Phase Ip		STRUCT
PTOC1_Str.Value		0	BOOL
PTOC1_Str.ST		16#FF	BYTE
PTOC2_Op	Tripping DMT I >		STRUCT
PTOC2_Op.Value		0	BOOL
PTOC2_Op.ST		16#FF	BYTE
PTOC2_Str	Excitation DMT I >		STRUCT
PTOC2_Str.Value		0	BOOL
PTOC2_Str.ST		16#FF	BYTE
PTOC3_Op	Tripping DMT I >>		STRUCT
PTOC3_Op.Value		0	BOOL
PTOC3_Op.ST		16#FF	BYTE
PTOC3_Str	Excitation DMT I >>		STRUCT
PTOC3_Str.Value		0	BOOL
PTOC3_Str.ST		16#FF	BYTE
PDIS1_Op	Tripping impedance protection Z1		STRUCT
PDIS1_Op.Value		0	BOOL
PDIS1_Op.ST		16#FF	BYTE
PDIS1_Str	Excitation impedance protection Z1		STRUCT
PDIS1_Str.Value		0	BOOL
PDIS1_Str.ST		16#FF	BYTE
PDIS2_Op	Tripping impedance protection Z2		STRUCT
PDIS2_Op.Value		0	BOOL
PDIS2_Op.ST		16#FF	BYTE
PDIS2_Str	Excitation impedance protection Z2		STRUCT
PDIS2_Str.Value		0	BOOL
PDIS2_Str.ST		16#FF	BYTE
PDIS3_Op	Tripping impedance protection Z1B		STRUCT
PDIS3_Op.Value		0	BOOL

I/O name	Description	Value	Data type
PDIS3_Op.ST		16#FF	BYTE
PDIS3_Str	Excitation impedance protection Z1B		STRUCT
PDIS3_Str.Value		0	BOOL
PDIS3_Str.ST		16#FF	BYTE
PDIS4_Op	General tripping impedance protection		STRUCT
PDIS4_Op.Value		0	BOOL
PDIS4_Op.ST		16#FF	BYTE
PDIS4_Str	General excitation impedance protection		STRUCT
PDIS4_Str.Value		0	BOOL
PDIS4_Str.ST		16#FF	BYTE
PDIF1_Op	Tripping differential protection IDIFF>		STRUCT
PDIF1_Op.Value		0	BOOL
PDIF1_Op.ST		16#FF	BYTE
PDIF1_Str	Excitation differential protection IDIFF>		STRUCT
PDIF1_Str.Value		0	BOOL
PDIF1_Str.ST		16#FF	BYTE
PDIF2_Op	Tripping differential protection IDIFF>>		STRUCT
PDIF2_Op.Value		0	BOOL
PDIF2_Op.ST		16#FF	BYTE
PDIF2_Str	Excitation differential protection IDIFF>>		STRUCT
PDIF2_Str.Value		0	BOOL
PDIF2_Str.ST		16#FF	BYTE
RBRF1_Str	Excitation disconnecter fault		STRUCT
RBRF1_Str.Value		0	BOOL
RBRF1_Str.ST		16#FF	BYTE
RBRF1_OpEx	Tripping disconnecter fault T1		STRUCT
RBRF1_OpEx.Value		0	BOOL
RBRF1_OpEx.ST		16#FF	BYTE
RBRF1_OpIn	Tripping disconnecter fault T2		STRUCT
RBRF1_OpIn.Value		0	BOOL
RBRF1_OpIn.ST		16#FF	BYTE
Loc	Local mode	0	BOOL
Error	Fault group message		STRUCT
Error.Value		0	BOOL
Error.ST		16#FF	BYTE
Warning	Warning group message		STRUCT
Warning.Value		0	BOOL
Warning.ST		16#FF	BYTE
PARSET_A	Parameter set A active	0	BOOL
PARSET_B	Parameter set B active	0	BOOL
PARSET_C	Parameter set C active	0	BOOL
PARSET_D	Parameter set D active	0	BOOL
ALM_sp1	Free		STRUCT

I/O name	Description	Value	Data type
ALM_sp1.Value		0	BOOL
ALM_sp1.ST		16#FF	BYTE
ALM_sp2	Free		STRUCT
ALM_sp2.Value		0	BOOL
ALM_sp2.ST		16#FF	BYTE
ALM_sp3	Free		STRUCT
ALM_sp3.Value		0	BOOL
ALM_sp3.ST		16#FF	BYTE
ALM_sp4	Free		STRUCT
ALM_sp4.Value		0	BOOL
ALM_sp4.ST		16#FF	BYTE
ALM_sp5	Free		STRUCT
ALM_sp5.Value		0	BOOL
ALM_sp5.ST		16#FF	BYTE
ALM_sp6	Free		STRUCT
ALM_sp6.Value		0	BOOL
ALM_sp6.ST		16#FF	BYTE
ALM_sp7	Free		STRUCT
ALM_sp7.Value		0	BOOL
ALM_sp7.ST		16#FF	BYTE
ALM_sp8	Free		STRUCT
ALM_sp8.Value		0	BOOL
ALM_sp8.ST		16#FF	BYTE
ALM_sp9	Free		STRUCT
ALM_sp9.Value		0	BOOL
ALM_sp9.ST		16#FF	BYTE
ALM_sp10	Free		STRUCT
ALM_sp10.Value		0	BOOL
ALM_sp10.ST		16#FF	BYTE
SwCount	Operating cycles Q0	0	DINT
OpTm	Operating hours	0	DINT
DINT_sp1	DINT value (free)		STRUCT
DINT_sp1.Value		0	DINT
DINT_sp1.ST		16#FF	BYTE
DINT_sp2	DINT value (free)		STRUCT
DINT_sp2.Value		0	DINT
DINT_sp2.ST		16#FF	BYTE
DINT_sp3	DINT value (free)		STRUCT
DINT_sp3.Value		0	DINT
DINT_sp3.ST		16#FF	BYTE
DINT_sp4	DINT value (free)		STRUCT
DINT_sp4.Value		0	DINT
DINT_sp4.ST		16#FF	BYTE

I/O name	Description	Value	Data type
DINT_sp5	DINT value (free)		STRUCT
DINT_sp5.Value		0	DINT
DINT_sp5.ST		16#FF	BYTE
DINT_sp6	DINT value (free)		STRUCT
DINT_sp6.Value		0	DINT
DINT_sp6.ST		16#FF	BYTE
BOOL_sp1	Free		STRUCT
BOOL_sp1.Value		0	BOOL
BOOL_sp1.ST		16#80	BYTE
BOOL_sp2	Free		STRUCT
BOOL_sp2.Value		0	BOOL
BOOL_sp2.ST		16#80	BYTE
BOOL_sp3	Free		STRUCT
BOOL_sp3.Value		0	BOOL
BOOL_sp3.ST		16#80	BYTE
BOOL_sp4	Free		STRUCT
BOOL_sp4.Value		0	BOOL
BOOL_sp4.ST		16#80	BYTE
BOOL_sp5	Free		STRUCT
BOOL_sp5.Value		0	BOOL
BOOL_sp5.ST		16#80	BYTE
BOOL_sp6	Free		STRUCT
BOOL_sp6.Value		0	BOOL
BOOL_sp6.ST		16#80	BYTE
BOOL_sp7	Free		STRUCT
BOOL_sp7.Value		0	BOOL
BOOL_sp7.ST		16#80	BYTE
BOOL_sp8	Free		STRUCT
BOOL_sp8.Value		0	BOOL
BOOL_sp8.ST		16#80	BYTE
BOOL_sp9	Free		STRUCT
BOOL_sp9.Value		0	BOOL
BOOL_sp9.ST		16#80	BYTE
BOOL_sp10	Free		STRUCT
BOOL_sp10.Value		0	BOOL
BOOL_sp10.ST		16#80	BYTE
OpTmWarning	1 = Operating time warning active	0	BOOL
OpTmAlarm	1 = Operating time alarm active	0	BOOL
SwCountWarning	1 = Operating cycle warning active	0	BOOL
SwCountAlarm	1 = Operating cycle alarm active	0	BOOL
LockAct			STRUCT
LockAct.Value		0	BOOL
LockAct.ST		16#80	BYTE

I/O name	Description	Value	Data type
LocalAct			STRUCT
LocalAct.Value		0	BOOL
LocalAct.ST		16#80	BYTE
AutAct			STRUCT
AutAct.Value		0	BOOL
AutAct.ST		16#80	BYTE
ManAct	1 = Manual: Manual value is active		STRUCT
ManAct.Value	Value	1	BOOL
ManAct.ST	Signal state	16#80	BYTE
OS_PermOut	Operator authorization: Output for OS	16#FFFFFFFF	DWORD
OS_PermLog	Operator authorization: Output for OS	16#FFFFFFFF	DWORD
ST_Worst	Worst signal state	16#80	BYTE
GrpErr			STRUCT
GrpErr.Value		0	BOOL
GrpErr.ST		16#80	BYTE
Status1	Word Status1	16#00000000	DWORD
Status3	Word Status3	16#00000000	DWORD
Status5	Word Status3	16#FFFFFFFF	DWORD
MsgErrlocal1	1 = Message error occurs	0	BOOL
MsgStatlocal1	Message status	16#0000	WORD
MsgAcknlocal1	Message acknowledgement status	16#0000	WORD
MsgErrlocal2	1 = Message error occurs	0	BOOL
MsgStatlocal2	Message status	16#0000	WORD
MsgAcknlocal2	Message acknowledgement status	16#0000	WORD
VisuMisc	Configuration of the units in the faceplate		STRUCT

10.2 PC_LINE block symbols

The following block symbols are available for the PC_LINE object:

PC_LINE/1: Symbol with group display and tag name:



Figure 10-1 PC_LINE/1

10.3 PC_LINE faceplate

Available views:

View	Description
Default window	Important measured values, device status
Message view	Messages from device (messages, warnings, etc.)
Maintenance window	Operating hours, switches, active parameter group
User-defined window 1	User-specific view 1
User-defined window 2	User-specific view 2
Memo view	Memo view

10.3.1 Default window

The default window offers the following functions:

- Display of important measured values of the device
- Display of auxiliary values
- Device status

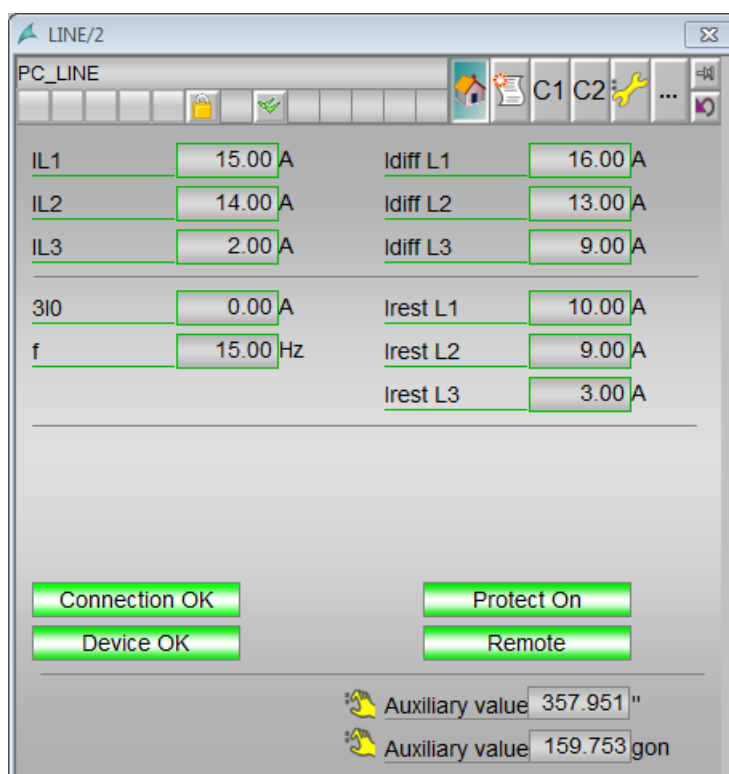


Figure 10-2 PC_LINE default window

Display

Note: The displays for the auxiliary values are only shown if at least one auxiliary value was interconnected to the corresponding block input.

Display	Meaning	Tag	Description
Connection status	Communication status of IED via Ethernet.	QCO-NERRO	QCONERRO = 0 (Connection OK) – GREEN QCONERRO = 1 (Comm. loss) – RED
Device status 1	Status of IED	Dev_OK	Dev_OK = 1 (Device OK) – GREEN Dev_OK = 2 (Device warning) – YELLOW Dev_OK = 3 (Device message) – RED
Device status 2	Behavior of IED	Beh	Beh = 1 (Protection on) – GREEN Beh = 2 (Protection locked) – YELLOW Beh = 3 (Protection test) – RED Beh = 4 (Protection test/locked) – RED Beh = 5 (Protection off) – RED
Device status 3	Local/Remote	Loc	Loc = 1: Local Loc = 0: Remote
IL1	Current L1	I_A#Value	Current L1
IL2	Current L2	I_B#Value	Current L2
IL3	Current L3	I_C#Value	Current L3
3I0	3I0	I30#Value	3I0
F	Frequency	F#Value	Frequency
Idiff L1	Idiff L1	IDIFFL1#Value	Idiff L1
Idiff L2	Idiff L2	IDIFFL2#Value	Idiff L2
Idiff L3	Idiff L3	IDIFFL3#Value	Idiff L3
Irest L1	Irest L1	IRESTL1#Value	Irest L1
Irest L2	Irest L2	IRESTL2#Value	Irest L2
Irest L3	Irest L3	IRESTL3#Value	Irest L3
Auxiliary value 1	User analog value 1	UserA-na1#Value	User analog value 1
Auxiliary value 2	User analog value 2	UserA-na2#Value	User analog value 2

10.3.2 Message view

This window displays every active message and warning of the IED. The APL default message window is used.

10.3.3 Maintenance window

The maintenance window displays the maintenance-specific information.

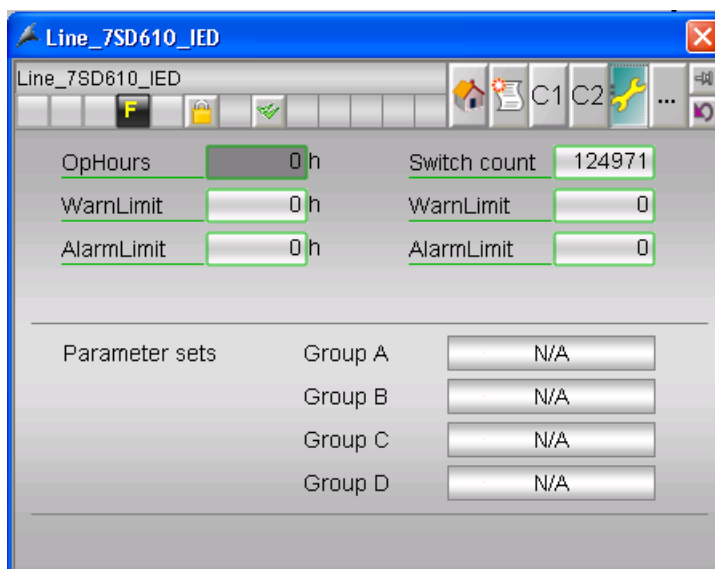


Figure 10-3 PC_LINE maintenance window

Display

Display	Meaning	Tag	Description
OpHours	Operating hours	OpTm	Operating hours
WarnLimit	Warning limit operating hours	OpTm-WarnLimit	Warning limit operating hours
AlarmLimit	Alarm limit operating hours	OpTmA-larmLimit	Alarm limit operating hours
Operating cycles	Control number Q0	SwCount	Control number Q0
WarnLimit	Warning limit operating cycles	SwCount-WarnLimit	Warning limit operating cycles
AlarmLimit	Alarm limit operating cycles	SwCountA-larmLimit	Alarm limit operating cycles
Group A	Parameter set A active	PARSET_A	Parameter set A active
Group B	Parameter set B active	PARSET_B	Parameter set B active
Group C	Parameter set C active	PARSET_C	Parameter set C active
Group D	Parameter set D active	PARSET_D	Parameter set D active

10.3.4 User view

The user-defined faceplates 1 and 2 can be customized by the customer. See also Creating a user view/templates (Page 31)

Example:

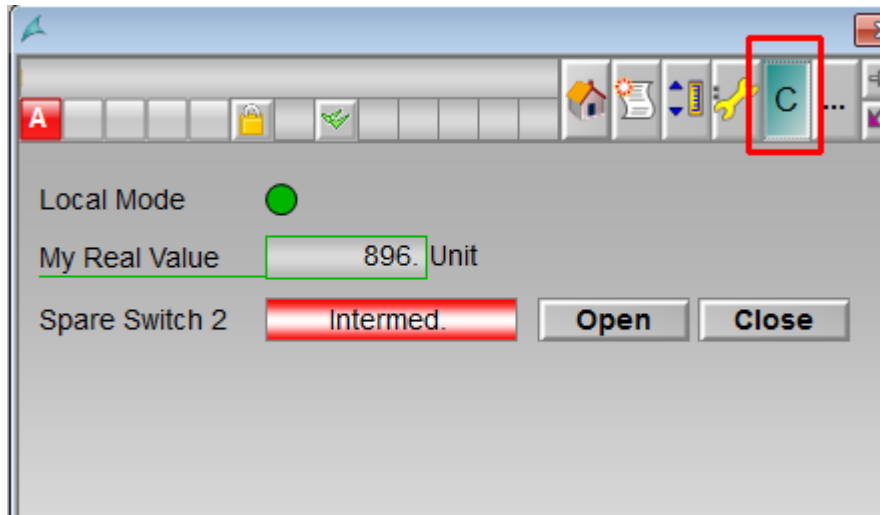


Figure 10-4 Example of user view

PC_BUSBAR

The PC_BUSBAR object represents a busbar. PC_BUSBAR does not represent a device of the switchgear; it only detects data from other CFC blocks. In addition a faceplate does not exist for this block.

PC_BUSBAR consists of the following objects:

- PLC block PC_BUSBAR
- WinCC block symbols

11.1 PLC block PC_BUSBAR

11.1.1 Description of PLC block PC_BUSBAR

The PC_BUSBAR (FB 2504) function block represents the busbar object in the PLC.

The block offers the following functions:

- Provision of data for the block symbol
- Various APL standard functions

11.1.2 Interfaces

The PLC block PC_BUSBAR offers the following interfaces:

1. **Block symbol**

The block symbol receives data from the block through the S7 connection between the PLC and WinCC.

2. **Logic of the S7 program (for controlling modes, switches, etc.)**

The PC_BUSBAR block can receive commands from the CFC logic and makes data available to other blocks.

11.1.3 Configuration

Use the CFC editor to install the block in a cyclic interrupt OB (OB30 to OB38). The block is also installed automatically in the startup OB (OB 100).

11.1.4 Group error

This block provides the standard APL function "Outputting group errors".

The following parameters are taken into consideration when forming the group error:

- CSF

The group error output is *GrpErr.Value*.

11.1.5 Groups of input values

Up to three value groups can be connected with the PC_BUSBAR block - usually depending on the connected input feed. Each group consists of the following values:

- Current (I)
- Voltage (U)
- Power (P)
- Reactive power (Q)
- Apparent power (S)
- Frequency (F)
- Power factor (PF)

The input groups can be activated with the "U_X_ON" inputs. Only one input group may be active at any given time. If several groups are activated, a message is triggered.

The output values of the block and the values displayed on the symbol follow the active group.

11.1.6 Labeling of the busbar

Labeling of the busbar is made with the structure attribute "Busbar_Desc" of the "Visu" input structure.

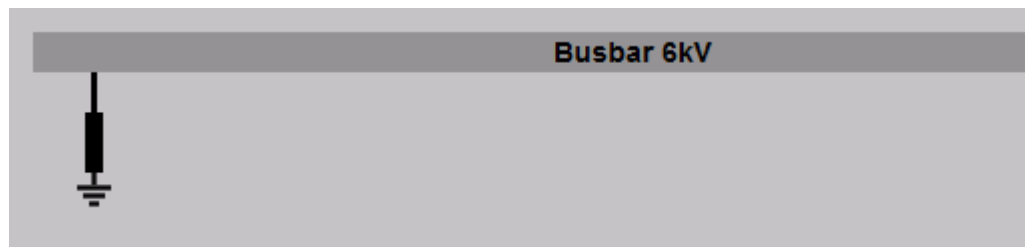
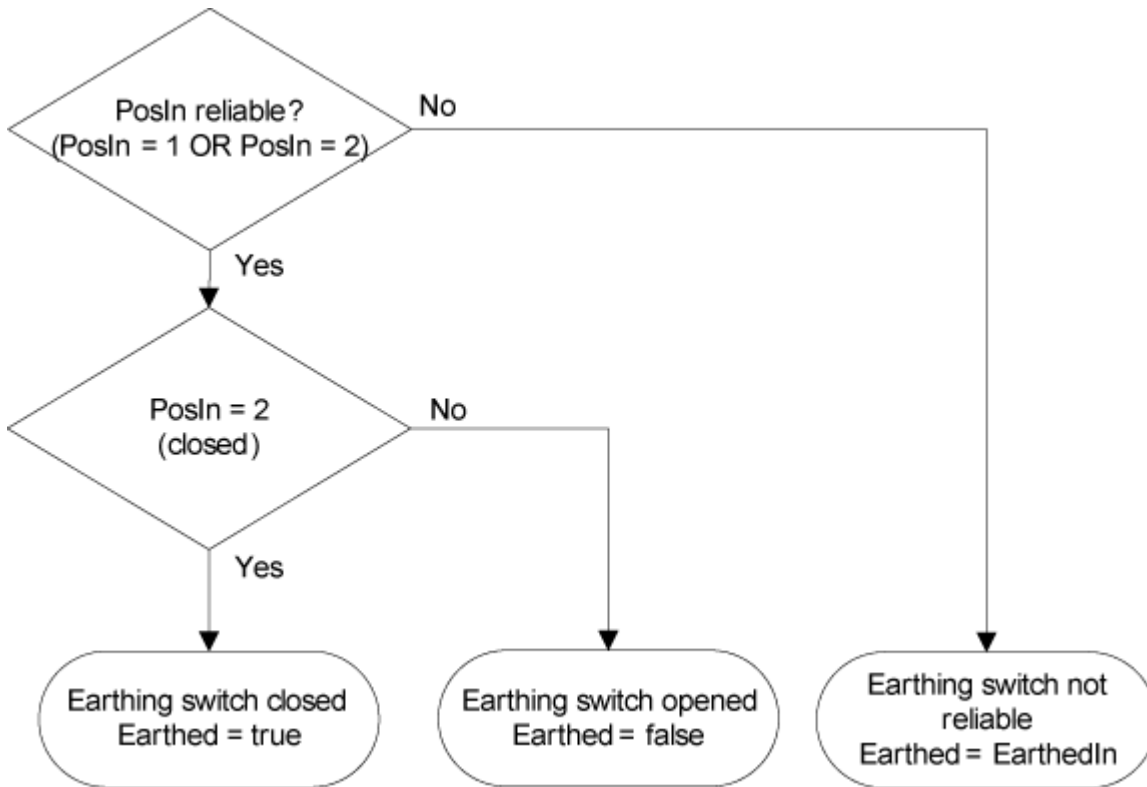


Figure 11-1 Labeling of the busbar

11.1.7 Ground display

The "Earthed" output indicates whether the busbar is grounded. This can be controlled through the position of the connected grounding switch (PosIn input) or through the Boolean input "EarthedIn".

The following logic applies:



11.1.8 Assignment of the status word for Status1 parameter

Status bit	Parameter
0-7	Not used
8	Not used (always true)
9-31	Not used

11.1.9 Messages

Message instance	Message identifier	Message class	Event
MsgEvId	SIG_1	Alarm - high	Several feed-ins connected simultaneously
	SIG_2	AS control system message - fault	AS process control message - external error
	SIG_3	Alarm - high	External message 1
	SIG_4	Alarm - high	External message 2
	SIG_5	Alarm - high	External message 3
	SIG_6	Alarm - high	External message 4
	SIG_7	Alarm - high	
	SIG_8	Alarm - high	

11.1.10 I/O list

11.1.10.1 Inputs

I/O name	Description	Value	Data type
U_1_ON	Voltage 1 active	On	BOOL
U_2_ON	Voltage 2 active	Off	BOOL
U_3_ON	Voltage 3 active	Off	BOOL
PosIn	Position of the grounding switch		DWORD
EarthedIn	Ground display when no switch feedback is available	0	BOOL
Visu	Configuration of the colors and labeling		STRUCT
I_1	Current feed-in 1		STRUCT
I_1.Value			REAL
I_1.ST			BYTE
U_1	Voltage feed-in 1		STRUCT
U_1.Value			REAL
U_1.ST			BYTE
P_1	Power feed-in 1		STRUCT
P_1.Value			REAL
P_1.ST			BYTE
Q_1	Reactive power feed-in 1		STRUCT
Q_1.Value			REAL
Q_1.ST			BYTE
S_1	Apparent power feed-in 1		STRUCT
S_1.Value			REAL
S_1.ST			BYTE

I/O name	Description	Value	Data type
F_1	Frequency feed-in 1		STRUCT
F_1.Value			REAL
F_1.ST			BYTE
PF_1	Cosine Phi feed-in 1		STRUCT
PF_1.Value			REAL
PF_1.ST			BYTE
I_2	Current feed-in 2		STRUCT
I_2.Value		0.0	REAL
I_2.ST		16#FF	BYTE
U_2	Voltage feed-in 2		STRUCT
U_2.Value		0.0	REAL
U_2.ST		16#FF	BYTE
P_2	Power feed-in 2		STRUCT
P_2.Value		0.0	REAL
P_2.ST		16#FF	BYTE
Q_2	Reactive power feed-in 2		STRUCT
Q_2.Value		0.0	REAL
Q_2.ST		16#FF	BYTE
S_2	Apparent power feed-in 2		STRUCT
S_2.Value		0.0	REAL
S_2.ST		16#FF	BYTE
F_2	Frequency feed-in 2		STRUCT
F_2.Value		0.0	REAL
F_2.ST		16#FF	BYTE
PF_2	Cosine Phi feed-in 2		STRUCT
PF_2.Value		0.0	REAL
PF_2.ST		16#FF	BYTE
I_3	Current feed-in 3		STRUCT
I_3.Value		0.0	REAL
I_3.ST		16#FF	BYTE
U_3	Voltage feed-in 3		STRUCT
U_3.Value		0.0	REAL
U_3.ST		16#FF	BYTE
P_3	Power feed-in 3		STRUCT
P_3.Value		0.0	REAL
P_3.ST		16#FF	BYTE
Q_3	Reactive power feed-in 3		STRUCT
Q_3.Value		0.0	REAL
Q_3.ST		16#FF	BYTE
S_3	Apparent power feed-in 3		STRUCT
S_3.Value		0.0	REAL
S_3.ST		16#FF	BYTE
F_3	Frequency feed-in 3		STRUCT

I/O name	Description	Value	Data type
F_3.Value		0.0	REAL
F_3.ST		16#FF	BYTE
PF_3	Cosine Phi feed-in 3		STRUCT
PF_3.Value		0.0	REAL
PF_3.ST		16#FF	BYTE
I_NOM	Rated current	0.0	REAL
V_NOM	Rated voltage	0.0	REAL
MsgLock			STRUCT
MsgLock.Value		0	BOOL
MsgLock.ST		16#80	BYTE
SampleTime	Sampling time [s]	0.1	REAL
MsgEvId	Message event ID	16#00000030	DWORD
CSF			STRUCT
CSF.Value		0	BOOL
CSF.ST		16#80	BYTE
ExtMsg1	External message 1		STRUCT
ExtMsg1.Value	Value	0	BOOL
ExtMsg1.ST	Signal state	16#80	BYTE
ExtMsg2	External message 2		STRUCT
ExtMsg2.Value	Value	0	BOOL
ExtMsg2.ST	Signal state	16#80	BYTE
ExtMsg3	External message 3		STRUCT
ExtMsg3.Value	Value	0	BOOL
ExtMsg3.ST	Signal state	16#80	BYTE
ExtMsg4	External message 4		STRUCT
ExtMsg4.Value	Value	0	BOOL
ExtMsg4.ST	Signal state	16#80	BYTE

11.1.10.2 Outputs

I/O name	Description	Value	Data type
F	Frequency		STRUCT
F.Value		0.0	REAL
F.ST		16#FF	BYTE
U	Voltage		STRUCT
U.Value		0.0	REAL
U.ST		16#FF	BYTE
I	Current requirement		STRUCT
I.Value		0.0	REAL
I.ST		16#FF	BYTE
P	Power		STRUCT
P.Value		0.0	REAL

I/O name	Description	Value	Data type
P.ST		16#FF	BYTE
Q	Reactive power		STRUCT
Q.Value		0.0	REAL
Q.ST		16#FF	BYTE
S	Apparent power		STRUCT
S.Value		0.0	REAL
S.ST		16#FF	BYTE
PF	Cosine Phi		STRUCT
PF.Value		0.0	REAL
PF.ST		16#FF	BYTE
Grounded	Ground display	0	BOOL
Pos	Grounding switch display	16#00000000	DWORD
BB_Color	Current color code for passing on to other blocks (e.g. PC_FEEDER)	16#0	DWORD
GrpErr			STRUCT
GrpErr.Value		0	BOOL
GrpErr.ST		16#80	BYTE
Status1	Word Status1	16#00000000	DWORD
Status3	Word Status3	16#00000000	DWORD
MsgErr	1 = Message error occurs	0	BOOL
MsgStat	Message status	16#0000	WORD
MsgAckn	Message acknowledgement status	16#0000	WORD
Visu_Out	Visualization for the icon		STRUCT

11.2 Color display of the busbar

You configure the color of the busbar with the structure "Visu" at the PC_BUSBAR function block. It is also possible to display the color differently depending on the status of the grounding switch and the voltage present. This feature uses the same hexadecimal color coding as in the WinCC Graphics Editor. The following structure attributes are available:

Attribute	Description	Start value
EarthColor	Busbar grounded, Earthed = 1	16#008500
ZeroVoltageColor	Voltage of the active value group is 0, Earthed = 0	16#919191
VoltageColor	Voltage of the active value group is greater than or less than 0	16#0000FF

The coloring is made according to the following logic:

11.2 Color display of the busbar

- When voltage is present ($U > 0$), the color VoltageColor is used.
- When no voltage is present and Earthed = 0, the ZeroVoltageColor is used.
- When no voltage is present and Earthed = 1, the EarthedColor is used.

The current color of the busbar can also be linked to other connected PCL blocks (PC_FEEDER, PC_FEEDER_2CB, PC_TRAFO). To do so, the output "BB_Color" at the color block PC_BUSBAR is linked to the input "BB_Color" at the other technological blocks.

Note

Only specific PC_FEEDER, PC_FEEDER_2CB and PC_TRAFO blocks support dynamic coloring. The following symbols can be used:

- PC_FEEDER/5
 - PC_FEEDER/6
 - PC_FEEDER/7
 - PC_FEEDER_2CB/5
 - PC_FEEDER_2CB/6
 - PC_FEEDER_2CB/7
 - PC_TRAFO/3
-

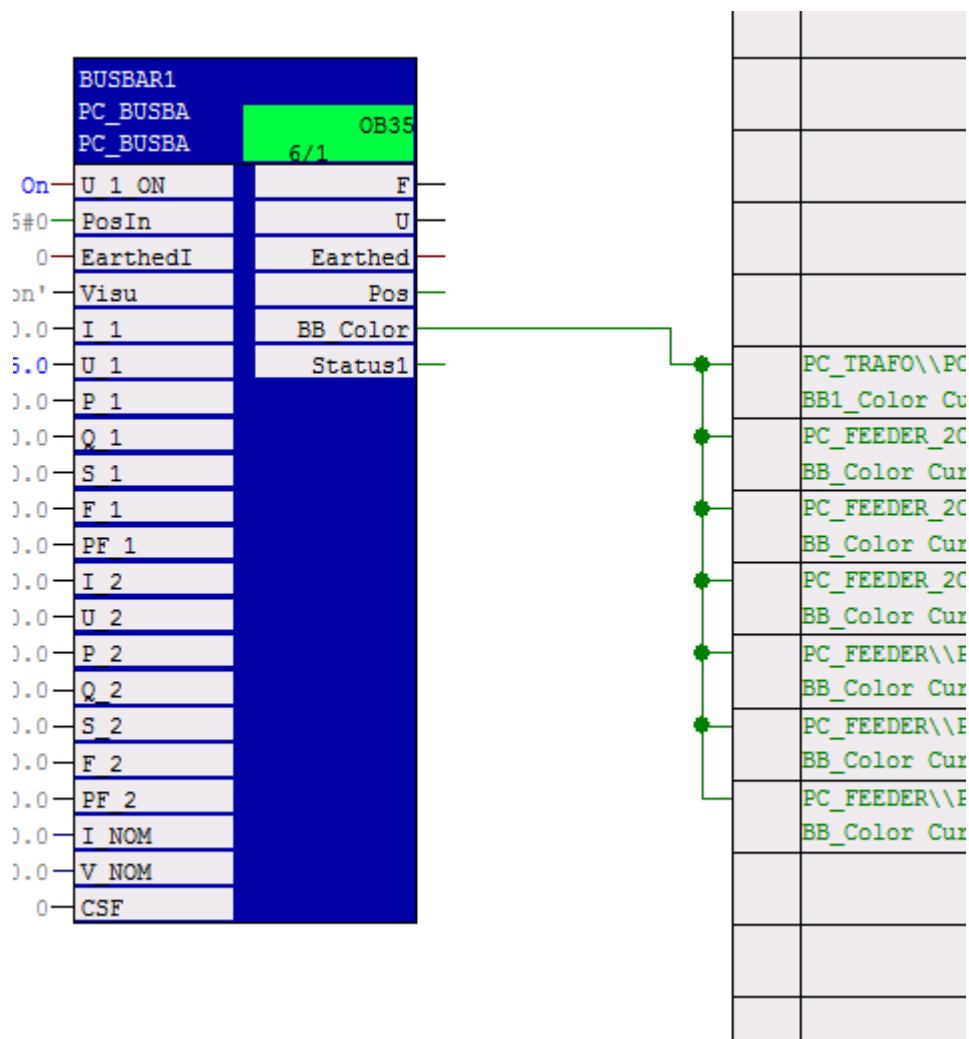


Figure 11-2 BB_Color

11.3 Representation of the grounding switch

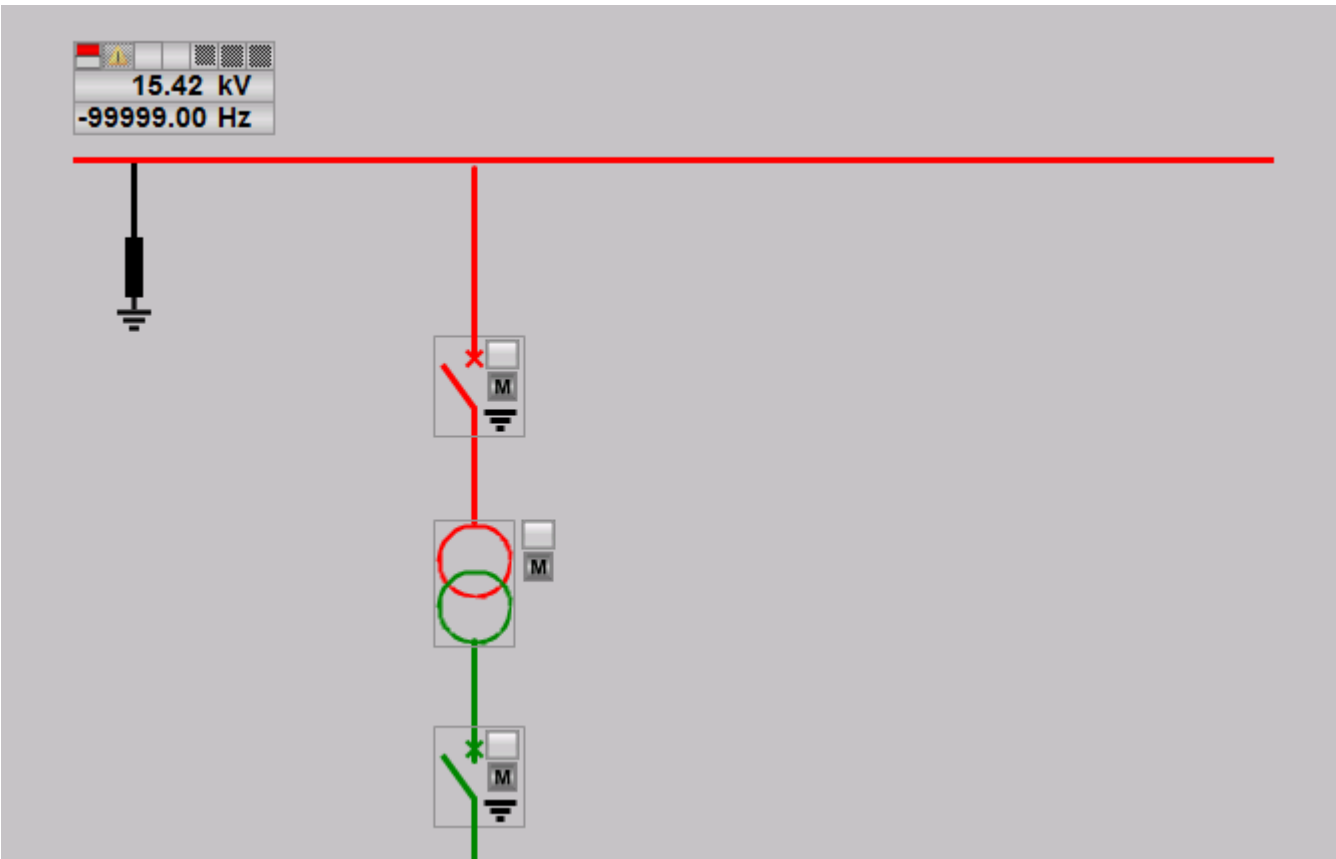


Figure 11-3 Color display

11.3 Representation of the grounding switch

The position of the ground switch is displayed in the busbar symbol when the symbol code "02" is configured on the symbol. The position follows the "PosIn" input of the PC_BUSBAR block.

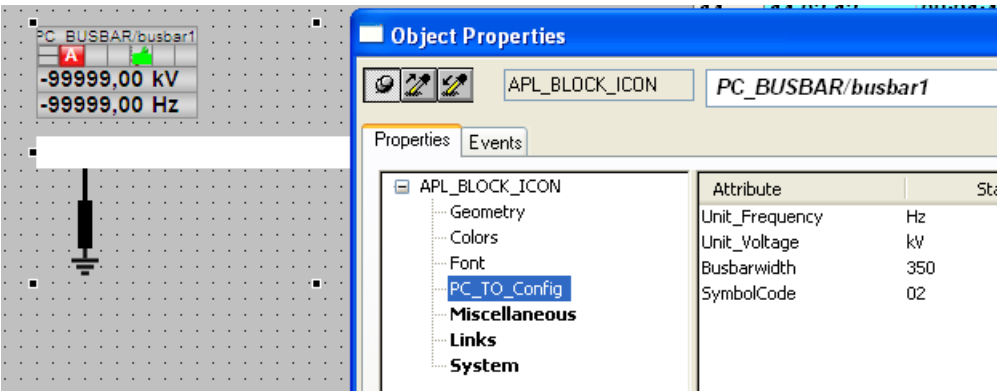


Figure 11-4 Symbol code configuration of the busbar

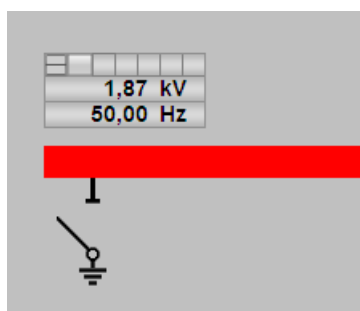


Figure 11-5 Representation of the grounding switch

11.4 PC_BUSBAR block symbols

The following block symbols are available for the PC_BUSBAR object:

PC_BUSBAR/1: Symbol with group display and tag name:

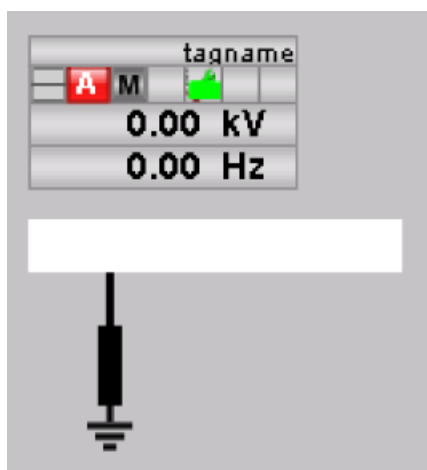


Figure 11-6 PC_BUSBAR/1

PC_BUSBAR/2: Symbol without group display and with tag name:

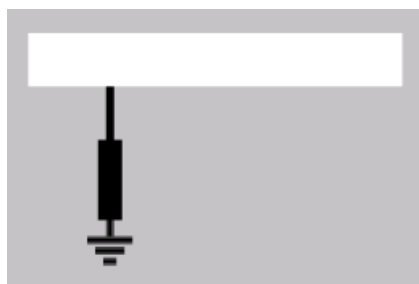


Figure 11-7 PC_BUSBAR/2

PC_MEAS

The PC_MEAS block is a generic block used to represent measuring points in an electrical system. It does not require special driver blocks; all data existing in the automation system can be connected and displayed. Examples are analog input drivers, fixed wired signals, other communication blocks, etc.

The following measured values can be displayed:

- Voltages phase-N-conductor
- Voltages phase-phase
- Currents
- Apparent power
- Active power
- Reactive power
- Total power
- Power factor (cos phi)
- Total cos phi
- Frequency
- THD-R voltage
- THD-R current

The PC-MEAS object consists of the following components:

- PLC block PC_MEAS
- WinCC block symbols
- WinCC faceplate PC-MEAS with several views

12.1 PLC block PC_MEAS

12.1.1 Description of PLC block PC_MEAS

The PC_MEAS (FB 2506) PLC block represents the PC_MEAS object in the PLC.

The block has the following functionalities:

- Collecting data
- Providing the data for the block symbols and the faceplate

- Standard APL functionalities
- Measured value monitoring and signaling

12.1.2 Interfaces

The PLC block PC_MEAS offers the following interfaces:

1. **Block symbol**
The block symbol receives data from the block through the S7 connection between the PLC and WinCC.
2. **Faceplate**
The faceplate receives data from the block through the S7 connection between the PLC and WinCC.
3. **Inputs for measured values**

12.1.3 Configuration

Use the CFC editor to install the block in a cyclic interrupt OB (OB30 to OB38). The block is also installed automatically in the startup OB (OB 100).

12.1.4 Group error

This block provides the standard APL function "Outputting group errors".

The following parameters are taken into consideration when forming the group error:

- CSF
- trip

The group error output is *GrpErr.Value*.

12.1.5 Measured value monitoring

The PC_MEAS block provides the functionality of measured value monitoring. The user has the possibility of defining limits for the analog values through the faceplate or the block inputs. If the limits are violated in either direction, a warning or an alarm is triggered through the Alarm_8P functionality.

The three phases of the currents and voltages trigger the same message; the corresponding phase is shown in the alarm text. The following warnings / alarms are generated:

Limit	Input at blocks	Monitored value(s)
Current high warning limit	I_HiLim	I_A, I_B, I_C
Current high alarm limit	I_HiHiLim	I_A, I_B, I_C
Current low warning limit	I_LoLim	I_A, I_B, I_C
Current low alarm limit	I_LoLoLim	I_A, I_B, I_C
Voltage high warning limit	U_HiLim	U_AB, U_BC, U_CA
Voltage high alarm limit	U_HiHiLim	U_AB, U_BC, U_CA
Voltage low warning limit	U_LoLim	U_AB, U_BC, U_CA
Voltage low alarm limit	U_LoLoLim	U_AB, U_BC, U_CA
Frequency high warning limit	F_HiLim	F
Frequency high alarm limit	F_HiHiLim	F
Frequency low warning limit	F_LoLim	F
Frequency low alarm limit	F_LoLoLim	F
Power factor low warning limit	PF_LoLim	PF
Power factor low alarm limit	PF_LoLoLim	PF

12.1.6 Assignment of the status word for Status1 parameter

Status bit	Parameter
0	Occupied
1	BatchEn
2-7	Not used
8	Not used (always true)
9-31	Not used

12.1.7 Assignment of the status word for Status5 parameter

The Status5 parameter is used to show / hide values in the faceplate. The display of the values depends on the respective quality code.

The values are displayed when the quality code is $\neq$ 0x00.

Status bit	Value	Description
0	I_A	Current phase 1
1	I_B	Current phase 2
2	I_C	Current phase 3
3	U_A	Voltage phase 1 – neutral
4	U_B	Voltage phase 2 – neutral
5	U_C	Voltage phase 3 – neutral
6	U_AB	Voltage phase 1 – phase 2
7	U_BC	Voltage phase 2 – phase 3
8	U_CA	Voltage phase 3 – phase 1
9	P	Active power
10	Q	Reactive power
11	S	Apparent power
12	P_TOT	Total power
13	F	Frequency
14	PF	Power factor (cos phi)
15	PF_TOT	Total cos phi
16	U_THD	THD-R voltage
17	I_THD	THD-R current
18	WpForw	Power counter P forwards
18	WqForw	Power counter Q forwards
18	WpRev	Power counter P reverse
18	WqRev	Power counter P reverse

12.1.8 Messages

Message instance	Message identifier	Message class	Event
MsgEvld_1	SIG_1	Alarm - high	\$\$\$BlockComment\$\$\$ External message 1
	SIG_2	Alarm - high	\$\$\$BlockComment\$\$\$ External message 2
	SIG_3	Alarm - high	\$\$\$BlockComment\$\$\$ External message 3
	SIG_4	Alarm - high	\$\$\$BlockComment\$\$\$ External message 4
	SIG_5	Alarm - high	\$\$\$BlockComment\$\$\$ External message 5
	SIG_6	Alarm - high	\$\$\$BlockComment\$\$\$ External message 6
	SIG_7	Alarm - high	\$\$\$BlockComment\$\$\$ External message 7
	SIG_8	Alarm - high	\$\$\$BlockComment\$\$\$ External message 8

MsgEvid_2	SIG_1	Warning - high	\$\$BlockComment\$\$ Current high warning limit - Phase @1% Value: @2%.2f@
	SIG_2	Alarm - high	\$\$BlockComment\$\$ Current high alarm limit - Phase @1% Value: @2%.2f@
	SIG_3	Warning - high	\$\$BlockComment\$\$ Current low warning limit - Phase @1% Value: @2%.2f@
	SIG_4	Alarm - high	\$\$BlockComment\$\$ Current low alarm limit - Phase @1% Value: @2%.2f@
	SIG_5	Warning - high	\$\$BlockComment\$\$ Voltage high warning limit - Phase @3% Value: @4%.2f@
	SIG_6	Alarm - high	\$\$BlockComment\$\$ Voltage high alarm limit - Phase @3% Value: @4%.2f@
	SIG_7	Warning - high	\$\$BlockComment\$\$ Voltage low warning limit - Phase @3% Value: @4%.2f@
	SIG_8	Alarm - high	\$\$BlockComment\$\$ Voltage low alarm limit - Phase @3% Value: @4%.2f@
MsgEvid_3	SIG_1	Warning - high	\$\$BlockComment\$\$ Frequency high warning limit - Value: @1%.3f@
	SIG_2	Alarm - high	\$\$BlockComment\$\$ Frequency high alarm limit - Value: @1%.3f@
	SIG_3	Warning - high	\$\$BlockComment\$\$ Frequency low warning limit - Value: @1%.3f@
	SIG_4	Alarm - high	\$\$BlockComment\$\$ Frequency low alarm limit - Value: @1%.3f@
	SIG_5	Warning - high	\$\$BlockComment\$\$ Power factor low warning limit - Value: @2%.3f@
	SIG_6	Alarm - high	\$\$BlockComment\$\$ Power factor low alarm limit - Value: @2%.3f@
	SIG_7	AS control system message - fault	\$\$BlockComment\$\$ Connection failure
	SIG_8	AS control system message - fault	\$\$BlockComment\$\$ AS process control message - external error

12.1.9 I/O list

12.1.9.1 Inputs

I/O name	Description	Value	Data type
I_A_IN	Current Phase 1		STRUCT
I_A_IN.Value			REAL

12.1 PLC block PC_MEAS

I_A_IN.ST			BYTE
I_B_IN	Current Phase 2		STRUCT
I_B_IN.Value			REAL
I_B_IN.ST			BYTE
I_C_IN	Current Phase 3		STRUCT
I_C_IN.Value			REAL
I_C_IN.ST			BYTE
U_A_IN	Voltage phase 1 – neutral		STRUCT
U_A_IN.Value			REAL
U_A_IN.ST			BYTE
U_B_IN	Voltage phase 2 – neutral		STRUCT
U_B_IN.Value			REAL
U_B_IN.ST			BYTE
U_C_IN	Voltage phase 3 – neutral		STRUCT
U_C_IN.Value			REAL
U_C_IN.ST			BYTE
U_AB_IN	Voltage phase 1 – phase 2		STRUCT
U_AB_IN.Value			REAL
U_AB_IN.ST			BYTE
U_BC_IN	Voltage phase 2 – phase 3		STRUCT
U_BC_IN.Value			REAL
U_BC_IN.ST			BYTE
U_CA_IN	Voltage Phase 3 - Phase 1		STRUCT
U_CA_IN.Value			REAL
U_CA_IN.ST			BYTE
P_IN	Active power		STRUCT
P_IN.Value			REAL
P_IN.ST			BYTE
Q_IN	Reactive power		STRUCT
Q_IN.Value			REAL
Q_IN.ST			BYTE
S_IN	Apparent power		STRUCT
S_IN.Value			REAL
S_IN.ST			BYTE
P_TOT_IN	Total power		STRUCT
P_TOT_IN.Value			REAL
P_TOT_IN.ST			BYTE
F_IN	Frequency		STRUCT
F_IN.Value			REAL
F_IN.ST		16#80	BYTE
PF_IN	Power factor		STRUCT
PF_IN.Value			REAL
PF_IN.ST		16#80	BYTE
PF_TOT_IN	Total power factor		STRUCT

PF_TOT_IN.Value			REAL
PF_TOT_IN.ST		16#80	BYTE
I_THD_IN	THD current		STRUCT
I_THD_IN.Value			REAL
I_THD_IN.ST		16#80	BYTE
U_THD_IN	THD voltage		STRUCT
U_THD_IN.Value			REAL
U_THD_IN.ST		16#80	BYTE
WpForward_IN	Real energy supply		STRUCT
WpForward_IN.Value			REAL
WpForward_IN.ST			BYTE
WpReverse_IN	Real energy demand		STRUCT
WpReverse_IN.Value			REAL
WpReverse_IN.ST			BYTE
WqForward_IN	Reactive energy supply		STRUCT
WqForward_IN.Value			REAL
WqForward_IN.ST			BYTE
WqReverse_IN	Reactive energy demand		STRUCT
WqReverse_IN.Value			REAL
WqReverse_IN.ST			BYTE
FP_Visu	Configuration of the faceplate		STRUCT
FP_Units	Configuration of the units in the faceplate		STRUCT
ConnErr	Linkable Connection Error	0	BOOL
FACTOR_ENERGY	Factor for display of energy values		REAL
MsgLock			STRUCT
MsgLock.Value		0	BOOL
MsgLock.ST		16#80	BYTE
SampleTime	Sampling time [s]		REAL
MsgEvid_1	Message event ID		DWORD
MsgEvid_2	Message event ID		DWORD
MsgEvid_3	Message event ID		DWORD
BatchEn	Enable remote operation of controller by Batch recipe	0	BOOL
BatchID	Current batch ID (number)		DWORD
BatchName	Current batch name	"	STRING[32]
StepNo	Batch step number		DWORD
Occupied	Occupied by batch	0	BOOL
CSF			STRUCT
CSF.Value		0	BOOL
CSF.ST		16#80	BYTE
ExtMsg1	External message 1		STRUCT
ExtMsg1.Value		0	BOOL
ExtMsg1.ST		16#80	BYTE
ExtMsg2	External message 2		STRUCT

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ExtMsg2.Value		0	BOOL
ExtMsg2.ST		16#80	BYTE
ExtMsg3	External message 3		STRUCT
ExtMsg3.Value		0	BOOL
ExtMsg3.ST		16#80	BYTE
ExtMsg4	External message 4		STRUCT
ExtMsg4.Value		0	BOOL
ExtMsg4.ST		16#80	BYTE
ExtMsg5	External message 5		STRUCT
ExtMsg5.Value		0	BOOL
ExtMsg5.ST		16#80	BYTE
ExtMsg6	External message 6		STRUCT
ExtMsg6.Value		0	BOOL
ExtMsg6.ST		16#80	BYTE
ExtMsg7	External message 7		STRUCT
ExtMsg7.Value		0	BOOL
ExtMsg7.ST		16#80	BYTE
ExtMsg8	External Message 8		STRUCT
ExtMsg8.Value		0	BOOL
ExtMsg8.ST		16#80	BYTE
UserAna1_IN	User analog input 1		STRUCT
UserAna1_IN.Value	Value	0.0	REAL
UserAna1_IN.ST	Signal state	16#FF	BYTE
UA1unit	Unit of measurement user analog input 1	0	INT
UserAna2_IN	User analog input 2		STRUCT
UserAna2_IN.Value	Value	0.0	REAL
UserAna2_IN.ST	Signal state	16#FF	BYTE
UA2unit	Unit of measurement user analog input 2	0	INT
UserAna3_IN	User analog input 3		STRUCT
UserAna3_IN.Value	Value	0.0	REAL
UserAna3_IN.ST	Signal state	16#FF	BYTE
UA3unit	Unit of measurement user analog input 3	0	INT
UserAna4_IN	User analog input 4		STRUCT
UserAna4_IN.Value	Value	0.0	REAL
UserAna4_IN.ST	Signal state	16#FF	BYTE
UA4unit	Unit of measurement user analog input 4	0	INT
UserAna5_IN	User analog input 5		STRUCT
UserAna5_IN.Value	Value	0.0	REAL
UserAna5_IN.ST	Signal state	16#FF	BYTE
UA5unit	Unit of measurement user analog input 5	0	INT
UserAna6_IN	User analog input 6		STRUCT
UserAna6_IN.Value	Value	0.0	REAL
UserAna6_IN.ST	Signal state	16#FF	BYTE
UA6unit	Unit of measurement user analog input 6	0	INT

UserAna7_IN	User analog input 7		STRUCT
UserAna7_IN.Value	Value	0.0	REAL
UserAna7_IN.ST	Signal state	16#FF	BYTE
UA7unit	Unit of measurement user analog input 7	0	INT
UserAna8_IN	User analog input 8		STRUCT
UserAna8_IN.Value	Value	0.0	REAL
UserAna8_IN.ST	Signal state	16#FF	BYTE
UA8unit	Unit of measurement user analog input 8	0	INT
UserStatus	User status bits	16#0	BYTE
SelFp1	Select faceplate 1		ANY
OS_Perm	Operator permissions		STRUCT
OS_Perm.Bit0	1 = Operator can switch to automatic mode	1	BOOL
OS_Perm.Bit1	1 = Operator can switch to manual mode	1	BOOL
OS_Perm.Bit2	1 = Operator can switch to local mode	1	BOOL
OS_Perm.Bit3	1 = Operator can switch to out of service mode	1	BOOL
OS_Perm.Bit4		1	BOOL
OS_Perm.Bit5		1	BOOL
OS_Perm.Bit6		1	BOOL
OS_Perm.Bit7		1	BOOL
OS_Perm.Bit8		1	BOOL
OS_Perm.Bit9		1	BOOL
OS_Perm.Bit10		1	BOOL
OS_Perm.Bit11	1 = Operator can enable function simulation	1	BOOL
OS_Perm.Bit12	1 = Operator can enable function maintenance release	1	BOOL
OS_Perm.Bit13		1	BOOL
OS_Perm.Bit14		1	BOOL
OS_Perm.Bit15		1	BOOL
OS_Perm.Bit16		1	BOOL
OS_Perm.Bit17		1	BOOL
OS_Perm.Bit18		1	BOOL
OS_Perm.Bit19		1	BOOL
OS_Perm.Bit20		1	BOOL
OS_Perm.Bit21		1	BOOL
OS_Perm.Bit22		1	BOOL
OS_Perm.Bit23		1	BOOL
OS_Perm.Bit24		1	BOOL
OS_Perm.Bit25		1	BOOL
OS_Perm.Bit26		1	BOOL
OS_Perm.Bit27		1	BOOL
OS_Perm.Bit28		1	BOOL
OS_Perm.Bit29		1	BOOL
OS_Perm.Bit30		1	BOOL

OS_Perm.Bit31		1	BOOL
Feature	Status of various features		STRUCT
Feature.Bit0	Reserved	0	BOOL
Feature.Bit1	1 = OosLi can switch to Out of Service	0	BOOL
Feature.Bit2	Reserved	0	BOOL
Feature.Bit3	Reserved	0	BOOL
Feature.Bit4	Reserved	0	BOOL
Feature.Bit5	Reserved	0	BOOL
Feature.Bit6	Reserved	0	BOOL
Feature.Bit7	Reserved	0	BOOL
Feature.Bit8	Reserved	0	BOOL
Feature.Bit9	Reserved	0	BOOL
Feature.Bit10	Reserved	0	BOOL
Feature.Bit11	Reserved	0	BOOL
Feature.Bit12	Hide OpTm	0	BOOL
Feature.Bit13	Hide SwCount	0	BOOL
Feature.Bit14	Reserved	0	BOOL
Feature.Bit15	Reserved	0	BOOL
Feature.Bit16	Reserved	0	BOOL
Feature.Bit17	Reserved	0	BOOL
Feature.Bit18	Reserved	0	BOOL
Feature.Bit19	Reserved	0	BOOL
Feature.Bit20	Reserved	0	BOOL
Feature.Bit21	Reserved	0	BOOL
Feature.Bit22	Reserved	0	BOOL
Feature.Bit23	Reserved	0	BOOL
Feature.Bit24	1 = Local authorization enabled	0	BOOL
Feature.Bit25	1 = Suppression of all messages if MsgLock = 1	0	BOOL
Feature.Bit26	Reserved	0	BOOL
Feature.Bit27	Reserved	0	BOOL
Feature.Bit28	Reserved	0	BOOL
Feature.Bit29	Reserved	0	BOOL
Feature.Bit30	Reserved	0	BOOL
Feature.Bit31	1 = Resetting interlock in manual mode	0	BOOL

12.1.9.2 Inputs / outputs

I/O name	Description	Value	Data type
I_HiLim	Current High Warning Limit		REAL

I_HiHiLim	Current High Alarm Limit		REAL
I_LoLim	Current Low Warning Limit		REAL
I_LoLoLim	Current Low Alarm Limit		REAL
U_HiLim	Voltage High Warning Limit		REAL
U_HiHiLim	Voltage High Alarm Limit		REAL
U_LoLim	Voltage Low Warning Limit		REAL
U_LoLoLim	Voltage Low Alarm Limit		REAL
F_HiLim	Frequency High Warning Limit		REAL
F_HiHiLim	Frequency High Alarm Limit		REAL
F_LoLim	Frequency Low Warning Limit		REAL
F_LoLoLim	Frequency Low Alarm Limit		REAL
PF_LoLim	Power factor Low Warning Limit		REAL
PF_LoLoLim	Power factor Low Alarm Limit		REAL

12.1.9.3 Outputs

I/O name	Description	Value	Data type
U_A	Voltage Phase A		STRUCT
U_A.Value		0.0	REAL
U_A.ST		16#FF	BYTE
U_B	Voltage Phase B		STRUCT
U_B.Value		0.0	REAL
U_B.ST		16#FF	BYTE
U_C	Voltage Phase C		STRUCT
U_C.Value		0.0	REAL
U_C.ST		16#FF	BYTE
U_AB	Voltage A-B		STRUCT
U_AB.Value		0.0	REAL
U_AB.ST		16#FF	BYTE
U_BC	Voltage B-C		STRUCT
U_BC.Value		0.0	REAL
U_BC.ST		16#FF	BYTE
U_CA	Voltage C-A		STRUCT
U_CA.Value		0.0	REAL
U_CA.ST		16#FF	BYTE
I_A	Current Phase A		STRUCT
I_A.Value		0.0	REAL
I_A.ST		16#FF	BYTE
I_B	Current Phase B		STRUCT

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I_B.Value		0.0	REAL
I_B.ST		16#FF	BYTE
I_C	Current Phase C		STRUCT
I_C.Value		0.0	REAL
I_C.ST		16#FF	BYTE
S	Apparent power		STRUCT
S.Value		0.0	REAL
S.ST		16#FF	BYTE
P	Power		STRUCT
P.Value		0.0	REAL
P.ST		16#FF	BYTE
Q	Reactive power		STRUCT
Q.Value		0.0	REAL
Q.ST		16#FF	BYTE
P_TOT	Total of apparent, active and reactive power		STRUCT
P_TOT.Value		0.0	REAL
P_TOT.ST		16#FF	BYTE
PF	cos phi - Power Factor		STRUCT
PF.Value		0.0	REAL
PF.ST		16#FF	BYTE
PF_TOT	Total power factor		STRUCT
PF_TOT.Value		0.0	REAL
PF_TOT.ST		16#FF	BYTE
F	Frequency		STRUCT
F.Value		0.0	REAL
F.ST		16#FF	BYTE
U_THD	THD-R voltage		STRUCT
U_THD.Value		0.0	REAL
U_THD.ST		16#FF	BYTE
I_THD	THD-R current		STRUCT
I_THD.Value		0.0	REAL
I_THD.ST		16#FF	BYTE
WpForward	Real energy supply		STRUCT
WpForward.Value		0.0	REAL
WpForward.ST		16#FF	BYTE
WpReverse	Real energy demand		STRUCT
WpReverse.Value		0.0	REAL
WpReverse.ST		16#FF	BYTE
WqForward	Reactive energy supply		STRUCT
WqForward.Value		0.0	REAL
WqForward.ST		16#FF	BYTE
WqReverse	Reactive energy demand		STRUCT
WqReverse.Value		0.0	REAL
WqReverse.ST		16#FF	BYTE

UserAna1	Additional Analog Value 1		STRUCT
UserAna1.Value		0.0	REAL
UserAna1.ST		16#FF	BYTE
UserAna2	Additional Analog Value 2		STRUCT
UserAna2.Value		0.0	REAL
UserAna2.ST		16#FF	BYTE
UserAna3	Additional Analog Value 3		STRUCT
UserAna3.Value		0.0	REAL
UserAna3.ST		16#FF	BYTE
UserAna4	Additional Analog Value 4		STRUCT
UserAna4.Value		0.0	REAL
UserAna4.ST		16#FF	BYTE
UserAna5	Additional Analog Value 5		STRUCT
UserAna5.Value		0.0	REAL
UserAna5.ST		16#FF	BYTE
UserAna6	Additional Analog Value 6		STRUCT
UserAna6.Value		0.0	REAL
UserAna6.ST		16#FF	BYTE
UserAna7	Additional Analog Value 7		STRUCT
UserAna7.Value		0.0	REAL
UserAna7.ST		16#FF	BYTE
UserAna8	Additional Analog Value 8		STRUCT
UserAna8.Value		0.0	REAL
UserAna8.ST		16#FF	BYTE
QCONNERR		0	BOOL
OS_PermOut	Operator permission: Output for OS	16#FFFFFFFF	DWORD
OS_PermLog	Operator permission: Output for OS	16#FFFFFFFF	DWORD
ST_Worst	Worst signal status	16#80	BYTE
GrpErr			STRUCT
GrpErr.Value		0	BOOL
GrpErr.ST		16#80	BYTE
Status1	Status1 Word	16#00000000	DWORD
Status3	Status3 Word	16#00000000	DWORD
Status5	Status3 Word	16#FFFFFFFF	DWORD
MsgErr1	1=Messaging Error Occurs	0	BOOL
MsgStat1	Message status	16#0000	WORD
MsgAckn1	Message acknowledgement status	16#0000	WORD
MsgErr2	1=Messaging Error Occurs	0	BOOL
MsgStat2	Message status	16#0000	WORD
MsgAckn2	Message acknowledgement status	16#0000	WORD
MsgErr3	1=Messaging Error Occurs	0	BOOL
MsgStat3	Message status	16#0000	WORD
MsgAckn3	Message acknowledgement status	16#0000	WORD
I_Hi	Current High Warning	0	BOOL

I_HiHi	Current High Alarm	0	BOOL
I_Lo	Current Low Warning	0	BOOL
I_LoLo	Current Low Alarm	0	BOOL
U_Hi	Voltage High Warning	0	BOOL
U_HiHi	Voltage High Alarm	0	BOOL
U_Lo	Voltage Low Warning	0	BOOL
U_LoLo	Voltage Low Alarm	0	BOOL
F_Hi	Frequency High Warning	0	BOOL
F_HiHi	Frequency High Alarm	0	BOOL
F_Lo	Frequency Low Warning	0	BOOL
F_LoLo	Frequency Low Alarm	0	BOOL
PF_Lo	Power factor Low Warning	0	BOOL
PF_LoLo	Power factor Low Alarm	0	BOOL
VisuMisc	Configuration of the units in the faceplate		STRUCT

12.2 PC_MEAS block symbols

The following block symbols are available for the PC_MEAS object:

PC_MEAS/1: Symbol with group display, tag name and current:

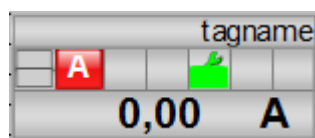


Figure 12-1 PC_MEAS_1

PC_MEAS/2: Symbol with group display, tag name, current, voltage and frequency:



Figure 12-2 PC_MEAS_2

12.3 PC_MEAS faceplate

Available views:

View	Description
Default window	Important measured values, device status
Message view	Messages from device (messages, warnings, etc.)
Measured value window	Measured value monitoring (limits)
User-defined window	User-specific view
Memo view	Memo view

12.3.1 Default window

The standard window offers the following functions:

- Display of the device status
- Display of important measured values
- Display of auxiliary values

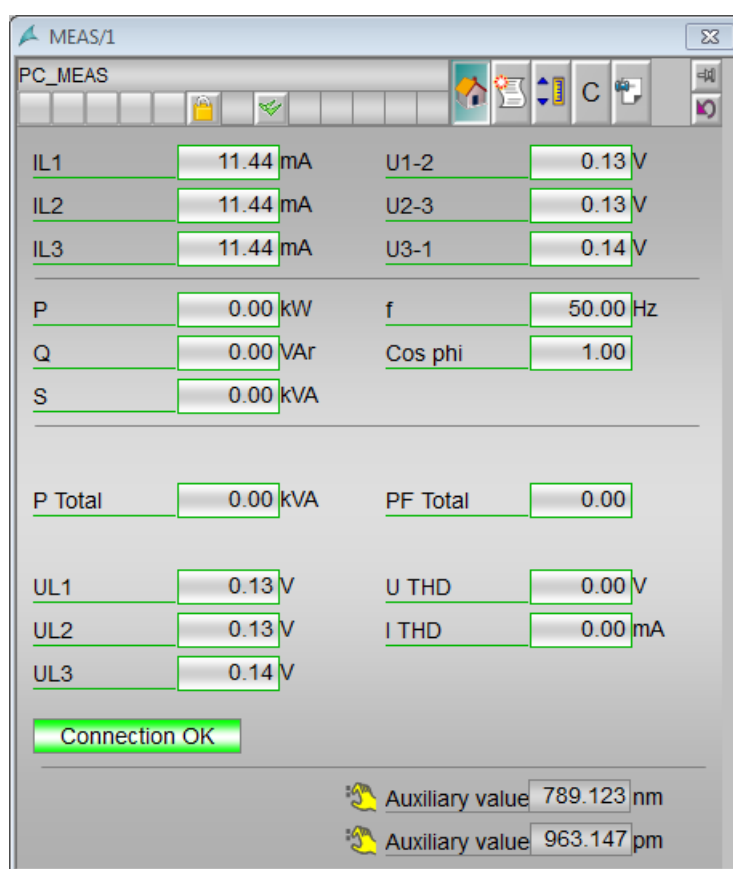


Figure 12-3 Standard window PC_MEAS

Display

Note: The displays for the auxiliary values are only shown if at least one auxiliary value was interconnected to the corresponding block input.

Display	Meaning	Tag	Description
Connect Status	Communication status of the IED via Ethernet.	QCONERR0	QCONERR0=0 (Connection OK) – GREEN QCONERR0=1 (Loss of Com) – RED
IL1	Current L1	I_A#Value	Current L1
IL2	Current L2	I_B#Value	Current L2
IL3	Current L3	I_C#Value	Current L3
U1-2	Voltage L1-L2	U_AB#Value	Voltage L1-L2
U2-3	Voltage L2-L3	U_BC#Value	Voltage L2-L3
U3-1	Voltage L3-L1	U_CA#Value	Voltage L3-L1
P	Power	P#Value	Power
Q	Reactive power	Q#Value	Reactive power
S	Apparent power	S#Value	Apparent power
F	Frequency	F#Value	Frequency

Cos phi	Power Factor	PF#Value	Power Factor
P Total	Total power	P_TOT#Value	Total power
PF Total	Total power factor	PF_TOT#Value	Total power factor
UL1	Voltage L1-Earth	U_A#Value	Voltage L1-Earth
UL2	Voltage L2-Earth	U_B#Value	Voltage L2-Earth
UL3	Voltage L3-Earth	U_C#Value	Voltage L3-Earth
U THD	THD Voltage	U_THD#Value	THD Voltage
I_THD	THD Current	I_THD#Value	THD Current
Auxiliary value 1	User analog value 1	UserA-na1#Value	User analog value 1
Auxiliary value 2	User analog value 2	UserA-na2#Value	User analog value 2

12.3.2 Message view

This window displays every active message and warning of the IED. The APL default message window is used.

12.3.3 Measured value window

The measured value windows displays the power counters and the warning / alarm limits for the measured values.

The screenshot shows the PC_MEAS/1 faceplate interface. It features a toolbar at the top with icons for home, settings, and other functions. The main area is divided into four sections: Current limits, Voltage limits, Frequency limits, and Power factor limits. Each section contains input fields for high and low alarm and warning values. At the bottom, there are four energy value displays: WpForw, WqForw, WpRev, and WqRev.

Current limits		Voltage limits	
I Hi Alarm	0,00 A	U Hi Alarm	0,00 kV
I Hi Warning	0,00 A	U Hi Warning	0,00 kV
I Lo Warning	0,00 A	U Lo Warning	0,00 kV
I Lo Alarm	0,00 A	U Lo Alarm	0,00 kV

Frequency limits		Power factor limits	
F Hi Alarm	0,00 Hz		
F Hi Warning	0,00 Hz		
F Lo Warning	0,00 Hz	PF Lo Warning	0,00
F Lo Alarm	0,00 Hz	PF Lo Alarm	0,00

WpForw	2566165 kWh
WqForw	0 kVarh
WpRev	0 kWh
WqRev	0 kVarh

Figure 12-4 PC_MEAS_Measurement

In the event that the signal status of all energy values (WpVorw, WqVorw, WpRückw, WqRückw) is transmitted with a value of 16#FF, the corresponding display elements are hidden. As soon as one of the energy values receives a status not equal to 16#FF, all display elements are re-displayed.

Display

Display	Meaning	Tag	Description
I Hi Warning	I_HiLim		Current High Warning Limit
I Hi Alarm	I_HiHiLim		Current High Alarm Limit
I Lo Warning	I_LoLim		Current Low Warning Limit
I Lo Alarm	I_LoLoLim		Current Low Alarm Limit
U Hi Warning	U_HiLim		Voltage High Warning Limit
U Hi Alarm	U_HiHiLim		Voltage High Alarm Limit
U Lo Warning	U_LoLim		Voltage Low Warning Limit
U Lo Alarm	U_LoLoLim		Voltage Low Alarm Limit

F Hi Warning	F_HiLim		Frequency High Warning Limit
F Hi Alarm	F_HiHiLim		Frequency High Alarm Limit
F Lo Warning	F_LoLim		Frequency Low Warning Limit
F Lo Alarm	F_LoLoLim		Frequency Low Alarm Limit
PF Lo Warning	PF_LoLim		Power factor Low Warning Limit
PF Lo Alarm	PF_LoLoLim		Power factor Low Alarm Limit
WpForw	Real energy supply	WpForward#Value	Real energy supply
WqForw	Real energy demand	WpForward#Value	Real energy demand
WpRev	Reactive energy supply	WpReverse#Value	Reactive energy supply
WqRev	Reactive energy demand	WqReverse#Value	Reactive energy demand

12.3.4 User view

The user-defined faceplate can be customized by the customer. See also Creating a user view/ templates (Page 31)

Example:

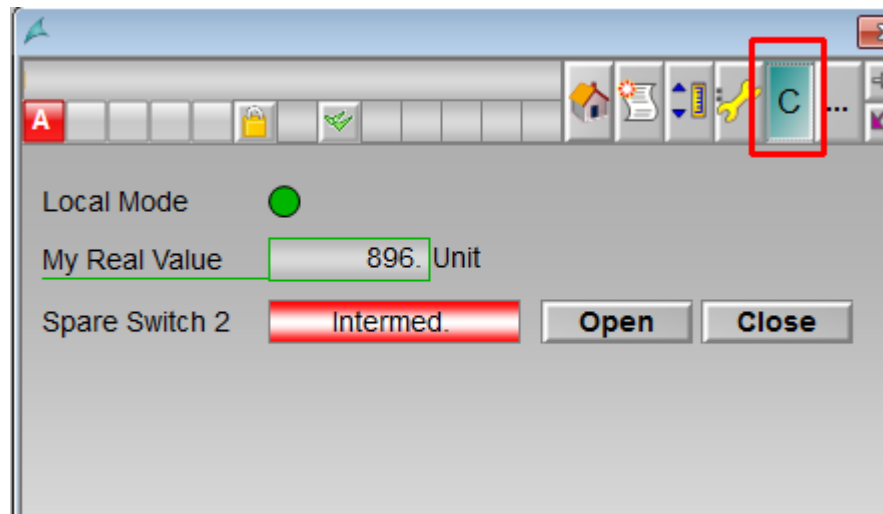


Figure 12-5 Example of user view

DataCollector blocks

13.1 Introduction

The DataCollector blocks are an extension of the PowerControl Library (PCL). They represent simulation and communication blocks and can be used for the following applications:

- Connection of I/O modules or other signal sources to the PCL. These can be used, for example, to read out hard-wired protection devices or protection devices with other interfaces.
- Simulation of values, messages and feedbacks for the technological blocks
- Preparation and final test before commissioning without needing the real hardware.

The following blocks are available:

Block name	Description	FB No.
DC_FEEDER	Connection of signals for the PC_FEEDER blocks	FB 2560
DC_TRAFO	Connection of signals for the PC_TRAFO blocks	FB 2562
DC_SYNC	Connection of signals for the PC_SYNC blocks	FB 2563
DC_LINE	Connection of signals for the PC_LINE blocks	FB 2564

13.2 Functionalities

The DataCollector blocks have the following functionalities:

- Collecting and transferring data to the technological blocks
- Generation of time stamps for the messages
- Simulation of the switch feedbacks in simulation mode

13.3 Configuration

Use the CFC editor to install the block in a cyclic interrupt OB (OB30 to OB38).

The DataCollector blocks are connected with the technological blocks through the "Snd_Data" and "Rcv_Data" structure connections.

The screenshot below shows a combination of I/O driver, DataCollector block and technological block (PC_FEEDER):

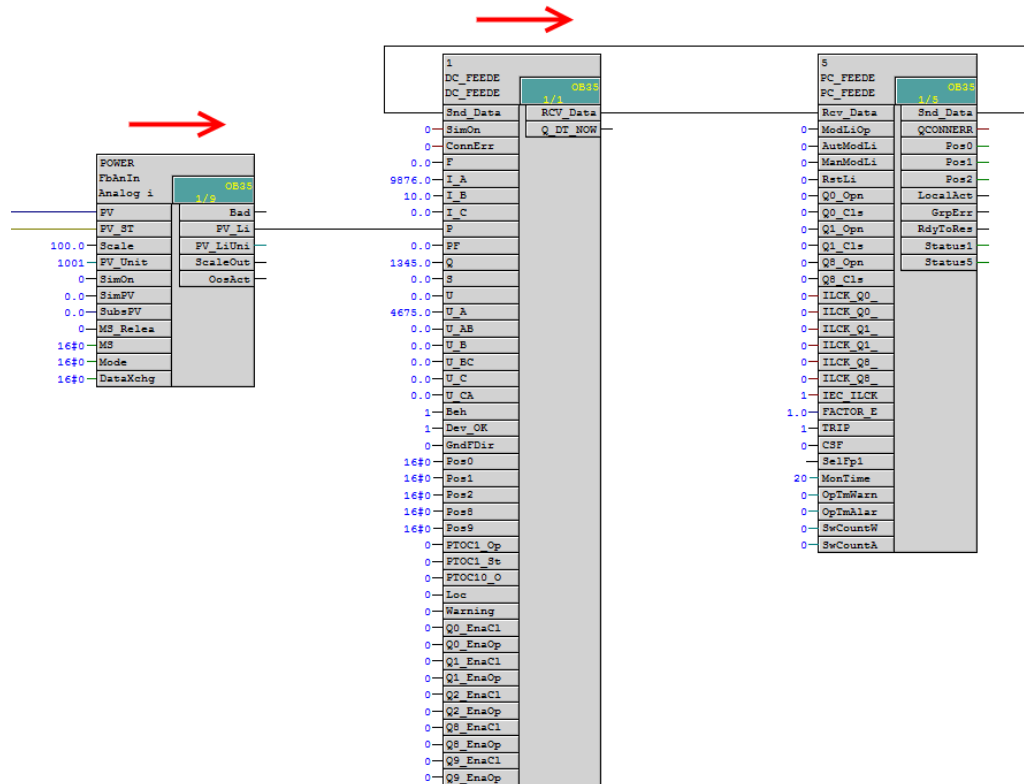


Figure 13-1 Example configuration DataCollector block

13.4 Simulation mode

The simulation mode is used to simulate feedbacks from electrical switches. This allows, for example, an application to be tested without connecting real switches.

The simulation mode is activated by the SimOn input.

When a command is sent by the technological block in simulation mode, the DataCollector block sends the corresponding feedback to the technological block.

If, for example, a switch is opened, the DataCollector block immediately gives the feedback "open".

13.5 Hiding non-existing values in the faceplate

Based on the signal status of the values at the block I/Os, it is possible to hide individual display elements in a faceplate. If the signal status is assigned the value 16#FF, then the associated display element is not shown in the faceplate.

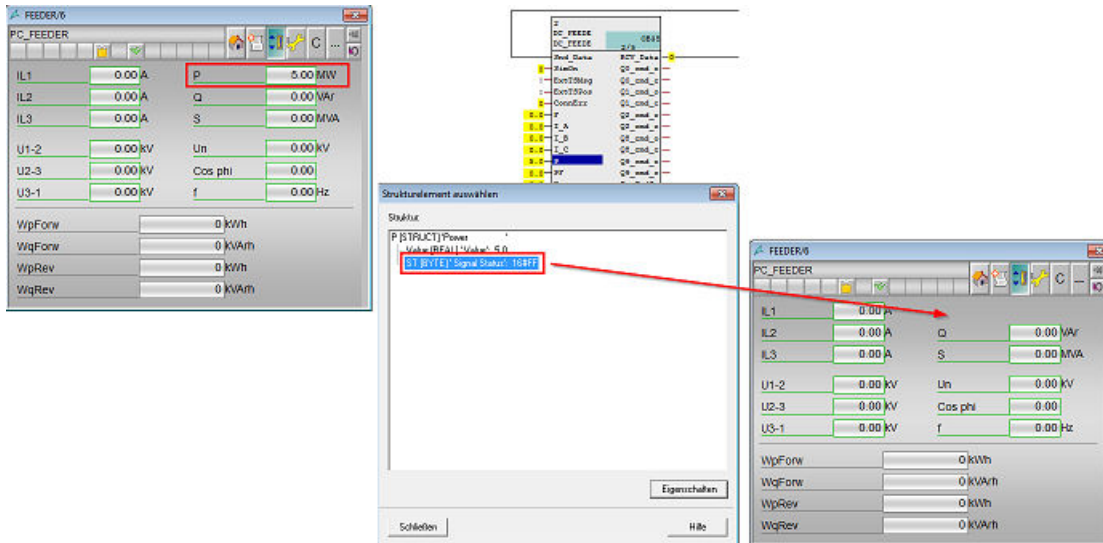


Figure 13-2 Hide DataCollector display

13.6 Generation of message time stamps

The DataCollector blocks offer the following options for using time stamps with alarm and switch position messages:

- System time of the automation system as basis for the time stamp
- Use of user-defined time stamps

13.6.1 Use of the system time

The DataCollector blocks automatically generate time stamps for the incoming messages. The system time is used as basis in the automation system.

To use the system time as time stamp for the messages to be generated, set the following values at the corresponding inputs:

- ExtTSMsg = 0 - Use internal system time for alarm messages
- ExtTSPos = 0 - Use internal system time for switch positions

13.6 Generation of message time stamps

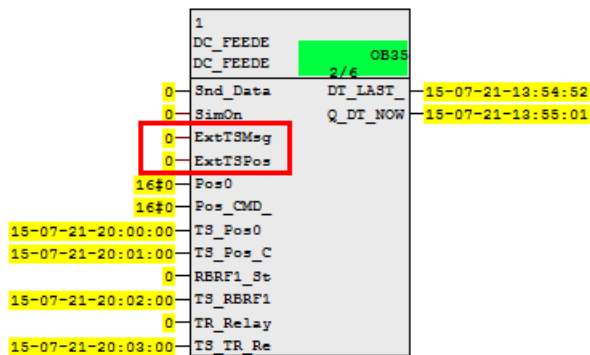


Figure 13-3 DC_FEEDER Internal time stamp

13.6.2 Use of user-defined time stamps

The DataCollector blocks give you the option of specifying your own time stamps for alarm and switch position messages.

To use these user-defined time stamps at the messages to be generated, set the following values at the corresponding inputs:

- ExtTSMsg = 1 - Use user-defined time stamps for alarm messages
- ExtTSPos = 1 - Use user-defined time stamps for switch positions

The time stamps for the corresponding messages can be specified at the associated TS_ (for example, TS_Pos0) inputs.

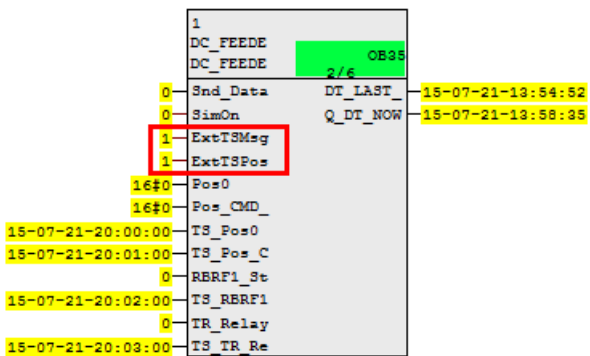


Figure 13-4 DC_FEEDER User-defined time stamp

13.7 DC_FEEDER

13.7.1 Description of DC_FEEDER

The DataCollector block DC_FEEDER (FB 2560) forms an interface to the technological block PC_FEEDER.

13.7.2 I/O lists

13.7.2.1 Inputs

I/O name	Description	Value	Data type
Snd_Data	Connect to DC_/SC_FEEDER		STRUCT
SimOn	Simulation on/off	1	BOOL
ExtTSMsg	1 = external time stamps used for messages 0 = creation of internal time stamps	FALSE	BOOL
ExtTSPos	1 = external time stamps used for switch positions 0 = creation of internal time stamps	FALSE	BOOL
ConnErr	Connection Error	0	BOOL
F	Frequency		STRUCT
F.Value		0.0	REAL
F.ST		16#80	BYTE
I_A	current phase A		STRUCT
I_A.Value		0.0	REAL
I_A.ST		16#80	BYTE
I_B	current phase		STRUCT
I_B.Value		0.0	REAL
I_B.ST		16#80	BYTE
I_C	current phase C		STRUCT
I_C.Value		0.0	REAL
I_C.ST		16#80	BYTE
P	Power		STRUCT
P.Value		0.0	REAL
P.ST		16#80	BYTE
PF	Power Factor		STRUCT
PF.Value		0.0	REAL
PF.ST		16#80	BYTE

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Q	Power Q		STRUCT
Q.Value		0.0	REAL
Q.ST		16#80	BYTE
S	Power S		STRUCT
S.Value		0.0	REAL
S.ST		16#80	BYTE
U	neutral voltage		STRUCT
U.Value		0.0	REAL
U.ST		16#80	BYTE
U_A	voltage phase A to ground		STRUCT
U_A.Value		0.0	REAL
U_A.ST		16#80	BYTE
U_AB	voltage phase A to phase B		STRUCT
U_AB.Value		0.0	REAL
U_AB.ST		16#80	BYTE
U_B	voltage phase B to ground		STRUCT
U_B.Value		0.0	REAL
U_B.ST		16#80	BYTE
U_BC	Voltage phase B to phase C		STRUCT
U_BC.Value		0.0	REAL
U_BC.ST		16#80	BYTE
U_C	voltage phase C to ground		STRUCT
U_C.Value		0.0	REAL
U_C.ST		16#80	BYTE
U_CA	voltage phase B to phase C		STRUCT
U_CA.Value		0.0	REAL
U_CA.ST		16#80	BYTE
REAL_sp1	Real Spare 1		STRUCT
REAL_sp1.Value		0.0	REAL
REAL_sp1.ST		16#80	BYTE
REAL_sp2	Real Spare 2		STRUCT
REAL_sp2.Value		0.0	REAL
REAL_sp2.ST		16#80	BYTE
REAL_sp3	Real Spare 3		STRUCT
REAL_sp3.Value		0.0	REAL
REAL_sp3.ST		16#80	BYTE
REAL_sp4	Real Spare 4		STRUCT
REAL_sp4.Value		0.0	REAL
REAL_sp4.ST		16#80	BYTE
REAL_sp5	Real Spare 5		STRUCT
REAL_sp5.Value		0.0	REAL
REAL_sp5.ST		16#80	BYTE
Beh	Behavior		STRUCT
Beh.Value		0	INT

Beh.ST		16#80	BYTE
Dev_OK	Device OK		STRUCT
Dev_OK.Value		0	INT
Dev_OK.ST		16#80	BYTE
GndFdir	Ground Fault direction		STRUCT
GndFDir.Value		2	INT
GndFDir.ST		16#80	BYTE
INT_sp1	Spare INT 1		STRUCT
INT_sp1.Value		0	INT
INT_sp1.ST		16#80	BYTE
INT_sp2	Spare INT 2		STRUCT
INT_sp2.Value		0	INT
INT_sp2.ST		16#80	BYTE
INT_sp3	Spare INT 3		STRUCT
INT_sp3.Value		0	INT
INT_sp3.ST		16#80	BYTE
INT_sp4	Spare INT 4		STRUCT
INT_sp4.Value		0	INT
INT_sp4.ST		16#80	BYTE
INT_sp5	Spare INT 5		STRUCT
INT_sp5.Value		0	INT
INT_sp5.ST		16#80	BYTE
WpForward	P energy forward		STRUCT
WpForward.Value		0	DINT
WpForward.ST		16#80	BYTE
WqForward	Q energy forward		STRUCT
WpForward.Value		0	DINT
WqForward.ST		16#80	BYTE
WpReverse	P energy reverse		STRUCT
WpReverse.Value		0	DINT
WpReverse.ST		16#80	BYTE
WqReverse	Q energy reverse		STRUCT
WpReverse.Value		0	DINT
WqReverse.ST		16#80	BYTE
SwCount	Switch Count Q0		STRUCT
SwCount.Value		0	DINT
SwCount.ST		16#80	BYTE
OpTm	Operating hours		STRUCT
OpTm.Value		0	DINT
OpTm.ST		16#80	BYTE
DINT_sp1	BS32 value 8 (spare)		STRUCT
DINT_sp1.Value		0	DINT
DINT_sp1.ST		16#80	BYTE
DINT_sp2	BS32 value 8 (spare)		STRUCT

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DINT_sp2.Value		0	DINT
DINT_sp2.ST		16#80	BYTE
Pos0	Position Q0 Circuit breaker		STRUCT
Pos0.Value		16#00000001	DWORD
Pos0.ST		16#80	BYTE
Pos1	Position Q1 Disconnect Switch		STRUCT
Pos1.Value		16#00000001	DWORD
Pos1.ST		16#80	BYTE
Pos2	Position Q2 Op		STRUCT
Pos2.Value		16#00000002	DWORD
Pos2.ST		16#80	BYTE
Pos8	Position Q8 Earth Switch		STRUCT
Pos8.Value		16#00000001	DWORD
Pos8.ST		16#80	BYTE
Pos9	Position Q9 Op		STRUCT
Pos9.Value		16#00000001	DWORD
Pos9.ST		16#80	BYTE
Pos_CMD_sp1	Position Q8 Earth switch		STRUCT
Pos_CMD_sp1.Value		16#00000000	DWORD
Pos_CMD_sp1.ST		16#80	BYTE
Pos_CMD_sp2	Position Q8 Earth switch		STRUCT
Pos_CMD_sp2.Value		16#00000000	DWORD
Pos_CMD_sp2.ST		16#80	BYTE
Pos_CMD_sp3	Position Q8 Ground Switch		STRUCT
Pos_CMD_sp3.Value		16#00000000	DWORD
Pos_CMD_sp3.ST		16#80	BYTE
TS_Pos0	Timestamp Position Q0 Circuit breaker		DT
TS_Pos1	Timestamp Position Q1 Disconnect Switch		DT
TS_Pos2	Timestamp Position Q2		DT
TS_Pos8	Timestamp Position Q8 Earth Switch		DT
TS_Pos9	Timestamp Position Q9		DT
TS_Pos_CMD_sp1	Timestamp Position Spare Switch 1		DT
TS_Pos_CMD_sp2	Timestamp Position Spare Switch 2		DT
TS_Pos_CMD_sp3	Timestamp Position Spare Switch 3		DT

AckCmd0	Acknowledge signal for Cmd0		STRUCT
AckCmd0.Value		0	BOOL
AckCmd0.ST		16#80	BYTE
AckCmd1	Acknowledge signal for Cmd1		STRUCT
AckCmd1.Value		0	BOOL
AckCmd1.ST		16#80	BYTE
AckCmd2	Acknowledge signal for Cmd2		STRUCT
AckCmd2.Value		0	BOOL
AckCmd2.ST		16#80	BYTE
AckCmd8	Acknowledge signal for Cmd8		STRUCT
AckCmd8.Value		0	BOOL
AckCmd8.ST		16#80	BYTE
AckCmd9	Acknowledge signal for Cmd9		STRUCT
AckCmd9.Value		0	BOOL
AckCmd9.ST		16#80	BYTE
AckCmdsp1	Acknowledge signal for Spare Command 1		STRUCT
AckCmdsp1.Value		0	BOOL
AckCmdsp1.ST		16#80	BYTE
AckCmdsp2	Acknowledge signal for Spare Command 2		STRUCT
AckCmdsp2.Value		0	BOOL
AckCmdsp2.ST		16#80	BYTE
AckCmdsp3	Acknowledge signal for Spare Command 3		STRUCT
AckCmdsp3.Value		0	BOOL
AckCmdsp3.ST		16#80	BYTE
PTOC1_Op	51 Overcurrent TRIP		STRUCT
PTOC1_Op.Value		1	BOOL
PTOC1_Op.ST		16#80	BYTE
PTOC1_Str	51 Overcurrent picked up		STRUCT
PTOC1_Str.Value		1	BOOL
PTOC1_Str.ST		16#80	BYTE
PTOC10_Op	67-1 Directional Overcurrent I > TRIP		STRUCT
PTOC10_Op.Value		0	BOOL
PTOC10_Op.ST		16#80	BYTE
PTOC10_Str	67-1 Directional Overcurrent I > picked up		STRUCT
PTOC10_Str.Value		0	BOOL
PTOC10_Str.ST		16#80	BYTE
PTOC11_Op	67-2 Directional Overcurrent I >>TRIP		STRUCT

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PTOC11_Op.Value		0	BOOL
PTOC11_Op.ST		16#80	BYTE
PTOC11_Str	67-2 Directional Overcurrent I >> picked up		STRUCT
PTOC11_Str.Value		0	BOOL
PTOC11_Str.ST		16#80	BYTE
PTOC12_Op	67N-1 Directional Overcurrent IE > TRIP		STRUCT
PTOC12_Op.Value		0	BOOL
PTOC12_Op.ST		16#80	BYTE
PTOC12_Str	67N-1 Directional Overcurrent IE > picked up		STRUCT
PTOC12_Str.Value		0	BOOL
PTOC12_Str.ST		16#80	BYTE
TS_PTOC1_Op	Timestamp EVID1_S1 51 Overcurrent TRIP		DT
TS_PTOC1_Str	Timestamp EVID1_S2 51 Overcurrent picked up		DT
TS_PTOC10_Op	Timestamp EVID1_S3 67-1 Directional Overcurrent I > TRIP		DT
TS_PTOC10_Str	Timestamp EVID1_S4 67-1 Directional Overcurrent I > picked up		DT
TS_PTOC11_Op	Timestamp EVID1_S5 67-2 Directional Overcurrent I >>TRIP		DT
TS_PTOC11_Str	Timestamp EVID1_S6 67-2 Directional Overcurrent I >> picked up		DT
TS_PTOC12_Op	Timestamp EVID1_S7 67N-1 Directional Overcurrent IE > TRIP		DT
TS_PTOC12_Str	Timestamp EVID1_S8 67N-1 Directional Overcurrent IE > picked up		DT
PTOC13_Op	67N-2 Directional Overcurrent IE >>TRIP		STRUCT
PTOC13_Op.Value		0	BOOL
PTOC13_Op.ST		16#80	BYTE
PTOC13_Str	67N-2 Directional Overcurrent IE >> picked up		STRUCT
PTOC13_Str.Value		0	BOOL
PTOC13_Str.ST		16#80	BYTE
PTOC18_Op	50-3 Overcurrent TRIP		STRUCT
PTOC18_Op.Value		0	BOOL
PTOC18_Op.ST		16#80	BYTE
PTOC18_Str	50-3 Overcurrent picked up		STRUCT
PTOC18_Str.Value		0	BOOL
PTOC18_Str.ST		16#80	BYTE
PTOC2_Op	51N Overcurrent TRIP		STRUCT
PTOC2_Op.Value		0	BOOL
PTOC2_Op.ST		16#80	BYTE

PTOC2_Str	51N Overcurrent picked up		STRUCT
PTOC2_Str.Value		0	BOOL
PTOC2_Str.ST		16#80	BYTE
PTOC3_Op	67-TOC Overcurrent TRIP		STRUCT
PTOC3_Op.Value		0	BOOL
PTOC3_Op.ST		16#80	BYTE
PTOC3_Str	67-TOC Overcurrent picked up		STRUCT
PTOC3_Str.Value		0	BOOL
PTOC3_Str.ST		16#80	BYTE
TS_PTOC13_Op	Timestamp EVID2_S1 67N-2 Directional Overcurrent IE >>TRIP		DT
TS_PTOC13_Str	Timestamp EVID2_S2 67N-2 Directional Overcurrent IE >> picked up		DT
TS_PTOC18_Op	Timestamp EVID2_S3 50-3 Overcurrent TRIP		DT
TS_PTOC18_Str	Timestamp EVID2_S4 50-3 Overcurrent picked up		DT
TS_PTOC2_Op	Timestamp EVID2_S5 51N Overcurrent TRIP		DT
TS_PTOC2_Str	Timestamp EVID2_S6 51N Overcurrent picked up		DT
TS_PTOC3_Op	Timestamp EVID2_S7 67-TOC Overcurrent TRIP		DT
TS_PTOC3_Str	Timestamp EVID2_S8 67-TOC Overcurrent picked up		DT
PTOC4_Op	67N-TOC Overcurrent TRIP		STRUCT
PTOC4_Op.Value		0	BOOL
PTOC4_Op.ST		16#80	BYTE
PTOC4_Str	67N-TOC Overcurrent picked up		STRUCT
PTOC4_Str.Value		0	BOOL
PTOC4_Str.ST		16#80	BYTE
PTOC5_Op	46-TOC Overcurrent TRIP		STRUCT
PTOC5_Op.Value		0	BOOL
PTOC5_Op.ST		16#80	BYTE
PTOC5_Str	46-TOC Overcurrent picked up		STRUCT
PTOC5_Str.Value		0	BOOL
PTOC5_Str.ST		16#80	BYTE
PTOC6_Op	50-1 Overcurrent I > TRIP		STRUCT
PTOC6_Op.Value		0	BOOL
PTOC6_Op.ST		16#80	BYTE
PTOC6_Str	50-1 Overcurrent I > picked up		STRUCT
PTOC6_Str.Value		0	BOOL
PTOC6_Str.ST		16#80	BYTE
PTOC7_Op	50-2 Overcurrent I >>TRIP		STRUCT
PTOC7_Op.Value		0	BOOL
PTOC7_Op.ST		16#80	BYTE

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PTOC7_Str	50-2 Overcurrent I >> picked up		STRUCT
PTOC7_Str.Value		0	BOOL
PTOC7_Str.ST		16#80	BYTE
TS_PTOC4_Op	Timestamp EVID3_S1 67N-TOC Overcurrent TRIP		DT
TS_PTOC4_Str	Timestamp EVID3_S2 67N-TOC Overcurrent picked up		DT
TS_PTOC5_Op	Timestamp EVID3_S3 46-TOC Overcurrent TRIP		DT
TS_PTOC5_Str	Timestamp EVID3_S4 46-TOC Overcurrent picked up		DT
TS_PTOC6_Op	Timestamp EVID3_S5 50-1 Overcurrent I > TRIP		DT
TS_PTOC6_Str	Timestamp EVID3_S6 50-1 Overcurrent I > picked up		DT
TS_PTOC7_Op	Timestamp EVID3_S7 50-2 Overcurrent I >>TRIP		DT
TS_PTOC7_Str	Timestamp EVID3_S8 50-2 Overcurrent I >> picked up		DT
PTOC8_Op	50N-1 Overcurrent IE > TRIP		STRUCT
PTOC8_Op.Value		0	BOOL
PTOC8_Op.ST		16#80	BYTE
PTOC8_Str	50N-1 Overcurrent IE > picked up		STRUCT
PTOC8_Str.Value		0	BOOL
PTOC8_Str.ST		16#80	BYTE
PTOC9_Op	50N-2 Overcurrent IE >> TRIP		STRUCT
PTOC9_Op.Value		0	BOOL
PTOC9_Op.ST		16#80	BYTE
PTOC9_Str	50N-2 Overcurrent IE >> picked up		STRUCT
PTOC9_Str.Value		0	BOOL
PTOC9_Str.ST		16#80	BYTE
PU_Relay	Trip signal		STRUCT
PU_Relay.Value		0	BOOL
PU_Relay.ST		16#80	BYTE
RBRF1_OpEx	Breaker failure TRIP T1		STRUCT
RBRF1_OpEx.Value		0	BOOL
RBRF1_OpEx.ST		16#80	BYTE
RBRF1_OpIn	Breaker failure TRIP T2		STRUCT
RBRF1_OpIn.Value		0	BOOL
RBRF1_OpIn.ST		16#80	BYTE
RBRF1_Str	Breaker failure picked up		STRUCT
RBRF1_Str.Value		0	BOOL
RBRF1_Str.ST		16#80	BYTE
TS_PTOC8_Op			DT
TS_PTOC8_Str	Timestamp EVID4_S1 50N-1 Overcurrent IE > TRIP		DT

TS_PTOC9_Op	Timestamp EVID4_S2 50N-1 Overcurrent IE > picked up		DT
TS_PTOC9_Str	Timestamp EVID4_S3 50N-2 Overcurrent IE >> TRIP		DT
TS_PU_Relay	Timestamp EVID4_S4 50N-2 Overcurrent IE >> picked up		DT
TS_RBRF1_OpEx	Timestamp EVID4_S5 trip signal		DT
TS_RBRF1_OpIn	Timestamp EVID4_S6 Breaker failure TRIP T1		DT
TS_RBRF1_Str	Timestamp EVID4_S6 Breaker failure TRIP T1		DT
TR_Relay	Trigger signal		STRUCT
TR_Relay.Value		0	BOOL
TR_Relay.ST		16#80	BYTE
PDIF1_Op	Differential protection IDIFF> TRIP		STRUCT
PDIF1_Op.Value		0	BOOL
PDIF1_Op.ST		16#80	BYTE
PDIF1_Str	Differential protection IDIFF> picked up		STRUCT
PDIF1_Str.Value		0	BOOL
PDIF1_Str.ST		16#80	BYTE
PDIF2_Op	Differential protection IDIFF>> TRIP		STRUCT
PDIF2_Op.Value		0	BOOL
PDIF2_Op.ST		16#80	BYTE
PDIF2_Str	Differential protection IDIFF>> picked up		STRUCT
PDIF2_Str.Value		0	BOOL
PDIF2_Str.ST		16#80	BYTE
PDIS1_Str	Impedance protection Z1 picked up		STRUCT
PDIS1_Str.Value		0	BOOL
PDIS1_Str.ST		16#80	BYTE
PDIS2_Str	Impedance protection Z2 picked up		STRUCT
PDIS2_Str.Value		0	BOOL
PDIS2_Str.ST		16#80	BYTE
PDIS3_Str	Impedance protection Z1B picked up		STRUCT
PDIS3_Str.Value		0	BOOL
PDIS3_Str.ST		16#80	BYTE
TS_TR_Relay	Timestamp EVID5_S1 trigger signal		DT
TS_PDIF1_Op	Timestamp EVID5_S2 Differential protection IDIFF> TRIP		DT
TS_PDIF1_Str	Timestamp EVID5_S3 Differential protection IDIFF> picked up		DT
TS_PDIF2_Op	Timestamp EVID5_S4 Differential protection IDIFF>> TRIP		DT
TS_PDIF2_Str	Timestamp EVID5_S5 Differential protection IDIFF>> picked up		DT

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TS_PDIS1_Str	Timestamp EVID5_S6 Impedance protection Z1 picked up		DT
TS_PDIS2_Str	Timestamp EVID5_S7 Impedance protection Z2 picked up		DT
TS_PDIS3_Str	Timestamp EVID5_S8 Impedance protection Z1B picked up		DT
PDIS4_Str	Impedance protection general picked up		STRUCT
PDIS4_Str.Value		0	BOOL
PDIS4_Str.ST		16#80	BYTE
PDIS1_Op	Impedance protection Z1 TRIP		STRUCT
PDIS1_Op.Value		0	BOOL
PDIS1_Op.ST		16#80	BYTE
PDIS2_Op	Impedance protection Z2 TRIP		STRUCT
PDIS2_Op.Value		0	BOOL
PDIS2_Op.ST		16#80	BYTE
PDIS3_Op	Impedance protection Z1B TRIP		STRUCT
PDIS3_Op.Value		0	BOOL
PDIS3_Op.ST		16#80	BYTE
PDIS4_Op	Impedance protection general TRIP		STRUCT
PDIS4_Op.Value		0	BOOL
PDIS4_Op.ST		16#80	BYTE
PDUP1_Op	Underexcitation protection Char. 1 TRIP		STRUCT
PDUP1_Op.Value		0	BOOL
PDUP1_Op.ST		16#80	BYTE
PDUP2_Op	Underexcitation protection Char. 2 TRIP		STRUCT
PDUP2_Op.Value		0	BOOL
PDUP2_Op.ST		16#80	BYTE
PDUP3_Op	Underexcitation protection Char. 3 TRIP		STRUCT
PDUP3_Op.Value		0	BOOL
PDUP3_Op.ST		16#80	BYTE
TS_PDIS4_Str	Timestamp EVID6_S1 Impedance protection general picked up		DT
TS_PDIS1_Op	Timestamp EVID6_S2 Impedance protection Z1 TRIP		DT
TS_PDIS2_Op	Timestamp EVID6_S3 Impedance protection Z2 TRIP		DT
TS_PDIS3_Op	Timestamp EVID6_S4 Impedance protection Z1B TRIP		DT
TS_PDIS4_Op	Timestamp EVID6_S5 Impedance protection general TRIP		DT
TS_PDUP1_Op	Timestamp EVID6_S6 Underexcitation protection Char. 1 TRIP		DT

TS_PDUP2_Op	Timestamp EVID6_S7 Underexcitation protection Char. 2 TRIP		DT
TS_PDUP3_Op	Timestamp EVID6_S8 Underexcitation protection Char. 3 TRIP		DT
PTOV1_Op	Overvoltage U> TRIP		STRUCT
PTOV1_Op.Value		0	BOOL
PTOV1_Op.ST		16#80	BYTE
PTOV1_Str	Overvoltage U> picked up		STRUCT
PTOV1_Str.Value		0	BOOL
PTOV1_Str.ST		16#80	BYTE
PTOV2_Op	Overvoltage U>> TRIP		STRUCT
PTOV2_Op.Value		0	BOOL
PTOV2_Op.ST		16#80	BYTE
PTOV2_Str	Overvoltage U>> picked up		STRUCT
PTOV2_Str.Value		0	BOOL
PTOV2_Str.ST		16#80	BYTE
PTUF1_Op	Frequency 1 Under Range TRIP		STRUCT
PTUF1_Op.Value		0	BOOL
PTUF1_Op.ST		16#80	BYTE
PTUF1_Str	Frequency 1 Under Range picked up		STRUCT
PTUF1_Str.Value		0	BOOL
PTUF1_Str.ST		16#80	BYTE
PTUF2_Op	Frequency 2 Under Range TRIP		STRUCT
PTUF2_Op.Value		0	BOOL
PTUF2_Op.ST		16#80	BYTE
PTUF2_Str	Frequency 2 Under Range picked up		STRUCT
PTUF2_Str.Value		0	BOOL
PTUF2_Str.ST		16#80	BYTE
TS_PTOV1_Op	Timestamp EVID7_S1 Overvoltage U> TRIP		DT
TS_PTOV1_Str	Timestamp EVID7_S2 Overvoltage U> picked up		DT
TS_PTOV2_Op	Timestamp EVID7_S3 Overvoltage U>> TRIP		DT
TS_PTOV2_Str	Timestamp EVID7_S4 Overvoltage U>> picked up		DT
TS_PTUF1_Op	Timestamp EVID7_S5 Frequency 1 Under Range TRIP		DT
TS_PTUF1_Str	Timestamp EVID7_S6 Frequency 1 Under Range picked up		DT
TS_PTUF2_Op	Timestamp EVID7_S7 Frequency 2 Under Range TRIP		DT
TS_PTUF2_Str	Timestamp EVID7_S8 Frequency 2 Under Range picked up		DT
PTUV1_Op	Undervoltage U< TRIP		STRUCT
PTUV1_Op.Value		0	BOOL

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PTUV1_Op.ST		16#80	BYTE
PTUV1_Str	Undervoltage U< picked up		STRUCT
PTUV1_Str.Value		0	BOOL
PTUV1_Str.ST		16#80	BYTE
PTUV2_Op	Undervoltage U<< TRIP		STRUCT
PTUV2_Op.Value		0	BOOL
PTUV2_Op.ST		16#80	BYTE
PTUV2_Str	Undervoltage U<< picked up		STRUCT
PTUV2_Str.Value		0	BOOL
PTUV2_Str.ST		16#80	BYTE
PVOC2_Op	Inadvert. Energ. Prot. TRIP		STRUCT
PVOC2_Op.Value		0	BOOL
PVOC2_Op.ST		16#80	BYTE
PVOC2_Str	Inadvert. Energ. Prot. Picked up		STRUCT
PVOC2_Str.Value		0	BOOL
PVOC2_Str.ST		16#80	BYTE
PDOP1_Op	Reverse power: TRIP		STRUCT
PDOP1_Op.Value		0	BOOL
PDOP1_Op.ST		16#80	BYTE
GAPC1_Op	External Trip 1 TRIP		STRUCT
GAPC1_Op.Value		0	BOOL
GAPC1_Op.ST		16#80	BYTE
TS_PTUV1_Op	Timestamp EVID8_S1 Undervoltage U< TRIP		DT
TS_PTUV1_Str	Timestamp EVID8_S2 Undervoltage U< picked up		DT
TS_PTUV2_Op	Timestamp EVID8_S3 Undervoltage U<< TRIP		DT
TS_PTUV2_Str	Timestamp EVID8_S4 Undervoltage U<< picked up		DT
TS_PVOC2_Op	Timestamp EVID8_S5 Inadvert. Energ. Prot. TRIP		DT
TS_PVOC2_Str	Timestamp EVID8_S6 Inadvert. Energ. Prot. Picked up		DT
TS_PDOP1_Op	Timestamp EVID8_S7 Reverse power: TRIP		DT
TS_GAPC1_Op	Timestamp EVID8_S8 External Trip 1 TRIP		DT
GAPC1_Str	External Trip 1 picked up		STRUCT
GAPC1_Str.Value		0	BOOL
GAPC1_Str.ST		16#80	BYTE
PARSET_A	Parameter Set A		STRUCT
PARSET_A.Value		0	BOOL
PARSET_A.ST		16#80	BYTE
PARSET_B	Parameter Set B		STRUCT
PARSET_B.Value		0	BOOL

PARSET_B.ST		16#80	BYTE
PARSET_C	Parameter Set C		STRUCT
PARSET_C.Value		0	BOOL
PARSET_C.ST		16#80	BYTE
PARSET_D	Parameter Set D		STRUCT
PARSET_D.Value		0	BOOL
PARSET_D.ST		16#80	BYTE
Error	Group Alarm		STRUCT
Error.Value		0	BOOL
Error.ST		16#80	BYTE
Loc	Local		STRUCT
Loc.Value		0	BOOL
Loc.ST		16#80	BYTE
Warning	Group Warning		STRUCT
Warning.Value		0	BOOL
Warning.ST		16#80	BYTE
TS_GAPC1_Str	Timestamp EVID9_S1 External Trip 1 picked up		DT
TS_PARSET_A	Timestamp EVID9_S2 Parameter Set A		DT
TS_PARSET_B	Timestamp EVID9_S2 Parameter Set B		DT
TS_PARSET_C	Timestamp EVID9_S2 Parameter Set C		DT
TS_PARSET_D	Timestamp EVID9_S2 Parameter Set D		DT
TS_Error	Timestamp EVID9_S6 Group Alarm		DT
TS_Loc	Timestamp EVID9_S7 Local		DT
TS_Warning	Timestamp EVID9_S8 Group Warning		DT
Q0_EnaCls	Interlocking: Enable Q0 Close command		STRUCT
Q0_EnaCls.Value		1	BOOL
Q0_EnaCls.ST		16#80	BYTE
Q0_EnaOpn	Interlocking: Enable Q0 Open command		STRUCT
Q0_EnaOpn.Value		1	BOOL
Q0_EnaOpn.ST		16#80	BYTE
Q1_EnaCls	Interlocking: Enable Q1 Close command		STRUCT
Q1_EnaCls.Value		1	BOOL
Q1_EnaCls.ST		16#80	BYTE
Q1_EnaOpn	Interlocking: Enable Q1 Open command		STRUCT
Q1_EnaOpn.Value		1	BOOL
Q1_EnaOpn.ST		16#80	BYTE
Q2_EnaCls	Interlocking: Enable Q2 Close command		STRUCT
Q2_EnaCls.Value		1	BOOL
Q2_EnaCls.ST		16#80	BYTE

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Q2_EnaOpn	Interlocking: Enable Q2 Open command		STRUCT
Q2_EnaOpn.Value		1	BOOL
Q2_EnaOpn.ST		16#80	BYTE
Q8_EnaCls	Interlocking: Enable Q8 Close command		STRUCT
Q8_EnaCls.Value		1	BOOL
Q8_EnaCls.ST		16#80	BYTE
Q8_EnaOpn	Interlocking: Enable Q8 Open command		STRUCT
Q8_EnaOpn.Value		1	BOOL
Q8_EnaOpn.ST		16#80	BYTE
TS_Q0_EnaCls	Timestamp EVID10_S1 Interlocking: Enable Q0 Close command		DT
TS_Q0_EnaOpn	Timestamp EVID10_S2 Interlocking: Enable Q0 Open command		DT
TS_Q1_EnaCls	Timestamp EVID10_S3 Interlocking: Enable Q1 Close command		DT
TS_Q1_EnaOpn	Timestamp EVID10_S4 Interlocking: Enable Q1 Open command		DT
TS_Q2_EnaCls	Timestamp EVID10_S5 Interlocking: Enable Q2 Close command		DT
TS_Q2_EnaOpn	Timestamp EVID10_S6 Interlocking: Enable Q2 Open command		DT
TS_Q8_EnaCls	Timestamp EVID10_S7 Interlocking: Enable Q8 Close command		DT
TS_Q8_EnaOpn	Timestamp EVID10_S8 Interlocking: Enable Q8 Open command		DT
Q9_EnaCls	Interlocking: Enable Q9 Close command		STRUCT
Q9_EnaCls.Value		1	BOOL
Q9_EnaCls.ST		16#80	BYTE
Q9_EnaOpn	Interlocking: Enable Q9 Open command		STRUCT
Q9_EnaOpn.Value		1	BOOL
Q9_EnaOpn.ST		16#80	BYTE
ALM_sp1	Spare Alarm 1		STRUCT
ALM_sp1.Value		0	BOOL
ALM_sp1.ST		16#80	BYTE
ALM_sp2	Spare Alarm 2		STRUCT
ALM_sp2.Value		0	BOOL
ALM_sp2.ST		16#80	BYTE
ALM_sp3	Spare Alarm 3		STRUCT
ALM_sp3.Value		0	BOOL
ALM_sp3.ST		16#80	BYTE
ALM_sp4	Spare Alarm 4		STRUCT
ALM_sp4.Value		0	BOOL

ALM_sp4.ST		16#80	BYTE
ALM_sp5	Spare Alarm 5		STRUCT
ALM_sp5.Value		0	BOOL
ALM_sp5.ST		16#80	BYTE
ALM_sp6	Spare Alarm 6		STRUCT
ALM_sp6.Value		0	BOOL
ALM_sp6.ST		16#80	BYTE
TS_Q9_EnaCls	Timestamp EVID11_S1 Interlocking: Enable Q9 Close command		DT
TS_Q9_EnaOpn	Timestamp EVID11_S2 Interlocking: Enable Q9 Open command		DT
TS_ALM_sp1	Timestamp EVID11_S3 Spare Alarm 1		DT
TS_ALM_sp2	Timestamp EVID11_S4 Spare Alarm 2		DT
TS_ALM_sp3	Timestamp EVID11_S5 Spare Alarm 3		DT
TS_ALM_sp4	Timestamp EVID11_S6 Spare Alarm 4		DT
TS_ALM_sp5	Timestamp EVID11_S7 Spare Alarm 5		DT
TS_ALM_sp6	Timestamp EVID11_S8 Spare Alarm 6		DT
ALM_sp7	Spare Alarm 7		STRUCT
ALM_sp7.Value		0	BOOL
ALM_sp7.ST		16#80	BYTE
ALM_sp8	Spare Alarm 8		STRUCT
ALM_sp8.Value		0	BOOL
ALM_sp8.ST		16#80	BYTE
ALM_sp9	Spare Alarm 9		STRUCT
ALM_sp9.Value		0	BOOL
ALM_sp9.ST		16#80	BYTE
ALM_sp10	Spare Alarm 10		STRUCT
ALM_sp10.Value		0	BOOL
ALM_sp10.ST		16#80	BYTE
ALM_sp11	Spare Alarm 11		STRUCT
ALM_sp11.Value		0	BOOL
ALM_sp11.ST		16#80	BYTE
ALM_sp12	Spare Alarm 12		STRUCT
ALM_sp12.Value		0	BOOL
ALM_sp12.ST		16#80	BYTE
ALM_sp13	Spare Alarm 13		STRUCT
ALM_sp13.Value		0	BOOL
ALM_sp13.ST		16#80	BYTE
ALM_sp14	Spare Alarm 14		STRUCT
ALM_sp14.Value		0	BOOL
ALM_sp14.ST		16#80	BYTE
TS_ALM_sp7	TS EVID12_S1 Spare Alarm 7		DT
TS_ALM_sp8	TS EVID12_S2 Spare Alarm 8		DT
TS_ALM_sp9	TS EVID12_S3 Spare Alarm 9		DT

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TS_ALM_sp10	TS EVID12_S4 Spare Alarm 10		DT
TS_ALM_sp11	TS EVID12_S5 Spare Alarm 11		DT
TS_ALM_sp12	TS EVID12_S6 Spare Alarm 12		DT
TS_ALM_sp13	TS EVID12_S7 Spare Alarm 13		DT
TS_ALM_sp14	TS EVID12_S8 Spare Alarm 14		DT

13.7.2.2 Outputs

I/O name	Description	Value	Data type
RCV_Data	Connect to DC_/SC_FEEDER		STRUCT
MSG_E_S_1	ALARM_8P Error	0	BOOL
MSG_U_S_1	1=Message Suppression active	0	BOOL
MSG_S_S_1	ALARM_8P: STATUS output	16#0000	WORD
MSG_A_S_1	ALARM_8P: ACK_STATE output	16#0000	WORD
MSG_E_S_2	ALARM_8P Error	0	BOOL
MSG_U_S_2	1=Message Suppression active	0	BOOL
MSG_S_S_2	ALARM_8P: STATUS output	16#0000	WORD
MSG_A_S_2	ALARM_8P: ACK_STATE output	16#0000	WORD
MSG_E_S_3	ALARM_8P Error	0	BOOL
MSG_U_S_3	1=Message Suppression active	0	BOOL
MSG_S_S_3	ALARM_8P: STATUS output	16#0000	WORD
MSG_A_S_3	ALARM_8P: ACK_STATE output	16#0000	WORD
MSG_E_S_4	ALARM_8P Error	0	BOOL
MSG_U_S_4	1=Message Suppression active	0	BOOL
MSG_S_S_4	ALARM_8P: STATUS output	16#0000	WORD
MSG_A_S_4	ALARM_8P: ACK_STATE output	16#0000	WORD
MSG_E_S_5	ALARM_8P Error	0	BOOL
MSG_U_S_5	1=Message Suppression active	0	BOOL
MSG_S_S_5	ALARM_8P: STATUS output	16#0000	WORD
MSG_A_S_5	ALARM_8P: ACK_STATE output	16#0000	WORD
MSG_E_S_6	ALARM_8P Error	0	BOOL
MSG_U_S_6	1=Message Suppression active	0	BOOL
MSG_S_S_6	ALARM_8P: STATUS output	16#0000	WORD
MSG_A_S_6	ALARM_8P: ACK_STATE output	16#0000	WORD
MSG_E_S_7	ALARM_8P Error	0	BOOL
MSG_U_S_7	1=Message Suppression active	0	BOOL
MSG_S_S_7	ALARM_8P: STATUS output	16#0000	WORD
MSG_A_S_7	ALARM_8P: ACK_STATE output	16#0000	WORD
MSG_E_S_8	ALARM_8P Error	0	BOOL

MSG_U_S_8	1=Message Suppression active	0	BOOL
MSG_S_S_8	ALARM_8P: STATUS output	16#0000	WORD
MSG_A_S_8	ALARM_8P: ACK_STATE output	16#0000	WORD
MSG_E_S_9	ALARM_8P Error	0	BOOL
MSG_U_S_9	1=Message Suppression active	0	BOOL
MSG_S_S_9	ALARM_8P: STATUS output	16#0000	WORD
MSG_A_S_9	ALARM_8P: ACK_STATE output	16#0000	WORD
MSG_E_S_10	ALARM_8P Error	0	BOOL
MSG_U_S_10	1=Message Suppression active	0	BOOL
MSG_S_S_10	ALARM_8P: STATUS output	16#0000	WORD
MSG_A_S_10	ALARM_8P: ACK_STATE output	16#0000	WORD
MSG_E_S_11	ALARM_8P Error	0	BOOL
MSG_U_S_11	1=Message Suppression active	0	BOOL
MSG_S_S_11	ALARM_8P: STATUS output	16#0000	WORD
MSG_A_S_11	ALARM_8P: ACK_STATE output	16#0000	WORD
MSG_E_S_12	ALARM_8P Error	0	BOOL
MSG_U_S_12	1=Message Suppression active	0	BOOL
MSG_S_S_12	ALARM_8P: STATUS output	16#0000	WORD
MSG_A_S_12	ALARM_8P: ACK_STATE output	16#0000	WORD
MSG_E_D_1	ALARM_8P Error	0	BOOL
MSG_U_D_1	1=Message Suppression active	0	BOOL
MSG_S_D_1	ALARM_8P: STATUS output	16#0000	WORD
MSG_A_D_1	ALARM_8P: ACK_STATE output	16#0000	WORD
MSG_E_D_2	ALARM_8P Error	0	BOOL
MSG_U_D_2	1=Message Suppression active	0	BOOL
MSG_S_D_2	ALARM_8P: STATUS output	16#0000	WORD
MSG_A_D_2	ALARM_8P: ACK_STATE output	16#0000	WORD
Q_DT_NOW	Last Error	1990-01-01-0:00:00	DATE_AND_TIME
Q0_cmd_op	Q0 Command open	0	BOOL
Q0_cmd_cl	Q0 Command close	0	BOOL
Q1_cmd_op	Q1 Command open	0	BOOL
Q1_cmd_cl	Q1 Command close	0	BOOL
Q2_cmd_op	Q2 Command open	0	BOOL
Q2_cmd_cl	Q2 Command close	0	BOOL
Q8_cmd_op	Q8 Command open	0	BOOL
Q8_cmd_cl	Q8 Command close	0	BOOL
Q9_cmd_op	Q9 Command open	0	BOOL
Q9_cmd_cl	Q9 Command close	0	BOOL
DT_LAST_MSG	Timestamp last error		DT
Q_DT_NOW	Timestamp date and time of PLC		DT

13.8 DC_TRAFO

13.8.1 Description of DC_TRAFO

The DataCollector block DC_TRAFO (FB 2562) forms an interface to the technological block PC_TRAFO.

13.8.2 I/O lists

13.8.2.1 Inputs

I/O name	Description	Value	Data type
Snd_Data	Connect to DC_/SC_FEEDER		STRUCT
SimOn	Simulation on/off	1	BOOL
ExtTSMsg	1 = external time stamps used for messages 0 = creation of internal time stamps	FALSE	BOOL
ExtTSPos	1 = external time stamps used for switch positions 0 = creation of internal time stamps	FALSE	BOOL
ConnErr	Connection Error	0	BOOL
F	Frequency		STRUCT
F.Value		0.0	REAL
F.ST		16#80	BYTE
P	Power		STRUCT
P.Value		0.0	REAL
P.ST		16#80	BYTE
Q	Power Q		STRUCT
Q.Value		0.0	REAL
Q.ST		16#80	BYTE
S	Power S		STRUCT
S.Value		0.0	REAL
S.ST		16#80	BYTE
PF	Power Factor		STRUCT
PF.Value		0.0	REAL
PF.ST		16#80	BYTE
U	Neutral voltage		STRUCT
U.Value		0.0	REAL
U.ST		16#80	BYTE

U_AB	Voltage phase A to phase B		STRUCT
U_AB.Value		0.0	REAL
U_AB.ST		16#80	BYTE
U_BC	Voltage phase B to phase C		STRUCT
U_BC.Value		0.0	REAL
U_BC.ST		16#80	BYTE
U_CA	voltage phase B to phase C		STRUCT
U_CA.Value		0.0	REAL
U_CA.ST		16#80	BYTE
I1_A	current 1 phase A		STRUCT
I1_A.Value		0.0	REAL
I1_A.ST		16#80	BYTE
I1_B	current 1 phase B		STRUCT
I1_B.Value		0.0	REAL
I1_B.ST		16#80	BYTE
I1_C	current 1 phase C		STRUCT
I1_C.Value		0.0	REAL
I1_C.ST		16#80	BYTE
I2_A	current 2 phase A		STRUCT
I2_A.Value		0.0	REAL
I2_A.ST		16#80	BYTE
I2_B	current 2 phase B		STRUCT
I2_B.Value		0.0	REAL
I2_B.ST		16#80	BYTE
I2_C	current 2 phase C		STRUCT
I2_C.Value		0.0	REAL
I2_C.ST		16#80	BYTE
I3_A	current 3 phase A		STRUCT
I3_A.Value		0.0	REAL
I3_A.ST		16#80	BYTE
I3_B	current 3 phase B		STRUCT
I3_B.Value		0.0	REAL
I3_B.ST		16#80	BYTE
I3_C	current 3 phase C		STRUCT
I3_C.Value		0.0	REAL
I3_C.ST		16#80	BYTE
THETA	THETA		STRUCT
THETA.Value		0.0	REAL
THETA.ST		16#80	BYTE
THETA_L1	THETA / Theta Trip L1		STRUCT
THETA_L1.Value		0.0	REAL
THETA_L1.ST		16#80	BYTE
THETA_L2	THETA / Theta Trip L2		STRUCT
THETA_L2.Value		0.0	REAL

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THETA_L2.ST		16#80	BYTE
THETA_L3	THETA / Theta Trip L3		STRUCT
THETA_L3.Value		0.0	REAL
THETA_L3.ST		16#80	BYTE
REAL_sp1	Real Spare 1		STRUCT
REAL_sp1.Value		0.0	REAL
REAL_sp1.ST		16#80	BYTE
REAL_sp2	Real Spare 2		STRUCT
REAL_sp2.Value		0.0	REAL
REAL_sp2.ST		16#80	BYTE
REAL_sp3	Real Spare 3		STRUCT
REAL_sp3.Value		0.0	REAL
REAL_sp3.ST		16#80	BYTE
REAL_sp4	Real Spare 4		STRUCT
REAL_sp4.Value		0.0	REAL
REAL_sp4.ST		16#80	BYTE
REAL_sp5	Real Spare 5		STRUCT
REAL_sp5.Value		0.0	REAL
REAL_sp5.ST		16#80	BYTE
Beh	Behavior		STRUCT
Beh.Value		0	INT
Beh.ST		16#80	BYTE
Dev_OK	Device OK		STRUCT
Dev_OK.Value		0	INT
Dev_OK.ST		16#80	BYTE
INT_sp1	Spare INT 1		STRUCT
INT_sp1.Value		0	INT
INT_sp1.ST		16#80	BYTE
INT_sp2	Spare INT 2		STRUCT
INT_sp2.Value		0	INT
INT_sp2.ST		16#80	BYTE
INT_sp3	Spare INT 3		STRUCT
INT_sp3.Value		0	INT
INT_sp3.ST		16#80	BYTE
INT_sp4	Spare INT 4		STRUCT
INT_sp4.Value		0	INT
INT_sp4.ST		16#80	BYTE
INT_sp5	Spare INT 5		STRUCT
INT_sp5.Value		0	INT
INT_sp5.ST		16#80	BYTE
INT_sp6	Spare INT 6		STRUCT
INT_sp6.Value		0	INT
INT_sp6.ST		16#80	BYTE
WpForward	P energy forward		STRUCT

WpForward.Value		0	DINT
WpForward.ST		16#80	BYTE
WqForward	Q energy forward		STRUCT
WpReverse.Value		0	DINT
WqForward.ST		16#80	BYTE
WpReverse	P energy reverse		STRUCT
WpReverse.Value		0	DINT
WpReverse.ST		16#80	BYTE
WqReverse	Q energy forward		STRUCT
WpReverse.Value		0	DINT
WqReverse.ST		16#80	BYTE
SwCount	Switch Count Q0		STRUCT
SwCount.Value		0	DINT
SwCount.ST		16#80	BYTE
OpTm	Operating hours		STRUCT
OpTm.Value		0	DINT
OpTm.ST		16#80	BYTE
DINT_sp1	BS32 value 8 (spare)		STRUCT
DINT_sp1.Value		0	DINT
DINT_sp1.ST		16#80	BYTE
DINT_sp2	BS32 value 8 (spare)		STRUCT
DINT_sp2.Value		0	DINT
DINT_sp2.ST		16#80	BYTE
Pos_CMD_sp1	Position Spare Command 1		STRUCT
Pos_CMD_sp1.Value		16#00000000 2	DWORD
Pos_CMD_sp1.ST		16#80	BYTE
Pos_CMD_sp2	Position Spare Command 2		STRUCT
Pos_CMD_sp2.Value		16#00000000 2	DWORD
Pos_CMD_sp2.ST		16#80	BYTE
Pos_CMD_sp3	Position Spare Command 3		STRUCT
Pos_CMD_sp3.Value		16#00000000 2	DWORD
Pos_CMD_sp3.ST		16#80	BYTE
Pos_CMD_sp4	Position Spare Command 4		STRUCT
Pos_CMD_sp4.Value		16#00000000 2	DWORD
Pos_CMD_sp4.ST		16#80	BYTE
TS_Pos_CMD_sp1	Timestamp Position Spare Switch 1		DT
TS_Pos_CMD_sp2	Timestamp Position Spare Switch 2		DT
TS_Pos_CMD_sp3	Timestamp Position Spare Switch 3		DT
TS_Pos_CMD_sp4	Timestamp Position Spare Switch 4		DT
AckCmdsp1	Acknowledge signal for Spare Command 1		STRUCT

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AckCmdsp1.Value		0	BOOL
AckCmdsp1.ST		16#80	BYTE
AckCmdsp2	Acknowledge signal for Spare Command 2		STRUCT
AckCmdsp2.Value		0	BOOL
AckCmdsp2.ST		16#80	BYTE
AckCmdsp3	Acknowledge signal for Spare Command 3		STRUCT
AckCmdsp3.Value		0	BOOL
AckCmdsp3.ST		16#80	BYTE
AckCmdsp4	Acknowledge signal for Spare Command 4		STRUCT
AckCmdsp4.Value		0	BOOL
AckCmdsp4.ST		16#80	BYTE
PTOC1_Str	Overcurrent I > picked up		STRUCT
PTOC1_Str.Value		0	BOOL
PTOC1_Str.ST		16#80	BYTE
PTOC1_Op	Overcurrent I > TRIP		STRUCT
PTOC1_Op.Value		0	BOOL
PTOC1_Op.ST		16#80	BYTE
PTOC2_Str	Overcurrent I >> picked up		STRUCT
PTOC2_Str.Value		0	BOOL
PTOC2_Str.ST		16#80	BYTE
PTOC2_Op	Overcurrent I >> TRIP		STRUCT
PTOC2_Op.Value		0	BOOL
PTOC2_Op.ST		16#80	BYTE
PTOC3_Str	Overcurrent Ip picked up		STRUCT
PTOC3_Str.Value		0	BOOL
PTOC3_Str.ST		16#80	BYTE
PTOC3_Op	Overcurrent Ip TRIP		STRUCT
PTOC3_Op.Value		0	BOOL
PTOC3_Op.ST		16#80	BYTE
PTOC7_Str	Overcurrent 3I0 > picked up		STRUCT
PTOC7_Str.Value		0	BOOL
PTOC7_Str.ST		16#80	BYTE
PTOC7_Op	Overcurrent 3I0 > TRIP		STRUCT
PTOC7_Op.Value		0	BOOL
PTOC7_Op.ST		16#80	BYTE
TS_PTOC1_Str	Timestamp EVID1_S1 Overcurrent I > picked up		DT
TS_PTOC1_Op	Timestamp EVID1_S2 Overcurrent I > TRIP		DT
TS_PTOC2_Str	Timestamp EVID1_S3 Overcurrent I >> picked up		DT

TS_PTOC2_Op	Timestamp EVID1_S4 Overcurrent I >> TRIP		DT
TS_PTOC3_Str	Timestamp EVID1_S5 Overcurrent Ip picked up		DT
TS_PTOC3_Op	Timestamp EVID1_S6 Overcurrent Ip TRIP		DT
TS_PTOC7_Str	Timestamp EVID1_S7 Overcurrent 3I0 > picked up		DT
TS_PTOC7_Op	Timestamp EVID1_S8 Overcurrent 3I0 > TRIP		DT
PTOC8_Str	Overcurrent 3I0 >> picked up		STRUCT
PTOC8_Str.Value		0	BOOL
PTOC8_Str.ST		16#80	BYTE
PTOC8_Op	Overcurrent 3I0 >> TRIP		STRUCT
PTOC8_Op.Value		0	BOOL
PTOC8_Op.ST		16#80	BYTE
PTOC9_Str	Overcurrent 3I0p picked up		STRUCT
PTOC9_Str.Value		0	BOOL
PTOC9_Str.ST		16#80	BYTE
PTOC9_Op	Overcurrent 3I0p TRIP		STRUCT
PTOC9_Op.Value		0	BOOL
PTOC9_Op.ST		16#80	BYTE
PTOC12_Str	Unbalanced Load I > picked up		STRUCT
PTOC12_Str.Value		0	BOOL
PTOC12_Str.ST		16#80	BYTE
PTOC12_Op	Unbalanced Load I > TRIP		STRUCT
PTOC12_Op.Value		0	BOOL
PTOC12_Op.ST		16#80	BYTE
TS_PTOC8_Str	Timestamp EVID2_S1 Overcurrent 3I0 >> picked up		DT
TS_PTOC8_Op	Timestamp EVID2_S2 Overcurrent 3I0 >> TRIP		DT
TS_PTOC9_Str	Timestamp EVID2_S3 Overcurrent 3I0p picked up		DT
TS_PTOC9_Op	Timestamp EVID2_S4 Overcurrent 3I0p TRIP		DT
TS_PTOC12_Str	Timestamp EVID2_S5 Unbalanced Load I > picked up		DT
TS_PTOC12_Op	Timestamp EVID2_S6 Unbalanced Load I > TRIP		DT
TS_PTOC13_Str	Timestamp EVID2_S7 Unbalanced Load I >> picked up		DT
TS_PTOC13_Op	Timestamp EVID2_S8 Unbalanced Load I >> TRIP		DT
PTOC13_Str	Unbalanced Load I >> picked up		STRUCT
PTOC13_Str.Value		0	BOOL
PTOC13_Str.ST		16#80	BYTE

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PTOC13_Op	Unbalanced Load I >> TRIP		STRUCT
PTOC13_Op.Value		0	BOOL
PTOC13_Op.ST		16#80	BYTE
PTOC15_Str	Unbalanced Load I2th picked up		STRUCT
PTOC15_Str.Value		0	BOOL
PTOC15_Str.ST		16#80	BYTE
PTOC15_Op	Unbalanced Load I2th TRIP		STRUCT
PTOC15_Op.Value		0	BOOL
PTOC15_Op.ST		16#80	BYTE
PTTR1_Str	Overload Theta picked up		STRUCT
PTTR1_Str.Value		0	BOOL
PTTR1_Str.ST		16#80	BYTE
PTTR1_Op	Overload Theta TRIP		STRUCT
PTTR1_Op.Value		0	BOOL
PTTR1_Op.ST		16#80	BYTE
PDIF1_Op	Differential protection IDIFF> TRIP		STRUCT
PDIF1_Op.Value		0	BOOL
PDIF1_Op.ST		16#80	BYTE
PDIF1_Str	Differential protection IDIFF> picked up		STRUCT
PDIF1_Str.Value		0	BOOL
PDIF1_Str.ST		16#80	BYTE
PDIF2_Op	Differential protection IDIFF>> TRIP		STRUCT
PDIF2_Op.Value		0	BOOL
PDIF2_Op.ST		16#80	BYTE
PDIF2_Str	Differential protection IDIFF>> picked up		STRUCT
PDIF2_Str.Value		0	BOOL
PDIF2_Str.ST		16#80	BYTE
TS_PTOC15_Str	Timestamp EVID3_S1 Unbalanced Load I2th picked up		DT
TS_PTOC15_Op	Timestamp EVID3_S2 Unbalanced Load I2th TRIP		DT
TS_PTTR1_Str	Timestamp EVID3_S3 Overload Theta picked up		DT
TS_PTTR1_Op	Timestamp EVID3_S4 Overload Theta TRIP		DT
TS_PDIF1_Op	Timestamp EVID3_S5 Differential protection IDIFF> TRIP		DT
TS_PDIF1_Str	Timestamp EVID3_S6 Differential protection IDIFF> picked up		DT
TS_PDIF2_Op	Timestamp EVID3_S7 Differential protection IDIFF>> TRIP		DT
TS_PDIF2_Str	Timestamp EVID3_S8 Differential protection IDIFF>> picked up		DT

PDIF3_Op	Differential protection I REF > picked up		STRUCT
PDIF3_Op.Value		0	BOOL
PDIF3_Op.ST		16#80	BYTE
PDIF3_Str	Differential protection I REF > TRIP		STRUCT
PDIF3_Str.Value		0	BOOL
PDIF3_Str.ST		16#80	BYTE
GAPC1_Op	External Trip 1 TRIP		STRUCT
GAPC1_Op.Value		0	BOOL
GAPC1_Op.ST		16#80	BYTE
GAPC1_Str	External Trip 1 picked up		STRUCT
GAPC1_Str.Value		0	BOOL
GAPC1_Str.ST		16#80	BYTE
RBRF1_Str	Breaker failure picked up		STRUCT
RBRF1_Str.Value		0	BOOL
RBRF1_Str.ST		16#80	BYTE
RBRF1_OpEx	Breaker failure TRIP T1		STRUCT
RBRF1_OpEx.Value		0	BOOL
RBRF1_OpEx.ST		16#80	BYTE
RBRF1_OpIn	Breaker failure TRIP T2		STRUCT
RBRF1_OpIn.Value		0	BOOL
RBRF1_OpIn.ST		16#80	BYTE
PTTR1_AlmThm	Overload Theta Alarm		STRUCT
PTTR1_AlmThm.Value		0	BOOL
PTTR1_AlmThm.ST		16#80	BYTE
TS_PDIF3_Op	Timestamp EVID4_S1 Differential protection I REF > picked up		DT
TS_PDIF3_Str	Timestamp EVID4_S2 Differential protection I REF > TRIP		DT
TS_GAPC1_Op	Timestamp EVID4_S3 External Trip 1 TRIP		DT
TS_GAPC1_Str	Timestamp EVID4_S4 External Trip 1 picked up		DT
TS_RBRF1_Str	Timestamp EVID4_S5 Breaker failure picked up		DT
TS_RBRF1_OpEx	Timestamp EVID4_S6 Breaker failure TRIP T1		DT
TS_RBRF1_OpIn	Timestamp EVID4_S7 Breaker failure TRIP T2		DT
TS_PTTR1_AlmThm	Timestamp EVID4_S8 Overload Theta Alarm		DT
PU_Relay	Trip signal		STRUCT
PU_Relay.Value		0	BOOL
PU_Relay.ST		16#80	BYTE
TR_Relay	Trigger signal		STRUCT
TR_Relay.Value		0	BOOL

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TR_Relay.ST		16#80	BYTE
PARSET_A	Parameter Set A		STRUCT
PARSET_A.Value		0	BOOL
PARSET_A.ST		16#80	BYTE
PARSET_B	Parameter Set B		STRUCT
PARSET_B.Value		0	BOOL
PARSET_B.ST		16#80	BYTE
PARSET_C	Parameter Set C		STRUCT
PARSET_C.Value		0	BOOL
PARSET_C.ST		16#80	BYTE
PARSET_D	Parameter Set D		STRUCT
PARSET_D.Value		0	BOOL
PARSET_D.ST		16#80	BYTE
Error	Group Alarm		STRUCT
Error.Value		0	BOOL
Error.ST		16#80	BYTE
Warning	Group Warning		STRUCT
Warning.Value		0	BOOL
Warning.ST		16#80	BYTE
TS_PU_Relay	Timestamp EVID5_S1 trip signal		DT
TS_TR_Relay	Timestamp EVID5_S2 trigger signal		DT
TS_PARSET_A	Timestamp EVID5_S3 Parameter Set A		DT
TS_PARSET_B	Timestamp EVID5_S4 Parameter Set B		DT
TS_PARSET_C	Timestamp EVID5_S5 Parameter Set C		DT
TS_PARSET_D	Timestamp EVID5_S6 Parameter Set D		DT
TS_Error	Timestamp EVID5_S7 Group Alarm		DT
TS_Warning	Timestamp	EVID5_S8 Group Warn- ing	
Loc	Local		STRUCT
Loc.Value		0	BOOL
Loc.ST		16#80	BYTE
ALM_sp1	Spare BOOL 1		STRUCT
ALM_sp1.Value		0	BOOL
ALM_sp1.ST		16#80	BYTE
ALM_sp2	Spare BOOL 2		STRUCT
ALM_sp2.Value		0	BOOL
ALM_sp2.ST		16#80	BYTE
ALM_sp3	Spare BOOL 3		STRUCT
ALM_sp3.Value		0	BOOL
ALM_sp3.ST		16#80	BYTE
ALM_sp4	Spare BOOL 4		STRUCT
ALM_sp4.Value		0	BOOL
ALM_sp4.ST		16#80	BYTE

ALM_sp5	Spare BOOL 5		STRUCT
ALM_sp5.Value		0	BOOL
ALM_sp5.ST		16#80	BYTE
ALM_sp6	Spare BOOL 6		STRUCT
ALM_sp6.Value		0	BOOL
ALM_sp6.ST		16#80	BYTE
ALM_sp7	Spare BOOL 7		STRUCT
ALM_sp7.Value		0	BOOL
ALM_sp7.ST		16#80	BYTE
TS_Loc	Timestamp EVID6_S1 Local		DT
TS_ALM_sp1	Timestamp EVID6_S2 Spare BOOL 1		DT
TS_ALM_sp2	Timestamp EVID6_S3 Spare BOOL 2		DT
TS_ALM_sp3	Timestamp EVID6_S4 Spare BOOL 3		DT
TS_ALM_sp4	Timestamp EVID6_S5 Spare BOOL 4		DT
TS_ALM_sp5	Timestamp EVID6_S5 Spare BOOL 5		DT
TS_ALM_sp6	Timestamp EVID6_S7 Spare BOOL 6		DT
TS_ALM_sp7	Timestamp EVID6_S8 Spare BOOL 7		DT

13.8.2.2 Outputs

I/O name	Description	Value	Data type
RCV_Data	Connect to DC/_SC_TRAFO		STRUCT
MSG_E_S_1	ALARM_8P Error	0	BOOL
MSG_U_S_1	1=Message Suppression active	0	BOOL
MSG_S_S_1	ALARM_8P: STATUS output	16#0000	WORD
MSG_A_S_1	ALARM_8P: ACK_STATE output	16#0000	WORD
MSG_E_S_2	ALARM_8P Error	0	BOOL
MSG_U_S_2	1=Message Suppression active	0	BOOL
MSG_S_S_2	ALARM_8P: STATUS output	16#0000	WORD
MSG_A_S_2	ALARM_8P: ACK_STATE output	16#0000	WORD
MSG_E_S_3	ALARM_8P Error	0	BOOL
MSG_U_S_3	1=Message Suppression active	0	BOOL
MSG_S_S_3	ALARM_8P: STATUS output	16#0000	WORD
MSG_A_S_3	ALARM_8P: ACK_STATE output	16#0000	WORD
MSG_E_S_4	ALARM_8P Error	0	BOOL
MSG_U_S_4	1=Message Suppression active	0	BOOL
MSG_S_S_4	ALARM_8P: STATUS output	16#0000	WORD
MSG_A_S_4	ALARM_8P: ACK_STATE output	16#0000	WORD
MSG_E_S_5	ALARM_8P Error	0	BOOL

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MSG_U_S_5	1=Message Suppression active	0	BOOL
MSG_S_S_5	ALARM_8P: STATUS output	16#0000	WORD
MSG_A_S_5	ALARM_8P: ACK_STATE output	16#0000	WORD
MSG_E_S_6	ALARM_8P Error	0	BOOL
MSG_U_S_6	1=Message Suppression active	0	BOOL
MSG_S_S_6	ALARM_8P: STATUS output	16#0000	WORD
MSG_A_S_6	ALARM_8P: ACK_STATE output	16#0000	WORD
MSG_E_D_1	ALARM_8P Error	0	BOOL
MSG_U_D_1	1=Message Suppression active	0	BOOL
MSG_S_D_1	ALARM_8P: STATUS output	16#0000	WORD
MSG_A_D_1	ALARM_8P: ACK_STATE output	16#0000	WORD
Q_DT_NOW	Last Error	1990-01-01 -0:00:00	DATE_AND_TIME
PosCmdSp1_op	Spare 1 Command open	0	BOOL
PosCmdSp1_cl	Spare 1 Command close	0	BOOL
PosCmdSp2_op	Spare 2 Command open	0	BOOL
PosCmdSp2_cl	Spare 2 Command close	0	BOOL
PosCmdSp3_op	Spare 3 Command open	0	BOOL
PosCmdSp3_cl	Spare 3 Command close	0	BOOL
PosCmdSp4_op	Spare 4 Command open	0	BOOL
PosCmdSp4_cl	Spare 4 Command close	0	BOOL
DT_LAST_MSG	Timestamp last error		DT
Q_DT_NOW	Timestamp date and time of PLC		DT

13.9 DC_SYNC

13.9.1 Description of DC_SYNC

The DataCollector block DC_SYNC (FB 2563) forms an interface to the technological block PC_SYNC.

13.9.2 I/O lists

13.9.2.1 Inputs

I/O name	Description	Value	Data type
Snd_Data	Connect to DC_/SC_SYNC		STRUCT
SimOn	Simulation on/off	1	BOOL
ExtTSMsg	1 = external time stamps used for messages 0 = creation of internal time stamps	FALSE	BOOL
ExtTSPos	1 = external time stamps used for switch positions 0 = creation of internal time stamps	FALSE	BOOL
ConnErr	Connection Error	0	BOOL
P1_U1	P1_U1		STRUCT
P1_U1.Value		0.0	REAL
P1_U1.ST		16#80	BYTE
P1_U2	P1_U2		STRUCT
P1_U2.Value		0.0	REAL
P1_U2.ST		16#80	BYTE
P2_U1	P2_U1		STRUCT
P2_U1.Value		0.0	REAL
P2_U1.ST		16#80	BYTE
P2_U2	P2_U2		STRUCT
P2_U2.Value		0.0	REAL
P2_U2.ST		16#80	BYTE
P3_U1	P3_U1		STRUCT
P3_U1.Value		0.0	REAL
P3_U1.ST		16#80	BYTE
P3_U2	P3_U2		STRUCT
P3_U2.Value		0.0	REAL
P3_U2.ST		16#80	BYTE
P4_U1	P4_U1		STRUCT
P4_U1.Value		0.0	REAL
P4_U1.ST		16#80	BYTE
P4_U2	P4_U2		STRUCT
P4_U2.Value		0.0	REAL
P4_U2.ST		16#80	BYTE
P5_U1	P5_U1		STRUCT
P5_U1.Value		0.0	REAL
P5_U1.ST		16#80	BYTE
P5_U2	P5_U2		STRUCT
P5_U2.Value		0.0	REAL

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P5_U2.ST		16#80	BYTE
P6_U1	P6_U1		STRUCT
P6_U1.Value		0.0	REAL
P6_U1.ST		16#80	BYTE
P6_U2	P6_U2		STRUCT
P6_U2.Value		0.0	REAL
P6_U2.ST		16#80	BYTE
P7_U1	P7_U1		STRUCT
P7_U1.Value		0.0	REAL
P7_U1.ST		16#80	BYTE
P7_U2	P7_U2		STRUCT
P7_U2.Value		0.0	REAL
P7_U2.ST		16#80	BYTE
P8_U1	P8_U1		STRUCT
P8_U1.Value		0.0	REAL
P8_U1.ST		16#80	BYTE
P8_U2	P8_U2		STRUCT
P8_U2.Value		0.0	REAL
P8_U2.ST		16#80	BYTE
P1_F1	P1_F1		STRUCT
P1_F1.Value		0.0	REAL
P1_F1.ST		16#80	BYTE
P1_F2	P1_F2		STRUCT
P1_F2.Value		0.0	REAL
P1_F2.ST		16#80	BYTE
P2_F1	P2_F1		STRUCT
P2_F1.Value		0.0	REAL
P2_F1.ST		16#80	BYTE
P2_F2	P2_F2		STRUCT
P2_F2.Value		0.0	REAL
P2_F2.ST		16#80	BYTE
P3_F1	P3_F1		STRUCT
P3_F1.Value		0.0	REAL
P3_F1.ST		16#80	BYTE
P3_F2	P3_F2		STRUCT
P3_F2.Value		0.0	REAL
P3_F2.ST		16#80	BYTE
P4_F1	P4_F1		STRUCT
P4_F1.Value		0.0	REAL
P4_F1.ST		16#80	BYTE
P4_F2	P4_F2		STRUCT
P4_F2.Value		0.0	REAL
P4_F2.ST		16#80	BYTE
P5_F1	P5_F1		STRUCT

P5_F1.Value		0.0	REAL
P5_F1.ST		16#80	BYTE
P5_F2	P5_F2		STRUCT
P5_F2.Value		0.0	REAL
P5_F2.ST		16#80	BYTE
P6_F1	P6_F1		STRUCT
P6_F1.Value		0.0	REAL
P6_F1.ST		16#80	BYTE
P6_F2	P6_F2		STRUCT
P6_F2.Value		0.0	REAL
P6_F2.ST		16#80	BYTE
P7_F1	P7_F1		STRUCT
P7_F1.Value		0.0	REAL
P7_F1.ST		16#80	BYTE
P7_F2	P7_F2		STRUCT
P7_F2.Value		0.0	REAL
P7_F2.ST		16#80	BYTE
P8_F1	P8_F1		STRUCT
P8_F1.Value		0.0	REAL
P8_F1.ST		16#80	BYTE
P8_F2	P8_F2		STRUCT
P8_F2.Value		0.0	REAL
P8_F2.ST		16#80	BYTE
P1_DU	P1_DU		STRUCT
P1_DU.Value		0.0	REAL
P1_DU.ST		16#80	BYTE
P2_DU	P2_DU		STRUCT
P2_DU.Value		0.0	REAL
P2_DU.ST		16#80	BYTE
P3_DU	P3_DU		STRUCT
P3_DU.Value		0.0	REAL
P3_DU.ST		16#80	BYTE
P4_DU	P4_DU		STRUCT
P4_DU.Value		0.0	REAL
P4_DU.ST		16#80	BYTE
P5_DU	P5_DU		STRUCT
P5_DU.Value		0.0	REAL
P5_DU.ST		16#80	BYTE
P6_DU	P6_DU		STRUCT
P6_DU.Value		0.0	REAL
P6_DU.ST		16#80	BYTE
P7_DU	P7_DU		STRUCT
P7_DU.Value		0.0	REAL
P7_DU.ST		16#80	BYTE

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P8_DU	P8_DU		STRUCT
P8_DU.Value		0.0	REAL
P8_DU.ST		16#80	BYTE
P1_DF	P1_DF		STRUCT
P1_DF.Value		0.0	REAL
P1_DF.ST		16#80	BYTE
P2_DF	P2_DF		STRUCT
P2_DF.Value		0.0	REAL
P2_DF.ST		16#80	BYTE
P3_DF	P3_DF		STRUCT
P3_DF.Value		0.0	REAL
P3_DF.ST		16#80	BYTE
P4_DF	P4_DF		STRUCT
P4_DF.Value		0.0	REAL
P4_DF.ST		16#80	BYTE
P5_DF	P5_DF		STRUCT
P5_DF.Value		0.0	REAL
P5_DF.ST		16#80	BYTE
P6_DF	P6_DF		STRUCT
P6_DF.Value		0.0	REAL
P6_DF.ST		16#80	BYTE
P7_DF	P7_DF		STRUCT
P7_DF.Value		0.0	REAL
P7_DF.ST		16#80	BYTE
P8_DF	P8_DF		STRUCT
P8_DF.Value		0.0	REAL
P8_DF.ST		16#80	BYTE
P1_DANG	P1_DANG		STRUCT
P1_DANG.Value		0.0	REAL
P1_DANG.ST		16#80	BYTE
P2_DANG	P2_DANG		STRUCT
P2_DANG.Value		0.0	REAL
P2_DANG.ST		16#80	BYTE
P3_DANG	P3_DANG		STRUCT
P3_DANG.Value		0.0	REAL
P3_DANG.ST		16#80	BYTE
P4_DANG	P4_DANG		STRUCT
P4_DANG.Value		0.0	REAL
P4_DANG.ST		16#80	BYTE
P5_DANG	P5_DANG		STRUCT
P5_DANG.Value		0.0	REAL
P5_DANG.ST		16#80	BYTE
P6_DANG	P6_DANG		STRUCT
P6_DANG.Value		0.0	REAL

P6_DANG.ST		16#80	BYTE
P7_DANG	P7_DANG		STRUCT
P7_DANG.Value		0.0	REAL
P7_DANG.ST		16#80	BYTE
P8_DANG	P8_DANG		STRUCT
P8_DANG.Value		0.0	REAL
P8_DANG.ST		16#80	BYTE
REAL_sp1	Spare Real 1		STRUCT
REAL_sp1.Value		0.0	REAL
REAL_sp1.ST		16#80	BYTE
REAL_sp2	Spare Real 2		STRUCT
REAL_sp2.Value		0.0	REAL
REAL_sp2.ST		16#80	BYTE
REAL_sp3	Spare Real 3		STRUCT
REAL_sp3.Value		0.0	REAL
REAL_sp3.ST		16#80	BYTE
REAL_sp4	Spare Real 4		STRUCT
REAL_sp4.Value		0.0	REAL
REAL_sp4.ST		16#80	BYTE
Beh	Behavior		STRUCT
Beh.Value		0	INT
Beh.ST		16#80	BYTE
Dev_OK	Device OK		STRUCT
Dev_OK.Value		3	INT
Dev_OK.ST		16#80	BYTE
INT_sp1	Spare INT 1		STRUCT
INT_sp1.Value		0	INT
INT_sp1.ST		16#80	BYTE
INT_sp2	Spare INT 2		STRUCT
INT_sp2.Value		0	INT
INT_sp2.ST		16#80	BYTE
INT_sp3	Spare INT 3		STRUCT
INT_sp3.Value		0	INT
INT_sp3.ST		16#80	BYTE
INT_sp4	Spare INT 4		STRUCT
INT_sp4.Value		0	INT
INT_sp4.ST		16#80	BYTE
INT_sp5	Spare INT 5		STRUCT
INT_sp5.Value		0	INT
INT_sp5.ST		16#80	BYTE
INT_sp6	Spare INT 6		STRUCT
INT_sp6.Value		0	INT
INT_sp6.ST		16#80	BYTE
SwCount	Switch Count Q0		STRUCT

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SwCount.Value		34	DINT
SwCount.ST		16#80	BYTE
OpTm	Operating hours		STRUCT
OpTm.Value		3432	DINT
OpTm.ST		16#80	BYTE
DINT_sp1	BS32 value 8 (spare)		STRUCT
DINT_sp1.Value		0	DINT
DINT_sp1.ST		16#80	BYTE
DINT_sp2	BS32 value 8 (spare)		STRUCT
DINT_sp2.Value		0	DINT
DINT_sp2.ST		16#80	BYTE
DINT_sp3	BS32 value 8 (spare)		STRUCT
DINT_sp3.Value		0	DINT
DINT_sp3.ST		16#80	BYTE
DINT_sp4	BS32 value 8 (spare)		STRUCT
DINT_sp4.Value		0	DINT
DINT_sp4.ST		16#80	BYTE
DINT_sp5	BS32 value 8 (spare)		STRUCT
DINT_sp5.Value		0	DINT
DINT_sp5.ST		16#80	BYTE
DINT_sp6	BS32 value 8 (spare)		STRUCT
DINT_sp6.Value		0	DINT
DINT_sp6.ST		16#80	BYTE
Pos_CMD_sp1	Position Spare Command 1		STRUCT
Pos_CMD_sp1.Value		16#00000003	DWORD
Pos_CMD_sp1.ST		16#80	BYTE
Pos_CMD_sp2	Position Spare Command 2		STRUCT
Pos_CMD_sp2.Value		16#00000003	DWORD
Pos_CMD_sp2.ST		16#80	BYTE
Pos_CMD_sp3	Position Spare Command 3		STRUCT
Pos_CMD_sp3.Value		16#00000003	DWORD
Pos_CMD_sp3.ST		16#80	BYTE
Pos_CMD_sp4	Position Spare Command 4		STRUCT
Pos_CMD_sp4.Value		16#00000003	DWORD
Pos_CMD_sp4.ST		16#80	BYTE
Pos_CMD_sp5	Position Spare Command 5		STRUCT
Pos_CMD_sp5.Value		16#00000003	DWORD
Pos_CMD_sp5.ST		16#80	BYTE
Pos_CMD_sp6	Position Spare Command 6		STRUCT

Pos_CMD_sp6.Value		16#00000003	DWORD
Pos_CMD_sp6.ST		16#80	BYTE
Pos_CMD_sp7	Position Spare Command 7		STRUCT
Pos_CMD_sp7.Value		16#00000003	DWORD
Pos_CMD_sp7.ST		16#80	BYTE
Pos_CMD_sp8	Position Spare Command 8		STRUCT
Pos_CMD_sp8.Value		16#00000003	DWORD
Pos_CMD_sp8.ST		16#80	BYTE
TS_Pos_CMD_sp1	Timestamp Position Spare Command 1		DT
TS_Pos_CMD_sp2	Timestamp Position Spare Command 2		DT
TS_Pos_CMD_sp3	Timestamp Position Spare Command 3		DT
TS_Pos_CMD_sp4	Timestamp Position Spare Command 4		DT
TS_Pos_CMD_sp5	Timestamp Position Spare Command 5		DT
TS_Pos_CMD_sp6	Timestamp Position Spare Command 6		DT
TS_Pos_CMD_sp7	Timestamp Position Spare Command 7		DT
TS_Pos_CMD_sp8	Timestamp Position Spare Command 8		DT
AckCmdsp1	Acknowledge signal for Spare command 1		STRUCT
AckCmdsp1.Value		0	BOOL
AckCmdsp1.ST		16#80	BYTE
AckCmdsp2	Acknowledge signal for Spare command 2		STRUCT
AckCmdsp2.Value		0	BOOL
AckCmdsp2.ST		16#80	BYTE
AckCmdsp3	Acknowledge signal for Spare command 3		STRUCT
AckCmdsp3.Value		0	BOOL
AckCmdsp3.ST		16#80	BYTE
AckCmdsp4	Acknowledge signal for Spare command 4		STRUCT
AckCmdsp4.Value		0	BOOL
AckCmdsp4.ST		16#80	BYTE
AckCmdsp5	Acknowledge signal for Spare command 5		STRUCT
AckCmdsp5.Value		0	BOOL
AckCmdsp5.ST		16#80	BYTE
AckCmdsp6	Acknowledge signal for Spare command 6		STRUCT
AckCmdsp6.Value		0	BOOL
AckCmdsp6.ST		16#80	BYTE
AckCmdsp7	Acknowledge signal for Spare command 7		STRUCT
AckCmdsp7.Value		0	BOOL

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AckCmdsp7.ST		16#80	BYTE
AckCmdsp8	Acknowledge signal for Spare command 8		STRUCT
AckCmdsp8.Value		0	BOOL
AckCmdsp8.ST		16#80	BYTE
TR_Relay	Relay PICKUP		STRUCT
TR_Relay.Value		0	BOOL
TR_Relay.ST		16#80	BYTE
PU_Relay	Relay General TRIP		STRUCT
PU_Relay.Value		0	BOOL
PU_Relay.ST		16#80	BYTE
GAPC1_Op	External Trip 1 TRIP		STRUCT
GAPC1_Op.Value		0	BOOL
GAPC1_Op.ST		16#80	BYTE
GAPC1_Str	External Trip 1 picked up		STRUCT
GAPC1_Str.Value		0	BOOL
GAPC1_Str.ST		16#80	BYTE
GAPC2_Op	External Trip 2 TRIP		STRUCT
GAPC2_Op.Value		0	BOOL
GAPC2_Op.ST		16#80	BYTE
GAPC2_Str	External Trip 2 picked up		STRUCT
GAPC2_Str.Value		0	BOOL
GAPC2_Str.ST		16#80	BYTE
GAPC3_Op	External Trip 3 TRIP		STRUCT
GAPC3_Op.Value		0	BOOL
GAPC3_Op.ST		16#80	BYTE
GAPC3_Str	External Trip 3 picked up		STRUCT
GAPC3_Str.Value		0	BOOL
GAPC3_Str.ST		16#80	BYTE
TS_TR_Relay	Timestamp EVID1_S1 Relay PICKUP		DT
TS_PU_Relay	Timestamp EVID1_S2 Relay General TRIP		DT
TS_GAPC1_Op	Timestamp EVID1_S3 External Trip 1 TRIP		DT
TS_GAPC1_Str	Timestamp EVID1_S4 External Trip 1 picked up		DT
TS_GAPC2_Op	Timestamp EVID1_S5 External Trip 2 TRIP		DT
TS_GAPC2_Str	Timestamp EVID1_S6 External Trip 2 picked up		DT
TS_GAPC3_Op	Timestamp EVID1_S7 External Trip 3 TRIP		DT
TS_GAPC3_Str	Timestamp EVID1_S8 External Trip 3 picked up		DT
GAPC4_Op	External Trip 4 TRIP		STRUCT
GAPC4_Op.Value		0	BOOL

GAPC4_Op.ST		16#80	BYTE
GAPC4_Str	External Trip 4 picked up		STRUCT
GAPC4_Str.Value		0	BOOL
GAPC4_Str.ST		16#80	BYTE
PTUF1_Op	Frequency 1 < TRIP		STRUCT
PTUF1_Op.Value		0	BOOL
PTUF1_Op.ST		16#80	BYTE
PTUF1_Str	Frequency 1 < picked up		STRUCT
PTUF1_Str.Value		0	BOOL
PTUF1_Str.ST		16#80	BYTE
PTUF2_Op	Frequency 2 < TRIP		STRUCT
PTUF2_Op.Value		0	BOOL
PTUF2_Op.ST		16#80	BYTE
PTUF2_Str	Frequency 2 < picked up		STRUCT
PTUF2_Str.Value		0	BOOL
PTUF2_Str.ST		16#80	BYTE
PTUF3_Op	Frequency 3 < TRIP		STRUCT
PTUF3_Op.Value		0	BOOL
PTUF3_Op.ST		16#80	BYTE
PTUF3_Str	Frequency 3 < picked up		STRUCT
PTUF3_Str.Value		0	BOOL
PTUF3_Str.ST		16#80	BYTE
TS_GAPC4_Op	Timestamp EVID2_S1 External Trip 4 TRIP		
TS_GAPC4_Str	Timestamp EVID2_S2 External Trip 4 picked up		
TS_PTUF1_Op	Timestamp EVID2_S3 Frequency 1 < TRIP		
TS_PTUF1_Str	Timestamp EVID2_S4 Frequency 1 < picked up		
TS_PTUF2_Op	Timestamp EVID2_S5 Frequency 2 < TRIP		
TS_PTUF2_Str	Timestamp EVID2_S6 Frequency 2 < picked up		
TS_PTUF3_Op	Timestamp EVID2_S7 Frequency 3 < TRIP		
TS_PTUF3_Str	Timestamp EVID2_S8 Frequency 3 < picked up		
PTUF4_Op	Frequency 4 < TRIP		STRUCT
PTUF4_Op.Value		0	BOOL
PTUF4_Op.ST		16#80	BYTE
PTUF4_Str	Frequency 4< picked up		STRUCT
PTUF4_Str.Value		0	BOOL
PTUF4_Str.ST		16#80	BYTE
RBRF1_Str	Breaker failure picked up		STRUCT
RBRF1_Str.Value		0	BOOL

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RBRF1_Str.ST		16#80	BYTE
RBRF1_OpEx	Breaker failure TRIP T1		STRUCT
RBRF1_OpEx.Value		0	BOOL
RBRF1_OpEx.ST		16#80	BYTE
RBRF1_OpIn	Breaker failure TRIP T2		STRUCT
RBRF1_OpIn.Value		0	BOOL
RBRF1_OpIn.ST		16#80	BYTE
Loc	Local operation		STRUCT
Loc.Value		0	BOOL
Loc.ST		16#80	BYTE
Error	Error Summary Alarm		STRUCT
Error.Value		0	BOOL
Error.ST		16#80	BYTE
Warning	Alarm Summary Event		STRUCT
Warning.Value		1	BOOL
Warning.ST		16#80	BYTE
TS_PTUF4_Op	Timestamp EVID3_S1 Frequency 4 < TRIP		DT
TS_PTUF4_Str	Timestamp EVID3_S2 Frequency 4< picked up		DT
TS_RBRF1_Str	Timestamp EVID3_S3 Breaker failure picked up		DT
TS_RBRF1_OpEx	Timestamp EVID3_S4 Breaker failure TRIP T1		DT
TS_RBRF1_OpIn	Timestamp EVID3_S5 Breaker failure TRIP T2		DT
TS_Loc	Timestamp EVID3_S6 Local Operation		DT
TS_Error	Timestamp EVID3_S7 Error Summary Alarm		DT
TS_Warning	Timestamp EVID3_S8 Alarm Summary Event		DT
P1_U_NOK	Point 1 Voltage Difference NOK		STRUCT
P1_U_NOK.Value		0	BOOL
P1_U_NOK.ST		16#80	BYTE
P2_U_NOK	Point 2 Voltage Difference NOK		STRUCT
P2_U_NOK.Value		0	BOOL
P2_U_NOK.ST		16#80	BYTE
P3_U_NOK	Point 3 Voltage Difference NOK		STRUCT
P3_U_NOK.Value		0	BOOL
P3_U_NOK.ST		16#80	BYTE
P4_U_NOK	Point 4 Voltage Difference NOK		STRUCT
P4_U_NOK.Value		0	BOOL
P4_U_NOK.ST		16#80	BYTE
P5_U_NOK	Point 5 Voltage Difference NOK		STRUCT
P5_U_NOK.Value		0	BOOL

P5_U_NOK.ST		16#80	BYTE
P6_U_NOK	Point 6 Voltage Difference NOK		STRUCT
P6_U_NOK.Value		0	BOOL
P6_U_NOK.ST		16#80	BYTE
P7_U_NOK	Point 7 Voltage Difference NOK		STRUCT
P7_U_NOK.Value		0	BOOL
P7_U_NOK.ST		16#80	BYTE
P8_U_NOK	Point 8 Voltage Difference NOK		STRUCT
P8_U_NOK.Value		0	BOOL
P8_U_NOK.ST		16#80	BYTE
TS_P1_U_NOK	Timestamp EVID4_S1 Point 1 Voltage Difference NOK		DT
TS_P2_U_NOK	Timestamp EVID4_S2 Point 2 Voltage Difference NOK		DT
TS_P3_U_NOK	Timestamp EVID4_S3 Point 3 Voltage Difference NOK		DT
TS_P4_U_NOK	Timestamp EVID4_S4 Point 4 Voltage Difference NOK		DT
TS_P5_U_NOK	Timestamp EVID4_S5 Point 5 Voltage Difference NOK		DT
TS_P6_U_NOK	Timestamp EVID4_S6 Point 6 Voltage Difference NOK		DT
TS_P7_U_NOK	Timestamp EVID4_S7 Point 7 Voltage Difference NOK		DT
TS_P8_U_NOK	Timestamp EVID4_S8 Point 8 Voltage Difference NOK		DT
P1_MEAS	Point 1 Measurement in Progress		STRUCT
P1_MEAS.Value		0	BOOL
P1_MEAS.ST		16#80	BYTE
P2_MEAS	Point 2 Measurement in Progress		STRUCT
P2_MEAS.Value		0	BOOL
P2_MEAS.ST		16#80	BYTE
P3_MEAS	Point 3 Measurement in Progress		STRUCT
P3_MEAS.Value		0	BOOL
P3_MEAS.ST		16#80	BYTE
P4_MEAS	Point 4 Measurement in Progress		STRUCT
P4_MEAS.Value		0	BOOL
P4_MEAS.ST		16#80	BYTE
P5_MEAS	Point 5 Measurement in Progress		STRUCT
P5_MEAS.Value		0	BOOL
P5_MEAS.ST		16#80	BYTE
P6_MEAS	Point 6 Measurement in Progress		STRUCT
P6_MEAS.Value		0	BOOL
P6_MEAS.ST		16#80	BYTE
P7_MEAS	Point 7 Measurement in Progress		STRUCT
P7_MEAS.Value		0	BOOL

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P7_MEAS.ST		16#80	BYTE
P8_MEAS	Point 8 Measurement in Progress		STRUCT
P8_MEAS.Value		0	BOOL
P8_MEAS.ST		16#80	BYTE
TS_P1_MEAS	Timestamp EVID5_S1 Point 1 Measurement in Progress		DT
TS_P2_MEAS	Timestamp EVID5_S2 Point 2 Measurement in Progress		DT
TS_P3_MEAS	Timestamp EVID5_S3 Point 3 Measurement in Progress		DT
TS_P4_MEAS	Timestamp EVID5_S4 Point 4 Measurement in Progress		DT
TS_P5_MEAS	Timestamp EVID5_S5 Point 5 Measurement in Progress		DT
TS_P6_MEAS	Timestamp EVID5_S6 Point 6 Measurement in Progress		DT
TS_P7_MEAS	Timestamp EVID5_S7 Point 7 Measurement in Progress		DT
TS_P8_MEAS	Timestamp EVID5_S8 Point 8 Measurement in Progress		DT
P1_F_NOK	Point 1 Frequency Difference NOK		STRUCT
P1_F_NOK.Value		0	BOOL
P1_F_NOK.ST		16#80	BYTE
P2_F_NOK	Point 2 Frequency Difference NOK		STRUCT
P2_F_NOK.Value		0	BOOL
P2_F_NOK.ST		16#80	BYTE
P3_F_NOK	Point 3 Frequency Difference NOK		STRUCT
P3_F_NOK.Value		0	BOOL
P3_F_NOK.ST		16#80	BYTE
P4_F_NOK	Point 4 Frequency Difference NOK		STRUCT
P4_F_NOK.Value		0	BOOL
P4_F_NOK.ST		16#80	BYTE
P5_F_NOK	Point 5 Frequency Difference NOK		STRUCT
P5_F_NOK.Value		0	BOOL
P5_F_NOK.ST		16#80	BYTE
P6_F_NOK	Point 6 Frequency Difference NOK		STRUCT
P6_F_NOK.Value		0	BOOL
P6_F_NOK.ST		16#80	BYTE
P7_F_NOK	Point 7 Frequency Difference NOK		STRUCT
P7_F_NOK.Value		0	BOOL
P7_F_NOK.ST		16#80	BYTE
P8_F_NOK	Point 8 Frequency Difference NOK		STRUCT
P8_F_NOK.Value		0	BOOL
P8_F_NOK.ST		16#80	BYTE
TS_P1_F_NOK	Timestamp EVID6_S1 Point 1 Frequency Difference NOK		DT

TS_P2_F_NOK	Timestamp EVID6_S2 Point 2 Frequency Difference NOK		DT
TS_P3_F_NOK	Timestamp EVID6_S3 Point 3 Frequency Difference NOK		DT
TS_P4_F_NOK	Timestamp EVID6_S4 Point 4 Frequency Difference NOK		DT
TS_P5_F_NOK	Timestamp EVID6_S5 Point 5 Frequency Difference NOK		DT
TS_P6_F_NOK	Timestamp EVID6_S6 Point 6 Frequency Difference NOK		DT
TS_P7_F_NOK	Timestamp EVID6_S7 Point 7 Frequency Difference NOK		DT
TS_P8_F_NOK	Timestamp EVID6_S8 Point 8 Frequency Difference NOK		DT
P1_ANG_NOK	Point 1 Angle Difference NOK		STRUCT
P1_ANG_NOK.Value		0	BOOL
P1_ANG_NOK.ST		16#80	BYTE
P2_ANG_NOK	Point 2 Angle Difference NOK		STRUCT
P2_ANG_NOK.Value		0	BOOL
P2_ANG_NOK.ST		16#80	BYTE
P3_ANG_NOK	Point 3 Angle Difference NOK		STRUCT
P3_ANG_NOK.Value		0	BOOL
P3_ANG_NOK.ST		16#80	BYTE
P4_ANG_NOK	Point 4 Angle Difference NOK		STRUCT
P4_ANG_NOK.Value		0	BOOL
P4_ANG_NOK.ST		16#80	BYTE
P5_ANG_NOK	Point 5 Angle Difference NOK		STRUCT
P5_ANG_NOK.Value		0	BOOL
P5_ANG_NOK.ST		16#80	BYTE
P6_ANG_NOK	Point 6 Angle Difference NOK		STRUCT
P6_ANG_NOK.Value		0	BOOL
P6_ANG_NOK.ST		16#80	BYTE
P7_ANG_NOK	Point 7 Angle Difference NOK		STRUCT
P7_ANG_NOK.Value		0	BOOL
P7_ANG_NOK.ST		16#80	BYTE
P8_ANG_NOK	Point 8 Angle Difference NOK		STRUCT
P8_ANG_NOK.Value		0	BOOL
P8_ANG_NOK.ST		16#80	BYTE
TS_P1_ANG_NOK	Timestamp EVID7_S1 Point 1 Angle Difference NOK		DT
TS_P2_ANG_NOK	Timestamp EVID7_S2 Point 2 Angle Difference NOK		DT
TS_P3_ANG_NOK	Timestamp EVID7_S3 Point 3 Angle Difference NOK		DT
TS_P4_ANG_NOK	Timestamp EVID7_S4 Point 4 Angle Difference NOK		DT

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TS_P5_ANG_NOK	Timestamp EVID7_S5 Point 5 Angle Difference NOK		DT
TS_P6_ANG_NOK	Timestamp EVID7_S6 Point 6 Angle Difference NOK		DT
TS_P7_ANG_NOK	Timestamp EVID7_S7 Point 7 Angle Difference NOK		DT
TS_P8_ANG_NOK	Timestamp EVID7_S8 Point 8 Angle Difference NOK		DT
P1_CL_EN	Point 1 Close Enable		STRUCT
P1_CL_EN.Value		0	BOOL
P1_CL_EN.ST		16#80	BYTE
P2_CL_EN	Point 2 Close Enable		STRUCT
P2_CL_EN.Value		0	BOOL
P2_CL_EN.ST		16#80	BYTE
P3_CL_EN	Point 3 Close Enable		STRUCT
P3_CL_EN.Value		0	BOOL
P3_CL_EN.ST		16#80	BYTE
P4_CL_EN	Point 4 Close Enable		STRUCT
P4_CL_EN.Value		0	BOOL
P4_CL_EN.ST		16#80	BYTE
P5_CL_EN	Point 5 Close Enable		STRUCT
P5_CL_EN.Value		0	BOOL
P5_CL_EN.ST		16#80	BYTE
P6_CL_EN	Point 6 Close Enable		STRUCT
P6_CL_EN.Value		0	BOOL
P6_CL_EN.ST		16#80	BYTE
P7_CL_EN	Point 7 Close Enable		STRUCT
P7_CL_EN.Value		0	BOOL
P7_CL_EN.ST		16#80	BYTE
P8_CL_EN	Point 8 Close Enable		STRUCT
P8_CL_EN.Value		0	BOOL
P8_CL_EN.ST		16#80	BYTE
TS_P1_CL_EN	Timestamp EVID8_S1 Point 1 Close Enable		DT
TS_P2_CL_EN	Timestamp EVID8_S2 Point 2 Close Enable		DT
TS_P3_CL_EN	Timestamp EVID8_S3 Point 3 Close Enable		DT
TS_P4_CL_EN	Timestamp EVID8_S4 Point 4 Close Enable		DT
TS_P5_CL_EN	Timestamp EVID8_S5 Point 5 Close Enable		DT
TS_P6_CL_EN	Timestamp EVID8_S6 Point 6 Close Enable		DT
TS_P7_CL_EN	Timestamp EVID8_S7 Point 7 Close Enable		DT

TS_P8_CL_EN	Timestamp EVID8_S8 Point 8 Close Enable		DT
PARSET_A	Parameter Set A active		STRUCT
PARSET_A.Value		0	BOOL
PARSET_A.ST		16#80	BYTE
PARSET_B	Parameter Set B active		STRUCT
PARSET_B.Value		0	BOOL
PARSET_B.ST		16#80	BYTE
PARSET_C	Parameter Set C active		STRUCT
PARSET_C.Value		0	BOOL
PARSET_C.ST		16#80	BYTE
PARSET_D	Parameter Set D active		STRUCT
PARSET_D.Value		0	BOOL
PARSET_D.ST		16#80	BYTE
ALM_sp1	Spare Alarm 1		STRUCT
ALM_sp1.Value		0	BOOL
ALM_sp1.ST		16#80	BYTE
ALM_sp2	Spare Alarm 2		STRUCT
ALM_sp2.Value		0	BOOL
ALM_sp2.ST		16#80	BYTE
ALM_sp3	Spare Alarm 3		STRUCT
ALM_sp3.Value		0	BOOL
ALM_sp3.ST		16#80	BYTE
ALM_sp4	Spare Alarm 4		STRUCT
ALM_sp4.Value		0	BOOL
ALM_sp4.ST		16#80	BYTE
TS_PARSET_A	Timestamp EVID9_S1 Parameter Set A active		DT
TS_PARSET_B	Timestamp EVID9_S2 Parameter Set B active		DT
TS_PARSET_C	Timestamp EVID9_S3 Parameter Set C active		DT
TS_PARSET_D	Timestamp EVID9_S4 Parameter Set D active		DT
TS_ALM_sp1	Timestamp EVID9_S5 Spare Alarm 1		DT
TS_ALM_sp2	Timestamp EVID9_S6 Spare Alarm 2		DT
TS_ALM_sp3	Timestamp EVID9_S7 Spare Alarm 3		DT
TS_ALM_sp4	Timestamp EVID9_S8 Spare Alarm 4		DT
ALM_sp5	Spare Alarm 5		STRUCT
ALM_sp5.Value		0	BOOL
ALM_sp5.ST		16#80	BYTE
ALM_sp6	Spare Alarm 6		STRUCT
ALM_sp6.Value		0	BOOL
ALM_sp6.ST		16#80	BYTE
ALM_sp7	Spare Alarm 7		STRUCT

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ALM_sp7.Value		0	BOOL
ALM_sp7.ST		16#80	BYTE
ALM_sp8	Spare Alarm 8		STRUCT
ALM_sp8.Value		0	BOOL
ALM_sp8.ST		16#80	BYTE
ALM_sp9	Spare Alarm 9		STRUCT
ALM_sp9.Value		0	BOOL
ALM_sp9.ST		16#80	BYTE
ALM_sp10	Spare Alarm 10		STRUCT
ALM_sp10.Value		0	BOOL
ALM_sp10.ST		16#80	BYTE
ALM_sp11	Spare Alarm 11		STRUCT
ALM_sp11.Value		0	BOOL
ALM_sp11.ST		16#80	BYTE
ALM_sp12	Spare Alarm 12		STRUCT
ALM_sp12.Value		0	BOOL
ALM_sp12.ST		16#80	BYTE
TS_ALM_sp5	Timestamp EVID10_S1 Spare Alarm 5		DT
TS_ALM_sp6	Timestamp EVID10_S2 Spare Alarm 6		DT
TS_ALM_sp7	Timestamp EVID10_S3 Spare Alarm 7		DT
TS_ALM_sp8	Timestamp EVID10_S4 Spare Alarm 8		DT
TS_ALM_sp9	Timestamp EVID10_S5 Spare Alarm 9		DT
TS_ALM_sp10	Timestamp EVID10_S6 Spare Alarm 10		DT
TS_ALM_sp11	Timestamp EVID10_S7 Spare Alarm 11		DT
TS_ALM_sp12	Timestamp EVID10_S8 Spare Alarm 12		DT
D_1A	Spare Double Point 1 A		STRUCT
D_1A.Value		0	BOOL
D_1A.ST		16#80	BYTE
D_1B	Spare Double Point 1 B		STRUCT
D_1B.Value		0	BOOL
D_1B.ST		16#80	BYTE
D_2A	Spare Double Point 2 A		STRUCT
D_2A.Value		0	BOOL
D_2A.ST		16#80	BYTE
D_2B	Spare Double Point 2 B		STRUCT
D_2B.Value		0	BOOL
D_2B.ST		16#80	BYTE
D_3A	Spare Double Point 3 A		STRUCT
D_3A.Value		0	BOOL
D_3A.ST		16#80	BYTE
D_3B	Spare Double Point 3 B		STRUCT
D_3B.Value		0	BOOL
D_3B.ST		16#80	BYTE
D_4A	Spare Double Point 4 A		STRUCT

D_4A.Value		0	BOOL
D_4A.ST		16#80	BYTE
D_4B	Spare Double Point 4 B		STRUCT
D_4B.Value		0	BOOL
D_4B.ST		16#80	BYTE
D_5A	Spare Double Point 5 A		STRUCT
D_5A.Value		0	BOOL
D_5A.ST		16#80	BYTE
D_5B	Spare Double Point 5 B		STRUCT
D_5B.Value		0	BOOL
D_5B.ST		16#80	BYTE
D_6A	Spare Double Point 6 A		STRUCT
D_6A.Value		0	BOOL
D_6A.ST		16#80	BYTE
D_6B	Spare Double Point 6 B		STRUCT
D_6B.Value		0	BOOL
D_6B.ST		16#80	BYTE
D_7A	Spare Double Point 7 A		STRUCT
D_7A.Value		0	BOOL
D_7A.ST		16#80	BYTE
D_7B	Spare Double Point 7 B		STRUCT
D_7B.Value		0	BOOL
D_7B.ST		16#80	BYTE
D_8A	Spare Double Point 8 A		STRUCT
D_8A.Value		0	BOOL
D_8A.ST		16#80	BYTE
D_8B	Spare Double Point 8 B		STRUCT
D_8B.Value		0	BOOL
D_8B.ST		16#80	BYTE
P1_U2_DOWN	Point 1 Decrease U2		STRUCT
P1_U2_DOWN.Value		1	BOOL
P1_U2_DOWN.ST		16#80	BYTE
P2_U2_DOWN	Point 2 Decrease U2		STRUCT
P2_U2_DOWN.Value		0	BOOL
P2_U2_DOWN.ST		16#80	BYTE
P3_U2_DOWN	Point 3 Decrease U2		STRUCT
P3_U2_DOWN.Value		0	BOOL
P3_U2_DOWN.ST		16#80	BYTE
P4_U2_DOWN	Point 4 Decrease U2		STRUCT
P4_U2_DOWN.Value		0	BOOL
P4_U2_DOWN.ST		16#80	BYTE
P5_U2_DOWN	Point 5 Decrease U2		STRUCT
P5_U2_DOWN.Value		0	BOOL
P5_U2_DOWN.ST		16#80	BYTE

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P6_U2_DOWN	Point 6 Decrease U2		STRUCT
P6_U2_DOWN.Value		0	BOOL
P6_U2_DOWN.ST		16#80	BYTE
P7_U2_DOWN	Point 7 Decrease U2		STRUCT
P7_U2_DOWN.Value		0	BOOL
P7_U2_DOWN.ST		16#80	BYTE
P8_U2_DOWN	Point 8 Decrease U2		STRUCT
P8_U2_DOWN.Value		0	BOOL
P8_U2_DOWN.ST		16#80	BYTE
P1_U2_UP	Point 1 Increase U2		STRUCT
P1_U2_UP.Value		0	BOOL
P1_U2_UP.ST		16#80	BYTE
P2_U2_UP	Point 2 Increase U2		STRUCT
P2_U2_UP.Value		0	BOOL
P2_U2_UP.ST		16#80	BYTE
P3_U2_UP	Point 3 Increase U2		STRUCT
P3_U2_UP.Value		0	BOOL
P3_U2_UP.ST		16#80	BYTE
P4_U2_UP	Point 4 Increase U2		STRUCT
P4_U2_UP.Value		0	BOOL
P4_U2_UP.ST		16#80	BYTE
P5_U2_UP	Point 5 Increase U2		STRUCT
P5_U2_UP.Value		0	BOOL
P5_U2_UP.ST		16#80	BYTE
P6_U2_UP	Point 6 Increase U2		STRUCT
P6_U2_UP.Value		0	BOOL
P6_U2_UP.ST		16#80	BYTE
P7_U2_UP	Point 7 Increase U2		STRUCT
P7_U2_UP.Value		0	BOOL
P7_U2_UP.ST		16#80	BYTE
P8_U2_UP	Point 8 Increase U2		STRUCT
P8_U2_UP.Value		0	BOOL
P8_U2_UP.ST		16#80	BYTE
P1_F2_DOWN	Point 1 Decrease F2		STRUCT
P1_F2_DOWN.Value		0	BOOL
P1_F2_DOWN.ST		16#80	BYTE
P2_F2_DOWN	Point 2 Decrease F2		STRUCT
P2_F2_DOWN.Value		0	BOOL
P2_F2_DOWN.ST		16#80	BYTE
P3_F2_DOWN	Point 3 Decrease F2		STRUCT
P3_F2_DOWN.Value		0	BOOL
P3_F2_DOWN.ST		16#80	BYTE
P4_F2_DOWN	Point 4 Decrease F2		STRUCT
P4_F2_DOWN.Value		0	BOOL

P4_F2_DOWN.ST		16#80	BYTE
P5_F2_DOWN	Point 5 Decrease F2		STRUCT
P5_F2_DOWN.Value		0	BOOL
P5_F2_DOWN.ST		16#80	BYTE
P6_F2_DOWN	Point 6 Decrease F2		STRUCT
P6_F2_DOWN.Value		0	BOOL
P6_F2_DOWN.ST		16#80	BYTE
P7_F2_DOWN	Point 7 Decrease F2		STRUCT
P7_F2_DOWN.Value		0	BOOL
P7_F2_DOWN.ST		16#80	BYTE
P8_F2_DOWN	Point 8 Decrease F2		STRUCT
P8_F2_DOWN.Value		0	BOOL
P8_F2_DOWN.ST		16#80	BYTE
P1_F2_UP	Point 1 Increase F2		STRUCT
P1_F2_UP.Value		0	BOOL
P1_F2_UP.ST		16#80	BYTE
P2_F2_UP	Point 2 Increase F2		STRUCT
P2_F2_UP.Value		0	BOOL
P2_F2_UP.ST		16#80	BYTE
P3_F2_UP	Point 3 Increase F2		STRUCT
P3_F2_UP.Value		0	BOOL
P3_F2_UP.ST		16#80	BYTE
P4_F2_UP	Point 4 Increase F2		STRUCT
P4_F2_UP.Value		0	BOOL
P4_F2_UP.ST		16#80	BYTE
P5_F2_UP	Point 5 Increase F2		STRUCT
P5_F2_UP.Value		0	BOOL
P5_F2_UP.ST		16#80	BYTE
P6_F2_UP	Point 6 Increase F2		STRUCT
P6_F2_UP.Value		0	BOOL
P6_F2_UP.ST		16#80	BYTE
P7_F2_UP	Point 7 Increase F2		STRUCT
P7_F2_UP.Value		0	BOOL
P7_F2_UP.ST		16#80	BYTE
P8_F2_UP	Point 8 Increase F2		STRUCT
P8_F2_UP.Value		0	BOOL
P8_F2_UP.ST		16#80	BYTE
BOOL_sp1	Spare		STRUCT
BOOL_sp1.Value		0	BOOL
BOOL_sp1.ST		16#80	BYTE
BOOL_sp2	Spare		STRUCT
BOOL_sp2.Value		0	BOOL
BOOL_sp2.ST		16#80	BYTE
BOOL_sp3	Spare		STRUCT

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BOOL_sp3.Value		0	BOOL
BOOL_sp3.ST		16#80	BYTE
BOOL_sp4	Spare		STRUCT
BOOL_sp4.Value		0	BOOL
BOOL_sp4.ST		16#80	BYTE
BOOL_sp5	Spare		STRUCT
BOOL_sp5.Value		0	BOOL
BOOL_sp5.ST		16#80	BYTE
BOOL_sp6	Spare		STRUCT
BOOL_sp6.Value		0	BOOL
BOOL_sp6.ST		16#80	BYTE
BOOL_sp7	Spare		STRUCT
BOOL_sp7.Value		0	BOOL
BOOL_sp7.ST		16#80	BYTE
BOOL_sp8	Spare		STRUCT
BOOL_sp8.Value		0	BOOL
BOOL_sp8.ST		16#80	BYTE
BOOL_sp9	Spare		STRUCT
BOOL_sp9.Value		0	BOOL
BOOL_sp9.ST		16#80	BYTE
BOOL_sp10	Spare		STRUCT
BOOL_sp10.Value		0	BOOL
BOOL_sp10.ST		16#80	BYTE
BOOL_sp11	Spare		STRUCT
BOOL_sp11.Value		0	BOOL
BOOL_sp11.ST		16#80	BYTE
BOOL_sp12	Spare		STRUCT
BOOL_sp12.Value		0	BOOL
BOOL_sp12.ST		16#80	BYTE
BOOL_sp13	Spare		STRUCT
BOOL_sp13.Value		0	BOOL
BOOL_sp13.ST		16#80	BYTE
BOOL_sp14	Spare		STRUCT
BOOL_sp14.Value		0	BOOL
BOOL_sp14.ST		16#80	BYTE
BOOL_sp15	Spare		STRUCT
BOOL_sp15.Value		0	BOOL
BOOL_sp15.ST		16#80	BYTE
BOOL_sp16	Spare		STRUCT
BOOL_sp16.Value		0	BOOL
BOOL_sp16.ST		16#80	BYTE
BOOL_sp17	Spare		STRUCT
BOOL_sp17.Value		0	BOOL
BOOL_sp17.ST		16#80	BYTE

BOOL_sp18	Spare		STRUCT
BOOL_sp18.Value		0	BOOL
BOOL_sp18.ST		16#80	BYTE

13.9.2.2 Outputs

I/O name	Description	Value	Data type
MSG_E_S_1	ALARM_8P Error	0	BOOL
MSG_U_S_1	1=Message Suppression active	0	BOOL
MSG_S_S_1	ALARM_8P: STATUS output	16#0000	WORD
MSG_A_S_1	ALARM_8P: ACK_STATE output	16#0000	WORD
MSG_E_S_2	ALARM_8P Error	0	BOOL
MSG_U_S_2	1=Message Suppression active	0	BOOL
MSG_S_S_2	ALARM_8P: STATUS output	16#0000	WORD
MSG_A_S_2	ALARM_8P: ACK_STATE output	16#0000	WORD
MSG_E_S_3	ALARM_8P Error	0	BOOL
MSG_U_S_3	1=Message Suppression active	0	BOOL
MSG_S_S_3	ALARM_8P: STATUS output	16#0000	WORD
MSG_A_S_3	ALARM_8P: ACK_STATE output	16#0000	WORD
MSG_E_S_4	ALARM_8P Error	0	BOOL
MSG_U_S_4	1=Message Suppression active	0	BOOL
MSG_S_S_4	ALARM_8P: STATUS output	16#0000	WORD
MSG_A_S_4	ALARM_8P: ACK_STATE output	16#0000	WORD
MSG_E_S_5	ALARM_8P Error	0	BOOL
MSG_U_S_5	1=Message Suppression active	0	BOOL
MSG_S_S_5	ALARM_8P: STATUS output	16#0000	WORD
MSG_A_S_5	ALARM_8P: ACK_STATE output	16#0000	WORD
MSG_E_S_6	ALARM_8P Error	0	BOOL
MSG_U_S_6	1=Message Suppression active	0	BOOL
MSG_S_S_6	ALARM_8P: STATUS output	16#0000	WORD
MSG_A_S_6	ALARM_8P: ACK_STATE output	16#0000	WORD
MSG_E_S_7	ALARM_8P Error	0	BOOL
MSG_U_S_7	1=Message Suppression active	0	BOOL
MSG_S_S_7	ALARM_8P: STATUS output	16#0000	WORD
MSG_A_S_7	ALARM_8P: ACK_STATE output	16#0000	WORD
MSG_E_S_8	ALARM_8P Error	0	BOOL
MSG_U_S_8	1=Message Suppression active	0	BOOL
MSG_S_S_8	ALARM_8P: STATUS output	16#0000	WORD
MSG_A_S_8	ALARM_8P: ACK_STATE output	16#0000	WORD

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MSG_E_S_9	ALARM_8P Error	0	BOOL
MSG_U_S_9	1=Message Suppression active	0	BOOL
MSG_S_S_9	ALARM_8P: STATUS output	16#0000	WORD
MSG_A_S_9	ALARM_8P: ACK_STATE output	16#0000	WORD
MSG_E_S_10	ALARM_8P Error	0	BOOL
MSG_U_S_10	1=Message Suppression active	0	BOOL
MSG_S_S_10	ALARM_8P: STATUS output	16#0000	WORD
MSG_A_S_10	ALARM_8P: ACK_STATE output	16#0000	WORD
MSG_E_S_11	ALARM_8P Error	0	BOOL
MSG_U_S_11	1=Message Suppression active	0	BOOL
MSG_S_S_11	ALARM_8P: STATUS output	16#0000	WORD
MSG_A_S_11	ALARM_8P: ACK_STATE output	16#0000	WORD
MSG_E_S_12	ALARM_8P Error	0	BOOL
MSG_U_S_12	1=Message Suppression active	0	BOOL
MSG_S_S_12	ALARM_8P: STATUS output	16#0000	WORD
MSG_A_S_12	ALARM_8P: ACK_STATE output	16#0000	WORD
MSG_E_D_1	ALARM_8P Error	0	BOOL
MSG_U_D_1	1=Message Suppression active	0	BOOL
MSG_S_D_1	ALARM_8P: STATUS output	16#0000	WORD
MSG_A_D_1	ALARM_8P: ACK_STATE output	16#0000	WORD
MSG_E_D_2	ALARM_8P Error	0	BOOL
MSG_U_D_2	1=Message Suppression active	0	BOOL
MSG_S_D_2	ALARM_8P: STATUS output	16#0000	WORD
MSG_A_D_2	ALARM_8P: ACK_STATE output	16#0000	WORD
Q_DT_NOW	Last Error	1990-01-01-0:00:00	DATE_AND_TIME
PosCmdSp1_op	Spare 1 Command open	0	BOOL
PosCmdSp1_cl	Spare 1 Command close	0	BOOL
PosCmdSp2_op	Spare 2 Command open	0	BOOL
PosCmdSp2_cl	Spare 2 Command close	0	BOOL
PosCmdSp3_op	Spare 3 Command open	0	BOOL
PosCmdSp3_cl	Spare 3 Command close	0	BOOL
PosCmdSp4_op	Spare 4 Command open	0	BOOL
PosCmdSp4_cl	Spare 4 Command close	0	BOOL
PosCmdSp5_op	Spare 5 Command open	0	BOOL
PosCmdSp5_cl	Spare 5 Command close	0	BOOL
PosCmdSp6_op	Spare 6 Command open	0	BOOL
PosCmdSp6_cl	Spare 6 Command close	0	BOOL
PosCmdSp7_op	Spare 7 Command open	0	BOOL
PosCmdSp7_cl	Spare 7 Command close	0	BOOL
PosCmdSp8_op	Spare 8 Command open	0	BOOL
PosCmdSp8_cl	Spare 8 Command close	0	BOOL
DT_LAST_MSG	Timestamp last error		DT
Q_DT_NOW	Timestamp date and time of PLC		DT

13.10 DC_LINE

13.10.1 Description of DC_LINE

The DataCollector block DC_LNE (FB 2564) forms an interface to the technological block PC_LINE.

13.10.2 I/O lists

13.10.2.1 Inputs

I/O name	Description	Value	Data type
Snd_Data	Connect to DC_/SC_LINE		STRUCT
SimOn	Simulation on/off	1	BOOL
ExtTSMsg	1 = external time stamps used for messages 0 = creation of internal time stamps	FALSE	BOOL
ExtTSPos	1 = external time stamps used for switch positions 0 = creation of internal time stamps	FALSE	BOOL
F	Frequency		STRUCT
F.Value		0.0	REAL
F.ST		16#80	BYTE
I_A	Current Phase A		STRUCT
I_A.Value		0.0	REAL
I_A.ST		16#80	BYTE
I_B	Current Phase B		STRUCT
I_B.Value		0.0	REAL
I_B.ST		16#80	BYTE
I_C	Current Phase C		STRUCT
I_C.Value		0.0	REAL
I_C.ST		16#80	BYTE
P	Power		STRUCT
P.Value		234.0	REAL
P.ST		16#80	BYTE

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PF	cos phi - Power Factor		STRUCT
PF.Value		0.0	REAL
PF.ST		16#80	BYTE
Q	Reactive power		STRUCT
Q.Value		0.0	REAL
Q.ST		16#80	BYTE
S	Apparent power		STRUCT
S.Value		0.0	REAL
S.ST		16#80	BYTE
U	Neutral voltage		STRUCT
U.Value		0.0	REAL
U.ST		16#80	BYTE
U_A	Voltage Phase A		STRUCT
U_A.Value		0.0	REAL
U_A.ST		16#80	BYTE
U_AB	Voltage A-B		STRUCT
U_AB.Value		0.0	REAL
U_AB.ST		16#80	BYTE
U_B	Voltage Phase B		STRUCT
U_B.Value		0.0	REAL
U_B.ST		16#80	BYTE
U_BC	Voltage B-C		STRUCT
U_BC.Value		0.0	REAL
U_BC.ST		16#80	BYTE
U_C	Voltage Phase C		STRUCT
U_C.Value		0.0	REAL
U_C.ST		16#80	BYTE
U_CA	Voltage C-A		STRUCT
U_CA.Value		0.0	REAL
U_CA.ST		16#80	BYTE
IDIFFL1	Idiff L1		STRUCT
IDIFFL1.Value		0.0	REAL
IDIFFL1.ST		16#80	BYTE
IDIFFL2	Idiff L2		STRUCT
IDIFFL2.Value		0.0	REAL
IDIFFL2.ST		16#80	BYTE
IDIFFL3	Idiff L3		STRUCT
IDIFFL3.Value		0.0	REAL
IDIFFL3.ST		16#80	BYTE
IDIFF3IO	diff 3IO		STRUCT
IDIFF3IO.Value		0.0	REAL
IDIFF3IO.ST		16#80	BYTE
IRESTL1	Irest L1		STRUCT
IRESTL1.Value		0.0	REAL

IRESTL1.ST		16#80	BYTE
IRESTL2	Irest L2		STRUCT
IRESTL2.Value		0.0	REAL
IRESTL2.ST		16#80	BYTE
IRESTL3	Irest L3		STRUCT
IRESTL3.Value		0.0	REAL
IRESTL3.ST		16#80	BYTE
I30	3I0		STRUCT
I30.Value		0.0	REAL
I30.ST		16#80	BYTE
REAL_sp1	Spare Real 1		STRUCT
REAL_sp1.Value		0.0	REAL
REAL_sp1.ST		16#80	BYTE
REAL_sp2	Spare Real 2		STRUCT
REAL_sp2.Value		0.0	REAL
REAL_sp2.ST		16#80	BYTE
REAL_sp3	Spare Real 3		STRUCT
REAL_sp3.Value		0.0	REAL
REAL_sp3.ST		16#80	BYTE
REAL_sp4	Spare Real 4		STRUCT
REAL_sp4.Value		0.0	REAL
REAL_sp4.ST		16#80	BYTE
REAL_sp5	Spare Real 5		STRUCT
REAL_sp5.Value		0.0	REAL
REAL_sp5.ST		16#80	BYTE
REAL_sp6	Spare Real 6		STRUCT
REAL_sp6.Value		0.0	REAL
REAL_sp6.ST		16#80	BYTE
REAL_sp7	Spare Real 7		STRUCT
REAL_sp7.Value		0.0	REAL
REAL_sp7.ST		16#80	BYTE
REAL_sp8	Spare Real 8		STRUCT
REAL_sp8.Value		0.0	REAL
REAL_sp8.ST		16#80	BYTE
Beh	Behavior		STRUCT
Beh.Value		0	INT
Beh.ST		16#80	BYTE
Dev_OK	Device OK		STRUCT
Dev_OK.Value		0	INT
Dev_OK.ST		16#80	BYTE
INT_sp1	Spare INT 1		STRUCT
INT_sp1.Value		0	INT
INT_sp1.ST		16#80	BYTE
INT_sp2	Spare INT 2		STRUCT

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INT_sp2.Value		0	INT
INT_sp2.ST		16#80	BYTE
INT_sp3	Spare INT 3		STRUCT
INT_sp3.Value		0	INT
INT_sp3.ST		16#80	BYTE
INT_sp4	Spare INT 4		STRUCT
INT_sp4.Value		0	INT
INT_sp4.ST		16#80	BYTE
INT_sp5	Spare INT 5		STRUCT
INT_sp5.Value		0	INT
INT_sp5.ST		16#80	BYTE
INT_sp6	Spare INT 6		STRUCT
INT_sp6.Value		0	INT
INT_sp6.ST		16#80	BYTE
SwCount			STRUCT
SwCount.Value		0	DINT
SwCount.ST		16#80	BYTE
OpTm			STRUCT
OpTm.Value		0	DINT
OpTm.ST		16#80	BYTE
DINT_sp1			STRUCT
DINT_sp1.Value		0	DINT
DINT_sp1.ST		16#80	BYTE
DINT_sp2			STRUCT
DINT_sp2.Value		0	DINT
DINT_sp2.ST		16#80	BYTE
DINT_sp3			STRUCT
DINT_sp3.Value		0	DINT
DINT_sp3.ST		16#80	BYTE
DINT_sp4			STRUCT
DINT_sp4.Value		0	DINT
DINT_sp4.ST		16#80	BYTE
DINT_sp5			STRUCT
DINT_sp5.Value		0	DINT
DINT_sp5.ST		16#80	BYTE
DINT_sp6			STRUCT
DINT_sp6.Value		0	DINT
DINT_sp6.ST		16#80	BYTE
Pos_CMD_sp1	Position Spare Command 1		STRUCT
Pos_CMD_sp1.Value		16#00000000 2	DWORD
Pos_CMD_sp1.ST		16#80	BYTE
Pos_CMD_sp2	Position Spare Command 2		STRUCT

Pos_CMD_sp2.Value		16#00000002	DWORD
Pos_CMD_sp2.ST		16#80	BYTE
Pos_CMD_sp3	Position Spare Command 3		STRUCT
Pos_CMD_sp3.Value		16#00000002	DWORD
Pos_CMD_sp3.ST		16#80	BYTE
Pos_CMD_sp4	Position Spare Command 4		STRUCT
Pos_CMD_sp4.Value		16#00000003	DWORD
Pos_CMD_sp4.ST		16#80	BYTE
TS_Pos_CMD_sp1	Timestamp Position Spare Command 1		DT
TS_Pos_CMD_sp2	Timestamp Position Spare Command 2		DT
TS_Pos_CMD_sp3	Timestamp Position Spare Command 3		DT
TS_Pos_CMD_sp4	Timestamp Position Spare Command 4		DT
AckCmdsp1	Acknowledge signal for Spare command 1		STRUCT
AckCmdsp1.Value		0	BOOL
AckCmdsp1.ST		16#80	BYTE
AckCmdsp2	Acknowledge signal for Spare command 2		STRUCT
AckCmdsp2.Value		0	BOOL
AckCmdsp2.ST		16#80	BYTE
AckCmdsp3	Acknowledge signal for Spare command 3		STRUCT
AckCmdsp3.Value		0	BOOL
AckCmdsp3.ST		16#80	BYTE
AckCmdsp4	Acknowledge signal for Spare command 4		STRUCT
AckCmdsp4.Value		0	BOOL
AckCmdsp4.ST		16#80	BYTE
TR_Relay	Relay PICKUP		STRUCT
TR_Relay.Value		0	BOOL
TR_Relay.ST		16#80	BYTE
PU_Relay	Relay General TRIP		STRUCT
PU_Relay.Value		0	BOOL
PU_Relay.ST		16#80	BYTE
PTOC1_Op	IDMT Phase Ip TRIP		STRUCT
PTOC1_Op.Value		0	BOOL
PTOC1_Op.ST		16#80	BYTE
PTOC1_Str	IDMT Phase Ip picked up		STRUCT
PTOC1_Str.Value		0	BOOL
PTOC1_Str.ST		16#80	BYTE
PTOC2_Op	DMT I > TRIP		STRUCT
PTOC2_Op.Value		0	BOOL

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PTOC2_Op.ST		16#80	BYTE
PTOC2_Str	DMT I > picked up		STRUCT
PTOC2_Str.Value		0	BOOL
PTOC2_Str.ST		16#80	BYTE
PTOC3_Op	DMT I >> TRIP		STRUCT
PTOC3_Op.Value		0	BOOL
PTOC3_Op.ST		16#80	BYTE
PTOC3_Str	DMT I >> picked up		STRUCT
PTOC3_Str.Value		0	BOOL
PTOC3_Str.ST		16#80	BYTE
TS_TR_Relay	Timestamp EVID1_S1 Relay PICKUP		DT
TS_PU_Relay	Timestamp EVID1_S2 Relay General TRIP		DT
TS_PTOC1_Op	Timestamp EVID1_S3 IDMT Phase Ip TRIP		DT
TS_PTOC1_Str	Timestamp EVID1_S4 IDMT Phase Ip picked up		DT
TS_PTOC2_Op	Timestamp EVID1_S5 DMT I > TRIP		DT
TS_PTOC2_Str	Timestamp EVID1_S6 DMT I > picked up		DT
TS_PTOC3_Op	Timestamp EVID1_S7 DMT I >> TRIP		DT
TS_PTOC3_Str	Timestamp EVID1_S8 DMT I >> picked up		DT
PDIS1_Op	Impedance protection Z1 TRIP		STRUCT
PDIS1_Op.Value		0	BOOL
PDIS1_Op.ST		16#80	BYTE
PDIS1_Str	Impedance protection Z1 picked up		STRUCT
PDIS1_Str.Value		0	BOOL
PDIS1_Str.ST		16#80	BYTE
PDIS2_Op	Impedance protection Z2 TRIP		STRUCT
PDIS2_Op.Value		0	BOOL
PDIS2_Op.ST		16#80	BYTE
PDIS2_Str	Impedance protection Z2 picked up		STRUCT
PDIS2_Str.Value		0	BOOL
PDIS2_Str.ST		16#80	BYTE
PDIS3_Op	Impedance protection Z1B TRIP		STRUCT
PDIS3_Op.Value		0	BOOL
PDIS3_Op.ST		16#80	BYTE
PDIS3_Str	Impedance protection Z1B picked up		STRUCT
PDIS3_Str.Value		0	BOOL
PDIS3_Str.ST		16#80	BYTE
PDIS4_Op	Impedance protection general TRIP		STRUCT
PDIS4_Op.Value		0	BOOL
PDIS4_Op.ST		16#80	BYTE

PDIS4_Str	Impedance protection general picked up		STRUCT
PDIS4_Str.Value		0	BOOL
PDIS4_Str.ST		16#80	BYTE
TS_PDIS1_Op	Timestamp EVID2_S1 Impedance protection Z1 TRIP		DT
TS_PDIS1_Str	Timestamp EVID2_S2 Impedance protection Z1 picked up		DT
TS_PDIS2_Op	Timestamp EVID2_S3 Impedance protection Z2 TRIP		DT
TS_PDIS2_Str	Timestamp EVID2_S4 Impedance protection Z2 picked up		DT
TS_PDIS3_Op	Timestamp EVID2_S5 Impedance protection Z1B TRIP		DT
TS_PDIS3_Str	Timestamp EVID2_S6 Impedance protection Z1B picked up		DT
TS_PDIS4_Op	Timestamp EVID2_S7 Impedance protection general TRIP		DT
TS_PDIS4_Str	Timestamp EVID2_S8 Impedance protection general picked up		DT
PDIF1_Op	Differential protection IDIFF> TRIP		STRUCT
PDIF1_Op.Value		0	BOOL
PDIF1_Op.ST		16#80	BYTE
PDIF1_Str	Differential protection IDIFF> picked up		STRUCT
PDIF1_Str.Value		0	BOOL
PDIF1_Str.ST		16#80	BYTE
PDIF2_Op	Differential protection IDIFF>> TRIP		STRUCT
PDIF2_Op.Value		0	BOOL
PDIF2_Op.ST		16#80	BYTE
PDIF2_Str	Differential protection IDIFF>> picked up		STRUCT
PDIF2_Str.Value		0	BOOL
PDIF2_Str.ST		16#80	BYTE
RBRF1_Str	Breaker failure picked up		STRUCT
RBRF1_Str.Value		0	BOOL
RBRF1_Str.ST		16#80	BYTE
RBRF1_OpEx	Breaker failure TRIP T1		STRUCT
RBRF1_OpEx.Value		0	BOOL
RBRF1_OpEx.ST		16#80	BYTE
RBRF1_OpIn	Breaker failure TRIP T2		STRUCT
RBRF1_OpIn.Value		0	BOOL
RBRF1_OpIn.ST		16#80	BYTE
Loc	Local operation		STRUCT
Loc.Value		0	BOOL
Loc.ST		16#80	BYTE

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TS_PDIF1_Op	Timestamp EVID3_S1 Differential protection IDIFF> TRIP		DT
TS_PDIF1_Str	Timestamp EVID3_S2 Differential protection IDIFF> picked up		DT
TS_PDIF2_Op	Timestamp EVID3_S3 Differential protection IDIFF>> TRIP		DT
TS_PDIF2_Str	Timestamp EVID3_S4 Differential protection IDIFF>> picked up		DT
TS_RBRF1_Str	Timestamp EVID3_S5 Breaker failure picked up		DT
TS_RBRF1_OpEx	Timestamp EVID3_S6 Breaker failure TRIP T1		DT
TS_RBRF1_OpIn	Timestamp EVID3_S7 Breaker failure TRIP T2		DT
TS_Loc	Timestamp EVID3_S8 Local Operation		DT
Error	Error Summary Alarm		STRUCT
Error.Value		0	BOOL
Error.ST		16#80	BYTE
Warning	Alarm Summary Event		STRUCT
Warning.Value		0	BOOL
Warning.ST		16#80	BYTE
PARSET_A	Parameter Set A active		STRUCT
PARSET_A.Value		0	BOOL
PARSET_A.ST		16#80	BYTE
PARSET_B	Parameter Set B active		STRUCT
PARSET_B.Value		0	BOOL
PARSET_B.ST		16#80	BYTE
PARSET_C	Parameter Set C active		STRUCT
PARSET_C.Value		0	BOOL
PARSET_C.ST		16#80	BYTE
PARSET_D	Parameter Set D active		STRUCT
PARSET_D.Value		0	BOOL
PARSET_D.ST		16#80	BYTE
ALM_sp1	Spare		STRUCT
ALM_sp1.Value		0	BOOL
ALM_sp1.ST		16#80	BYTE
ALM_sp2	Spare		STRUCT
ALM_sp2.Value		0	BOOL
ALM_sp2.ST		16#80	BYTE
TS_Error	Timestamp EVID4_S1 Error Summary Alarm		DT
TS_Warning	Timestamp EVID4_S2 Alarm Summary Event		DT
TS_PARSET_A	Timestamp EVID4_S3 Parameter Set A active		DT

TS_PARSET_B	Timestamp EVID4_S4 Parameter Set B active		DT
TS_PARSET_C	Timestamp EVID4_S5 Parameter Set C active		DT
TS_PARSET_D	Timestamp EVID4_S6 Parameter Set D active		DT
TS_ALM_sp1	Timestamp EVID4_S7 Spare		DT
TS_ALM_sp2	Timestamp EVID4_S8 Spare		DT
ALM_sp3	Spare		STRUCT
ALM_sp3.Value		0	BOOL
ALM_sp3.ST		16#80	BYTE
ALM_sp4	Spare		STRUCT
ALM_sp4.Value		0	BOOL
ALM_sp4.ST		16#80	BYTE
ALM_sp5	Spare		STRUCT
ALM_sp5.Value		0	BOOL
ALM_sp5.ST		16#80	BYTE
ALM_sp6	Spare		STRUCT
ALM_sp6.Value		0	BOOL
ALM_sp6.ST		16#80	BYTE
ALM_sp7	Spare		STRUCT
ALM_sp7.Value		0	BOOL
ALM_sp7.ST		16#80	BYTE
ALM_sp8	Spare		STRUCT
ALM_sp8.Value		0	BOOL
ALM_sp8.ST		16#80	BYTE
ALM_sp9	Spare		STRUCT
ALM_sp9.Value		0	BOOL
ALM_sp9.ST		16#80	BYTE
ALM_sp10	Spare		STRUCT
ALM_sp10.Value		0	BOOL
ALM_sp10.ST		16#80	BYTE
TS_ALM_sp3	Timestamp EVID5_S1 Spare		DT
TS_ALM_sp4	Timestamp EVID5_S2 Spare		DT
TS_ALM_sp5	Timestamp EVID5_S3 Spare		DT
TS_ALM_sp6	Timestamp EVID5_S4 Spare		DT
TS_ALM_sp7	Timestamp EVID5_S5 Spare		DT
TS_ALM_sp8	Timestamp EVID5_S6 Spare		DT
TS_ALM_sp9	Timestamp EVID5_S7 Spare		DT
TS_ALM_sp10	Timestamp EVID5_S8 Spare		DT
BOOL_sp1	Spare 1		STRUCT
BOOL_sp1.Value		0	BOOL
BOOL_sp1.ST		16#80	BYTE
BOOL_sp2	Spare 2		STRUCT

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BOOL_sp2.Value		0	BOOL
BOOL_sp2.ST		16#80	BYTE
BOOL_sp3	Spare 3		STRUCT
BOOL_sp3.Value		0	BOOL
BOOL_sp3.ST		16#80	BYTE
BOOL_sp4	Spare 4		STRUCT
BOOL_sp4.Value		0	BOOL
BOOL_sp4.ST		16#80	BYTE
BOOL_sp5	Spare 5		STRUCT
BOOL_sp5.Value		0	BOOL
BOOL_sp5.ST		16#80	BYTE
BOOL_sp6	Spare 6		STRUCT
BOOL_sp6.Value		0	BOOL
BOOL_sp6.ST		16#80	BYTE
BOOL_sp7	Spare 7		STRUCT
BOOL_sp7.Value		0	BOOL
BOOL_sp7.ST		16#80	BYTE
BOOL_sp8	Spare 8		STRUCT
BOOL_sp8.Value		0	BOOL
BOOL_sp8.ST		16#80	BYTE
BOOL_sp9	Spare 9		STRUCT
BOOL_sp9.Value		0	BOOL
BOOL_sp9.ST		16#80	BYTE
BOOL_sp10	Spare 10		STRUCT
BOOL_sp10.Value		0	BOOL
BOOL_sp10.ST		16#80	BYTE

13.10.2.2 Outputs

I/O name	Description	Value	Data type
RCV_Data	Connect to DC_/SC_LINE		STRUCT
MSG_E_S_1	ALARM_8P Error	0	BOOL
MSG_U_S_1	1=Message Suppression active	0	BOOL
MSG_S_S_1	ALARM_8P: STATUS output	16#0000	WORD
MSG_A_S_1	ALARM_8P: ACK_STATE output	16#0000	WORD
MSG_E_S_2	ALARM_8P Error	0	BOOL
MSG_U_S_2	1=Message Suppression active	0	BOOL
MSG_S_S_2	ALARM_8P: STATUS output	16#0000	WORD
MSG_A_S_2	ALARM_8P: ACK_STATE output	16#0000	WORD

MSG_E_S_3	ALARM_8P Error	0	BOOL
MSG_U_S_3	1=Message Suppression active	0	BOOL
MSG_S_S_3	ALARM_8P: STATUS output	16#0000	WORD
MSG_A_S_3	ALARM_8P: ACK_STATE output	16#0000	WORD
MSG_E_S_4	ALARM_8P Error	0	BOOL
MSG_U_S_4	1=Message Suppression active	0	BOOL
MSG_S_S_4	ALARM_8P: STATUS output	16#0000	WORD
MSG_A_S_4	ALARM_8P: ACK_STATE output	16#0000	WORD
MSG_E_S_5	ALARM_8P Error	0	BOOL
MSG_U_S_5	1=Message Suppression active	0	BOOL
MSG_S_S_5	ALARM_8P: STATUS output	16#0000	WORD
MSG_A_S_5	ALARM_8P: ACK_STATE output	16#0000	WORD
MSG_E_D_1	ALARM_8P Error	0	BOOL
MSG_U_D_1	1=Message Suppression active	0	BOOL
MSG_S_D_1	ALARM_8P: STATUS output	16#0000	WORD
MSG_A_D_1	ALARM_8P: ACK_STATE output	16#0000	WORD
Q_DT_NOW	Last Error	1990-01-01-0:00:00	DATE_AND_TIME
PosCmdSp1_op	Spare 1 Command open	0	BOOL
PosCmdSp1_cl	Spare 1 Command close	0	BOOL
PosCmdSp2_op	Spare 2 Command open	0	BOOL
PosCmdSp2_cl	Spare 2 Command close	0	BOOL
PosCmdSp3_op	Spare 3 Command open	0	BOOL
PosCmdSp3_cl	Spare 3 Command close	0	BOOL
PosCmdSp4_op	Spare 4 Command open	0	BOOL
PosCmdSp4_cl	Spare 4 Command close	0	BOOL
DT_LAST_MSG	Timestamp last error		DT
Q_DT_NOW	Timestamp date and time of PLC		DT

14.1 Station Gateway

Basic information about engineering the Station Gateway communication is available in the Station Gateway manual.

Proceed as follows to insert a technological object:

14.1.1 Inserting the CFC blocks SG_FEEDER and PC_FEEDER

For PC_FEEDER insert an SG_FEEDER and a PC_FEEDER block and interconnect them through the inputs / outputs Snd_Data and Rcv_Data:

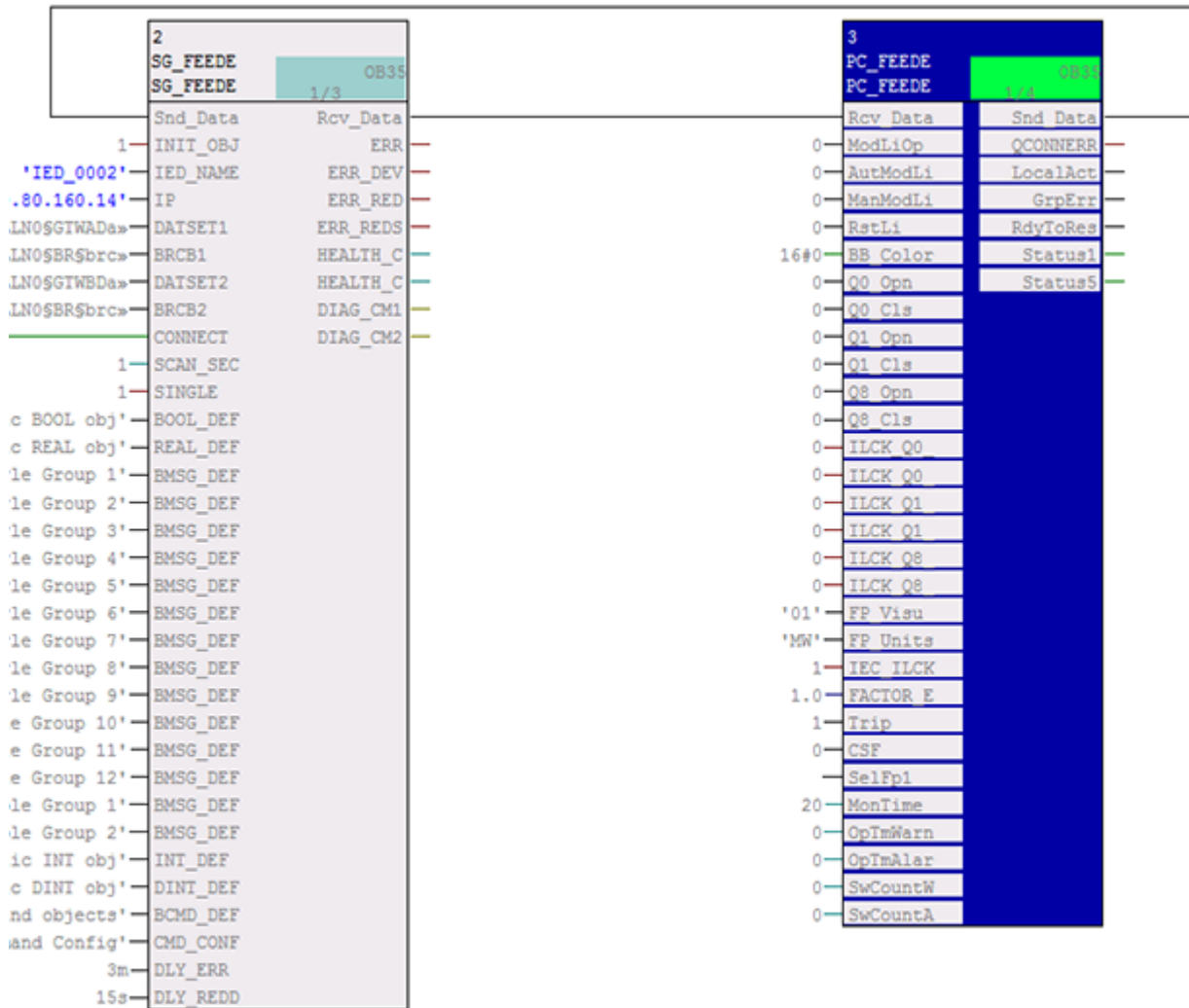


Figure 14-1 SG_FEEDER and PC_FEEDER blocks

14.1.2 Configuring the SG_FEEDER block

The following inputs of the SG_XX block have to be customized:

Name of the input	Description	Example
IED_NAME	IED name of the device	'IED_0002'
IP	IP address of the device	'140.80.160.14'
CONNECT	Structure for connecting with the I61_LINK block	
SINGLE	1 = only one CM, 0 = redundant pair	1

14.1.3 Configuring PC_FEEDER

The following inputs of the PC_XX block have to be customized:

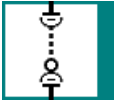
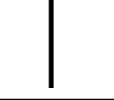
Name of the input	Description	Example
QX_Opn/QX_Cls	Open / Close command	Connect with generator / rising edge
ILCK_QX_Opn/ILCK_QX_Cls	Interlocks	
FP_Visu#Ilck_Dis	Interlocks are used in the faceplate -> the commands are locked in the faceplate	
IEC_ILCK_EN	Interlock signals from IEC protocol are used (for example, CTRL/Q0CILO1\$ST\$EnacIs\$stVal)	
TRIP	Command for external tripping, sends a command to open to the switch Q0	
OpTmWarnLimit	Warning limit for operating hours in hours. 0 = no message generated	
OpTmAlarmLimit	Alarm limit for operating hours in hours. 0 = no message generated	
SwCountWarnLimit	Warning limit for operating cycles. 0 = no message generated	
SwCountAlarmLimit	Alarm limit for operating cycles. 0 = no message generated	

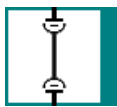
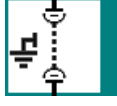
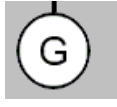
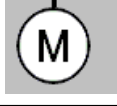
14.1 Station Gateway

Appendix

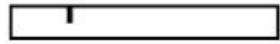
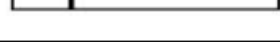
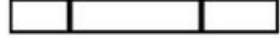
A.1 Symbol codes

A.1.1 PC_FEEDER symbol codes

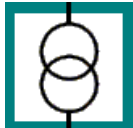
Symbol	Code	Description
	00	Empty field
	01	Circuit breaker with disconnect and truck (2 switches!)
	02	Circuit breaker
	03	Disconnect and grounding switch (2 switches!)
	04	Disconnect
	05	Simple connection
	06	Disconnect
	07	Disconnect
	08	Grounding switch
	09	

Symbol	Code	Description
	10	Disconnecter
	11	Grounding switch
	12	Disconnecter with truck and grounding switch (2 switches!)
	14	Generator
	15	Motor

A.1.2 Symbol codes for connection fields of the PC_FEEDER faceplate

Symbol	Code	Description
	00	Blank
	01	1 connection under field 5 (for field 10)
	02	Connection between 2 rows (for fields 8, 9, 10)
	03	Connection via field 1 and 2 (for field 7)
	04	Connection between 2 rows (for fields 8, 9)
	05	Connection via field 1 (for field 7)
	06	Connection via field 1 (for field 7)
	07	Connection (for fields 8, 9, 10)
	08	Connection between 2 rows (for fields 8, 9)
	09	Connection between 2 rows (for fields 8, 9)
	10	Connection via field 1 and 2 (for field 7)

A.1.3 Symbol codes for PC_TRAFO symbol and faceplate

Symbol	Code	Description
	01	Transformer with two windings
	02	Transformer with two windings and on-load tap changer on primary side
	03	Transformer with two windings and on-load tap changer on secondary side
	04	Transformer with three windings
	05	Transformer with three windings and on-load tap changer on primary side
	06	Transformer with three windings and on-load tap changer on secondary side
	07	Power transducer
	08	Transformer with three windings and three connectors

A.1.4 Symbol codes PC_BUSBAR

Symbol	Code	Description
	00	No grounding
	01	Low-resistance grounding
	02	Grounding switch

A.2 References

- [1] PCS 7 Advanced Process Library 9.0 Function Manual
- [3] APL Online Help
- [4] Station Gateway documentation
- [5] Station Controller documentation